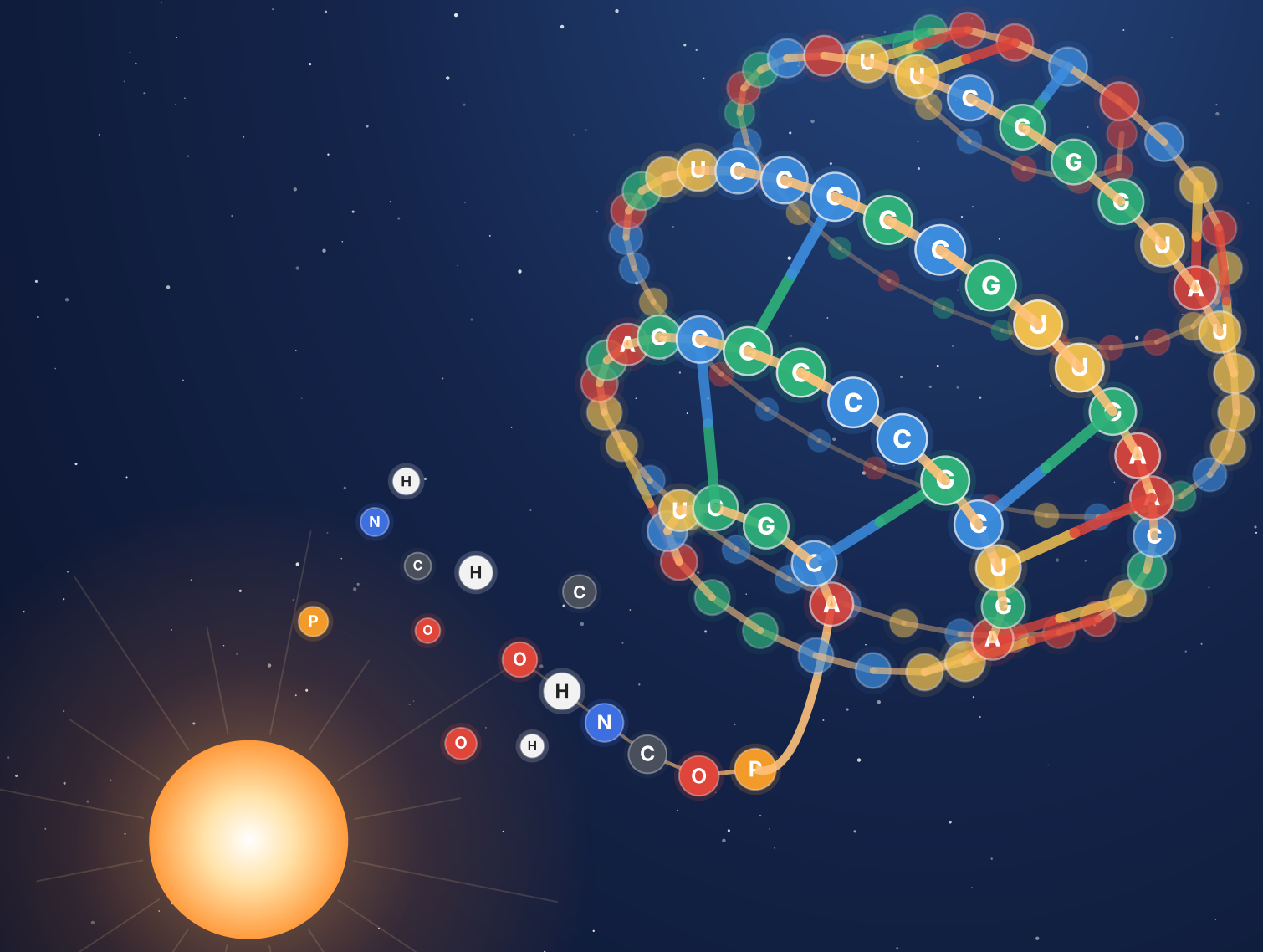
From Atoms to RNA

An Introduction to the Chemical Origins of Life



GOKYONA WAHIRO

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To my wife, Julie,
and to our three children

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Preface

This book follows a single chemical path: from the atoms made in stars to RNA, the first molecule known to be able both to carry genetic information and to act as a catalyst.

No prior knowledge of chemistry or biology is assumed. Every technical term is defined precisely on first use and collected in the Glossary. Essential equations are included, and every symbol in them is explained.

Throughout the book, statements are distinguished by their status:

- **Established**: supported by reproducible experiments or observations.
- **Hypothesis**: a mainstream explanation that is consistent with the evidence but not yet demonstrated.
- **Open question**: a problem for which no satisfactory answer currently exists.

Each chapter begins with learning objectives and ends with a summary and exercises. The appendices give a glossary of terms, units and constants, and advice on reading the research literature.

1. Where Atoms Come From

Learning objectives

After reading this chapter you should be able to:

1. Describe the structure of an atom in terms of protons, neutrons and electrons, and define atomic number, mass number and isotope.
2. Explain how hydrogen and helium formed in the first minutes after the Big Bang, and how heavier elements, including carbon, nitrogen, oxygen and phosphorus, are produced in stars.
3. Compare the abundance of the elements in the Universe with their abundance in RNA, and explain why phosphorus is the scarcest of the five elements RNA requires.
4. Outline how material made in stars became part of the solar system, and state the age of the solar system and the time scale of the Earth's formation.

1.1. Matter, elements and atoms

1.1.1. The question of this book

Every molecule of RNA in every living cell is built from atoms of only five elements: hydrogen (H), carbon (C), nitrogen (N), oxygen (O) and phosphorus (P) (Figure 1.1). This book follows those five elements from their origin to the point at which they are joined into a molecule that can direct the synthesis of copies of itself. The first step is to define precisely what an element and an atom are.

1.1.2. The atom and its three particles

An **atom** is the smallest unit of a chemical element. It consists of a small, dense, positively charged **nucleus** surrounded by a much larger region occupied by **electrons** (Figure 1.2).

The nucleus is made of two kinds of particle. A **proton** carries one positive elementary charge, +e, where $e = 1.602\,176\,634 \times 10^{-19}$ coulombs, a value that is exact by definition in the International System of Units (National Institute of Standards and Technology,

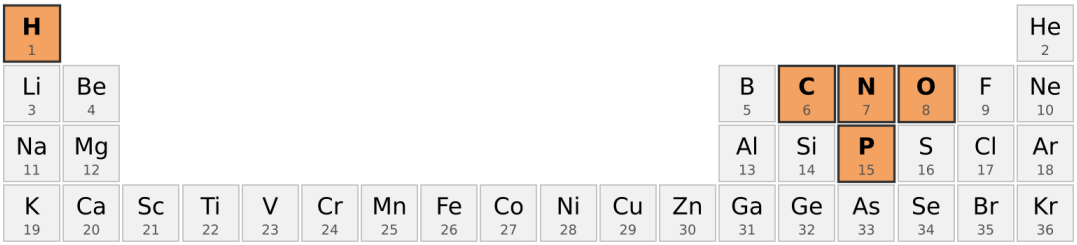


Figure 1.1.: The first 36 elements of the periodic table, arranged by atomic number. The five elements that make up the canonical RNA molecule are highlighted.

n.d.a). A **neutron** carries no charge and has almost the same mass as a proton: 1.6726 $\times$ 10^{-27} kg for the proton and 1.6749 $\times$ 10^{-27} kg for the neutron (National Institute of Standards and Technology, n.d.a). Protons and neutrons are known collectively as **nucleons**. An **electron** carries one negative elementary charge, $-$e, and has a mass about 1/1836 that of a proton (National Institute of Standards and Technology, n.d.a). Consequently more than 99.9 percent of the mass of an atom resides in its nucleus.

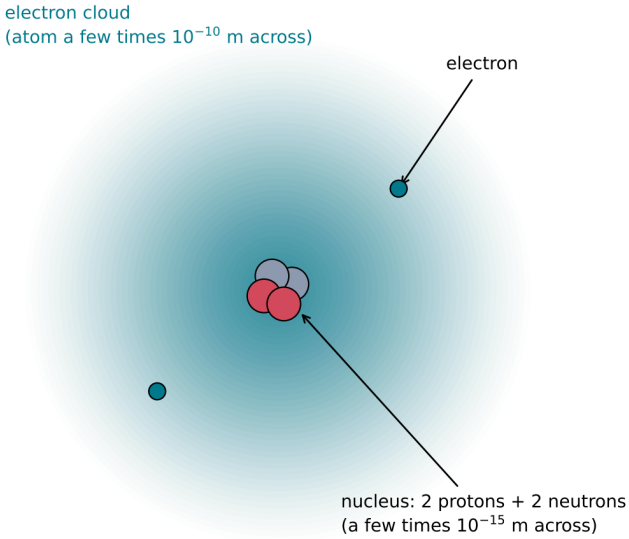


Figure 1.2.: Schematic of a helium-4 atom. The nucleus contains two protons and two neutrons; two electrons occupy the surrounding region. The diagram is not to scale: if the nucleus were drawn 1 mm across, the atom would be tens of metres across.

The size difference between the nucleus and the atom is extreme. An atom has a radius of 1 to 2 $\times$ 10^{-10} m, whereas a nucleus ranges from about 4 $\times$ 10^{-15} m across for carbon

to about 15×10^{-15} m for lead (Encyclopaedia Britannica, n.d.). The atom is therefore tens of thousands of times wider than its nucleus. The electrons do not move in fixed orbits, as Figure 1.2 might suggest. Quantum mechanics describes each electron by a probability distribution, and the shaded region in the figure represents where an electron is likely to be found. Chapter 2 develops this description, because it is the arrangement of electrons that determines how atoms bond to one another.

An atom is electrically neutral when it contains equal numbers of protons and electrons. An atom that has gained or lost electrons carries a net charge and is called an **ion**. Ions will be important in later chapters: the phosphate group of RNA, for example, carries negative charge in water, and positively charged magnesium ions play a central role in how RNA folds and reacts.

1.1.3. Atomic number and the definition of an element

The number of protons in a nucleus is the **atomic number**, symbol Z. The atomic number alone determines which element an atom belongs to: every atom with Z = 1 is hydrogen, every atom with Z = 6 is carbon, and every atom with Z = 15 is phosphorus. An **element** is therefore a substance made of atoms that all share the same atomic number. At present 118 elements have been identified and named, the most recent being elements 113, 115, 117 and 118, whose names were approved in 2016 (International Union of Pure and Applied Chemistry, n.d.).

The periodic table (Figure 1.1) arranges the elements in order of increasing atomic number. Elements in the same column have similar arrangements of their outermost electrons and therefore similar chemical behaviour, a relationship explained in Chapter 2.

1.1.4. Mass number and isotopes

The total number of nucleons in a nucleus is the **mass number**, symbol A:

$$A = Z + N$$

where Z is the number of protons and N is the number of neutrons. Atoms of the same element can contain different numbers of neutrons. Such variants are called **isotopes**. They have the same atomic number, and therefore almost identical chemistry, but different mass numbers.

An isotope is written with its mass number as a superscript before the element symbol. Hydrogen provides the simplest example (Figure 1.3). Ordinary hydrogen, ^{1}H or protium, has a nucleus consisting of a single proton. Deuterium, ^{2}H, has one proton and one neutron; in the international reference standard for ocean water it accounts for 0.0156

percent of hydrogen atoms (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.c). Tritium, ^{3}H, has one proton and two neutrons; it is radioactive, with a half-life of 12.3 years (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.c), and is therefore present only in trace amounts.

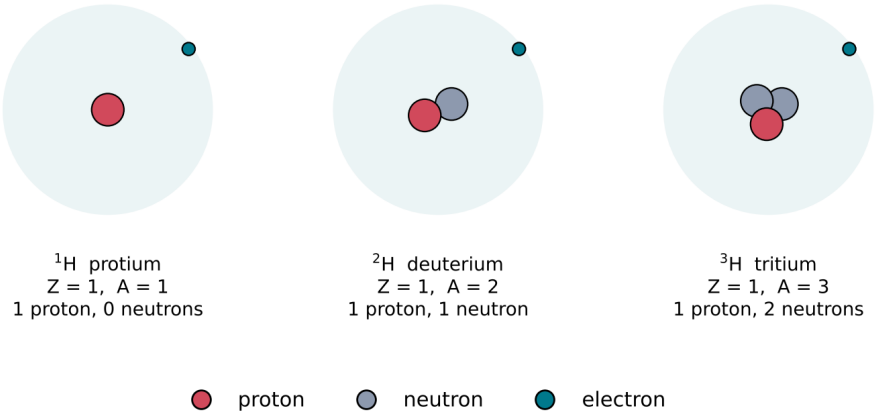


Figure 1.3.: The three isotopes of hydrogen. All have atomic number Z = 1 and differ only in the number of neutrons, and therefore in mass number A.

Carbon offers a second example that recurs throughout this book. Carbon-12 (six protons, six neutrons) makes up about 99 percent of natural carbon (98.84 to 99.04 percent, depending on the source), and carbon-13 (six protons, seven neutrons) about 1 percent (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.a). Carbon-14, with eight neutrons, is radioactive, with a half-life of 5,730 years (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.a). Because living organisms take up carbon-12 slightly more readily than carbon-13, the ratio of the two isotopes in ancient rocks is one of the lines of evidence used to search for the earliest traces of life (Section 3.3).

1.1.5. Stability of nuclei

Not every combination of protons and neutrons forms a stable nucleus. Protons repel one another electrically; they are held together, along with neutrons, by the **strong nuclear force**, which is far stronger than the electrical repulsion but acts only over distances comparable to the size of a nucleus. A nucleus with too many or too few neutrons relative to its protons is unstable and transforms, or **decays**, into a different nucleus, emitting radiation in the process. The time required for half of the nuclei in a sample to decay is the **half-life**. Radioactive decay with known half-lives provides the clocks by which the ages of the Earth and its oldest rocks are measured (Section 3.1).

1.1.6. Where the elements come from

The definitions above raise the question that occupies the rest of this chapter. If every element is defined by the number of protons in its nucleus, then making a new element requires changing that number, which requires a nuclear reaction. Ordinary chemistry, which rearranges electrons, cannot do this. The hydrogen in water, the carbon and nitrogen in a nucleobase and the phosphorus in a phosphate group were therefore all produced by nuclear processes, either in the first minutes of the Universe or later inside stars. Sections 1.2 to 1.5 describe those processes in turn.

1.2. The first three minutes

1.2.1. An expanding, cooling Universe

The Universe has been expanding since its beginning about 13.8 billion years ago (Planck Collaboration, Aghanim, et al. 2020). The evidence for this comes from two independent observations. First, light from distant galaxies is shifted to longer wavelengths in proportion to their distance, which shows that space between galaxies is stretching. Second, the sky is filled with a nearly uniform microwave radiation, the **cosmic microwave background**, which is the cooled remnant of radiation from a time when the Universe was hot and dense. The model that accounts for both observations, in which the Universe evolved from an extremely hot, dense initial state, is called the **Big Bang** model.

An expanding gas cools. Extrapolating backwards, the early Universe was hotter the earlier one looks. Temperature determines which particles can exist as separate entities. Above about 10^{10} K, collisions are so energetic that any nucleus formed is immediately broken apart again, and matter consists of free protons, neutrons, electrons, photons and neutrinos. The formation of the first nuclei therefore had to wait until the Universe had cooled sufficiently. This process is called **Big Bang nucleosynthesis** (Figure 1.4).

1.2.2. Fixing the ratio of neutrons to protons

The outcome of Big Bang nucleosynthesis is controlled by one number: the ratio of neutrons to protons at the moment nuclei begin to form. At temperatures above about 10^{10} K (a thermal energy of about 1 MeV), weak nuclear reactions involving neutrinos and electrons convert neutrons into protons and protons into neutrons fast enough to keep the two in balance. Because a neutron is slightly heavier than a proton (by 1.293 MeV, about 0.14 percent of the proton mass (Fields, Molaro, and Sarkar 2024)), the balance favours protons, and increasingly so as the temperature falls.

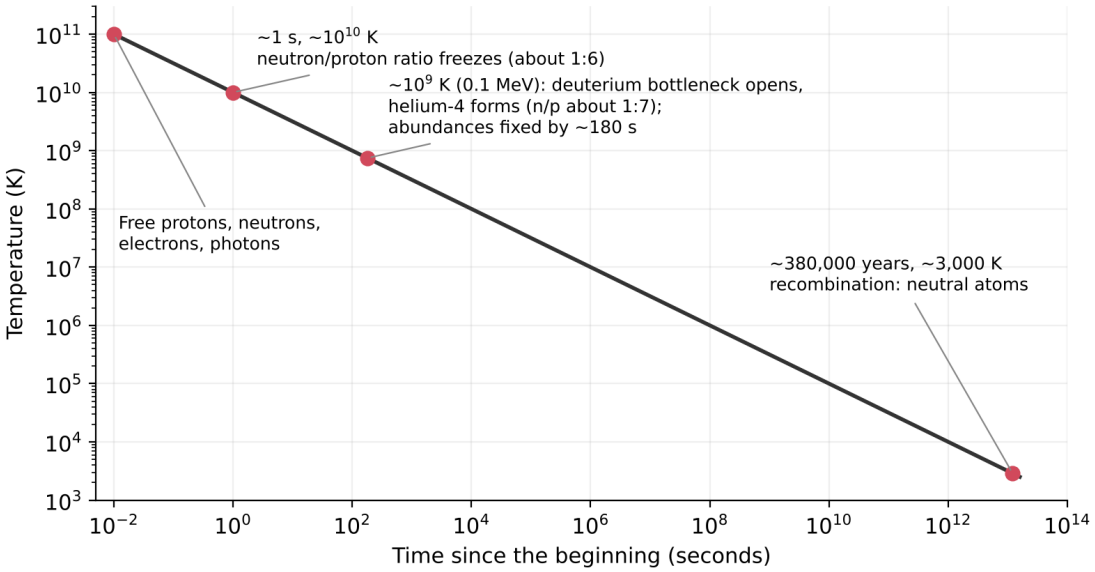


Figure 1.4.: The main events of the first few hundred thousand years, plotted against the temperature of the Universe. The curve uses the simple approximation $T \approx 10^{10}\ \mathrm{K} \times (t\ /\ 1\ \mathrm{s})^{-1/2}$, valid while radiation dominates; times and temperatures are approximate.

About one second after the beginning, the Universe had cooled to roughly 10^{10} K and these conversion reactions became too slow to keep up with the expansion. The neutron-to-proton ratio then stopped adjusting to temperature, or **froze out**, at approximately 1 neutron for every 6 protons (Fields, Molaro, and Sarkar 2024). After freeze-out, the only process that changed the ratio was the decay of free neutrons. A free neutron is unstable: it decays into a proton, an electron and an antineutrino, with a mean lifetime of 879.4 ± 0.6 seconds (Fields, Molaro, and Sarkar 2024). Neutrons were therefore steadily lost until they were locked inside nuclei, where they are stable.

1.2.3. The deuterium bottleneck

The first step in building any nucleus heavier than hydrogen is the fusion of a proton and a neutron into a deuterium nucleus (a **deuteron**):

$$\mathrm{p} + \mathrm{n} \rightarrow {}^{2}\mathrm{H} + \gamma$$

where γ denotes a high-energy photon (a gamma ray) that carries away the energy released. The deuteron is only weakly bound, so in the early Universe it was destroyed by energetic photons as fast as it formed. Photons vastly outnumbered nucleons, by about 1.6 billion to one (Fields, Molaro, and Sarkar 2024), so even when the average

photon energy fell below the deuteron binding energy, enough photons in the high-energy tail of the distribution remained to break deuterons apart. Nucleosynthesis was held up at this first step, a delay known as the **deuterium bottleneck**.

The bottleneck opened only when the temperature fell to about 10^9 K (a thermal energy of about 0.1 MeV), within the first few minutes (Fields, Molaro, and Sarkar 2024). By then neutron decay had reduced the ratio to about 1 neutron for every 7 protons (Fields, Molaro, and Sarkar 2024).

1.2.4. From deuterium to helium

Once deuterons survived, further fusion reactions followed rapidly. Deuterons combined with protons, neutrons and each other to form tritium (^{3}H), helium-3 (^{3}He) and finally helium-4 (^{4}He). Helium-4, with two protons and two neutrons, is an exceptionally tightly bound nucleus. Almost every neutron that existed when the bottleneck opened therefore ended up in helium-4.

This allows a simple estimate of how much helium the Big Bang produced. Suppose there are n neutrons for every p protons. Each helium-4 nucleus contains 2 neutrons and 2 protons, so if all neutrons end up in helium-4, the mass of helium is the mass of 2n nucleons, out of a total of n + p nucleons. The **helium mass fraction**, symbol Y, is therefore

$$Y = \frac{2n}{n+p} = \frac{2\,(n/p)}{1+n/p}$$

where n/p is the neutron-to-proton ratio. With n/p = 1/7,

$$Y = \frac{2 \times (1/7)}{1+1/7} = \frac{2}{8} = 0.25$$

About one quarter of the mass of ordinary matter should therefore have emerged from the first few minutes as helium-4, and the remaining three quarters as hydrogen. Measurements of helium in the oldest stars and in gas clouds that have undergone little stellar processing give Y = 0.245 ± 0.003, close to this estimate (Fields, Molaro, and Sarkar 2024). This agreement is one of the principal observational supports of the Big Bang model.

1.2.5. Why nucleosynthesis stopped at lithium

Big Bang nucleosynthesis produced almost nothing heavier than helium. The reason lies in nuclear stability: there is no stable nucleus with mass number 5 or mass number 8. A helium-4 nucleus that captures a proton or a neutron forms a mass-5 nucleus that falls apart almost instantly, and two helium-4 nuclei that fuse form beryllium-8, which also falls apart almost instantly. Only small amounts of lithium-7 (and beryllium-7, which later decays to lithium-7) formed through less direct routes. At the same time, the expansion was lowering the temperature and density, and by about 180 seconds after the beginning the abundances of the light nuclei were essentially fixed (Fields, Molaro, and Sarkar 2024).

The products of Big Bang nucleosynthesis are summarised in Figure 1.5. Hydrogen and helium-4 dominate. Deuterium, helium-3 and lithium-7 are present in trace amounts. Carbon, nitrogen, oxygen and phosphorus are absent.

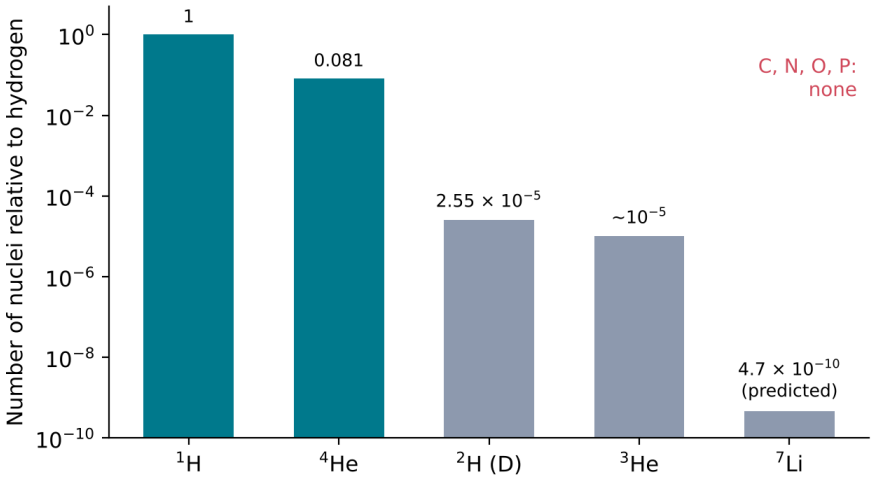


Figure 1.5.: Abundances of the light nuclei produced by Big Bang nucleosynthesis, expressed as number of nuclei relative to hydrogen (logarithmic scale). Values from the Particle Data Group review: deuterium and the predicted lithium-7 as given there, helium-4 converted from the mass fraction Y = 0.245 (He/H = Y / [4(1 − Y)]); the helium-3 bar shows only its order of magnitude (Fields, Molaro, and Sarkar 2024).

1.2.6. From nuclei to atoms

The nuclei formed in the first minutes were not yet atoms. The Universe was still hot enough to strip electrons from any nucleus, so matter remained an ionised gas, or **plasma**, of nuclei and free electrons. Only about 380,000 years later, when the temperature had fallen to about 3,000 K, could electrons bind permanently to nuclei to form neutral

hydrogen and helium atoms (European Space Agency, n.d.). This event is called **recombination**. Once matter became neutral, photons could travel freely through space; the cosmic microwave background observed today is this released radiation, now cooled to 2.7255 K by the continued expansion (Fixsen 2009).

1.2.7. Consequence for this book

Of the five elements that make up RNA, the Big Bang supplied only one: hydrogen. Every atom of carbon, nitrogen, oxygen and phosphorus had to be made later, inside stars. The barrier that stopped Big Bang nucleosynthesis, the absence of stable nuclei of mass 5 and 8, is the same barrier that stars must overcome to make carbon. Section 1.3 describes how they do it.

1.3. Stars as element factories

1.3.1. Why fusion releases energy

The mass of a nucleus is always slightly less than the total mass of the free protons and neutrons that compose it. The difference, multiplied by the square of the speed of light, is the **binding energy**: the energy that would be required to separate the nucleus completely into its nucleons, and equally the energy released when the nucleus is assembled from them. The relation between mass and energy is Einstein's

$$E = mc^2$$

where E is energy, m is mass and c is the speed of light in vacuum (299 792 458 m s^{-1}). In nuclear physics, masses are usually expressed in **atomic mass units** (u), defined as one twelfth of the mass of a carbon-12 atom; one atomic mass unit is equivalent to 931.494 MeV of energy (National Institute of Standards and Technology, n.d.a).

The binding energy of a nucleus with Z protons and N neutrons can be calculated from measured atomic masses:

$$B = \left[Z\, m(^1\mathrm{H}) + N\, m_\mathrm{n} - m(\mathrm{atom})\right] c^2$$

where m(^{1}H) is the mass of a hydrogen atom, m_n is the mass of a neutron and m(atom) is the mass of the atom in question. Using the hydrogen atom rather than the bare proton keeps the electron masses balanced on both sides of the equation.

What matters for the release of energy is the binding energy *per nucleon*, B/A. A nuclear reaction releases energy when its products have a higher binding energy per nucleon than its reactants. Figure 1.6 shows B/A for a selection of nuclei, calculated with the

equation above from atomic masses tabulated by the IUPAC Commission on Isotopic Abundances and Atomic Weights (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.c, n.d.b, n.d.f, n.d.d, n.d.e, n.d.j) and the neutron mass from CODATA 2022 (National Institute of Standards and Technology, n.d.a). The value rises steeply from deuterium (1.11 MeV) to helium-4 (7.07 MeV), continues to rise more slowly through carbon-12 (7.68 MeV) and oxygen-16 (7.98 MeV), reaches a maximum near iron-56 (8.790 MeV) and nickel-62 (8.795 MeV), and then declines slowly towards the heaviest nuclei, such as uranium-238 (7.57 MeV). Fusing light nuclei into heavier ones therefore releases energy, up to the region of iron and nickel. Beyond that region, fusion consumes energy instead.

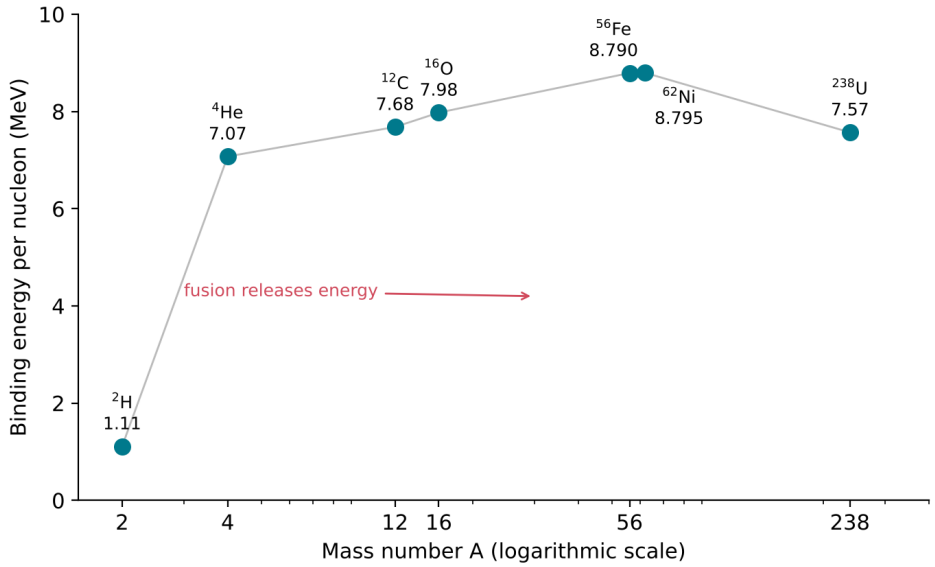


Figure 1.6.: Binding energy per nucleon for selected nuclei, calculated from IUPAC atomic masses and the CODATA 2022 neutron mass. The maximum lies near iron-56 and nickel-62. Fusion of nuclei lighter than this region releases energy.

The steep rise from hydrogen to helium is the reason why the conversion of hydrogen into helium is the main energy source of stars, and the reason why helium-4, once formed, is so difficult to break apart.

1.3.2. Hydrogen burning: the proton-proton chain

A **star** is a self-gravitating ball of gas in which nuclear fusion in the core supplies the energy radiated from the surface. Gravity compresses and heats the core until fusion begins; the energy released then supports the star against further collapse. The fusion of hydrogen into helium is called **hydrogen burning**. The term is conventional: it refers to a nuclear process, not to chemical combustion.

In stars of the Sun's mass, hydrogen burning proceeds mainly through the **proton-proton chain** (pp chain), which produces about 99 percent of the Sun's energy (Adelberger et al. 2011). Its net result is

$$4\,\mathrm{p} \rightarrow {}^{4}\mathrm{He} + 2\mathrm{e}^{+} + 2\nu_e$$

where p is a proton, e^{+} is a **positron** (the antiparticle of the electron, with the same mass and opposite charge) and ν_e is an electron **neutrino**, an electrically neutral particle of very small mass that interacts only weakly with matter (Adelberger et al. 2011). The most common sequence of steps (Figure 1.7) begins with the fusion of two protons into a deuteron, during which one proton is converted into a neutron. The deuteron then captures a further proton to form helium-3, and two helium-3 nuclei fuse to form helium-4, releasing two protons.

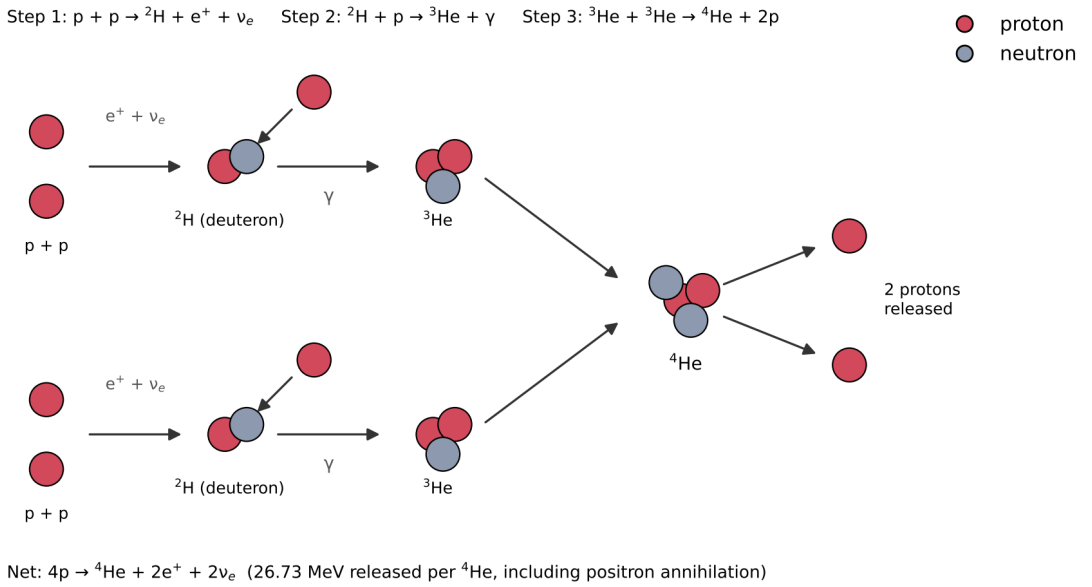


Figure 1.7.: The main branch of the proton-proton chain. Two protons fuse to form a deuteron, a positron and a neutrino; the deuteron captures a proton to form helium-3; two helium-3 nuclei, each produced by the first two steps, fuse into helium-4 and release two protons. The net result is the conversion of four protons into one helium-4 nucleus.

The energy released can be calculated from the atomic masses of hydrogen-1 (1.007 825 03 u) and helium-4 (4.002 603 25 u) (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.c, n.d.b). Four hydrogen atoms have a mass of 4.031 300 13 u; one helium-4 atom has a mass 0.028 696 88 u smaller. Multiplying by 931.494 MeV per atomic mass unit gives an energy release of 26.73 MeV for each helium-4 nucleus formed, which includes the energy from the annihilation of the two positrons with electrons. A

small part of this energy is carried away by the neutrinos, which escape from the star. The mass lost is 0.71 percent of the original mass of hydrogen.

This small fraction, multiplied over the mass of a star, is enough to power it for billions of years. The Sun radiates energy at a rate of 3.828×10^{26} W (International Astronomical Union 2015). Dividing by c^2 shows that it converts about 4.26×10^9 kg of mass into energy every second, roughly four million tonnes. The temperature at the centre of the Sun is about 15.5 million K (Adelberger et al. 2011), and the Sun has been burning hydrogen in its core for about 4.6 billion years (Adelberger et al. 2011; NASA, n.d.).

1.3.3. Helium burning and the beryllium-8 problem

When the hydrogen in a star's core is exhausted, fusion there stops and the core, now made mostly of helium, contracts under gravity and heats up. Helium nuclei carry two positive charges each, so they repel each other more strongly than protons do, and fusing them requires a higher temperature: near 10^8 K (José and Iliadis 2011). At such temperatures **helium burning** begins.

The obvious first step, the fusion of two helium-4 nuclei, leads to a dead end. The product, beryllium-8, is the unstable mass-8 nucleus already encountered in Section 1.2. Its ground state has an energy width of 5.57 eV (Tilley et al. 2004), which corresponds to a half-life of about 8×10^{-17} s: it falls apart into two helium-4 nuclei almost as soon as it forms. In a hot, dense stellar core, however, beryllium-8 is formed continuously, so a tiny equilibrium population is always present. If a third helium-4 nucleus strikes a beryllium-8 nucleus during its brief existence, carbon-12 can form. The overall process,

$$3\,{}^4\mathrm{He} \rightarrow {}^{12}\mathrm{C} + \gamma$$

is called the **triple-alpha process**, because a helium-4 nucleus is also called an alpha particle.

1.3.4. The Hoyle state

Even with a population of beryllium-8 present, the capture of a third helium-4 nucleus would be far too slow to explain the observed abundance of carbon unless the reaction were greatly enhanced. Such enhancement occurs when the combined energy of the colliding nuclei matches the energy of an **excited state** of the product nucleus, an arrangement of its nucleons with more energy than the lowest-energy arrangement, or **ground state**. Reactions at such a matching energy, called a **resonance**, proceed far faster than reactions at other energies.

In the early 1950s Fred Hoyle argued that, because carbon is abundant, carbon-12 must possess an excited state just above the combined energy of beryllium-8 and helium-4,

and he predicted it at about 7.68 MeV (Freer and Fynbo 2014). In 1953 Dunbar, Pixley, Wenzel and Whaling observed a level at 7.68 ± 0.03 MeV in carbon-12, and in their report thanked Hoyle for pointing out its astrophysical significance (Dunbar et al. 1953). Its properties were confirmed in subsequent measurements (Freer and Fynbo 2014). The state is now known as the **Hoyle state**. It lies at 7.65 MeV above the ground state of carbon-12, 285 keV above the energy of beryllium-8 plus helium-4 (Figure 1.8) (Freer and Fynbo 2014).

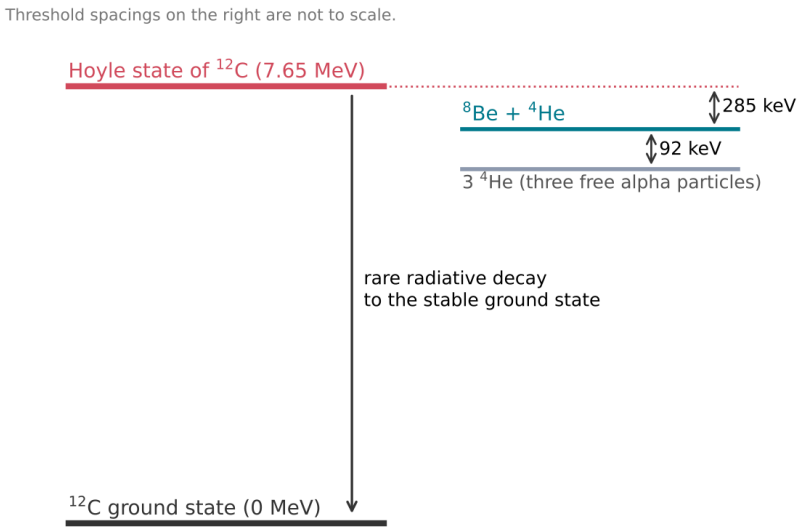


Figure 1.8.: Energy levels relevant to the triple-alpha process. The Hoyle state of carbon-12 lies 285 keV above the combined energy of beryllium-8 and helium-4, which itself lies 92 keV above three separate helium-4 nuclei. A small fraction of carbon-12 nuclei formed in the Hoyle state decay to the ground state by emitting radiation instead of breaking apart again.

Most carbon-12 nuclei formed in the Hoyle state break up again into three helium-4 nuclei. A small fraction instead lose their excess energy by emitting radiation and settle into the stable ground state. That small fraction is the origin of essentially all the carbon in the Universe, including the carbon in every RNA molecule.

Key experiment: predicting a nuclear state from the existence of carbon

Hoyle's prediction is a rare example in physics of a property of a nucleus being inferred from an astronomical observation, the abundance of carbon, before it was confirmed in the laboratory. Hoyle predicted a state near 7.68 MeV; Dunbar and colleagues, studying the reaction ^{14}N(d,α)^{12}C with a high-resolution spectrometer, observed a state at that energy (Dunbar et al. 1953; Freer and Fynbo 2014). Later

work refined the energy to 7.653 ± 0.008 MeV and indicated a spin and parity of 0^+ (Freer and Fynbo 2014).

1.3.5. From carbon to oxygen

Once carbon-12 is present, it can capture a further helium-4 nucleus:

$$^{12}\mathrm{C} + {}^{4}\mathrm{He} \rightarrow {}^{16}\mathrm{O} + \gamma$$

This reaction converts part of the carbon into oxygen-16 (Silva Aguirre 2017). Unlike the triple-alpha process, it has no resonance that would make it fast enough to destroy most of the carbon as it is made (Freer and Fynbo 2014). Helium burning therefore leaves behind a core rich in both carbon and oxygen (José and Iliadis 2011).

1.3.6. Consequence for this book

Stars produce two of the five elements of RNA, carbon and oxygen, through helium burning, and they convert hydrogen into the helium needed to make them. Nitrogen and phosphorus require further processes, described in Sections 1.4 and 1.5.

1.4. Beyond carbon: nitrogen, massive stars and supernovae

1.4.1. The CNO cycle and the origin of nitrogen

Once a star contains carbon, nitrogen and oxygen, these nuclei open a second route for hydrogen burning, the **CNO cycle**. In its main branch, a carbon-12 nucleus captures four protons in succession, with two positron emissions along the way, and finally emits a helium-4 nucleus, regenerating carbon-12 (J. W. Johnson et al. 2022):

$$^{12}\mathrm{C}(\mathrm{p},\gamma)\,{}^{13}\mathrm{N}(\beta^{+}\nu_e)\,{}^{13}\mathrm{C}(\mathrm{p},\gamma)\,{}^{14}\mathrm{N}(\mathrm{p},\gamma)\,{}^{15}\mathrm{O}(\beta^{+}\nu_e)\,{}^{15}\mathrm{N}(\mathrm{p},\alpha)\,{}^{12}\mathrm{C}$$

This compact notation is standard in nuclear astrophysics. In A(b,c)D, nucleus A captures particle b and emits particle c, becoming nucleus D. β^+ denotes **beta-plus decay**, in which a proton inside a nucleus turns into a neutron and a positron and a neutrino are emitted. The cycle is shown in Figure 1.9.

The carbon, nitrogen and oxygen nuclei act as **catalysts**: they are not consumed, and the net result is the same as that of the proton-proton chain, four protons converted into one helium-4 nucleus. The steps, however, proceed at very different rates. The slowest

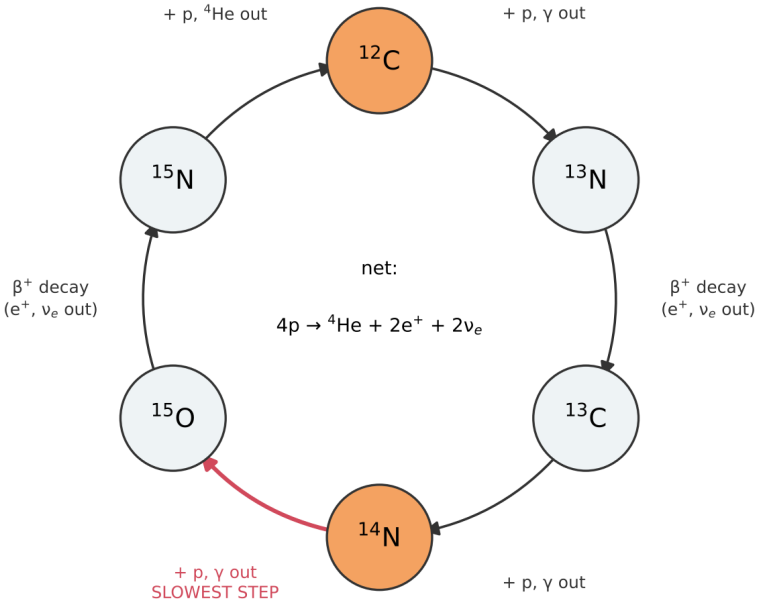


Figure 1.9.: The main branch of the CNO cycle. Carbon, nitrogen and oxygen act as catalysts: they are regenerated at the end of each cycle, and the net effect, like that of the proton-proton chain, is to convert four protons into one helium-4 nucleus. The slowest step, proton capture by nitrogen-14, is marked in red.

by far is the capture of a proton by nitrogen-14, $^{14}\mathrm{N(p,\gamma)^{15}O}$. Nuclei therefore pile up at the stage just before this step, and the first-order effect of the cycle is to convert most of the carbon in a stellar core into nitrogen-14 (J. W. Johnson et al. 2022). This is the principal way in which stars make nitrogen.

The CNO cycle is very sensitive to temperature. In the Sun, with a central temperature of about 15.5 million K, it produces only about 1 percent of the energy (Adelberger et al. 2011). In stars with central temperatures above about 20 million K the CNO cycle dominates (Ray 2009), which corresponds to main-sequence stars with initial masses above about 1.3 times that of the Sun (J. W. Johnson et al. 2022). Nitrogen made in this way is later returned to interstellar space, both by massive stars when they explode and by lower-mass stars, which shed their outer layers late in their lives (J. W. Johnson et al. 2022).

1.4.2. Burning stages in massive stars

A star with more than about eight times the mass of the Sun (8 ± 1 solar masses, a value that depends to some extent on stellar models) does not stop at helium burning (Smartt 2009). Each time a fuel is exhausted in the core, the core contracts and heats until the ashes of the previous stage ignite. Carbon burns to neon and magnesium, neon to oxygen and magnesium, oxygen to silicon and sulfur, and finally silicon to iron. Table 1.1 gives typical central temperatures and durations for a star of 20 solar masses, computed with stellar evolution models (Heger et al. 2002).

Table 1.1.: Nuclear burning stages in the core of a 20-solar-mass star, from stellar evolution models (Heger et al. 2002).

Fuel	Main products	Central temperature (10^9 K)	Duration
Hydrogen	Helium	0.037	8.1 million years
Helium	Oxygen, carbon	0.19	1.2 million years
Carbon	Neon, magnesium	0.87	980 years
Neon	Oxygen, magnesium	1.6	0.6 years
Oxygen	Silicon, sulfur	2.0	1.3 years
Silicon	Iron	3.3	0.031 years (about 11 days)

Two features of the table stand out. Each stage requires a higher temperature than the last, because heavier nuclei carry more positive charge and repel each other more strongly. And each stage is shorter than the last, ending with a silicon-burning stage that lasts about eleven days. Because burning continues in shells around the core after it has stopped at the centre, a massive star near the end of its life has a layered structure, with the products of successive burning stages arranged in concentric shells (Figure 1.10).

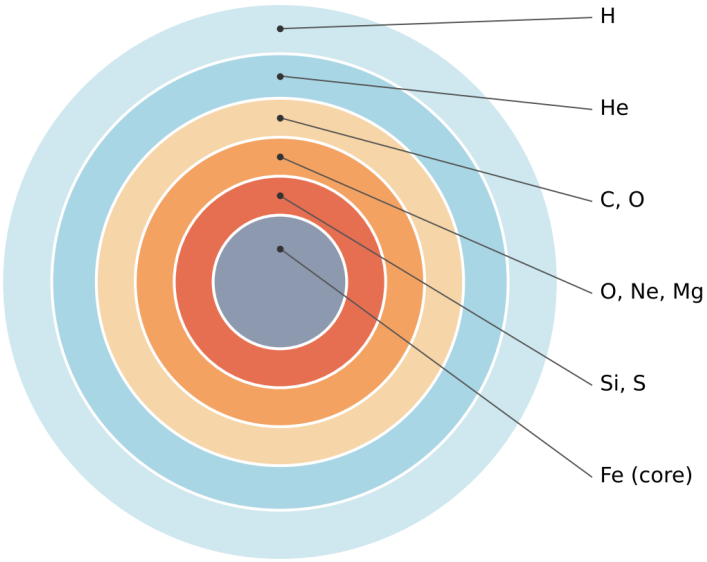


Figure 1.10.: Schematic structure of a massive star shortly before core collapse (not to scale). Successive burning stages leave concentric shells, with hydrogen in the outer envelope and iron at the centre. The labels give the main constituent of each shell.

1.4.3. The end at iron: core collapse

The sequence stops at iron because of the binding-energy curve (Figure 1.6). Iron-56 and nickel-62 lie at its maximum, so fusing iron-group nuclei into heavier ones does not release energy. Once the core is made of iron, nuclear burning no longer provides the pressure that supports it against gravity (Smartt 2009). When the iron core grows beyond the **Chandrasekhar mass**, the largest mass that the pressure of its electrons can support, it collapses (Heger et al. 2002).

The collapse of an iron core of about 1.4 solar masses releases an enormous amount of gravitational energy (Smartt 2009). Most of it, around 3×10^{46} J (3×10^{53} erg), is carried away by neutrinos. A small fraction drives an explosion that ejects the outer layers of the star with a kinetic energy of typically about 10^{44} J (10^{51} erg); the light emitted is usually 1 to 10 percent of this kinetic energy (Smartt 2009). This event is a **core-collapse supernova**. The ejected material carries the products of all the burning stages, including carbon, oxygen and nitrogen, into interstellar space, where it mixes with the gas from which later generations of stars and planets form.

1.4.4. Consequence for this book

Three of the five elements of RNA are now accounted for: carbon and oxygen from helium burning, and nitrogen from the CNO cycle. Hydrogen dates from the first minutes of the Universe. The remaining element, phosphorus, is the subject of the next section.

1.5. Phosphorus: the scarce element

1.5.1. One stable isotope

Phosphorus has atomic number 15. Of its isotopes, only one is stable: phosphorus-31, with 15 protons and 16 neutrons. An element with a single stable isotope is called **monoisotopic**. Every phosphorus atom in a rock, in sea water or in an RNA molecule is therefore an atom of ^{31}P, and the standard atomic weight of phosphorus, 30.973 761 998(5), is simply the atomic mass of that isotope (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.g). Other isotopes, such as ^{32}P and ^{33}P, are radioactive; they are made in small amounts by cosmic rays and in nuclear reactors and do not persist (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.g).

In RNA, phosphorus occurs in the phosphate groups that link one nucleotide to the next, one phosphorus atom per nucleotide (Chapters 5 and 6). An RNA molecule cannot be built without it.

1.5.2. Measuring abundances: the log epsilon scale

The abundance of an element in a star is measured from the strength of the absorption lines that its atoms produce in the star's spectrum. Astronomers report the result on a logarithmic scale defined relative to hydrogen:

$$\log \varepsilon_{\mathrm{X}} = \log_{10}\left(\frac{N_{\mathrm{X}}}{N_{\mathrm{H}}}\right) + 12$$

Here N_{X} is the number of atoms of element X per unit volume and N_{H} is the number of hydrogen atoms in the same volume. The constant 12 is added so that hydrogen has $\log \varepsilon_{\mathrm{H}} = 12.00$ by definition and all other elements have positive values (Asplund et al. 2009). A difference of 1 on this scale is a factor of 10 in number. The scale counts atoms, not mass.

Table 1.2 gives the present-day abundances in the photosphere of the Sun, the layer from which its visible light escapes, for the five elements of RNA and for helium, silicon, sulfur and iron. These values are used in the next section as the reference against which the composition of the Earth and of living matter is compared.

Table 1.2.: Abundances in the solar photosphere (Asplund et al. 2009). The helium value is not measured from photospheric lines but inferred from helioseismology, the study of oscillations of the Sun, and the iron value is the one obtained from lines of singly ionised iron. The last column is computed as $10^{\log \varepsilon - 5.41}$ and rounded to two significant figures.

Element	Z	$\log \varepsilon$	Atoms per phosphorus atom
Hydrogen	1	12.00	3.9×10^6
Helium	2	10.93 ± 0.01	3.3×10^5
Oxygen	8	8.69 ± 0.05	1900
Carbon	6	8.43 ± 0.05	1000
Nitrogen	7	7.83 ± 0.05	260
Silicon	14	7.51 ± 0.04	130
Iron	26	7.50 ± 0.04	120
Sulfur	16	7.12 ± 0.03	51
Phosphorus	15	5.41 ± 0.03	1

1.5.3. How scarce phosphorus is

The last column of Table 1.2 is obtained directly from the definition of $\log \varepsilon$. For carbon, for example,

$$\frac{N_{\mathrm{C}}}{N_{\mathrm{P}}} = 10^{8.43-5.41} = 10^{3.02} \approx 1050.$$

In the Sun there are therefore about 1000 carbon atoms, 1900 oxygen atoms, 260 nitrogen atoms and nearly four million hydrogen atoms for every atom of phosphorus (Figure 1.11). Of the five elements of RNA, phosphorus is by far the scarcest: about 260 times less common than nitrogen, the next scarcest.

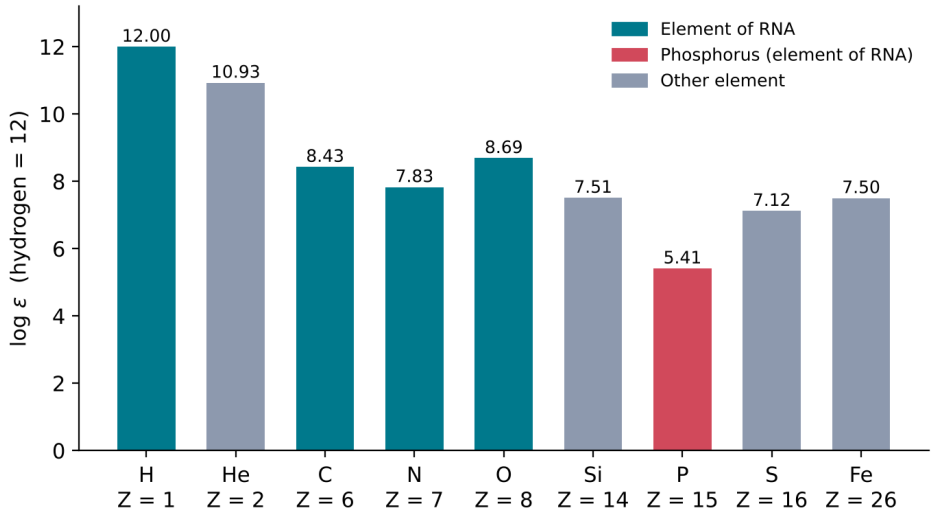


Figure 1.11.: Abundances of selected elements in the solar photosphere on the log ε scale, in order of atomic number (Asplund et al. 2009). The five elements of RNA are coloured. Each unit on the vertical axis is a factor of 10 in the number of atoms. Phosphorus lies about two units, a factor of about 100, below its neighbours silicon and sulfur.

The comparison with its neighbours in the periodic table is equally striking. Silicon (Z = 14) is about 130 times and sulfur (Z = 16) about 50 times more abundant than phosphorus (Z = 15), which lies between them. One feature of nuclear structure distinguishes phosphorus from these neighbours. The dominant isotopes of silicon and sulfur are ^{28}Si (about 92 percent of natural silicon) and ^{32}S (about 95 percent of natural sulfur) (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.h, n.d.i). Like ^{12}C and ^{16}O, they contain equal numbers of protons and neutrons, and their mass numbers are multiples of 4. A reaction between such a nucleus and a helium-4 nucleus conserves both the number of protons and the number of neutrons, so it can only produce another nucleus of the same kind. Phosphorus-31, with 15 protons and 16 neutrons, is not of this kind: no sequence of helium-4 captures starting from ^{12}C or ^{16}O can reach it. Its formation requires reactions in which single protons or neutrons are gained or lost. This structural difference is consistent with the low abundance of phosphorus, but a quantitative explanation requires detailed calculations of nuclear reaction networks, which are

beyond the scope of this chapter.

1.5.4. Where phosphorus is made

Phosphorus is thought to be synthesised in massive stars and dispersed into interstellar space when these stars explode as supernovae (Koo et al. 2013). Until 2013, this conclusion rested on stellar models. It was then tested by direct observation of the debris of a supernova.

Key experiment: phosphorus in the remnant of a supernova

A **supernova remnant** is the expanding cloud of gas ejected by a supernova explosion, together with the interstellar gas it sweeps up. Cassiopeia A is a young supernova remnant. In 2013, Koo, Lee, Moon, Yoon and Raymond reported near-infrared spectroscopic observations of Cassiopeia A, from which they determined the amounts of phosphorus and iron in the material ejected by the supernova. They found that the ratio of phosphorus to iron-56, the main product of the final burning stage, was up to 100 times the average ratio in the Milky Way (Koo et al. 2013). This confirmed that phosphorus is produced in supernovae. The measured range of ratios agreed with predictions from supernova nucleosynthesis models, but not with a scenario in which the inner layers of the star are completely mixed during the explosion (Koo et al. 2013).

1.5.5. Consequence for this book

All five elements of RNA have now been traced to their sources: hydrogen to the first minutes of the Universe, carbon and oxygen to helium burning, nitrogen to the CNO cycle, and phosphorus to massive stars and their supernovae. Phosphorus is made in the smallest amount, about one atom for every thousand carbon atoms in the Sun. Its scarcity, and the chemistry of the minerals in which it is locked on a planet, will return as a central problem in Chapter 4.

1.6. Cosmic abundances and the elements of life

1.6.1. Three compositions

Section 1.5 described the composition of the Sun. This section compares it with two other compositions: the rocks at the surface of the Earth and living cells. The comparison shows which elements a planet makes easily available, and which elements living matter concentrates.

Two different measures of composition are used, and they should not be confused. A **mass fraction** is the mass of an element divided by the total mass of the sample. An **atom ratio** is the number of atoms of one element divided by the number of atoms of another. To convert one into the other, each mass fraction is divided by the relative atomic mass of the element, the average mass of its atoms in atomic mass units (Section 1.3), as listed in the periodic table of the IUPAC (International Union of Pure and Applied Chemistry, n.d.). For an element X,

$$n_{\mathrm{X}} = \frac{w_{\mathrm{X}}}{A_r(\mathrm{X})}$$

where w_{X} is the mass fraction of X, $A_r(\mathrm{X})$ its relative atomic mass, and n_{X} is proportional to the number of atoms of X in the sample. Because the atom ratio is independent of the units used, this section compares compositions by the number of atoms of each element per atom of phosphorus, as in Table 1.2.

1.6.2. The Earth's upper crust

The **continental crust** is the outer layer of rock that forms the continents. Its upper part is the most accessible region of the solid Earth, and its composition can be estimated by averaging chemical analyses of rocks sampled over large areas of the surface (Rudnick and Gao 2003). Geochemists conventionally report the major elements as oxides, because in rock-forming minerals these elements are combined with oxygen. Table 1.3 gives the recommended composition of the upper continental crust (Rudnick and Gao 2003).

Table 1.3.: Major-element composition of the Earth's upper continental crust. The oxide values are the recommended values of Rudnick and Gao (2003), recast to a water-free basis; all iron is reported as FeO. The element mass percentages and atom ratios are calculated from the oxide values in this book, using the atomic masses of the elements, and are rounded.

Oxide	Mass percent	Element	Mass percent of element	Atoms per phosphorus atom
SiO_2	66.6	Si	31.1	520
Al_2O_3	15.4	Al	8.2	140
FeO (total Fe)	5.04	Fe	3.9	33
CaO	3.59	Ca	2.6	30
Na_2O	3.27	Na	2.4	50
K_2O	2.80	K	2.3	28
MgO	2.48	Mg	1.5	29
TiO_2	0.64	Ti	0.38	4
MnO	0.10	Mn	0.08	0.7

Oxide	Mass percent	Element	Mass percent of element	Atoms per phosphorus atom
P_2O_5	0.15	P	0.065	1
(oxygen in all oxides)		O	47.5	1400

As an example of the calculation, P_2O_5 has a molar mass of $2 \times 30.974 + 5 \times 15.999 = 141.94$ g mol^{-1}, of which phosphorus contributes 61.95 g mol^{-1}. A mass fraction of 0.15 percent P_2O_5 therefore contains $0.15 \times 61.95 / 141.94 = 0.065$ percent phosphorus.

Three features of Table 1.3 matter for this book. First, the upper crust is about half oxygen by mass, and silicon and oxygen together account for about four-fifths of it; rock is chiefly a framework of silicon and oxygen. Second, phosphorus is a minor element of the crust, at about 0.065 percent by mass, but it is present in every analysis. Third, carbon and nitrogen are absent from the list of major elements altogether. Nitrogen is a trace element, at 83 parts per million by mass in the upper crust (Rudnick and Gao 2003). Converted to atoms, this is about 0.3 nitrogen atoms for every phosphorus atom. In the Sun there are about 260 nitrogen atoms per phosphorus atom (Table 1.2).

The ratio of nitrogen to phosphorus is therefore lower in the crust than in the Sun by a factor of nearly 1000. Relative to phosphorus, most of the nitrogen of the material from which the solar system formed is missing from the Earth's crust. Where it went, and how much nitrogen and carbon the early Earth retained, are questions for Section 1.7 and Chapter 3.

1.6.3. Living matter

The composition of a cell depends on the organism and on its growth conditions, so only representative values can be given. For the bacterium *Escherichia coli*, a standard laboratory organism, the dry mass is about 47 percent carbon (Heldal, Norland, and Tumyr 1985), 20 percent oxygen, 14 percent nitrogen, 8 percent hydrogen and 3 percent phosphorus (Neidhardt, Ingraham, and Schaechter, n.d.). (**Dry mass** is the mass remaining after all water has been removed.) Converted to atoms, these values give about 40 carbon atoms, 10 nitrogen atoms, 13 oxygen atoms and 80 hydrogen atoms for each phosphorus atom.

A second, widely used reference is the composition of marine **phytoplankton**, the microscopic photosynthetic organisms of the surface ocean. Biologists use an atom ratio of carbon to nitrogen to phosphorus of C:N:P = 106:16:1, based on an analysis of the average elemental composition of marine organisms; it is known as the **Redfield ratio**,

after A. C. Redfield (Geider and La Roche 2002). The composition of marine particulate matter can deviate significantly from this ratio (Geider and La Roche 2002), but it remains a useful reference value.

1.6.4. Comparing the three

Figure 1.12 compares the number of carbon and nitrogen atoms per phosphorus atom in the Sun, the upper crust and living matter.

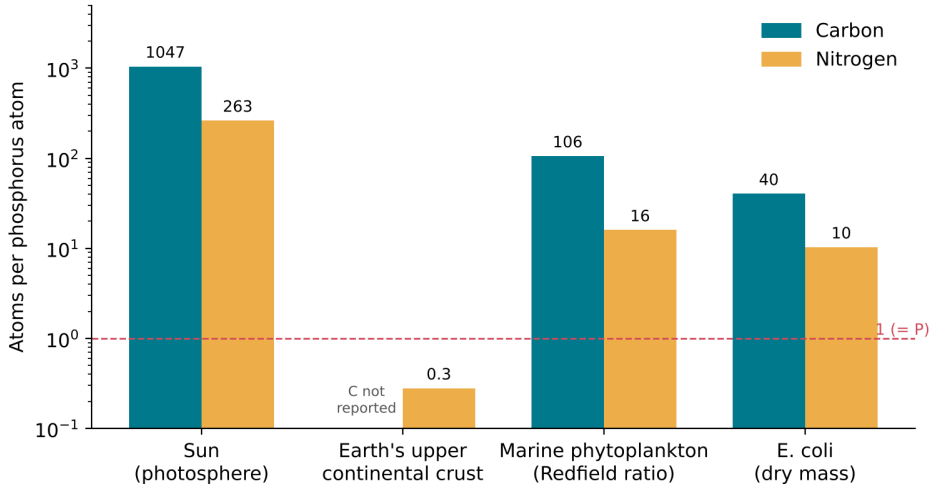


Figure 1.12.: Number of carbon atoms (left bar of each pair) and nitrogen atoms (right bar) per phosphorus atom in the solar photosphere (Asplund et al. 2009), the Earth's upper continental crust (Rudnick and Gao 2003), marine phytoplankton (Redfield ratio) (Geider and La Roche 2002) and the bacterium *E. coli* (from dry-mass composition) (Heldal, Norland, and Tumyr 1985; Neidhardt, Ingraham, and Schaechter, n.d.). The vertical axis is logarithmic. Carbon is not among the elements reported for the upper crust in the source used here, so no carbon bar is shown for it.

The figure shows two contrasts. Compared with the crust, living matter is enormously enriched in nitrogen, and in carbon, which is not even a major element of rock. Compared with the Sun, living matter is enriched in phosphorus: there are about 1000 carbon atoms per phosphorus atom in the Sun but only about 100 in phytoplankton and 40 in *E. coli*. Relative to carbon, living cells therefore contain roughly 10 to 25 times more phosphorus than the Sun.

1.6.5. Consequence for this book

The elements of RNA are not distributed as living matter requires. Carbon and nitrogen are abundant in the Universe but scarce in rock. Phosphorus is present in rock but scarce in the Universe, and in rock it is bound in minerals; how it can be brought into solution is taken up in Chapter 4. Any chemical route to RNA must therefore have drawn carbon and nitrogen from the atmosphere, the oceans or material delivered from space, and phosphorus from minerals. Where these supplies came from is the subject of Section 1.7 and Chapter 3.

1.7. From stellar ejecta to the forming Earth

1.7.1. Recycling between stars and gas

The elements made in stars are not lost when the stars die. Supernovae (Section 1.4) and the winds of stars return newly made nuclei to the gas between the stars, where they mix with gas that has not yet been processed. Stars that form later can therefore contain elements made by earlier stars. The Sun is an example: its photosphere contains carbon, nitrogen, oxygen and phosphorus (Table 1.2), none of which was made in the Big Bang (Section 1.2).

Stars form in **molecular clouds**: cold, dense clouds of interstellar gas in which hydrogen is present mainly as H_2 molecules, together with dust grains. The Sun, like most stars, is assumed to have been born in such a cloud, within a cluster of stars; from the size of the solar system and from the radioactive isotopes discussed below, it has been estimated that this birth cluster contained at least 2000 stars (Pfalzner et al. 2015).

1.7.2. The age of the solar system

The oldest solids known to have formed in the solar system are found in **chondrites**, meteorites that consist of unprocessed early solar-system material (Pfalzner et al. 2015). Chondrites are made of three kinds of component cemented by a fine-grained matrix: **calcium-aluminium-rich inclusions** (CAIs), which are rare; **chondrules**, which are abundant, rounded objects rich in magnesium and silicon; and grains of iron-nickel metal (Pfalzner et al. 2015).

CAIs have been dated by the uranium-lead method, which uses the radioactive decay of uranium isotopes to lead isotopes (the method is explained in Section 3.1), to 4.567 to 4.568 billion years (Pfalzner et al. 2015). This is taken as the age of the solar system, and later events are dated as times after CAI formation. Chondrules formed up to about 4 million years later (Pfalzner et al. 2015).

1.7.3. The protoplanetary disc

A cloud core that collapses under its own gravity does not fall straight onto the forming star. Because the core rotates, much of its material settles into a flattened, rotating disc of gas and dust around the young star, a **protoplanetary disc**, from which planets form. Such discs are observed around young stars today. Their masses are typically between 0.2 and 20 percent of the mass of the central star (Pfalzner et al. 2015). Their lifetimes can be estimated from the fraction of young stars that still show the infrared emission of a disc: this fraction falls exponentially with an e-folding time of about 3 million years (Pfalzner et al. 2015). Most stars therefore lose their discs within a few million years, although longer lifetimes, of 5 to 10 million years, have also been argued (Pfalzner et al. 2015).

The temperature of a disc falls with distance from the star, and this determines which substances can exist as solids. The **snow line** (or ice line) is the distance from the star beyond which ice grains can form (Pfalzner et al. 2015). For water, the corresponding temperature is about 150 K; for carbon monoxide, about 20 K (Pfalzner et al. 2015). Inside the water snow line, only rock-forming and metallic materials, such as the oxides and silicates of Table 1.3 and iron-nickel metal, are solid. Water, carbon monoxide and other volatile substances remain gases there.

1.7.4. Radioactive clues to the birth environment

Chondrites also contain the decay products of radioactive isotopes that no longer exist in the solar system. These **short-lived radionuclides** are radioactive isotopes with half-lives of less than 200 million years that were present in the early solar system (Pfalzner et al. 2015). Their presence shows that the material of the solar system was enriched in freshly made nuclei shortly before the solar system formed.

Key experiment: extinct radioactivity in meteorites

Analyses of the isotopes in meteorites show that they contain the decay products of aluminium-26 and iron-60, which have half-lives of 0.72 million and 2.6 million years respectively (Pfalzner et al. 2015). Neither isotope survives today, so they must have been incorporated into solids within a few half-lives of their production. Iron-60 is too rich in neutrons to be produced by irradiation, and it is absent from the winds of massive stars; its only possible source in the early solar system is at least one supernova (Pfalzner et al. 2015). Aluminium-26 cannot come from the same single supernova, because supernovae produce far more iron-60 than aluminium-26; its most likely origin is the wind of a single massive star nearby (Pfalzner et al. 2015). The isotopes of meteorites therefore record that material from massive stars, the same stars that make carbon, oxygen and phosphorus (Section 1.4, Section 1.5), was being added to the solar system's raw material close to the time of its formation.

The decay of aluminium-26 also released heat. Because its half-life is short, this heat source was confined to the first 5 million years of solar-system history (Pfalzner et al. 2015).

1.7.5. Building the Earth

Within the disc, dust grains stick together and grow into larger bodies, which collide and merge to form planets. The process took different lengths of time for different bodies. Some meteorites come from asteroids that melted and separated internally into a metallic core and a rocky mantle, a process called **differentiation**. The parent bodies of the magmatic iron meteorites, one group of iron meteorites, differentiated within 1 to 2 million years of CAI formation, and those of most other differentiated meteorites within 3 to 5 million years (Pfalzner et al. 2015). Mars probably formed within 2 to 10 million years (Pfalzner et al. 2015). The Earth grew more slowly: measurements of the hafnium-tungsten system in terrestrial rocks and in chondrites show that its growth lasted at least 38 million years, and probably as long as 120 million years, after the formation of the solar system (Pfalzner et al. 2015). The giant impact that formed the Moon is an important event for dating the late stages of the Earth's growth (Pfalzner et al. 2015).

There is growing evidence that much of the inventory of volatile elements in the inner solar system was added late, during the last stages of planetary growth (Pfalzner et al. 2015). Volatile substances, including water and compounds of carbon and nitrogen, are the raw materials of the chemistry described in the rest of this book; how and when the Earth acquired them is discussed in Chapter 3. Figure 1.13 summarises the time scales.

1.7.6. Consequence for this book

The five elements of RNA were made in the Big Bang and in stars, were dispersed by stellar winds and supernovae, mixed into a molecular cloud, and became part of the solid and gaseous material of the protoplanetary disc from which the Earth formed over several tens of millions of years. The Earth's crust that resulted is rich in oxygen, silicon and metals but poor in carbon and nitrogen, and it holds its phosphorus in minerals (Section 1.6). Chapter 2 turns from nuclei to the chemical bonds that join atoms into molecules; Chapter 3 returns to the young Earth and the materials it offered for chemistry.

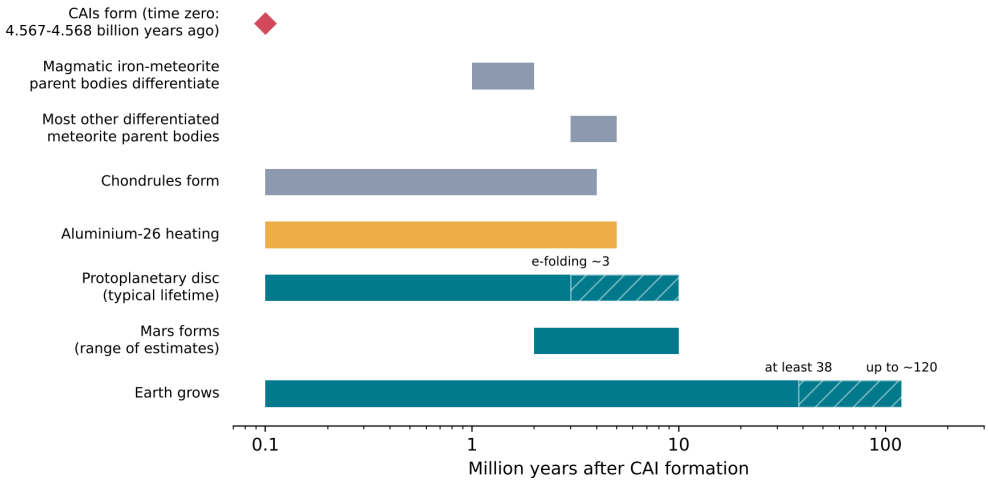


Figure 1.13.: Time scales of the formation of the solar system, measured from the formation of the first solids (CAIs), 4.567 to 4.568 billion years ago (Pfalzner et al. 2015). The horizontal axis is logarithmic; because such an axis cannot show time zero, CAI formation and bars that begin at time zero are drawn from 0.1 million years. Bars show the ranges given in the text. Hatched portions mark the uncertain upper part of a range: for the disc, the longer lifetimes of 5 to 10 million years that have been argued; for the Earth, the span between the firm lower limit of 38 million years and the probable end near 120 million years.

Summary

- An atom consists of a nucleus of protons and neutrons, surrounded by electrons. The nucleus holds almost all the mass but occupies a tiny fraction of the volume.
- The atomic number Z (number of protons) defines the element. The mass number A is the number of protons plus neutrons.
- Isotopes are atoms of the same element with different numbers of neutrons. They behave almost identically in chemistry but can differ in stability.
- RNA is built from only five elements: H, C, N, O and P. Each was produced by nuclear reactions, not by chemistry.
- The early Universe was too hot for nuclei to survive. Nuclei formed only after it cooled, in the first few minutes.
- The neutron-to-proton ratio (about 1 to 7 when nucleosynthesis began) fixed the helium mass fraction at about 0.25.
- Big Bang nucleosynthesis stopped at lithium because no stable nuclei exist with mass number 5 or 8. It produced hydrogen but none of C, N, O or P.
- Fusion of light nuclei releases energy because binding energy per nucleon rises from hydrogen up to the region of iron and nickel.
- Stars like the Sun convert hydrogen into helium through the proton-proton chain, releasing 26.7 MeV per helium-4 nucleus formed.
- In helium burning near 10^8 K, three helium-4 nuclei form carbon-12 through unstable beryllium-8 and the Hoyle state at 7.65 MeV; further helium capture turns part of the carbon into oxygen-16.
- In the CNO cycle, carbon, nitrogen and oxygen catalyse hydrogen burning; its slowest step converts most of the carbon in a stellar core into nitrogen-14, the main stellar source of nitrogen.
- Stars of more than about 8 solar masses burn successively heavier fuels up to iron, then their iron cores collapse and the stars explode as core-collapse supernovae, ejecting newly made elements into space.
- Phosphorus has a single stable isotope, ^{31}P. It is made in massive stars and released by supernovae, as confirmed by observations of the supernova remnant Cassiopeia A.
- On the log ε scale, abundances are counted in atoms relative to 10^{12} hydrogen atoms. In the Sun there are about 1000 carbon atoms for every phosphorus atom; phosphorus is the scarcest of the five elements of RNA.
- The Earth's upper crust is mostly oxygen and silicon. Phosphorus is a minor element of rock, carbon is not a major element at all, and there are fewer nitrogen atoms than phosphorus atoms.
- Living matter is rich in carbon and nitrogen compared with rock, and rich in phosphorus compared with the Sun: about 106 carbon atoms per phosphorus atom in marine phytoplankton (the Redfield ratio), against about 1000 in the Sun.
- The solar system formed from a molecular cloud enriched by earlier stars. Its oldest solids, CAIs in chondritic meteorites, are 4.567 to 4.568 billion years old.

- Extinct aluminium-26 and iron-60 in meteorites show that material from massive stars, including at least one supernova, was added shortly before the solar system formed.
- Planets formed in a protoplanetary disc that lasted a few million years. The Earth grew over at least 38 and probably up to about 120 million years.

Exercises

1. A nucleus contains 7 protons and 8 neutrons. State its atomic number, mass number and element, and write its isotope symbol.
2. Explain why ^{12}C and ^{14}C behave almost identically in chemical reactions but differently in a radiation detector.
3. If the nucleus of an atom were enlarged to the size of a marble (1 cm across), estimate the diameter of the whole atom.
4. Use the equation for Y to calculate the helium mass fraction if the neutron-to-proton ratio had been 1/5 when the deuterium bottleneck opened. Would a longer bottleneck produce more or less helium? Explain.
5. Explain why the Big Bang produced almost no carbon, even though carbon-12 is a very stable nucleus.
6. Using the atomic masses given in Section 1.3, calculate the fraction of mass converted into energy when four hydrogen atoms become one helium-4 atom.
7. The Sun loses about 4.26×10^9 kg of mass per second through hydrogen burning. Assuming this rate has been constant, estimate the total mass the Sun has converted into energy over 4.6 billion years. (One year is about 3.16×10^7 s.)
8. Explain why the existence of the Hoyle state was predicted from astronomical observations before it was found in the laboratory.
9. Write out the main branch of the CNO cycle as six separate reactions, and show that its net effect is $4\mathrm{p} \rightarrow {}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e$.
10. Using Table 1.1, calculate the ratio of the durations of hydrogen burning and silicon burning in a 20-solar-mass star.
11. Explain, with reference to Figure 1.6, why a star cannot obtain energy by fusing iron.
12. Using Table 1.2, calculate how many oxygen atoms and how many nitrogen atoms there are in the Sun for every phosphorus atom, and check your answers against the last column of the table.
13. Show that a nucleus with equal numbers of protons and neutrons that captures a helium-4 nucleus again has equal numbers of protons and neutrons. Use this result to explain why phosphorus-31 cannot be made from carbon-12 by helium-4 captures alone.
14. Explain why the standard atomic weight of phosphorus is known far more precisely than that of sulfur.

15. Using the oxide composition in Table 1.3, calculate the mass percent of silicon in the upper crust. (Relative atomic masses: Si 28.085, O 15.999.)
16. A sample of dry biomass contains 50 percent carbon and 3 percent phosphorus by mass. Calculate the number of carbon atoms per phosphorus atom, and compare it with the Redfield ratio and with the Sun.
17. Explain why comparisons of composition in this chapter are made with atom ratios rather than with mass fractions.
18. Aluminium-26 has a half-life of 0.72 million years. What fraction of the aluminium-26 present when CAIs formed remained after 5 million years? Use this to explain why aluminium-26 heating was confined to the earliest solar system.
19. Explain why the presence of iron-60 decay products in meteorites is taken as evidence that at least one supernova contributed material to the early solar system.
20. Using Figure 1.13, compare the lifetime of a typical protoplanetary disc with the time the Earth took to form. What does this imply about the amount of gas available during the late stages of the Earth's growth?

2. How Atoms Join into Molecules

Learning objectives

After reading this chapter you should be able to:

1. Describe how the electrons of an atom are arranged in shells, subshells and orbitals, and write the electron configurations of hydrogen, carbon, nitrogen, oxygen and phosphorus.
2. Explain how atoms form covalent bonds by sharing electrons, and predict the number of bonds formed by each of the five elements of RNA.
3. Explain the geometry of simple molecules, the polarity of bonds, and the forces that act between molecules, with water as the central example.
4. Use the concepts of enthalpy, entropy and Gibbs free energy to decide whether a reaction can occur, and explain how temperature and catalysts affect its rate.

2.1. Electrons and orbitals

2.1.1. From nuclei to chemistry

Chapter 1 followed the nuclei of hydrogen, carbon, nitrogen, oxygen and phosphorus from their origin in the Big Bang and in stars. Nuclei, however, do not take part in chemical reactions. In every process described from here onwards, nuclei remain unchanged, and only the arrangement of electrons changes. Chemistry is the study of these rearrangements. To understand why atoms combine as they do, it is first necessary to know how the electrons of an isolated atom are arranged.

2.1.2. Energy levels and the electronvolt

The electrons of an atom cannot have any energy whatever. They occupy a set of discrete **energy levels**, determined by the attraction of the positive nucleus and the repulsion between electrons. This is a consequence of quantum mechanics, the physical theory that describes particles on the scale of atoms; its mathematical form is beyond the scope of this book, but its results can be stated precisely.

Energies on the atomic scale are conveniently measured in **electronvolts** (eV). One electronvolt is the energy gained by a particle with one elementary charge when it moves through an electric potential difference of one volt. Because the elementary charge is exact (Section 1.1), so is the electronvolt: 1 eV = 1.602 176 634 × 10^{-19} J (National Institute of Standards and Technology, n.d.a).

2.1.3. Orbitals, shells and subshells

The state of an electron in an atom is described by an **orbital**: a region around the nucleus, described by a mathematical function called a wavefunction, in which an electron of a given energy is likely to be found. An electron in an atom does not move along a fixed path; the orbital gives only the probability of finding it at each point in space.

Orbitals are grouped as follows.

- A **shell** is a set of orbitals with the same **principal quantum number** n, which takes the values 1, 2, 3, … Electrons in shells with larger n are, on average, farther from the nucleus and less tightly bound.
- Within a shell, orbitals are divided into **subshells**, labelled s, p, d and f. The shell with n = 1 contains only an s subshell; n = 2 contains s and p; n = 3 contains s, p and d.
- An s subshell contains one orbital, a p subshell three, a d subshell five and an f subshell seven.

A subshell is written with its shell number followed by its letter: 1s, 2s, 2p, 3s, 3p, 3d and so on. An s orbital is spherically symmetric about the nucleus. The three p orbitals of a subshell each have two lobes on opposite sides of the nucleus and are directed along three mutually perpendicular axes, conventionally labelled x, y and z (Figure 2.1).

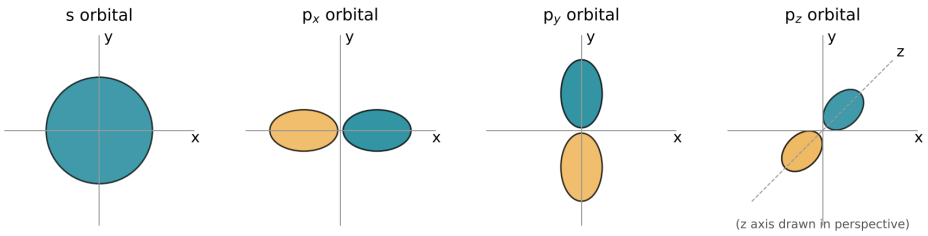


Figure 2.1.: Schematic shapes of s and p orbitals, drawn as two-dimensional outlines. Each outline marks the region in which an electron in that orbital is most likely to be found. An s orbital is spherical; the three p orbitals of a subshell are directed along perpendicular axes. The two lobes of a p orbital have opposite signs of the wavefunction, indicated by the two colours.

2.1.4. The Pauli exclusion principle

The number of electrons that each orbital can hold is fixed by the **Pauli exclusion principle**: no two electrons in an atom can be in exactly the same quantum state. Electrons possess an intrinsic property called **spin**, which can take one of two values, conventionally written as "up" and "down". An orbital can therefore hold at most two electrons, and only if their spins are opposite.

It follows that an s subshell holds at most 2 electrons, a p subshell 6, a d subshell 10 and an f subshell 14. Shell n contains n^2 orbitals and can therefore hold at most $2n^2$ electrons: 2 in the first shell, 8 in the second and 18 in the third.

2.1.5. Electron configurations

The **electron configuration** of an atom lists how many electrons occupy each subshell. It is written as the subshell labels with the number of electrons as a superscript. In the lowest-energy state of an atom, its ground state (Section 1.3), the subshells are filled approximately in order of increasing energy: 1s, 2s, 2p, 3s, 3p, then 4s and 3d. Table 2.1 gives the measured ground-state configurations of the five elements of RNA, together with some of their neighbours in the periodic table (Kramida et al. 2024).

Table 2.1.: Ground-state electron configurations and first ionisation energies of selected atoms (Kramida et al. 2024). [Ne] stands for the configuration of neon, $1s^2\ 2s^2\ 2p^6$. Ionisation energies are rounded to three decimal places.

Element	Z	Electron configuration	Valence electrons	First ionisation energy (eV)
Hydrogen	1	$1s^1$	1	13.598
Helium	2	$1s^2$	(2, full shell)	24.587
Carbon	6	$1s^2\ 2s^2\ 2p^2$	4	11.260
Nitrogen	7	$1s^2\ 2s^2\ 2p^3$	5	14.534
Oxygen	8	$1s^2\ 2s^2\ 2p^4$	6	13.618
Neon	10	$1s^2\ 2s^2\ 2p^6$	(8, full shell)	21.565
Sodium	11	[Ne] $3s^1$	1	5.139
Phosphorus	15	[Ne] $3s^2\ 3p^3$	5	10.487
Sulfur	16	[Ne] $3s^2\ 3p^4$	6	10.360

The notation [Ne] $3s^2\ 3p^3$ for phosphorus means that phosphorus has the ten electrons of a neon atom in its first two shells, plus two electrons in the 3s subshell and three in the 3p subshell: 15 electrons in all, equal to its atomic number.

Figure 2.2 shows the configurations of the RNA elements as an energy-level diagram. Each box is one orbital, and each arrow is one electron, pointing up or down according

to its spin. Where electrons occupy a p subshell that is less than half full, they spread over separate orbitals with parallel spins before any orbital receives a second electron; this is known as **Hund's rule**. Nitrogen and phosphorus, with three p electrons each, have one electron in each of their three p orbitals.

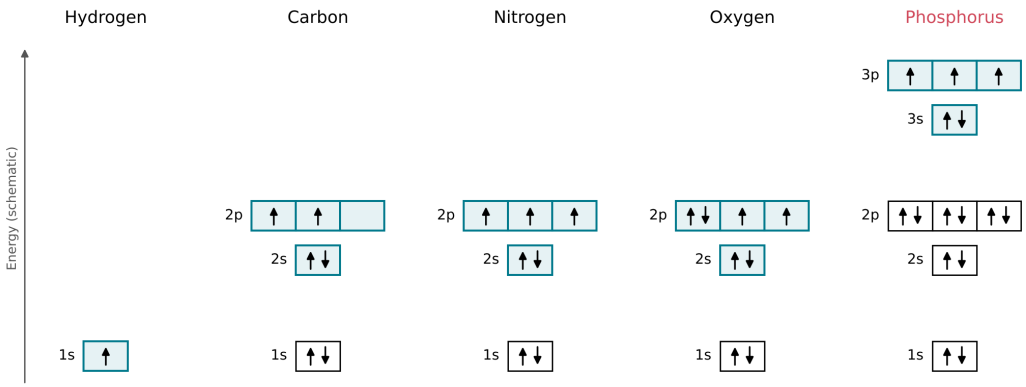


Figure 2.2.: Ground-state electron configurations of hydrogen, carbon, nitrogen, oxygen and phosphorus. Each box represents an orbital and each arrow an electron; the direction of the arrow indicates the spin. The outermost shell, which contains the valence electrons, is highlighted. Energies are schematic and not to scale.

2.1.6. Valence electrons

The electrons in the outermost occupied shell are the **valence electrons**. They are the least tightly bound and the ones that take part in chemical bonding. Electrons in inner shells, called core electrons, are held more tightly and do not take part in bonding. Carbon has four valence electrons ($2s^2\ 2p^2$), nitrogen five ($2s^2\ 2p^3$), oxygen six ($2s^2\ 2p^4$) and phosphorus five ($3s^2\ 3p^3$). Hydrogen has a single electron, which is its valence electron.

Nitrogen and phosphorus lie in the same column, or group, of the periodic table, and both have five valence electrons in the configuration $s^2\ p^3$. They differ in that the valence electrons of phosphorus are in the third shell rather than the second. This shared configuration is the reason for many chemical similarities between the two elements, and the difference in shell is the reason for some of the differences that will appear in later chapters.

2.1.7. Ionisation energy: a measure of how tightly electrons are held

The arrangement of electrons into shells is not merely a bookkeeping convention; it can be measured. The **first ionisation energy** of an atom is the minimum energy required

to remove one electron from the neutral atom in its ground state, leaving a positive ion. Figure 2.3 shows the measured values for the first eighteen elements (Kramida et al. 2024).

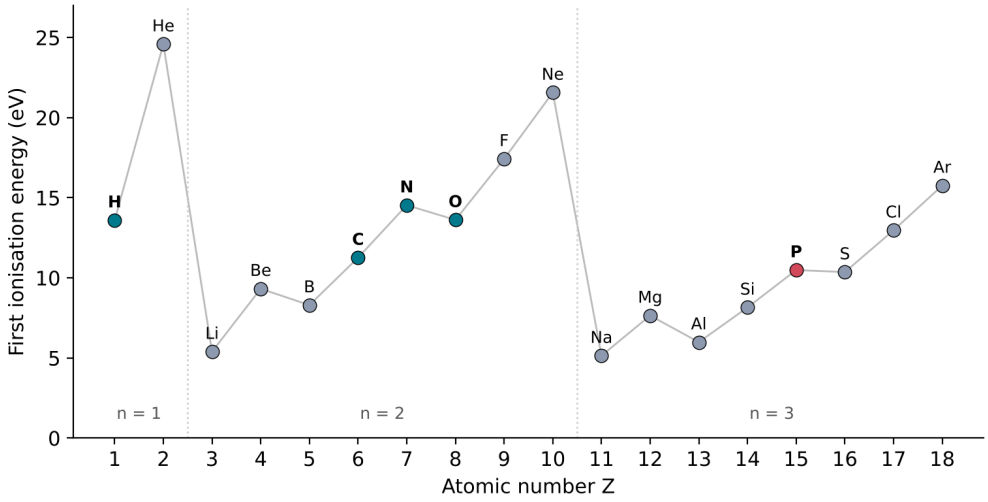


Figure 2.3.: First ionisation energies of the elements from hydrogen (Z = 1) to argon (Z = 18), from the NIST Atomic Spectra Database (Kramida et al. 2024). The elements of RNA are coloured. The maxima at helium, neon and argon correspond to completed shells; the sharp drops at lithium and sodium mark the start of a new shell.

Two features of the figure confirm the shell structure. First, the noble gases helium (24.59 eV), neon (21.56 eV) and argon (15.76 eV) have the highest ionisation energies in their rows: their outermost shells or subshells are full, and their electrons are held most tightly. Second, the next elements, lithium (5.39 eV) and sodium (5.14 eV), have the lowest values: each has a single electron that begins a new shell, farther from the nucleus and shielded from it by the inner electrons (Kramida et al. 2024). Within a row, the ionisation energy tends to rise from left to right, because the nuclear charge increases while the electrons being removed remain in the same shell.

The ionisation energies of the RNA elements lie between about 10 and 15 eV. These values are large compared with the energy of thermal motion at the temperatures relevant to the origin of life, which is of the order of a few hundredths of an electronvolt per particle (the product of the Boltzmann constant, $k_B = 8.617\ 333\ 262 \times 10^{-5}$ eV K^{-1}, and the absolute temperature: about 0.026 eV at 25 °C, or 298.15 K) (National Institute of Standards and Technology, n.d.a). Isolated atoms of these elements therefore keep their electrons. They do, however, react with each other, by sharing electrons rather than losing them. This sharing, the covalent bond, is the subject of the next section.

2.1.8. Consequence for this book

The chemistry of RNA is determined by the valence electrons of five elements: one for hydrogen, four for carbon, five for nitrogen, six for oxygen and five for phosphorus. Every bond in the molecules described in the rest of this book is formed from these electrons.

2.2. The covalent bond

2.2.1. Sharing a pair of electrons

A **covalent bond** is a chemical bond formed by a pair of electrons shared between two atoms. The simplest example is the hydrogen molecule, H_2. Each hydrogen atom brings one electron in a 1s orbital. When the two atoms approach, the two electrons, with opposite spins, come to occupy a region between and around both nuclei, where each electron is attracted by both nuclei at once. The resulting molecule has lower energy than two separate atoms, and so it is stable.

The distance between the nuclei of two bonded atoms is the **bond length**. In H_2 it is 0.741 Å (National Institute of Standards and Technology 2022), where the ångström (Å) is a unit of length equal to 10^{-10} m. Bond lengths between the atoms of the RNA elements are all of the order of 1 Å. This is the scale of molecules, tens of thousands of times the diameter of a nucleus (Section 1.1).

2.2.2. Completing a shell

Section 2.1 showed that atoms with completely filled shells, the noble gases, hold their electrons most tightly. Bonding between the elements of RNA can be described by a simple counting rule that follows from this. By sharing electrons, each atom tends to acquire the electron configuration of the nearest noble gas:

- Hydrogen shares electrons until it has two electrons around it, the configuration of helium.
- Carbon, nitrogen and oxygen share electrons until they have eight valence electrons, the configuration of neon. This is the **octet rule**.

Electrons shared in a bond count towards the shells of both atoms. An atom with v valence electrons therefore needs $8 - v$ shared electrons from its partners, and so forms $8 - v$ covalent bonds. This number is the **valence** of the atom (Table 2.2).

Table 2.2.: The usual number of covalent bonds formed by hydrogen, carbon, nitrogen and oxygen, from the octet rule (duet rule for hydrogen).

Element	Valence electrons	Electrons needed	Usual number of covalent bonds
Hydrogen	1	1 (to reach 2)	1
Carbon	4	4	4
Nitrogen	5	3	3
Oxygen	6	2	2

The octet rule is a counting rule, not a law of nature, but it predicts correctly the bonding of carbon, nitrogen and oxygen in almost all the molecules in this book. Phosphorus, whose valence electrons are in the third shell, is an important exception: in phosphate, the form in which phosphorus occurs in RNA, each phosphorus atom is bonded to four oxygen atoms. How these bonds are best described is taken up when phosphate is introduced in Chapter 4.

2.2.3. Lewis structures

A **Lewis structure** represents the valence electrons of a molecule. Each line between two atom symbols stands for one shared pair of electrons, that is, one covalent bond. Pairs of valence electrons that belong to one atom only and are not shared are called **lone pairs**, and are drawn as pairs of dots. Figure 2.4 shows Lewis structures of some molecules that will appear repeatedly in this book.

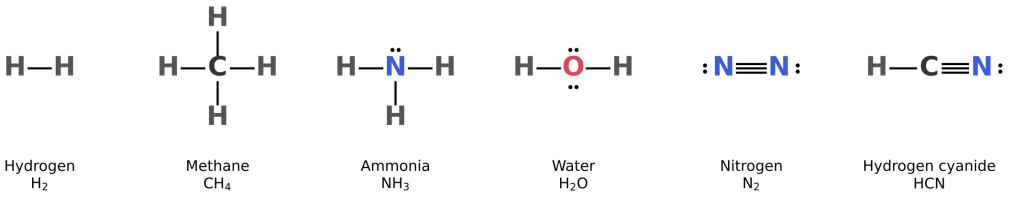


Figure 2.4.: Lewis structures of hydrogen (H_2), methane (CH_4), ammonia (NH_3), water (H_2O), nitrogen (N_2) and hydrogen cyanide (HCN). Each line is a shared pair of electrons; each pair of dots is a lone pair. In every structure, hydrogen has two electrons around it and carbon, nitrogen and oxygen have eight. The drawings show which atoms are bonded, not the shapes of the molecules, which are the subject of Section 2.3.

In methane, CH_4, carbon shares one pair with each of four hydrogen atoms; its four valence electrons and the four from hydrogen give carbon an octet. In ammonia, NH_3, nitrogen forms three bonds and keeps one lone pair. In water, H_2O, oxygen forms two

bonds and keeps two lone pairs. The measured lengths of these bonds are 1.087 Å for C–H in methane, 1.012 Å for N–H in ammonia and 0.958 Å for O–H in water (National Institute of Standards and Technology 2022). The lone pairs of nitrogen and oxygen are not inert: they are responsible for much of the chemistry of ammonia and water, including the hydrogen bonds discussed in Section 2.5.

2.2.4. Double and triple bonds

Two atoms can share more than one pair of electrons. Two shared pairs form a **double bond**, written with two lines, and three shared pairs form a **triple bond**, written with three. The number of shared pairs is called the **bond order**. In the nitrogen molecule, N_2, the two nitrogen atoms share three pairs, so each nitrogen has three bonds and one lone pair, and an octet. In hydrogen cyanide, HCN, carbon forms one bond to hydrogen and a triple bond to nitrogen: four bonds in all, as the octet rule requires. Hydrogen cyanide appears again in Chapters 3 and 4, as a starting material in experiments on the synthesis of the building blocks of RNA.

The bond order has a measurable effect on the bond length. The three simplest compounds of carbon and hydrogen with two carbon atoms show this directly (Figure 2.5). In ethane, C_2H_6, the carbon atoms are joined by a single bond of length 1.536 Å; in ethylene, C_2H_4, by a double bond of 1.339 Å; and in acetylene, C_2H_2, by a triple bond of 1.203 Å (National Institute of Standards and Technology 2022). Each additional shared pair draws the two nuclei closer together. The same trend holds for nitrogen: the triple bond in N_2 is 1.098 Å long (National Institute of Standards and Technology 2022), and the carbon-nitrogen triple bond in HCN 1.156 Å (National Institute of Standards and Technology 2022).

2.2.5. Consequence for this book

Every molecule on the path to RNA is held together by covalent bonds between hydrogen, carbon, nitrogen, oxygen and phosphorus. The octet rule predicts how many bonds each atom forms: four for carbon, three for nitrogen, two for oxygen and one for hydrogen. These numbers determine which molecules can exist, and therefore which structures are possible for the sugars, bases and phosphates that make up a nucleotide.

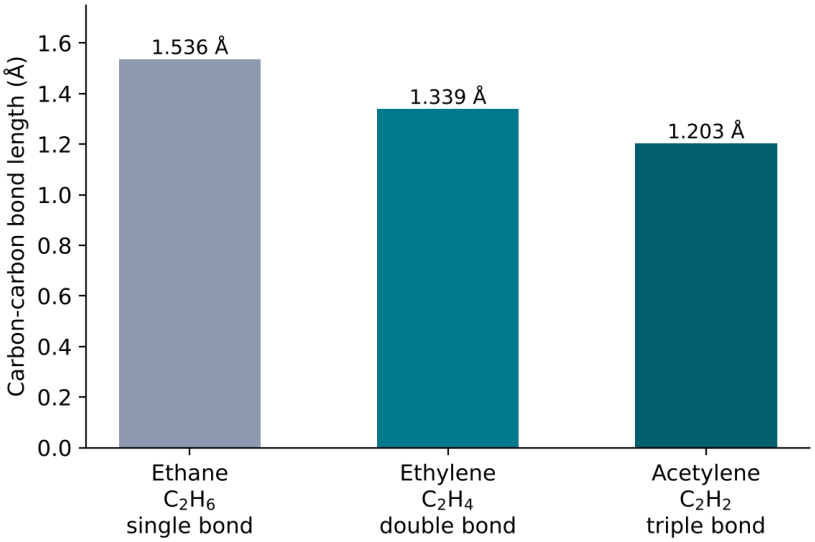


Figure 2.5.: Length of the carbon-carbon bond in ethane (single bond), ethylene (double bond) and acetylene (triple bond), from experimental data compiled in the NIST CCCBDB (National Institute of Standards and Technology 2022). The bond becomes shorter as the number of shared electron pairs increases.

2.3. Molecular shape and why carbon forms four bonds

2.3.1. Molecules have shapes

A Lewis structure shows which atoms are bonded, but not how the atoms are arranged in space. Molecules have definite three-dimensional shapes, which can be measured by spectroscopy and diffraction methods. Shape matters for everything that follows in this book: whether two molecules fit together, whether a chain can fold, and whether a base pair can form all depend on the angles between bonds.

The shape of a molecule is described by its bond lengths (Section 2.2) and its **bond angles**, the angles between two bonds that share an atom. For the three simplest hydrides of the second-row elements, the measured angles are 109.47° for H–C–H in methane, 106.67° for H–N–H in ammonia and 104.48° for H–O–H in water (National Institute of Standards and Technology 2022).

2.3.2. Electron pairs repel each other

These angles can be understood with a simple model, **valence-shell electron-pair repulsion** (VSEPR). Its premise is that the electron pairs in the valence shell of a

central atom, both bonding pairs and lone pairs, repel each other, and therefore arrange themselves around the atom as far apart as possible.

For four electron pairs, the arrangement that keeps them farthest apart points them towards the four corners of a regular **tetrahedron**, a solid with four triangular faces, with the central atom at its centre. The angle between any two of the four directions is

$$\theta = \arccos\left(-\tfrac{1}{3}\right) \approx 109.47^\circ$$

where arccos is the inverse cosine function. The value follows from geometry alone. Methane, with four identical C–H bonds and no lone pairs, is a regular tetrahedron, and its measured angle, 109.471° (National Institute of Standards and Technology 2022), agrees with the geometric value.

Ammonia and water also have four electron pairs around the central atom, but some of them are lone pairs: one in ammonia and two in water (Figure 2.4). The four pairs still point roughly towards the corners of a tetrahedron, but the shape of the molecule is described by the positions of the atoms only. Ammonia is therefore a **trigonal pyramid**, with nitrogen at the apex and the three hydrogen atoms forming the base, and water is **bent** (Figure 2.6). The angles in both molecules are smaller than the tetrahedral angle, and smaller in water than in ammonia. VSEPR attributes this to the lone pairs, which are held by only one nucleus, spread more widely around it and push the bonding pairs closer together: the more lone pairs, the smaller the angle.

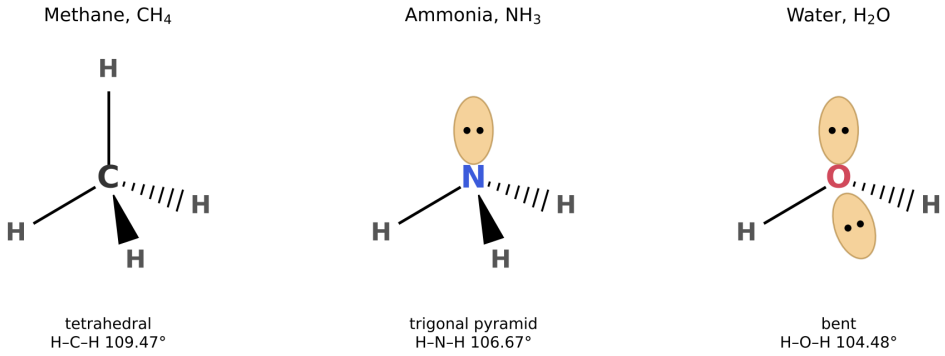


Figure 2.6.: Shapes of methane, ammonia and water, with the measured bond angles (National Institute of Standards and Technology 2022). In all three, four electron pairs around the central atom point approximately towards the corners of a tetrahedron. Lone pairs (shaded lobes) are not counted in the shape of the molecule, which is tetrahedral for methane, a trigonal pyramid for ammonia and bent for water. Wedge-shaped bonds point towards the viewer and dashed bonds away.

2.3.3. Hybrid orbitals

The ground-state configuration of carbon, $1s^2\ 2s^2\ 2p^2$, has only two unpaired electrons (Figure 2.2), yet carbon forms four bonds, and in methane all four are identical. Neither the spherical 2s orbital nor the three perpendicular 2p orbitals point towards the corners of a tetrahedron. These facts are reconciled by the concept of **hybridisation**: mathematical combinations of an atom's s and p orbitals that produce a new set of equivalent orbitals, called **hybrid orbitals**, pointing in the directions of the bonds.

- Combining the 2s orbital with all three 2p orbitals gives four equivalent **sp^3 hybrid orbitals**, pointing to the corners of a tetrahedron, 109.47° apart. One valence electron is placed in each, and each forms a bond. This describes carbon in methane and ethane, and the carbon atoms of the sugar ribose, as Chapter 4 will show.
- Combining the 2s orbital with two 2p orbitals gives three **sp^2 hybrid orbitals**, lying in a plane at 120° to each other. The remaining 2p orbital stands perpendicular to that plane. This describes carbon in ethylene: each carbon forms three bonds in the plane, and the side-by-side overlap of the two remaining p orbitals forms the second bond of the double bond. The measured angles in ethylene, 117.6° for H–C–H and 121.2° for H–C–C (National Institute of Standards and Technology 2022), are close to 120°, and the molecule is flat.
- Combining the 2s orbital with one 2p orbital gives two **sp hybrid orbitals**, pointing in opposite directions, 180° apart. The two remaining p orbitals form the second and third bonds of a triple bond. This describes carbon in acetylene and in hydrogen cyanide, both of which are linear: the measured H–C–C angle in acetylene and H–C–N angle in HCN are both 180° (National Institute of Standards and Technology 2022).

A bond formed by direct, head-on overlap of orbitals along the line between two nuclei is called a **sigma (σ) bond**. A bond formed by side-by-side overlap of two parallel p orbitals, above and below that line, is called a **pi (π) bond**. A single bond is a σ bond; a double bond consists of one σ and one π bond; a triple bond of one σ and two π bonds (Figure 2.7).

Hybridisation is a model, not a separate physical process: an atom does not first rearrange its orbitals and then bond. It is a way of describing the electron distribution of a bonded atom with orbitals that match the observed geometry. Within this book, its value is practical. From the number of bonds and lone pairs on an atom, it predicts the local shape of a molecule.

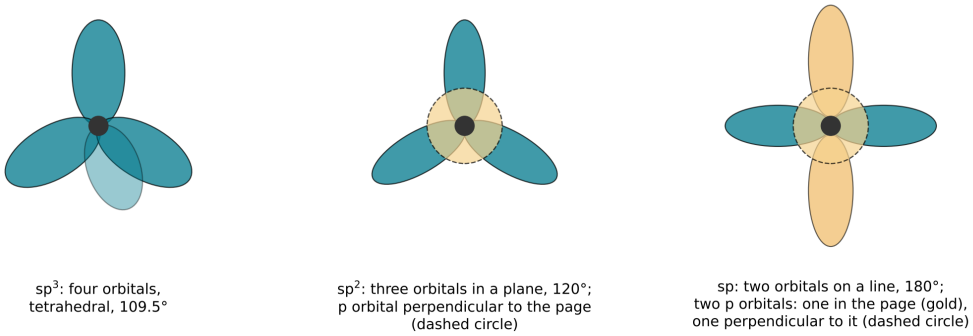


Figure 2.7.: Hybridisation of carbon. Left: four sp^3 orbitals pointing to the corners of a tetrahedron (109.5°). The pale lobe points behind the page. Centre: three sp^2 orbitals in the plane of the page (120°); the remaining p orbital is perpendicular to the page, drawn as a dashed circle. Right: two sp orbitals on a line (180°), with two remaining p orbitals perpendicular to the line and to each other, one in the page and one perpendicular to it. Schematic, not to scale.

Table 2.3.: Hybridisation and geometry. For counting electron pairs, a double or triple bond counts as one pair, because all of its shared electrons lie between the same two atoms.

Electron pairs around atom (bonds + lone pairs)	Hybridisation	Arrangement of pairs	Ideal angle	Examples
4	sp^3	tetrahedral	109.5°	CH_4, NH_3, H_2O
3	sp^2	trigonal planar	120°	C_2H_4
2	sp	linear	180°	C_2H_2, HCN

2.3.4. Consequence for this book

Carbon can form four single bonds in a tetrahedral arrangement, or combine single bonds with double or triple bonds in flat or linear arrangements. This versatility allows carbon atoms to form chains, rings and branched skeletons of almost unlimited variety. In RNA, both forms occur side by side: the carbon atoms of the sugar ribose are sp^3 and tetrahedral, whereas the atoms of the bases are sp^2 and form flat rings. The flatness of the bases is what allows them to stack on top of each other and to pair with each other, as Chapter 7 will show.

2.4. Electronegativity and polar bonds

2.4.1. Unequal sharing

In the hydrogen molecule, the two atoms are identical and share their electron pair equally. When two different atoms are bonded, this is generally not so: one atom attracts the shared electrons more strongly than the other, and the electron density of the bond is shifted towards it. The tendency of an atom in a bond to attract the shared electrons towards itself is called its **electronegativity**, usually written χ (the Greek letter chi).

Electronegativity is not measured directly, like a mass or an ionisation energy. It is a derived quantity, defined through a chosen procedure, and several scales exist. The most widely used is the one introduced by Linus Pauling.

Key experiment: Pauling's electronegativity scale

In 1932, Pauling observed that a bond between two different atoms A and B is usually stronger than the average of the A–A and B–B bonds (Pauling 1932). He attributed the extra strength to the unequal sharing of electrons, which adds an electrostatic attraction between the partly charged atoms, and used it to define a difference in electronegativity:

$$|\chi_{\mathrm{A}} - \chi_{\mathrm{B}}| = (\mathrm{eV})^{-1/2} \sqrt{E_{\mathrm{d}}(\mathrm{AB}) - \tfrac{1}{2}\left[E_{\mathrm{d}}(\mathrm{AA}) + E_{\mathrm{d}}(\mathrm{BB})\right]}$$

Here E_{d} is the dissociation energy of the bond indicated, the energy needed to break it, in electronvolts; the factor $(\mathrm{eV})^{-1/2}$ makes the result a pure number. The formula gives only differences, so one element must be assigned a reference value. Using this method, electronegativity values for many elements were later recalculated from improved thermochemical data by Allred (Allred 1961). These are the values used in this book.

2.4.2. Electronegativities of the RNA elements

Table 2.4 and Figure 2.8 give the Pauling electronegativities of the five elements of RNA and some others for comparison (Allred 1961).

Table 2.4.: Pauling electronegativities from thermochemical data (Allred 1961).

Element	χ (Pauling scale)	Element	χ (Pauling scale)
Hydrogen	2.20	Fluorine	3.98
Carbon	2.55	Sodium	0.93

Element	χ (Pauling scale)	Element	χ (Pauling scale)
Nitrogen	3.04	Magnesium	1.31
Oxygen	3.44	Sulfur	2.58
Phosphorus	2.19	Chlorine	3.16

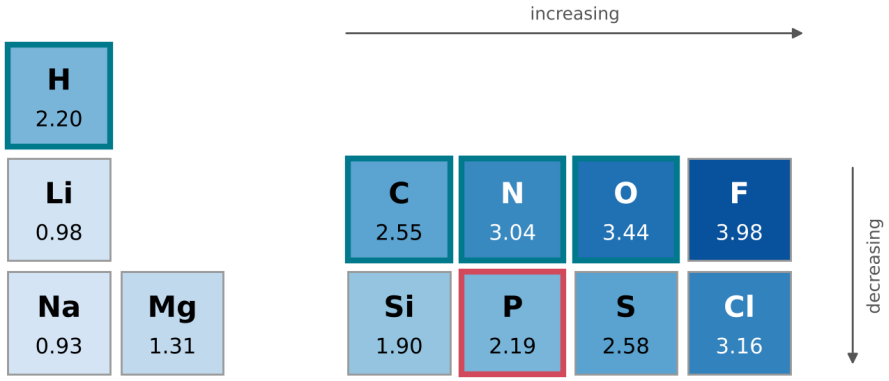


Figure 2.8.: Pauling electronegativities of selected elements (Allred 1961), arranged by position in the periodic table. The elements of RNA are coloured. Electronegativity increases from left to right across a row and decreases down a column.

Two trends are visible. Across a row of the periodic table, electronegativity increases from left to right, as the nuclear charge increases while the valence electrons remain in the same shell; this parallels the trend in ionisation energy (Figure 2.3). Down a column, electronegativity decreases, because the valence electrons lie in a shell farther from the nucleus: phosphorus (2.19) is much less electronegative than nitrogen (3.04), although both have five valence electrons. Oxygen and nitrogen are the most electronegative of the RNA elements; carbon, hydrogen and phosphorus lie close together, between 2.19 and 2.55.

2.4.3. Polar bonds

A covalent bond in which the electrons are shared unequally is a **polar bond**. Its more electronegative end carries a small excess of negative charge and the other end a small deficit. These are written $\delta-$ and $\delta+$ (δ, delta, meaning "a fraction of"), to distinguish them from the full charges of ions. The larger the difference in electronegativity, the more polar the bond. From Table 2.4, the differences for some bonds that occur in RNA and its precursors are:

Table 2.5.: Electronegativity differences $\Delta\chi$ for bonds between the RNA elements, calculated from Table 2.4.

Bond	$\Delta\chi$	Negative end
C–H	0.35	C
C–N	0.49	N
N–H	0.84	N
C–O	0.89	O
O–H	1.24	O
P–O	1.25	O

C–H bonds are nearly non-polar. O–H and P–O bonds are strongly polar, with the negative charge on oxygen. When the difference becomes very large, as between sodium (0.93) and chlorine (3.16), the electron is not shared but effectively transferred, and the result is a pair of ions held together by electrostatic attraction, an **ionic bond**. Covalent and ionic bonding are the two ends of a continuous range, and most bonds in biological molecules lie between them, at the covalent end.

2.4.4. Dipole moments

Separated charges $+q$ and $-q$ at a distance d form an electric **dipole**, whose strength is measured by the **dipole moment**

$$\mu = q\,d$$

in units of coulomb-metres (C m). For molecules the customary unit is the **debye** (D): 1 D = 3.335 64 $\times$ 10^{-30} C m, the dipole moment of charges of $\pm 10^{-10}$ statcoulomb (the unit of charge of the older centimetre-gram-second system) separated by 1 Å.

A molecule as a whole has a dipole moment if the dipoles of its polar bonds do not cancel. Water, with two polar O–H bonds at an angle of 104.48°, has a dipole moment of 1.857 D; ammonia has 1.476 D; and hydrogen cyanide 2.980 D (National Institute of Standards and Technology 2022). Methane has four slightly polar C–H bonds, but they point to the corners of a regular tetrahedron and cancel exactly: its dipole moment is zero (National Institute of Standards and Technology 2022). A molecule with a dipole moment is called a **polar molecule**. The shape of a molecule (Section 2.3) and the polarity of its bonds therefore together determine whether it is polar (Figure 2.9).

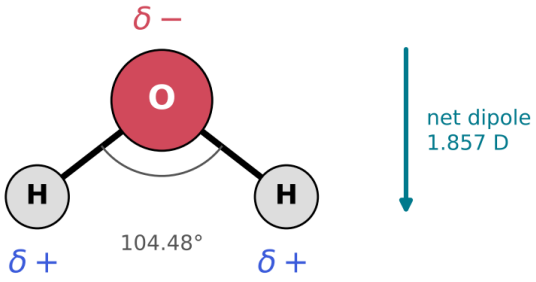


Figure 2.9.: Polarity of the water molecule. Each O–H bond is polar, with partial negative charge ($\delta-$) on oxygen and partial positive charge ($\delta+$) on hydrogen. Because the molecule is bent, the two bond dipoles add to a net dipole moment of 1.857 D (National Institute of Standards and Technology 2022), directed along the bisector of the H–O–H angle. The arrow is drawn from the negative towards the positive end, the convention used in physics; some chemistry texts draw it the opposite way.

2.4.5. Consequence for this book

Oxygen and nitrogen, the most electronegative elements of RNA, make every O–H, N–H, C–O and P–O bond polar. Polar bonds are the sites at which most chemical reactions in this book take place, and polar molecules attract each other and dissolve in water. The polarity of water itself, the subject of Section 2.6, underlies the hydrogen bonds that hold the two strands of a nucleic acid together.

2.5. Forces between molecules

2.5.1. Bonds within molecules and forces between them

Covalent bonds hold the atoms of a molecule together. Molecules also attract one another, by much weaker **intermolecular forces**. These forces do not join atoms into new molecules, and they are broken and re-formed continually at ordinary temperatures. They nevertheless determine whether a substance is a gas, a liquid or a solid, whether it dissolves in water, and, in RNA, whether two strands pair and how a single strand folds. All intermolecular forces are electrostatic in origin: they arise from the attraction between regions of positive and negative charge on neighbouring molecules.

2.5.2. Dipole-dipole and dispersion forces

Polar molecules (Section 2.4) attract each other because the positive end of one is drawn towards the negative end of another. This is a **dipole-dipole interaction**.

Non-polar molecules attract each other too, more weakly. The electron cloud of any atom or molecule fluctuates, so that at any instant it may have a small, temporary dipole. That dipole distorts the electron cloud of a neighbouring molecule, inducing a dipole in it, and the two temporary dipoles attract. This is the **dispersion force** (also called the London force). It acts between all atoms and molecules, and it is larger for larger molecules, whose electron clouds contain more electrons and are more easily distorted. Dipole-dipole and dispersion interactions, together with related interactions between permanent and induced dipoles, are collectively called **van der Waals forces**.

2.5.3. The hydrogen bond

One intermolecular interaction is of particular importance for water and for nucleic acids. The International Union of Pure and Applied Chemistry (IUPAC) defines it as follows: "The hydrogen bond is an attractive interaction between a hydrogen atom from a molecule or a molecular fragment X–H in which X is more electronegative than H, and an atom or a group of atoms in the same or a different molecule, in which there is evidence of bond formation" (Arunan et al. 2011).

A hydrogen bond is written X–H $\cdot\cdot\cdot$ Y, where the dotted line marks the hydrogen bond and the solid line the covalent bond. X is the **hydrogen-bond donor** (the atom to which hydrogen is covalently bonded) and Y is the **hydrogen-bond acceptor**. In the molecules of this book, X and Y are usually oxygen or nitrogen, the most electronegative of the RNA elements (Table 2.4), and Y carries a lone pair. According to the IUPAC recommendations, the forces involved include electrostatic attraction, together with contributions from charge transfer and dispersion; the X–H bond is polarised; and the X–H $\cdot\cdot\cdot$ Y angle is usually close to 180°, and should preferably be above 110°, the bond being stronger the closer the angle is to 180° (Arunan et al. 2011). Hydrogen bonds are therefore directional, unlike dispersion forces. The earlier IUPAC glossary of 1994 gave the associated energies as usually less than 20 to 25 kJ mol^{-1}, with a few exceptions involving fluorine (IUPAC, n.d.l) — weaker than covalent bonds, but strong enough to dominate the behaviour of water.

2.5.4. Evidence from boiling points

Intermolecular forces must be overcome when a liquid boils, so a higher boiling point indicates stronger forces between the molecules. Table 2.6 compares the normal boiling points (at a pressure of one standard atmosphere) of four small molecules containing hydrogen (National Institute of Standards and Technology, n.d.b).

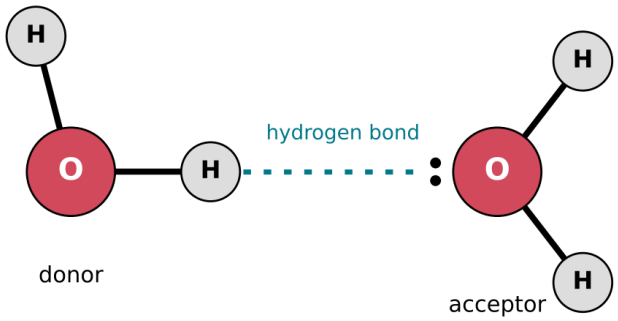
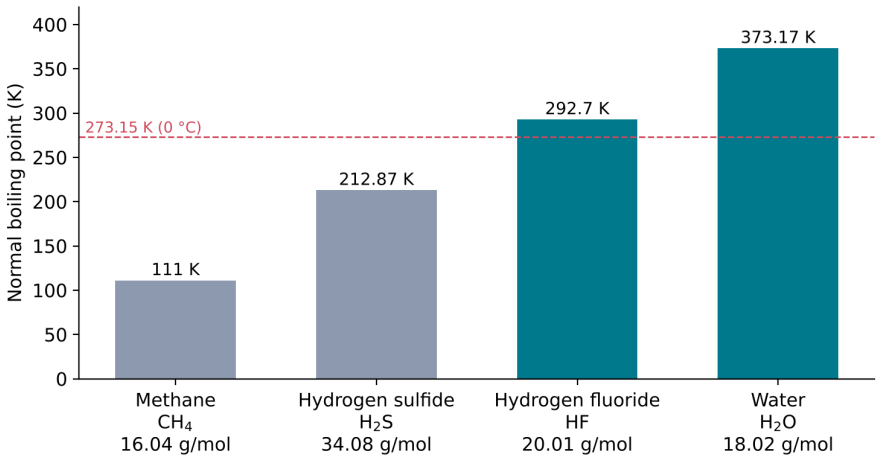


Figure 2.10.: A hydrogen bond between two water molecules. The O–H group of the left-hand molecule (donor) points towards a lone pair on the oxygen of the right-hand molecule (acceptor). The covalent O–H bonds are drawn as solid lines and the hydrogen bond as a dotted line. Schematic; distances not to scale.

Table 2.6.: Normal boiling points from the NIST Chemistry WebBook (National Institute of Standards and Technology, n.d.b). Molar masses are calculated from standard atomic weights; Celsius values are obtained by subtracting 273.15.

Substance	Formula	Molar mass (g mol^{-1})	Normal boiling point (K)	(°C)
Methane	CH_4	16.04	111 ± 2	−162
Water	H_2O	18.02	373.17 ± 0.04	100.02
Hydrogen fluoride	HF	20.01	292.7	19.6
Hydrogen sulfide	H_2S	34.08	212.87	−60.28

Methane and water have almost the same molar mass, and therefore similar dispersion forces, yet water boils 262 K higher. Methane is non-polar and forms no hydrogen bonds; water forms extensive hydrogen bonds through its two O–H groups and two lone pairs. The comparison with hydrogen sulfide is equally instructive. Sulfur lies directly below oxygen in the periodic table, and H_2S is nearly twice as heavy as H_2O, so its dispersion forces are stronger; yet it boils 160 K lower. Sulfur (electronegativity 2.58) is much less electronegative than oxygen (3.44), and the hydrogen bonds formed by H_2S are much weaker than those of water. Hydrogen fluoride, with the most electronegative element of all, also has an anomalously high boiling point for its size. Figure 2.11 summarises the comparison.



2.5.5. The hydrophobic effect

Non-polar molecules, such as hydrocarbons, dissolve poorly in water and tend to cluster together when placed in it; oil and water separate. This is the **hydrophobic effect** (from the Greek for "water-fearing"). It is not caused by a special attraction between the non-polar molecules. Rather, a non-polar molecule cannot form hydrogen bonds, and the water molecules around it are restricted in the hydrogen bonds they can form with each other. Clustering the non-polar molecules together reduces the area of contact with water and so the number of water molecules affected. Molecules or parts of molecules that interact favourably with water, such as polar and charged groups, are called **hydrophilic** ("water-loving"); non-polar ones are **hydrophobic**.

2.5.6. Consequence for this book

Two strands of RNA are held together by hydrogen bonds between their bases, and the stacking of the flat bases on each other, discussed in Chapter 7, also depends on interactions with the surrounding water. Hydrogen bonds and the hydrophobic effect will therefore return in Chapter 7, where they determine how RNA pairs and folds. They are equally central to the next section, because they give water its unusual properties as a solvent.

2.6. Water as a solvent

2.6.1. Why water matters here

Most of the chemical steps described in the rest of this book are thought to have taken place in water or at its surface: in ponds, oceans, hydrothermal fluids or thin films on minerals. Water is not an inert background. It dissolves the reactants, it carries electric charges between them, and it takes part in reactions as a reactant and a product. This section collects the properties of water that later chapters will rely on.

2.6.2. A small, polar molecule with four hydrogen-bonding sites

A water molecule is bent (H–O–H angle 104.48°) and polar (dipole moment 1.857 D) (National Institute of Standards and Technology 2022). Its oxygen atom carries two lone pairs and is bonded to two hydrogen atoms (Figure 2.4). Each molecule can therefore donate two hydrogen bonds, through its two O–H groups, and accept two, through its two lone pairs. In liquid water, each molecule is linked to its neighbours by a network of hydrogen bonds that are constantly broken and re-formed. This network is responsible for the high boiling point of water (Table 2.6), and also for its ability to dissolve other

polar molecules: a molecule that can form hydrogen bonds with water, such as an alcohol, a sugar or a nucleobase, can take a place in the network.

2.6.3. Dissolving ions

Water also dissolves ionic compounds, such as sodium chloride, which do not dissolve in non-polar liquids. Two properties of water are responsible.

First, water strongly weakens the electrostatic force between charges. In a vacuum, two charges q_1 and q_2 at a distance r exert on each other the force given by Coulomb's law,

$$F = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}$$

where ε_0 is the electric constant (the permittivity of vacuum). In a medium, the force is reduced by a factor ε_r, the **relative permittivity** (also called the dielectric constant) of the medium:

$$F = \frac{q_1 q_2}{4\pi\varepsilon_0\varepsilon_\mathrm{r} r^2}$$

For liquid water at 25 °C and 0.1 MPa, $\varepsilon_\mathrm{r} = 78.375$ (International Association for the Properties of Water and Steam 2011). The attraction between a sodium ion and a chloride ion in water is therefore almost eighty times weaker than at the same distance in a vacuum, and the ions can separate.

Second, water molecules arrange themselves around each dissolved ion, forming a **hydration shell**. Around a positive ion, such as Na^+, the water molecules turn their negative oxygen ends towards the ion; around a negative ion, such as Cl^-, they turn their positive hydrogen atoms towards it (Figure 2.12). The attraction between the ion and the oriented water dipoles stabilises the ion in solution.

2.6.4. Self-ionisation of water

Water itself ionises to a small extent. A proton (a hydrogen nucleus, H^+) passes from one water molecule to another:

$$2\,H_2O \rightleftharpoons H_3O^+ + OH^-$$

The products are the **hydronium ion**, H_3O^+, and the **hydroxide ion**, OH^-. The double arrow indicates that the reaction runs in both directions and reaches an equilibrium, in which forward and reverse reactions occur at equal rates. At equilibrium, the product of the concentrations of the two ions (strictly, of their activities, which are

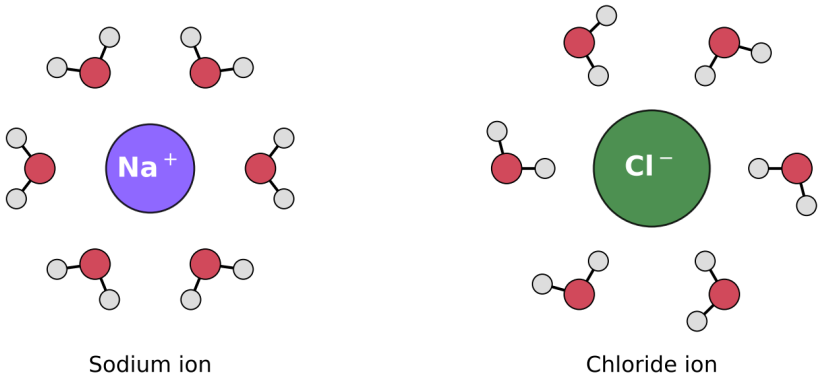


Figure 2.12.: Hydration of ions (schematic). Water molecules around a sodium ion (left) point their partially negative oxygen atoms towards the positive ion; those around a chloride ion (right) point their partially positive hydrogen atoms towards the negative ion. Only one layer of water molecules is shown, in two dimensions.

effective concentrations corrected for interactions between ions) is constant at a given temperature. This constant is the **ion product of water**, $\mathrm{K_w}$:

$$K_\mathrm{w} = a(\mathrm{H_3O^+})\, a(\mathrm{OH^-})$$

where a denotes activity, expressed relative to a standard concentration of 1 mol per litre. Because $\mathrm{K_w}$ is very small, it is usually given as $\mathrm{p}K_\mathrm{w} = -\log_{10} \mathrm{K_w}$. At 25 °C, $\mathrm{p}K_\mathrm{w} = 13.99$ (International Association for the Properties of Water and Steam 2024): in pure water at that temperature, the concentrations of hydronium and hydroxide ions are each about 10^{-7} mol per litre, and, since liquid water at 25 °C contains about 55 mol of water molecules per litre (its density is 997.05 kg $\mathrm{m^{-3}}$ (International Association for the Properties of Water and Steam 2011)), only about two water molecules in a thousand million are ionised at any moment.

2.6.5. Acids, bases and pH

An **acid** is a substance that donates a proton ($\mathrm{H^+}$) to another molecule, and a **base** is a substance that accepts one. In water, an acid raises the concentration of $\mathrm{H_3O^+}$ and a base raises that of $\mathrm{OH^-}$. The acidity of a solution is expressed by its **pH**:

$$\mathrm{pH} = -\log_{10} a(\mathrm{H_3O^+})$$

Since the two activities multiply to K_w, a solution is neutral, with equal amounts of H_3O^+ and OH^-, when pH = $pK_w/2$. A solution with a lower pH is acidic, and one with a higher pH is basic (or alkaline). Each unit of pH corresponds to a factor of ten in the hydronium-ion activity.

The familiar statement that neutral water has pH 7 is true only near 25 °C. The ion product of water depends strongly on temperature (Table 2.7). Neutral water has pH 7.48 at 0 °C and 6.13 at 100 °C. This matters for the origin of life, because the settings considered in Chapter 3 range from cold pools to hot springs and hydrothermal vents: a pH measured at one temperature cannot be compared with one at another without taking this into account.

Table 2.7.: Ion product of water at saturation pressure (0.1 MPa for liquid water below 100 °C), from the IAPWS release on the ionisation constant of water (International Association for the Properties of Water and Steam 2024). Neutral pH values are calculated as $pK_w/2$ and rounded.

Temperature (°C)	pK_w	pH of neutral water ($pK_w/2$)
0	14.95	7.48
25	13.99	7.00
100	12.25	6.13

2.6.6. Consequence for this book

Water dissolves polar molecules and ions, weakens the forces between charges by a factor of nearly eighty, and supplies and removes protons. Many of the molecules on the way to RNA, including the phosphate groups of nucleotides, carry charges whose state depends on pH, and many of the reactions that join them are reversible reactions with water itself. Chapter 5 will show that joining building blocks by removing water is one of the central difficulties of making RNA in water.

2.7. Energy in chemistry

2.7.1. Bonds store energy

Breaking a covalent bond always requires energy, and forming one always releases energy. The **bond enthalpy** of a bond is the energy needed to break it in a molecule in the gas phase, at constant pressure, per mole of bonds. (One **mole** is 6.022 140 76 × 10^{23} particles, the Avogadro number (National Institute of Standards and Technology, n.d.a); chemical energies are usually given in kilojoules per mole, kJ mol^{-1}.)

Bond enthalpies can be calculated from measured enthalpies of formation, defined below. The values for a few bonds, calculated in this way from CODATA reference data compiled in the NIST Chemistry WebBook (National Institute of Standards and Technology, n.d.b; Cox, Wagman, and Medvedev 1984), are:

Table 2.8.: Bond enthalpies at 298.15 K, calculated in this book from standard enthalpies of formation of the gaseous atoms and molecules (National Institute of Standards and Technology, n.d.b; Cox, Wagman, and Medvedev 1984). The C–H value uses the enthalpy of formation of methane from the NIST-JANAF tables (Chase 1998).

Bond	Molecule	Bond enthalpy (kJ mol^{-1})
H–H	H_2	436.0
C–H	CH_4 (average of four)	415.9
O–H	H_2O (average of two)	463.5
N≡N	N_2	945.4

The triple bond of N_2 is more than twice as strong as any of the single bonds in the table. This is why nitrogen gas, which makes up most of the Earth's atmosphere today, is chemically so unreactive, and why converting it into compounds that can enter the chemistry of RNA is difficult, a problem that returns in Chapter 3.

2.7.2. Enthalpy

The heat exchanged by a chemical reaction carried out at constant pressure equals the change in a quantity called **enthalpy**, H. A reaction that releases heat to its surroundings has a negative enthalpy change, $\Delta H < 0$, and is called **exothermic**; one that absorbs heat has $\Delta H > 0$ and is **endothermic**.

Enthalpy changes are tabulated as **standard enthalpies of formation**, $\Delta_f H^\circ$: the enthalpy change when one mole of a compound is formed from its elements in their most stable forms, with all substances in their standard states (here, a pressure of 1 bar and a temperature of 298.15 K). For liquid water, $\Delta_f H^\circ = -285.830$ kJ mol^{-1}; for water vapour, -241.826 kJ mol^{-1} (National Institute of Standards and Technology, n.d.b; Cox, Wagman, and Medvedev 1984). Because enthalpy depends only on the state of a system and not on how it got there, the enthalpy change of any reaction can be calculated as the sum of the enthalpies of formation of the products minus that of the reactants. This is **Hess's law**. For example, the bond enthalpy of O–H in Table 2.8 was obtained by calculating the enthalpy needed to break a mole of water vapour into isolated atoms:

$$\Delta H = 2\Delta_f H^\circ(\mathrm{H}) + \Delta_f H^\circ(\mathrm{O}) - \Delta_f H^\circ(\mathrm{H_2O,\ g}) = 2(217.998) + 249.18 - (-241.826) = 927.0\,\mathrm{kJ\,mol^{-1}}$$

and dividing by the two O–H bonds.

2.7.3. Entropy

Whether a process occurs does not depend on energy alone. Water evaporates from an open dish at room temperature although evaporation absorbs heat (its ΔH is positive, as calculated below). The second quantity that decides the direction of change is **entropy**, S. Entropy measures the number of microscopic arrangements of a system's particles and energy that are consistent with its macroscopic state. A gas, whose molecules move freely through a large volume, has a higher entropy than the liquid from which it formed: the standard molar entropy of water vapour is 188.835 J K^{-1} mol^{-1}, and that of liquid water 69.95 J K^{-1} mol^{-1} (National Institute of Standards and Technology, n.d.b; Cox, Wagman, and Medvedev 1984). The **second law of thermodynamics** states that in any spontaneous process, the total entropy of a system and its surroundings increases.

2.7.4. Gibbs energy

For a process at constant temperature and pressure, both criteria can be combined into a single quantity, the **Gibbs energy** (also called the Gibbs free energy):

$$G = H - TS, \qquad \Delta G = \Delta H - T\Delta S$$

where T is the absolute temperature. A process at constant temperature and pressure can occur spontaneously only if it lowers the Gibbs energy of the system, ΔG < 0. If ΔG > 0, the reverse process is the spontaneous one. If ΔG = 0, the system is at equilibrium.

The boiling of water illustrates the balance between the two terms. For the evaporation of one mole of water at 298.15 K, using the values above,

$$\Delta H = -241.826 - (-285.830) = +44.004\,\mathrm{kJ\,mol^{-1}}, \qquad \Delta S = 188.835 - 69.95 = +118.89\,\mathrm{J\,K^{-1}\,mol^{-1}}$$

$$\Delta G = 44.004 - 298.15 \times 0.11889 = +8.56\ \mathrm{kJ\,mol^{-1}}$$

ΔG is positive: at 25 °C and 1 bar, liquid water does not turn spontaneously into vapour at 1 bar. The enthalpy term favours the liquid, the entropy term favours the gas, and the entropy term grows with temperature. The two balance (ΔG = 0) when T = ΔH/ΔS = 44 004 / 118.89 ≈ 370 K. Figure 2.13 shows ΔG as a function of temperature. This simple estimate, which assumes that ΔH and ΔS do not change with temperature, comes within 3 K of the measured normal boiling point of water, 373.17 K (Table 2.6). (The

pressures differ slightly: the standard state is 1 bar, the normal boiling point refers to 1 atmosphere, 1.013 25 bar.)

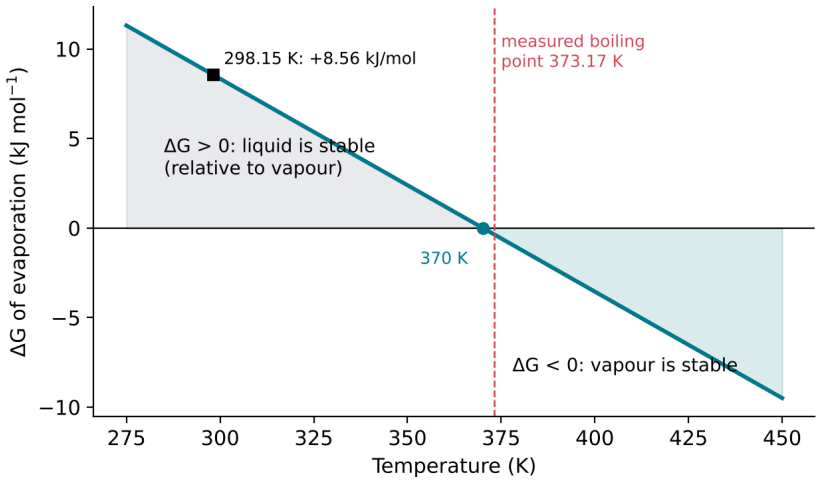


Figure 2.13.: Gibbs energy change for the evaporation of water as a function of temperature, $\Delta G = \Delta H - T\Delta S$, calculated with the standard values at 298.15 K ($\Delta H = 44.004$ kJ mol^{-1}, $\Delta S = 118.89$ J K^{-1} mol^{-1}) (National Institute of Standards and Technology, n.d.b; Cox, Wagman, and Medvedev 1984) and assuming both are constant. ΔG crosses zero near 370 K; the measured normal boiling point, 373.17 K, is marked.

2.7.5. Thermodynamically favourable is not the same as fast

The formation of liquid water from hydrogen and oxygen,

$$\mathrm{H_2(g)} + \tfrac{1}{2}\,\mathrm{O_2(g)} \rightarrow \mathrm{H_2O(l)}$$

has $\Delta H = -285.830$ kJ mol^{-1} and $\Delta S = 69.95 - 130.680 - \tfrac{1}{2}(205.152) = -163.31$ J K^{-1} mol^{-1} (National Institute of Standards and Technology, n.d.b; Cox, Wagman, and Medvedev 1984), so that at 298.15 K $\Delta G = -285.830 + 298.15 \times 0.16331 = -237.14$ kJ mol^{-1}. The reaction is strongly favourable. Yet a mixture of hydrogen and oxygen can be kept at room temperature without reacting appreciably until it is ignited. Thermodynamics states whether a reaction *can* occur; it says nothing about how fast it occurs. That is the subject of the next section, and the distinction is central to the origin of life: many reactions needed to build RNA are favourable but slow, and some that would destroy it, such as its reaction with water, are favourable as well.

2.7.6. Consequence for this book

Every step on the path to RNA must satisfy $\Delta G < 0$ under the conditions where it takes place, or must be coupled to another process that supplies Gibbs energy. Joining building blocks into a chain while releasing water is generally unfavourable in water, as Chapter 5 will discuss; how this obstacle may have been overcome is one of the central questions of the book.

2.8. Reaction rates

2.8.1. Rate and rate constant

Section 2.7 ended with a mixture of hydrogen and oxygen that does not react although the reaction is strongly favourable. The speed of a reaction is described by its **rate of reaction**: for a reaction a A + b B → p P + q Q, the rate is the rate of change of the concentration of any reactant or product, divided by its coefficient in the equation (IUPAC, n.d.o):

$$v = -\frac{1}{a}\frac{\mathrm{d[A]}}{\mathrm{d}t} = -\frac{1}{b}\frac{\mathrm{d[B]}}{\mathrm{d}t} = \frac{1}{p}\frac{\mathrm{d[P]}}{\mathrm{d}t} = \frac{1}{q}\frac{\mathrm{d[Q]}}{\mathrm{d}t}$$

where square brackets denote concentration and d/dt denotes the rate of change with time. The rate usually depends on the concentrations of the reactants. For many reactions it is proportional to a product of concentrations, for example $v = k[\mathrm{A}][\mathrm{B}]$, where the proportionality factor k is the **rate constant**. The rate constant does not depend on concentration, but it depends strongly on temperature.

2.8.2. Activation energy

For two molecules to react, existing bonds must be stretched and partly broken before new ones can form. The arrangement of atoms at the point of highest energy along this path is called the **transition state**. The energy that the reactants must acquire to reach it is the **activation energy**, E_a (Figure 2.14). Molecules gain this energy through collisions, and at any temperature only a fraction of them have enough. The activation energy is a barrier: it does not affect the overall Gibbs energy change of the reaction, but it determines how fast the reaction proceeds.

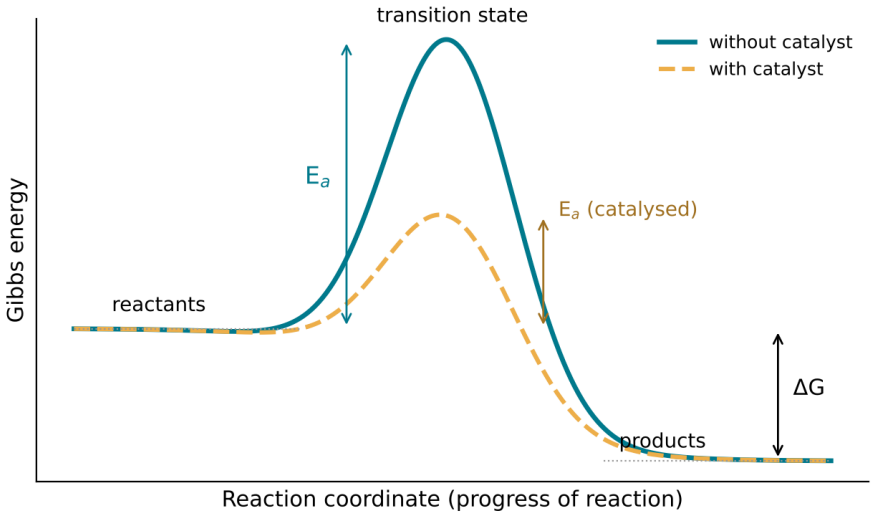


Figure 2.14.: Energy profile of a reaction (schematic). The reactants must pass over an energy barrier, the activation energy E_a, at the transition state before forming the products. A catalyst (dashed curve) provides an alternative path with a lower barrier. The difference in energy between reactants and products, and therefore the position of the equilibrium, is the same with and without the catalyst.

2.8.3. The Arrhenius equation

The dependence of a rate constant on temperature is described by the **Arrhenius equation** (IUPAC, n.d.d):

$$k = A\,\mathrm{e}^{-E_a/RT}$$

Here A is the pre-exponential factor, which has the same units as k; E_a is the activation energy in J mol^{-1}; T is the absolute temperature; and R is the molar gas constant, 8.314 462 618 J K^{-1} mol^{-1} (National Institute of Standards and Technology, n.d.a). In its original form, A and E_a are taken as independent of temperature (IUPAC, n.d.d). More generally, IUPAC defines the activation energy through the temperature dependence of the rate constant itself, $E_a = RT^2\,\mathrm{d}(\ln k)/\mathrm{d}T$ (IUPAC, n.d.i).

The exponential makes rates extremely sensitive to temperature and to the size of the barrier. Taking the ratio of the rate constants at two temperatures, A cancels:

$$\frac{k(T_2)}{k(T_1)} = \exp\left[\frac{E_a}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)\right]$$

For a reaction with E_a = 50 kJ mol^{-1}, warming from 25 °C (298.15 K) to 35 °C (308.15 K) increases the rate constant by a factor of 1.92, and warming to 100 °C by a factor of about 58. For E_a = 100 kJ mol^{-1}, the corresponding factors are 3.70 and about 3300 (Figure 2.15). The same equation shows that a reaction at 0 °C is slower than at 25 °C by a factor of 6.3 (for 50 kJ mol^{-1}) or 40 (for 100 kJ mol^{-1}). Temperature is therefore one of the most powerful controls on prebiotic chemistry: it speeds up the reactions that build molecules, and equally those that destroy them.

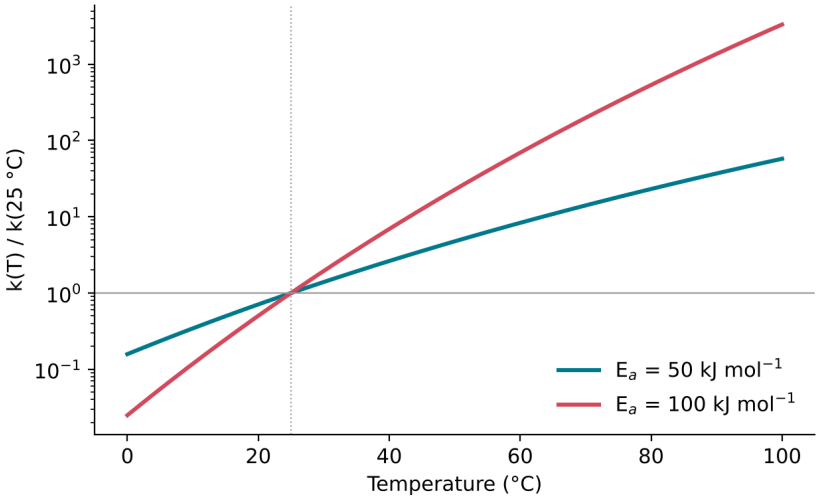


Figure 2.15.: Relative rate constant k(T)/k(298.15 K) calculated from the Arrhenius equation for activation energies of 50 and 100 kJ mol^{-1}, from 0 °C to 100 °C. The vertical axis is logarithmic. The higher the activation energy, the more strongly the rate depends on temperature.

2.8.4. Catalysts

A **catalyst** is a substance that increases the rate of a reaction without modifying the overall standard Gibbs energy change of the reaction (IUPAC, n.d.e). It does so by providing an alternative reaction path with a lower activation energy (Figure 2.14), and it is regenerated at the end of each cycle, so that a small amount can convert a large amount of reactant. Because a catalyst does not change ΔG, it speeds up the forward and reverse reactions alike and does not change the position of equilibrium; it only allows equilibrium to be reached sooner. The effect of lowering the barrier is large: by the Arrhenius equation, lowering E_a by 20 kJ mol^{-1} at 25 °C, with A unchanged, increases the rate constant by a factor of about 3200.

Catalysis by one of the products of a reaction is called **autocatalysis** (IUPAC, n.d.e). In an autocatalytic reaction, the more product there is, the faster more of it is made. A

molecule that catalyses the formation of copies of itself is the chemical core of replication, and autocatalysis will return in Section 8.6.

The catalysts of living cells are mostly proteins, called enzymes, but some are RNA molecules, called ribozymes. The discovery that RNA can act as a catalyst, as well as carry genetic information, is one of the foundations of the hypothesis that RNA preceded proteins; it is described in Chapter 7.

2.8.5. Consequence for this book

Whether a molecule accumulates on the early Earth depends on a competition between rates: the rate at which it is formed, and the rates at which it is destroyed, for example by reaction with water. Thermodynamics (Section 2.7) determines which direction is downhill; kinetics determines which processes are fast enough to matter. Both will be used throughout the remaining chapters.

Summary

- Chemical reactions rearrange electrons; nuclei are unchanged.
- Electrons in atoms occupy orbitals, grouped into shells (n = 1, 2, 3, …) and subshells (s, p, d, f). An s subshell has one orbital, p three, d five.
- By the Pauli exclusion principle, each orbital holds at most two electrons of opposite spin; shell n holds at most $2n^2$ electrons.
- The ground-state configurations of the RNA elements are H $1s^1$, C $1s^2\ 2s^2\ 2p^2$, N $1s^2\ 2s^2\ 2p^3$, O $1s^2\ 2s^2\ 2p^4$ and P [Ne] $3s^2\ 3p^3$.
- Valence electrons, in the outermost shell, take part in bonding: H 1, C 4, N 5, O 6, P 5.
- Measured ionisation energies confirm the shell structure: they peak at the noble gases and drop sharply at the start of each new shell.
- A covalent bond is a shared pair of electrons. By sharing, H reaches two electrons and C, N and O reach eight (the octet rule), so they usually form 1, 4, 3 and 2 bonds respectively.
- Lewis structures show bonds as lines and lone pairs as dots. Two or three shared pairs form double or triple bonds; higher bond order gives shorter bonds (C–C 1.536 Å, C=C 1.339 Å, C≡C 1.203 Å).
- Electron pairs around an atom repel each other (VSEPR). Four pairs point to the corners of a tetrahedron (109.47°): methane is tetrahedral, ammonia pyramidal (106.67°) and water bent (104.48°).
- Hybrid orbitals describe bonding geometry: sp^3 (tetrahedral), sp^2 (planar, 120°) and sp (linear, 180°). Double and triple bonds contain one σ bond plus one or two π bonds.

- Electronegativity measures how strongly an atom attracts shared electrons (Pauling scale: H 2.20, C 2.55, N 3.04, O 3.44, P 2.19). Unequal sharing makes bonds polar; O–H and P–O bonds are strongly polar, C–H bonds nearly non-polar.
- A molecule is polar if its bond dipoles do not cancel: water (1.857 D) and ammonia (1.476 D) are polar; methane, by symmetry, is not.
- Molecules attract each other through dipole-dipole and dispersion (van der Waals) forces, and, when an X–H group with electronegative X meets an electronegative acceptor, through hydrogen bonds X–H $\cdot\cdot\cdot$ Y, which are directional.
- Hydrogen bonds explain why water (373.17 K) boils far above methane (111 K) and hydrogen sulfide (212.87 K). Non-polar molecules cluster in water: the hydrophobic effect.
- Water dissolves polar molecules through hydrogen bonds and ions through its high relative permittivity (78.4 at 25 °C) and hydration shells.
- Water self-ionises: 2 $H_2O \rightleftharpoons H_3O^+ + OH^-$, with $pK_w = 13.99$ at 25 °C. $pH = -\log_{10} a(H_3O^+)$; neutral pH is $pK_w/2$, which is 7.00 at 25 °C but 7.48 at 0 °C and 6.13 at 100 °C.
- Bond enthalpies measure bond strength: H–H 436.0, C–H 415.9, O–H 463.5 and N≡N 945.4 kJ mol^{-1}. The very strong N≡N bond makes N_2 unreactive.
- A process at constant T and p is spontaneous only if $\Delta G = \Delta H - T\Delta S < 0$. ΔG predicts whether a reaction can occur, not how fast: $H_2 + \frac{1}{2}O_2 \rightarrow H_2O$ has $\Delta G = -237.14$ kJ mol^{-1} but does not proceed without ignition.
- Reaction rates depend on an activation energy barrier. The Arrhenius equation, $k = A\, e^{-E_a/RT}$, makes rates very sensitive to temperature: for $E_a = 100$ kJ mol^{-1}, warming from 25 °C to 100 °C speeds a reaction about 3300-fold.
- A catalyst lowers the activation energy without changing ΔG or the equilibrium. Autocatalysis, in which a product catalyses its own formation, is the chemical core of replication.

Exercises

1. How many electrons can the n = 4 shell hold? List its subshells and the number of electrons each can hold.
2. Write the full electron configuration (without the [Ne] abbreviation) of phosphorus and of sulfur, and state the number of valence electrons of each.
3. Draw box-and-arrow diagrams for the 2p electrons of carbon, nitrogen and oxygen, following Hund's rule.
4. Convert the first ionisation energy of hydrogen (13.598 eV) into joules, and into kilojoules per mole of atoms (Avogadro constant $6.022\,140\,76 \times 10^{23}$ mol^{-1}, exact (National Institute of Standards and Technology, n.d.a)).
5. Explain why the first ionisation energy of sodium is much lower than that of neon, although sodium has one more proton.

6. Nitrogen and phosphorus have the same number of valence electrons. Using Table 2.1, explain why the first ionisation energy of phosphorus is lower than that of nitrogen.
7. Draw Lewis structures for formaldehyde (CH_2O, with carbon as the central atom) and for carbon dioxide (CO_2). State the bond order of each carbon-oxygen bond.
8. Using the octet rule, predict the formula of the simplest compound of nitrogen with hydrogen and of oxygen with hydrogen, and check your answers against Figure 2.4.
9. By what percentage is the carbon-carbon bond in acetylene shorter than that in ethane?
10. Predict the shape of each molecule and the hybridisation of its central atom: formaldehyde (CH_2O), carbon dioxide (CO_2), and the ammonium ion (NH_4^+).
11. Explain why the H–O–H angle in water (104.48°) is smaller than the H–C–H angle in methane (109.47°), although both central atoms are surrounded by four electron pairs.
12. How many σ bonds and how many π bonds are there in a molecule of hydrogen cyanide?
13. Using Table 2.4, rank the bonds C–H, N–H, O–H, P–O and C–N in order of increasing polarity, and mark the $\delta-$ end of each.
14. Carbon dioxide (CO_2) is linear and contains two polar C=O bonds. Predict whether it has a dipole moment, and explain.
15. Express the dipole moment of water in C m. If the charges were $\pm e$ (the elementary charge), how far apart would they have to be to give this dipole moment? Compare your answer with the O–H bond length.
16. Identify the hydrogen-bond donors and acceptors in a molecule of ammonia, and draw a hydrogen bond between an ammonia molecule and a water molecule.
17. Using Table 2.6, explain why H_2S boils at a lower temperature than H_2O although its molar mass is almost twice as large.
18. Explain why methane dissolves poorly in water, using the idea of the hydrophobic effect.
19. By what factor is the electrostatic force between two ions at a fixed distance reduced when they are moved from a vacuum into water at 25 °C?
20. A solution at 25 °C has pH 5. Calculate the activity of H_3O^+ and of OH^-. Is the solution acidic or basic?
21. A hot spring at 100 °C has pH 6.5. Is it acidic, neutral or basic? Use Table 2.7.
22. Using the standard enthalpies of formation of the gaseous atoms in this section, calculate the bond enthalpy of N≡N, and compare it with that of H–H.
23. Using the data for water in this section, calculate ΔG for the evaporation of water at 350 K and at 390 K, assuming ΔH and ΔS are constant. At which temperature is evaporation spontaneous at 1 bar?
24. A reaction has $\Delta H = -20$ kJ mol^{-1} and $\Delta S = -100$ J K^{-1} mol^{-1}. Above what temperature does it stop being spontaneous?
25. A reaction has $E_a = 75$ kJ mol^{-1}. By what factor does its rate constant change between 25 °C and 50 °C?
26. Explain why a catalyst cannot make a reaction with $\Delta G > 0$ proceed sponta-

neously.

27. RNA is both formed and destroyed by reactions that are faster at higher temperature. Explain, using the Arrhenius equation, why the net effect of heating on the amount of RNA depends on the activation energies of the two processes.

3. Feedstock on the Early Earth

Learning objectives

After reading this chapter you should be able to:

1. Explain how radioactive decay is used to measure the ages of rocks and meteorites, and calculate an age from a measured ratio of daughter to parent atoms.
2. Describe the formation of the Earth and the Moon and the evidence for liquid water on the Earth during the Hadean eon.
3. Compare the models proposed for the early atmosphere, and explain why its composition matters for the synthesis of organic molecules.
4. Identify the main sources of small organic molecules on the early Earth, the key feedstock molecules of prebiotic chemistry, and the settings in which they may have accumulated.

3.1. Geological time and radiometric dating

3.1.1. The problem of measuring deep time

Chapter 1 gave the age of the oldest solids in the solar system as 4.567 to 4.568 billion years (Section 1.7). No observer was present to record this, and no physical record of a calendar survives. Ages of this kind are obtained from the decay of radioactive isotopes, which acts as a clock built into the minerals themselves. This section explains how such a clock works, which isotopes are used, and how the age of the Earth was determined.

Geological ages are expressed in years before the present. The year used in science is the **annum** (symbol a), and multiples are written with the usual SI prefixes: 1 ka = 10^3 a, 1 Ma = 10^6 a (a million years) and 1 Ga = 10^9 a (a billion years). These are the units used, for example, in the joint recommendations of IUPAC and IUGS on half-lives (Villa et al. 2022). Thus 4.567 Ga means 4.567 billion years ago.

3.1.2. The law of radioactive decay

Radioactive decay was introduced in Section 1.1 as the spontaneous transformation of an unstable nucleus. The decay of an individual nucleus is a random event: it is impossible to predict when a given nucleus will decay. For a large number of identical nuclei, however, the number that decay in a short time interval is proportional to the number present. This is expressed by the equation

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -\lambda N, \tag{3.1}$$

where N is the number of radioactive nuclei at time t, dN/dt is the rate at which this number changes, and λ is the **decay constant**, the probability per unit time that a nucleus decays. The decay constant is a property of the nuclide; it is not measurably affected by temperature, pressure or chemical bonding under geological conditions, which is why radioactive decay can serve as a clock. The minus sign indicates that N decreases.

The solution of Equation 3.1 is

$$N = N_0\,\mathrm{e}^{-\lambda t}, \tag{3.2}$$

where $\mathrm{N_0}$ is the number of radioactive nuclei at t = 0 and e = 2.718… is the base of natural logarithms. The **half-life**, $\mathrm{t_{1/2}}$, is the time after which half of the original nuclei remain. Setting $\mathrm{N = N_0/2}$ in Equation 3.2 gives

$$t_{1/2} = \frac{\ln 2}{\lambda} \approx \frac{0.693}{\lambda}. \tag{3.3}$$

After one half-life a fraction 1/2 of the original nuclei remains, after two half-lives 1/4, and after n half-lives $(1/2)^{\mathrm{n}}$. The decay curve has the same shape for every radioactive nuclide; only the time scale differs (Figure 3.1).

3.1.3. Parent, daughter and age

The decaying nuclide is called the **parent** and the nuclide it decays into the **daughter**. If a mineral contained $\mathrm{N_0}$ parent atoms and no daughter atoms when it formed, and if no atoms of either have entered or left it since, the number of daughter atoms produced by decay, D*, equals the number of parent atoms that have decayed:

$$D^* = N_0 - N = N\left(\mathrm{e}^{\lambda t} - 1\right), \tag{3.4}$$

where the second form follows from Equation 3.2 by writing $\mathrm{N_0 = N\ e^{\lambda t}}$. Solving for t gives the **radiometric age**:

$$t = \frac{1}{\lambda} \ln\left(1 + \frac{D^*}{N}\right). \tag{3.5}$$

The advantage of Equation 3.5 is that it requires only the present ratio of daughter to parent atoms, D*/N, which can be measured with a mass spectrometer, and the decay constant. The original number of parent atoms need not be known. For example, when $D^*/N = 1$, half of the parent has decayed, and $t = \ln 2/\lambda = t_{1/2}$.

Equation 3.5 rests on two assumptions, and much of the practice of geochronology consists of testing them.

- **Closed system.** No parent or daughter atoms have been added to or removed from the sample since the time being dated. Heating, chemical alteration or the action of water can break this condition.
- **Known initial daughter.** The daughter atoms measured must be corrected for any that were present when the mineral formed. D* in Equation 3.5 is the radiogenic part only, that is, the part produced by decay within the sample.

The event that a radiometric age records is the moment at which the clock was set: usually the crystallisation of a mineral, after which parent and daughter atoms could no longer move freely in or out of it.

3.1.4. Radioactive systems used for dating

To measure an age with good precision, a parent must be chosen whose half-life is not much shorter than the age to be measured; otherwise too little parent would remain. Carbon-14, with a half-life of 5,730 years (IUPAC Commission on Isotopic Abundances and Atomic Weights, n.d.a), is useful for archaeological samples but not for the early Earth: 4 billion years correspond to about 700 000 half-lives, after which no carbon-14 atom of the original material would remain. The systems used for ancient rocks and meteorites have half-lives of hundreds of millions to tens of billions of years (Table 3.1).

Table 3.1.: Radioactive systems used to date ancient rocks and meteorites. The uncertainties of the uranium, rubidium and samarium half-lives are given with a coverage factor of 2, that is, as approximately 95 percent confidence intervals (Villa et al. 2022). For potassium-40, the percentages give the fractions of decays leading to each daughter (Ruedas 2017).

Parent	Daughter	Half-life	Source
^{238}U	^{206}Pb	(4.4683 ± 0.0096) Ga	Villa et al. (2022)

Parent	Daughter	Half-life	Source
^{235}U	^{207}Pb	(703.81 ± 1.92) Ma	measurement of 1971, as reported by Villa et al. (2022)
^{40}K	^{40}Ar (10.72 %), ^{40}Ca (89.28 %)	1.248 Ga	Ruedas (2017)
^{87}Rb	^{87}Sr	(49.61 ± 0.16) Ga	Villa et al. (2015)
^{147}Sm	^{143}Nd	(106.25 ± 0.38) Ga	Villa et al. (2022)

The decay constants follow from Equation 3.3. For uranium-238, $\lambda = \ln 2/(4.4683 \times 10^9$ a) = 1.551 25 $\times$ 10^{-10} a^{-1}, the value recommended by the International Union of Pure and Applied Chemistry (IUPAC) and the International Union of Geological Sciences (IUGS) (Villa et al. 2022). For uranium-235, $\lambda = 9.8485 \times 10^{-10}$ a^{-1}.

Uranium-238 and uranium-235 do not decay directly to lead. Each starts a chain of alpha and beta decays through a series of shorter-lived intermediate nuclides; the chains end at the stable isotopes lead-206 and lead-207 respectively. Because the intermediate nuclides have half-lives very much shorter than those of the parents, each chain behaves, on time scales of millions of years and longer, as a single decay from uranium to lead with the half-life of the parent.

3.1.5. Two clocks in one mineral

The uranium-lead system is the most widely used for the oldest materials because a uranium-bearing mineral contains two independent clocks. Applying Equation 3.5 separately to the pairs ^{238}U-^{206}Pb and ^{235}U-^{207}Pb gives two ages. If the mineral has remained a closed system, the two ages agree, and such an age is called **concordant**. If lead has been lost, the two ages disagree, and the disagreement itself reveals that the closed-system assumption has failed.

The two parents decay at very different rates. Over the age of the solar system, 4.567 Ga, Equation 3.2 shows that 49 percent of the original uranium-238 survives, but only 1.1 percent of the uranium-235 (Figure 3.1). The ratio of radiogenic lead-207 to radiogenic lead-206 therefore changes with time in a known way. It depends only on the age and on the present ratio of the two uranium isotopes, which is nearly the same in all natural materials, and not on the amount of uranium in the sample. The age obtained from this ratio is called a **lead-lead age**. This is the method by which the CAIs of Chapter 1 were dated to 4.567 to 4.568 Ga (Pfalzner et al. 2015).

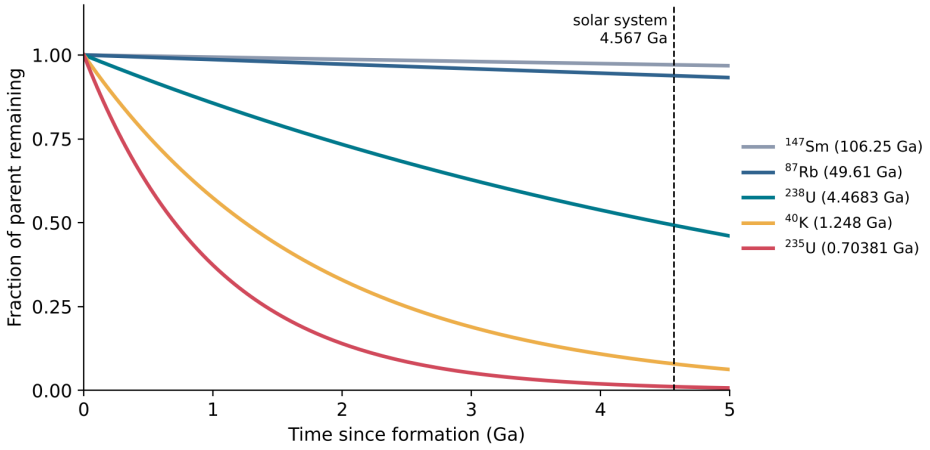


Figure 3.1.: Fraction of the original parent nuclei that remain as a function of time, from Equation 3.2 with the half-lives of Table 3.1. The vertical line marks the age of the solar system, 4.567 Ga. Rubidium-87 and samarium-147 decay so slowly that most of the original amount is still present; nearly all of the uranium-235 present at the formation of the solar system has decayed.

Key experiment: the age of the Earth from meteorite lead

In 1956 Clair Patterson published an age of 4.55 ± 0.07 billion years for the Earth and for meteorites, obtained from the isotopic composition of lead (Patterson 1956). The rocks of the Earth's surface have been melted and reworked many times, so no rock survives from the time of the Earth's formation. Patterson's approach was to assume that the Earth and meteorites formed at the same time from material with the same initial isotopic composition of lead, so that an age measured on meteorites would also be the age of the Earth.

The key sample was lead from the iron meteorite Canyon Diablo, which contains the iron sulfide mineral troilite (FeS) as well as iron-nickel metal and silicate minerals (Patterson 1956). Lead is concentrated in sulfide minerals, whereas uranium is not, so the lead in troilite has received very little radiogenic lead since the meteorite formed; it records the isotopic composition of lead at the beginning of the solar system. Meteorites with more uranium contain lead enriched in the radiogenic isotopes ^{206}Pb and ^{207}Pb. Comparing the two kinds of lead with the decay of uranium-238 and uranium-235 gives the time elapsed since they shared a common composition.

The value obtained, 4.55 billion years, is close to the lead-lead ages of CAIs, 4.567 to 4.568 Ga (Pfalzner et al. 2015). As described in Section 1.7, the Earth itself grew over the first tens of millions of years after CAI formation, so its age is a few tens of millions of years less than that of the solar system.

3.1.6. The geological time scale

The history of the Earth is divided into units of time. The largest are **eons**, which are subdivided into eras, periods and smaller units. The boundaries of these units are defined by the International Commission on Stratigraphy (ICS), which assigns them numerical ages in the International Chronostratigraphic Chart (Cohen et al. 2024). The four eons are listed in Table 3.2 and shown in Figure 3.2.

Table 3.2.: The eons of Earth history in the International Chronostratigraphic Chart, version 2024/12 (Cohen et al. 2024). The beginning of the Hadean is placed at the formation of the solar system.

Eon	Begins (Ma)	Ends (Ma)
Hadean	about 4567	4031 ± 3
Archean	4031 ± 3	2500
Proterozoic	2500	538.8 ± 0.6
Phanerozoic	538.8 ± 0.6	present

The **Hadean** eon is the first part of Earth history, from the formation of the solar system to 4031 ± 3 Ma (Cohen et al. 2024). It is followed by the **Archean** eon, which lasted until 2500 Ma. The events of interest to this book, the formation of the oceans and atmosphere and the chemistry that preceded life, belong to the Hadean and the early Archean. These eons are also those for which the geological record is least complete, which is why the evidence considered in the rest of this chapter comes largely from single mineral grains, from meteorites, and from models.

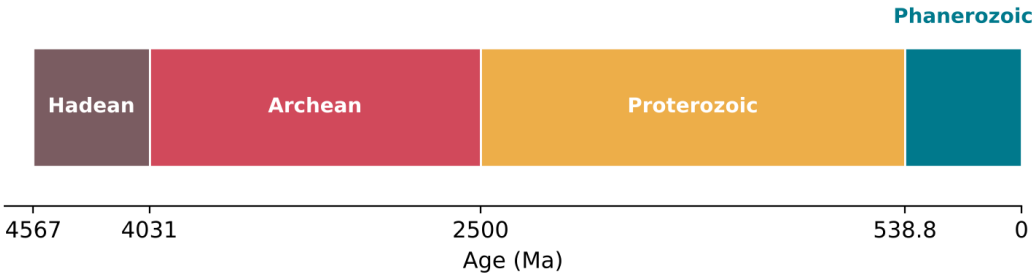


Figure 3.2.: The four eons of Earth history, with boundary ages from the International Chronostratigraphic Chart, version 2024/12 (Cohen et al. 2024). The Hadean and Archean together make up nearly half of the age of the Earth. The time axis runs from the formation of the solar system on the left to the present on the right.

3.2. The formation of the Earth and Moon; the Hadean eon

3.2.1. The Earth and its Moon

The Earth is the largest of the four rocky planets of the solar system. Its mass is 5.9722 $\times$ 10^{24} kg, its mean radius 6371 km and its mean density 5513 kg m^{-3} (NASA Space Science Data Coordinated Archive 2024a). As described in Section 1.7, it grew by the collision and merging of smaller bodies over at least 38 million years, and probably as long as 120 million years, after the formation of the first solids (Pfalzner et al. 2015).

The Moon is unusually large in relation to its planet. Its mass is 7.346 $\times$ 10^{22} kg, 1.23 percent of the mass of the Earth, and its mean radius is 1737.4 km, 27 percent of the Earth's (NASA Space Science Data Coordinated Archive 2024b). Its mean density, 3344 kg m^{-3}, is much lower than that of the Earth (NASA Space Science Data Coordinated Archive 2024b). Because iron is much denser than rock, this low density shows that the Moon contains proportionally far less metallic iron than the Earth. The mean distance between the two bodies is 384 400 km (NASA Space Science Data Coordinated Archive 2024b).

3.2.2. The giant impact

The origin of the Moon is explained by the **giant impact hypothesis**. In its most widely held form, a planet about the size of Mars collided with the proto-Earth when the latter had grown to approximately 90 percent of its present mass (K. Zahnle et al. 2007). The impact threw debris into orbit around the Earth. This debris rapidly accreted to form the Moon, which was almost completely molten when it formed (Barboni et al. 2017).

Two lines of evidence support this model.

- **Isotopic similarity.** Different bodies of the solar system have slightly different proportions of the three stable isotopes of oxygen, ^{16}O, ^{17}O and ^{18}O, so that the oxygen isotope composition of a rock identifies the reservoir from which it came. Measurements on lunar and terrestrial rocks show that the Earth and Moon are indistinguishable in this respect: the difference in a measure of the ^{17}O excess is -1 ± 5 parts per million (two standard errors) (Young et al. 2016). This indicates that the material of the impactor and of the proto-Earth was thoroughly mixed during the impact (Young et al. 2016).
- **Melting.** The formation of the Moon from hot debris implies that it began as a body covered by a global ocean of molten rock, the lunar magma ocean. As this ocean crystallised, dense minerals sank and a buoyant crust rich in the mineral plagioclase floated to the top; this crust forms the pale highlands of the Moon (Barboni et al. 2017).

The time of the impact has been debated: some estimates placed it within about 100 million years of the formation of the solar system (4.47 to 4.45 Ga), and others between 4.42 and 4.35 Ga, 150 to 200 million years after it (Barboni et al. 2017).

Key experiment: dating the Moon with zircons from Apollo 14

Barboni and co-workers dated small fragments of the mineral zircon ($ZrSiO_4$) from rock samples collected by the Apollo 14 mission (Barboni et al. 2017). Zircon contains uranium and is widely used for uranium-lead dating; it is discussed further in Section 3.3. The fragments gave precise, concordant uranium-lead ages.
The researchers also measured the isotopes of hafnium in the same fragments. Hafnium-176 is produced by the radioactive decay of lutetium-176, so the ratio $^{176}Hf/^{177}Hf$ in the solar system has increased with time; a low initial ratio recorded in a zircon indicates that its source material was separated early from the lutetium that would have raised it. The ratios were very low, near the initial value of the solar system (Barboni et al. 2017). Together, the two isotope systems showed that the Moon had differentiated, that is, its magma ocean had largely solidified and separated into layers, by 4.51 billion years ago (Barboni et al. 2017). The four oldest model ages average 4.51 ± 0.01 billion years (one standard deviation) (Barboni et al. 2017).
Because the Moon must have formed before it solidified, this is a minimum age. The authors concluded that the giant impact, and hence the formation of the Earth-Moon system, took place within the first approximately 60 million years of solar-system history, with an uncertainty of about 10 million years (Barboni et al. 2017).

3.2.3. After the impact: from magma ocean to water ocean

The energy released by the giant impact was sufficient to raise the temperature of the Earth by thousands of degrees (K. Zahnle et al. 2007). Zahnle and co-workers reconstructed the sequence of events that followed (K. Zahnle et al. 2007):

1. The Earth was surrounded by a hot atmosphere of vaporised silicate rock, which cooled and condensed over about 1000 years.
2. As it cooled, the Earth released volatile substances, including water and carbon dioxide, from its interior into the atmosphere. Its surface remained a **magma ocean**, a global layer of molten rock, for about another 2 million years. Tidal heating by the newly formed Moon was a major source of energy for this magma ocean.
3. Once the surface had solidified, heat flowing from the interior no longer had a significant effect on the climate. Water vapour in the atmosphere could then condense to form oceans. What remained above them was an atmosphere of about 100 bar of carbon dioxide, at a surface temperature of about 500 K.

4. The further cooling of the surface was controlled by how quickly carbon dioxide was removed from the atmosphere, by reaction with rocks to form carbonate minerals and the transport of these carbonates into the interior. If this removal was efficient, it could have taken as little as 10 million years; if it was inefficient, the surface could have remained near 500 K for many tens of millions of years.

Here the **bar** is a unit of pressure equal to 10^5 Pa, close to the average atmospheric pressure at sea level today. The conclusion of this reconstruction is that the Earth could have had liquid water oceans and conditions suitable for life as early as 10 to 20 million years after the Moon-forming impact, if carbon dioxide was removed efficiently (K. Zahnle et al. 2007). The evidence that liquid water was indeed present early in the Hadean is the subject of Section 3.3.

3.2.4. The faint young Sun

The Sun has not always been as bright as it is now. The gradual brightening of a star while it burns hydrogen in its core is a consequence of basic physics and of the way thermonuclear reaction rates depend on density, temperature and composition (Feulner 2012). According to standard solar models, the total power output of the Sun, its **luminosity**, was about 30 percent lower 4.57 billion years ago than at present (Feulner 2012). Its evolution can be approximated by

$$\frac{L(t)}{L_\odot} = \frac{1}{1 + \frac{2}{5}\left(1 - \frac{t}{t_\odot}\right)}, \tag{3.6}$$

where L(t) is the luminosity at time t after the Sun began to burn hydrogen, $L_\odot = 3.85 \times 10^{26}$ W is the present luminosity and $t_\odot = 4.57$ Ga is the age of the Sun (Feulner 2012). At t = 0, Equation 3.6 gives 1/1.4 = 0.714. At the end of the Hadean, about 4.0 Ga (t = 0.57 Ga), the Sun emitted 74 percent of its present luminosity (Figure 3.3).

Less sunlight implies a colder surface. Without the addition of potent greenhouse gases, such as carbon dioxide, the early Earth should have been frozen over at most times and places (K. Zahnle et al. 2007). The difficulty of reconciling a fainter Sun with evidence for liquid water on the early Earth is known as the **faint young Sun problem** (Feulner 2012). The composition of the early atmosphere is discussed in Section 3.4.

3.2.5. Impacts after the Moon

The giant impact was the last of the major collisions that built the Earth, but smaller bodies continued to strike the Earth and Moon. The surfaces of the Moon, Mars and Mercury are heavily cratered, which shows that the rocky planets were struck by an intense bombardment during their first billion years or more (Bottke and Norman 2017).

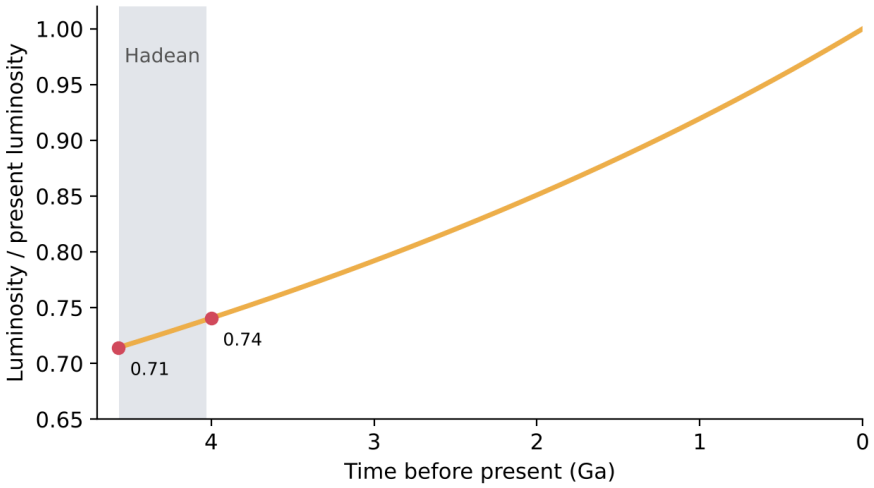


Figure 3.3.: Luminosity of the Sun relative to its present value, from Equation 3.6, as a function of time before the present. The shaded band marks the Hadean eon.

On the Earth, whose surface has been continually reworked, this record has been largely erased, so the history of the bombardment is read mainly from the Moon and from meteorites.

The **Late Heavy Bombardment** refers to impacts that occurred after the outer layers of the planets had become rigid enough to preserve craters and basins (Bottke and Norman 2017). Its timing has been controversial. Ages of rocks melted by impacts on the Moon, and of shock events recorded in meteorites, point to a distinct episode of elevated impact rate between about 4.2 to 4.0 Ga and 3.5 Ga, preceded by a relatively quiet interval between about 4.4 Ga and 4.2 to 4.0 Ga (Bottke and Norman 2017). Layers of impact debris in ancient terrestrial rocks show that a long tail of impacts on the Earth continued until about 2.5 to 2.0 Ga (Bottke and Norman 2017). Impacts are relevant to the origin of life in two opposite ways: large impacts could heat and sterilise the surface, whereas impacts also delivered material and energy that may have driven prebiotic chemistry (Section 3.5).

Figure 3.4 summarises the events of the first few hundred million years.

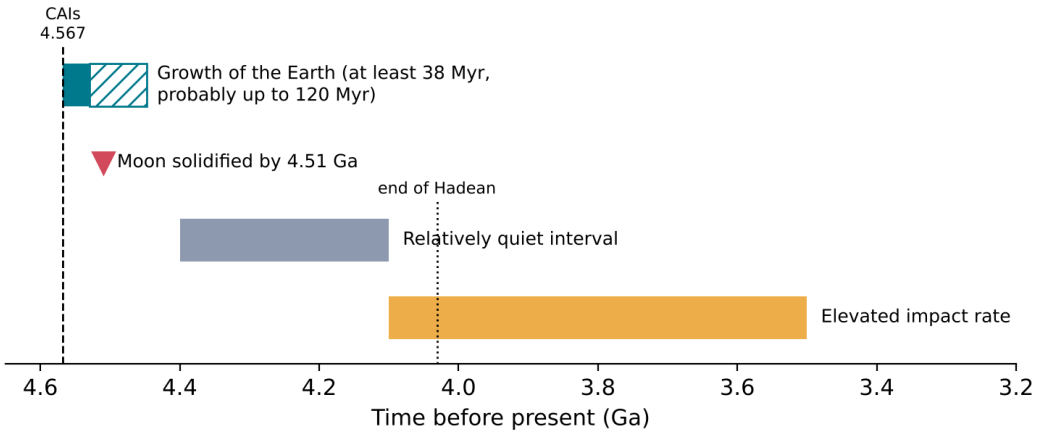


Figure 3.4.: Events of the Hadean eon and the early Archean. The bar for the growth of the Earth is solid up to the firm lower limit of 38 million years after CAI formation and hatched up to the probable end near 120 million years (Pfalzner et al. 2015). The triangle marks the time by which the Moon had solidified, 4.51 Ga (Barboni et al. 2017); the giant impact occurred before this. The relatively quiet interval and the episode of elevated impact rate follow Bottke and Norman (2017); where the source gives a range for a boundary (4.2 to 4.0 Ga), the bars meet at its midpoint. The end of the Hadean is at 4031 ± 3 Ma (Cohen et al. 2024).

3.3. The earliest oceans: evidence from zircons

3.3.1. A record without rocks

No rock is known to have survived from the first 500 million years of Earth history. The oldest known components of a rock body are those of the Acasta Gneiss of northwestern Canada, with an age of 4030 million years (Wilde et al. 2001), close to the end of the Hadean eon (Table 3.2). Information about the Hadean Earth itself therefore comes from individual mineral grains that are older than any surviving rock. Nearly all of these grains are crystals of zircon.

Zircon is a mineral of composition $ZrSiO_4$ (zirconium silicate). It crystallises from silicate melts, including the melts that solidify to form granite (Wilde et al. 2001). Three properties make it the principal witness to the Hadean.

- It contains trace amounts of uranium, so each crystal can be dated individually by the uranium-lead method (Section 3.1).
- It is hard and chemically resistant. When the rock in which it grew is weathered and eroded, zircon crystals survive as grains of sediment, and they can survive later burial, heating and deformation.
- It preserves inside its crystal structure, and in small inclusions of other minerals, chemical information about the melt from which it crystallised (Harrison 2009).

A grain that has been eroded from its original rock and incorporated into a sediment is called **detrital**. A detrital zircon may therefore be much older than the sedimentary rock in which it is found: its age records the crystallisation of its parent rock, which may have been destroyed long ago.

3.3.2. Oxygen isotopes and the delta notation

The oxygen in a mineral consists of the stable isotopes ^{16}O, ^{17}O and ^{18}O. Their proportions vary slightly from one material to another, because physical and chemical processes separate isotopes of different mass to a small degree. This separation is called **isotope fractionation**. Because the variations are small, they are expressed relative to a standard, using the **delta notation**:

$$\delta^{18}O = \left(\frac{R_{\text{sample}}}{R_{\text{standard}}} - 1 \right) \times 1000, \tag{3.7}$$

where R is the atom ratio $^{18}O/^{16}O$ in the sample and in the standard. The factor 1000 expresses the result in parts per thousand, or **per mil** (symbol ‰). For oxygen the standard is Vienna Standard Mean Ocean Water (VSMOW), the same international reference for ocean water that was used for the isotopes of hydrogen in Section 1.1. A

sample with $\delta^{18}O = +5$ ‰ contains 0.5 percent more ^{18}O, relative to ^{16}O, than the standard.

Fractionation of oxygen isotopes between minerals and water is larger at low temperatures than at high temperatures. Rocks that have reacted with liquid water near the Earth's surface, for example in weathering, therefore become enriched in ^{18}O. If such rocks are later buried and melted, the melt inherits their high $\delta^{18}O$ value, and so do zircons that crystallise from it.

Key experiment: the Jack Hills zircons

The Jack Hills lie in the Narryer Gneiss Terrane of the Yilgarn Craton, Western Australia. Metamorphosed sedimentary rocks there contain detrital zircons that had previously been dated to as old as about 4276 million years (Wilde et al. 2001).

In 2001, Wilde, Valley, Peck and Graham reported a detailed micro-analytical study of Jack Hills zircons, measuring uranium-lead ages, rare earth elements and oxygen isotope ratios within single crystals (Wilde et al. 2001). One grain gave an age of 4404 ± 8 million years, about 130 million years older than any previously identified on Earth (Wilde et al. 2001). It was formed some 160 million years after the first solids of the solar system (Section 1.7), and about 100 million years after the Moon had solidified (Section 3.2).

The grain was zoned: its composition varied from core to rim, with $\delta^{18}O$ values from 7.4 to 5.0 ‰, indicating that it grew from a melt whose composition was changing (Wilde et al. 2001). Its chemistry, its high $\delta^{18}O$ values and tiny inclusions of quartz (SiO_2) are consistent with growth from a granitic melt with $\delta^{18}O$ between 8.5 and 9.5 ‰ (Wilde et al. 2001). Values this high indicate that the melt incorporated material from the Earth's surface that had interacted with liquid water at low temperature (Wilde et al. 2001).

The authors concluded that this zircon is the earliest evidence for continental crust and oceans on the Earth (Wilde et al. 2001): by 4.4 Ga, less than 200 million years after the formation of the solar system, liquid water existed at the surface.

The single grain of 2001 has since been placed in a wider context. Studies of many Jack Hills zircons older than 4 Ga show that they crystallised at relatively low temperatures, that some are enriched in ^{18}O, and that they contain mineral inclusions similar to those formed by modern crustal processes (Harrison 2009). These observations have been interpreted as evidence of an early hydrosphere, and of early crust rich in silica that was produced and then weathered in the presence of abundant water (Harrison 2009). Together with the reconstruction in Section 3.2, they support the view that the Hadean Earth, after its first few tens of millions of years, had liquid water oceans for much of its history.

3.3.3. Carbon in a Hadean zircon

Zircons can also preserve other minerals as inclusions. Bell and co-workers examined more than 10 000 Jack Hills zircons and found one, older than 3.8 Ga, that contains inclusions of graphite, a crystalline form of carbon (Bell et al. 2015). The zircon gave a concordant uranium-lead age of 4.10 ± 0.01 Ga (Bell et al. 2015). Images made by transmission X-ray microscopy showed that the host crystal was free of cracks, so the graphite was interpreted as having been enclosed when the zircon grew, not introduced later (Bell et al. 2015).

Carbon has two stable isotopes, ^{12}C and ^{13}C (Section 1.1). Their ratio is expressed as $\delta^{13}C$, defined as in Equation 3.7 with the ratio $^{13}C/^{12}C$ and a carbonate reference standard (PDB). Living organisms take up ^{12}C slightly more readily than ^{13}C, so carbon that has passed through organisms tends to have negative $\delta^{13}C$ values. The graphite inclusions have $\delta^{13}C = -24 \pm 5$ ‰, a value consistent with a biological origin (Bell et al. 2015). If this interpretation is correct, a biosphere existed on the Earth by 4.1 Ga, about 300 million years earlier than had been proposed (Bell et al. 2015).

This result must be weighed with care. It rests on inclusions in a single grain, and a value of $\delta^{13}C$ consistent with biology does not by itself exclude a non-biological origin; the authors themselves describe the carbon as potentially biogenic (Bell et al. 2015). For comparison, the oldest microscopic fossils are about 3.5 billion years old, and chemical traces of life in rocks are argued to extend to about 3.8 Ga (Bell et al. 2015). Figure 3.5 shows these dates together.

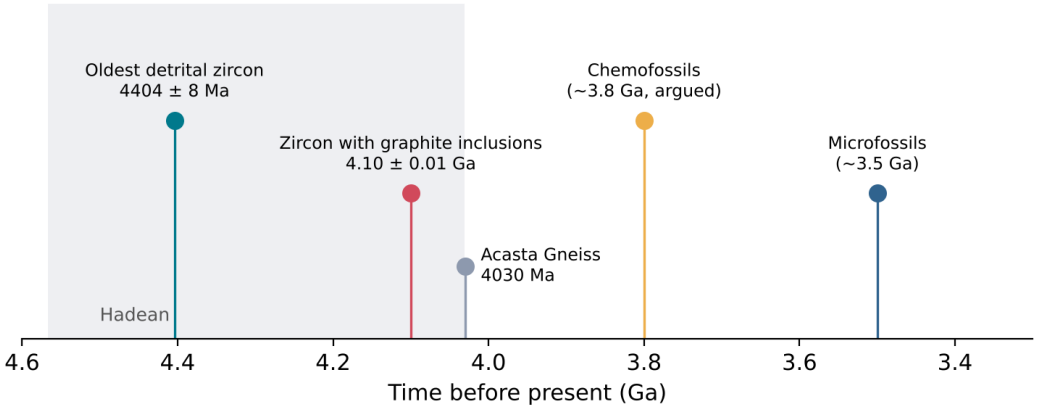


Figure 3.5.: The earliest record of the Earth and of possible life. Shown are the oldest detrital zircon (Wilde et al. 2001); the zircon containing graphite with $\delta^{13}C = -24 \pm 5$ ‰ (Bell et al. 2015); the oldest components of the Acasta Gneiss, the oldest known rock (Wilde et al. 2001); and the approximate ages to which the chemical record of life (chemofossils) and the fossil record of microscopic organisms extend (Bell et al. 2015).

3.4. The early atmosphere: reducing or neutral?

3.4.1. Why the atmosphere matters

The composition of the early atmosphere determines which small organic molecules could be made in it. Experiments have shown that an electric discharge, acting on a mixture of water, methane and ammonia (or nitrogen), produces several biologically important organic compounds, including amino acids (Cleaves et al. 2008). Such experiments are discussed in Chapter 4. Their success depends strongly on the composition of the gas mixture, and this composition is the subject of this section.

3.4.2. Oxidation and reduction

The relevant property of the atmosphere is its oxidation state. **Oxidation** is the loss of electrons by an atom, ion or molecule; **reduction** is the gain of electrons. The two always occur together, in a **redox reaction**, because the electrons lost by one species are gained by another. For atoms in molecules, the bookkeeping is done with **oxidation numbers**: in a covalent bond, the shared electrons are assigned formally to the more electronegative atom (Section 2.4), and the oxidation number is the charge that each atom would carry if this assignment were real. Hydrogen is normally assigned +1 and oxygen −2.

With these rules, the carbon atom in methane (CH_4) has oxidation number −4, in carbon monoxide (CO) and hydrogen cyanide (HCN) +2, and in carbon dioxide (CO_2) +4. Nitrogen has −3 in ammonia (NH_3) and 0 in N_2 (Figure 3.6). An atmosphere is called **reducing** when carbon and nitrogen are present mainly in their low oxidation states, as CH_4 and NH_3, together with H_2. It is called **neutral** (or weakly reducing) when they are present mainly as CO_2 and N_2, with water but without much H_2 or free oxygen. The present atmosphere, which contains free O_2, is **oxidising**.

3.4.3. Gases from the interior

The earliest atmosphere was produced largely by the release of gases from the Earth's interior. The elements available were hydrogen, carbon, nitrogen, oxygen and sulfur, and the molecules they form in volcanic gases depend on how oxidising the source magma is (Trail, Watson, and Tailby 2011). This is measured by the **oxygen fugacity** of the magma: the effective pressure of O_2 in equilibrium with it, which is extremely small. Geologists describe it relative to reference **buffers**, assemblages of minerals that fix the oxygen fugacity at a known value at each temperature. Two buffers bracket the relevant range (Trail, Watson, and Tailby 2011):

- the **iron-wüstite buffer**, defined by the coexistence of metallic iron and wüstite (FeO), which is strongly reducing;

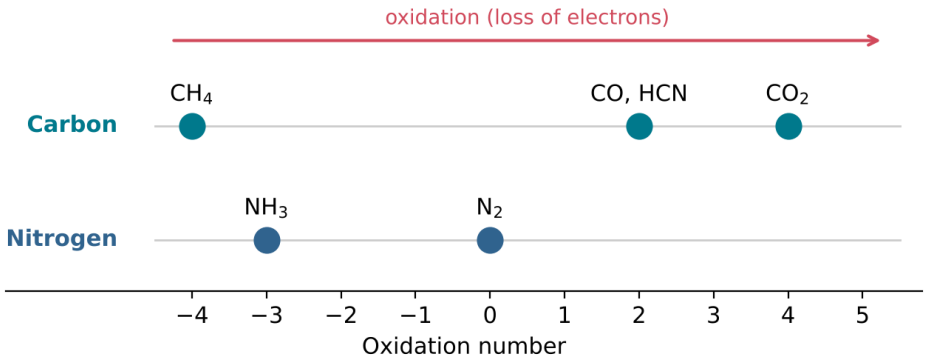


Figure 3.6.: Oxidation numbers of carbon and nitrogen in the gases discussed in this section. Moving to the right corresponds to oxidation (loss of electrons); moving to the left, to reduction. A reducing atmosphere contains the species on the left, a neutral atmosphere those on the right.

- the **fayalite-magnetite-quartz buffer**, defined by the coexistence of fayalite (Fe_2SiO_4), magnetite (Fe_3O_4) and quartz (SiO_2), which corresponds to the oxidation state of present-day magmas.

The gases released under the two conditions are compared in Table 3.3.

Table 3.3.: Volcanic gases expected from magmas of different oxidation state (Trail, Watson, and Tailby 2011).

Oxidation state of magma	Main volcanic gases
Near the iron-wüstite buffer (reducing)	CH_4, H_2, H_2S, NH_3, CO
Near the fayalite-magnetite-quartz buffer (present-day)	H_2O, CO_2, SO_2, N_2

Because the Hadean rock record is sparse or absent, the oxidation state of Hadean magmas has been estimated from zircons. Trail, Watson and Tailby used the amount of cerium incorporated into zircon crystals, which depends on the oxidation state of the melt, to estimate the oxygen fugacity of the melts from which Hadean zircons grew (Trail, Watson, and Tailby 2011). They found average values consistent with the fayalite-magnetite-quartz buffer, similar to present-day conditions, and values similar to those of present-day lavas from the mantle as early as about 4350 million years ago (Trail, Watson, and Tailby 2011). They concluded that gases released from the Earth's interior later than about 200 million years after the formation of the solar system would not have produced a reducing atmosphere (Trail, Watson, and Tailby 2011).

This result agrees with the widely held view that the early atmosphere was not reducing but was dominated by N_2 and CO_2 (Cleaves et al. 2008). It is also consistent

with the dense carbon dioxide atmosphere that followed the condensation of the oceans (Section 3.2).

3.4.4. Transient reducing atmospheres after impacts

A neutral atmosphere is less favourable for the synthesis of organic molecules than a reducing one. The action of electric discharges on neutral gas mixtures has been found to be much less efficient in producing organic compounds (Cleaves et al. 2008). This creates a problem: the origin of life has been argued to demand a highly reduced early atmosphere, rich in CH_4, H_2 and NH_3, whereas the geological evidence suggests that the mantle has always been relatively oxidised and its gases dominated by CO_2, H_2O and N_2 (K. J. Zahnle et al. 2020).

One proposed resolution involves the material that struck the Earth after the Moon-forming impact, called the **late veneer** (K. J. Zahnle et al. 2020). Isotopic evidence indicates that this material was extremely dry and highly reduced, and that it contained metallic iron (K. J. Zahnle et al. 2020). Zahnle and co-workers modelled what happens when such material reacts with the water of the oceans (K. J. Zahnle et al. 2020):

1. The main product of reaction between the iron of the impactor and the Earth's water is H_2.
2. The largest impacts, which vaporise the oceans, produce high pressures and long cooling times that favour the formation of CH_4 and NH_3. Smaller impacts are much less productive, unless catalysts were available or additional reducing power came from the Earth's crust or mantle.
3. The resulting atmospheres rich in H_2 and CH_4 are transient. Through photochemistry, reactions driven by sunlight, they generate organic compounds containing nitrogen over periods of up to tens of millions of years, in total amounts that could form a layer up to half a kilometre thick, while hydrogen escapes to space.
4. When the methane has gone, the atmosphere is typically rich in H_2 and CO, and is eventually oxidised to CO_2.

In this picture the long-term state of the Hadean atmosphere was neutral, but it was interrupted by episodes, each lasting up to tens of millions of years, during which a reducing atmosphere could exist and organic molecules were produced efficiently.

3.4.5. Neutral atmospheres reconsidered

The yields of organic compounds from neutral atmospheres may also have been underestimated. Cleaves and co-workers repeated electric-discharge experiments with neutral gas mixtures and found that significant amounts of amino acids are produced (Cleaves et al. 2008). The low yields previously reported appear to have been caused by the destruction of the organic compounds, during the chemical treatment of the products,

by nitrite and nitrate formed in the discharge. When an inhibitor of this oxidation, such as dissolved ferrous iron (Fe^{2+}), was added, the yield of amino acids increased greatly (Cleaves et al. 2008). The authors concluded that synthesis in a neutral atmosphere may have depended on the availability of such oxidation inhibitors in the oceans, and may have been more important than previously thought (Cleaves et al. 2008).

The question of the composition of the early atmosphere is therefore not settled in a single answer. The evidence favours a long-term atmosphere of N_2 and CO_2, but leaves open the possibility of reducing conditions for limited times, or in limited places, which may have been enough for prebiotic chemistry.

3.5. Sources of small molecules

3.5.1. Three categories of source

The small organic molecules from which the building blocks of RNA could be made had to come from somewhere. Chyba and Sagan divided the possible sources on the early Earth into three categories (Chyba and Sagan 1992):

1. **Exogenous delivery**: organic molecules already present in extraterrestrial objects, such as meteorites and interplanetary dust, and delivered intact to the Earth.
2. **Impact-shock synthesis**: organic molecules formed from simpler gases by the shock waves and heating produced when objects enter the atmosphere or strike the surface.
3. **Endogenous synthesis by other energy sources**: organic molecules formed on the Earth itself, using energy from ultraviolet light, electrical discharges such as lightning, or other sources.

To these may be added the chemistry of **hydrothermal systems**, in which water heated by contact with hot rock reacts with the rock and with dissolved gases.

Each category requires two things: a supply of carbon, nitrogen and other elements in a suitable chemical form, and a source of energy to drive reactions that would not otherwise proceed (Section 2.7). The relative importance of the categories therefore depends on the composition of the atmosphere discussed in Section 3.4.

3.5.2. Energy for atmospheric synthesis

Three energy sources can drive organic synthesis in the atmosphere.

- **Ultraviolet light** is electromagnetic radiation with wavelengths shorter than those of visible light, that is, below about 400 nm. A photon of ultraviolet light carries enough energy to break chemical bonds; the energy of a photon is E =

hc/λ, where h = 6.626 070 15 × 10^{-34} J s is the Planck constant, c = 299 792 458 m s^{-1} the speed of light and λ the wavelength (National Institute of Standards and Technology, n.d.a), so a photon of wavelength 200 nm has an energy of 6.2 eV, comparable to the bond enthalpies of Table 2.8 (a C–H bond enthalpy of 415.9 kJ mol^{-1} corresponds to 4.31 eV per bond). The fragments produced can then recombine to form new molecules. This is photochemistry (Section 3.4).

- **Electrical discharges**, such as lightning, heat a narrow channel of gas to high temperatures and break molecules into atoms, ions and fragments, which recombine as the gas cools.
- **Shocks** from objects entering the atmosphere at high speed heat and compress gas in a similar way.

Chyba and Sagan estimated the amount of organic material produced by each source for different compositions of the early atmosphere (Chyba and Sagan 1992). Their main conclusions were:

- In a strongly reducing atmosphere, production by atmospheric shocks seems to have exceeded that by electrical discharges. Production by ultraviolet light may have exceeded both, but only if a gas such as H_2S, which absorbs ultraviolet light of long wavelength, was supplied fast enough (Chyba and Sagan 1992).
- In an atmosphere of intermediate oxidation state, which they considered more likely, shocks were probably of little importance. For a ratio of H_2 to CO_2 of about 0.1, the net production of organic molecules was some three orders of magnitude (a factor of about 1000) lower than for reducing atmospheres; the most important sources were ultraviolet light and the delivery of intact organic molecules in interplanetary dust particles (Chyba and Sagan 1992).
- At still lower ratios of H_2 to CO_2, interplanetary dust particles may have been the dominant source (Chyba and Sagan 1992).

The general conclusion is that the more oxidised the atmosphere, the less organic material it produces, and the more important delivery from space becomes. Endogenous, exogenous and impact-shock sources could each have contributed significantly to the origin of life (Chyba and Sagan 1992).

3.5.3. Delivery from space

Interplanetary dust particles (IDPs) are small particles of extraterrestrial dust that fall on the Earth from space; Chyba and Sagan considered them the most likely carrier of intact organic molecules from space (Chyba and Sagan 1992). **Meteorites** are larger bodies that survive passage through the atmosphere and reach the ground. Some meteorites contain a remarkable variety of organic compounds. An analysis of the organic fraction of the Murchison meteorite (Section 3.7) found tens of thousands of different molecular compositions, corresponding to probably millions of different structures (Schmitt-Kopplin et al. 2010).

Meteorites and interplanetary dust also delivered molecules of biological interest, including nucleobases (Pearce et al. 2017), the components of nucleotides that carry genetic information (Chapter 5). Pearce and co-workers modelled the fate of meteoritic nucleobases delivered into small ponds that were repeatedly filled with water and dried (Pearce et al. 2017). In their model the nucleobases came mainly from meteorites rather than from interplanetary dust particles, and were quickly lost, by seepage of water out of the ponds during wet periods and by destruction by ultraviolet light during dry periods (Pearce et al. 2017). Any chemistry that used them must therefore have been fast. The model is discussed further in Section 3.7.

3.5.4. Impacts as sources of reducing power

Impacts can also change the chemistry of the atmosphere as a whole. As described in Section 3.4, the impact of large bodies containing metallic iron on an ocean-covered Earth may have produced transient atmospheres rich in H_2 and CH_4, in which nitrogen-containing organic compounds were generated photochemically over periods of up to tens of millions of years (K. J. Zahnle et al. 2020). In this picture impacts do not merely deliver or synthesise organic molecules directly; they also supply the reducing conditions that make atmospheric synthesis efficient.

3.5.5. Hydrothermal systems

Hydrothermal vents are sites on the sea floor where water heated within the crust flows out into the ocean. They are geochemically reactive environments, and today they support rich communities of microorganisms (Martin et al. 2008). Hydrothermal fluids contain hydrogen (H_2) and carbon dioxide (CO_2) (Martin et al. 2008), which form a **redox couple**: a reducing and an oxidising substance that can react with each other by the transfer of electrons. There are striking parallels between the chemistry of this couple and the central energy-yielding reactions of some modern microorganisms that make their organic matter from CO_2 (autotrophs) (Martin et al. 2008). This has led to the proposal that the chemistry of life began in such settings (Martin et al. 2008).

Hydrothermal systems therefore offer an endogenous source of reduced carbon that does not depend on the composition of the atmosphere. Their limitations for the synthesis of RNA, in particular the instability of RNA and its components in hot water, are discussed in later chapters.

3.5.6. Feedstock and energy in one system

The categories above are not exclusive. One influential set of experiments by Patel and co-workers showed that precursors of ribonucleotides (the units of RNA), amino acids and lipids can all be made from hydrogen cyanide (HCN) and some of its derivatives, by

a sequence of reactions in which ultraviolet light provides energy and hydrogen sulfide (H_2S) acts as the reducing agent (Patel et al. 2015). Such chemistry combines an atmospheric or exogenous source of carbon and nitrogen (HCN), a geochemical source of reducing power (H_2S), and sunlight. The key molecules of this and other routes are introduced in Section 3.6.

Figure 3.7 summarises the sources described in this section.

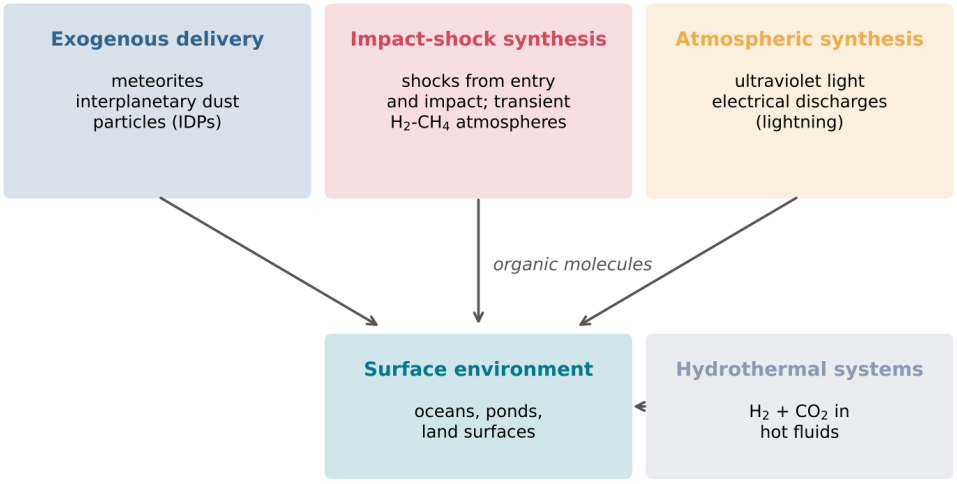


Figure 3.7.: Sources of small organic molecules on the early Earth, following the three categories of Chyba and Sagan (1992), with hydrothermal systems added (Martin et al. 2008). Arrows show the flow of material to the surface environment in which prebiotic chemistry may have taken place. The relative importance of the sources depends on the oxidation state of the atmosphere.

3.6. Key feedstock molecules

3.6.1. What is a feedstock molecule?

A **feedstock molecule** is a small, simple molecule from which more complex molecules can be built in a sequence of reactions. In prebiotic chemistry the term refers to molecules that could plausibly have formed or accumulated on the early Earth by the processes of Section 3.5, and that react, under suitable conditions, to give the building blocks of biological molecules. This section introduces four of them: hydrogen cyanide, formaldehyde, cyanamide and phosphate, together with the phosphorus mineral schreibersite. How they are converted into the building blocks of RNA is the subject of Chapters 4 and 5.

The first three are compounds of hydrogen, carbon, nitrogen and oxygen. Their structures and some of their properties are given in Figure 3.8 and Table 3.4.

Table 3.4.: Three carbon-containing feedstock molecules. Boiling points and the triple-point temperature of cyanamide are from the NIST Chemistry WebBook (National Institute of Standards and Technology, n.d.b); molar masses are calculated from standard atomic weights.

Molecule	Formula	Molar mass (g mol^{-1})	Normal boiling point (K)
Hydrogen cyanide	HCN	27.03	299.0 ± 0.5
Formaldehyde	H_2CO	30.03	253.85
Cyanamide	H_2NCN	54.05	not tabulated; solid below 318.71 K

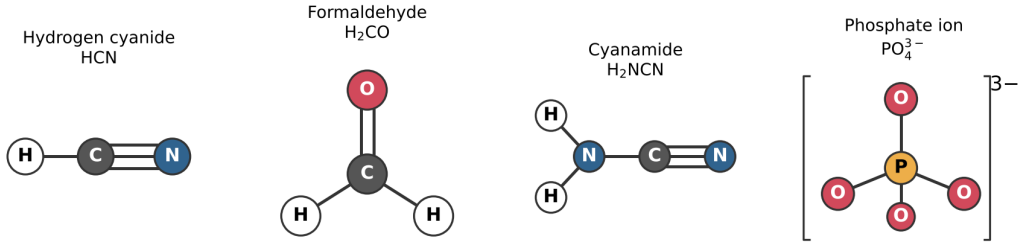


Figure 3.8.: Structures of the feedstock molecules discussed in this section. Lines represent covalent bonds (Section 2.2); double and triple bonds are drawn as two and three lines. Hydrogen cyanide is linear and contains a carbon-nitrogen triple bond; formaldehyde is planar and contains a carbon-oxygen double bond; cyanamide contains an amino group (NH_2) bonded to a nitrile group (C≡N). The phosphate ion is tetrahedral (drawn here in projection); its charge of −3 is spread over the four oxygen atoms.

3.6.2. Hydrogen cyanide

Hydrogen cyanide (HCN) is a linear molecule in which the carbon atom forms a single bond to hydrogen and a triple bond to nitrogen (Section 2.3). It boils at 299.0 K (25.9 °C) (National Institute of Standards and Technology, n.d.b), so on a warm early Earth it could exist as a gas in the atmosphere and dissolve in water. In the classification of Section 3.4, its carbon atom has oxidation number +2, intermediate between that of methane and that of carbon dioxide. In the transient reducing atmospheres that may have followed large impacts, nitrogen-containing organic compounds are generated

photochemically (K. J. Zahnle et al. 2020); HCN is the simplest compound of this kind.

HCN is of central importance because it contains carbon and nitrogen bonded together, as they are in the nucleobases of RNA. Concentrated solutions of ammonium cyanide (NH_4CN), the salt formed from HCN and ammonia, polymerise. The products include the nucleobases adenine and guanine; this chemistry goes back to original studies by Oró and Kimball in 1961 and has been examined in later work (Borquez et al. 2005). The yields are low: about 0.1 percent of adenine after hydrolysis of the soluble products at pH 8, and about 0.05 percent from the insoluble black polymer that also forms (Borquez et al. 2005). Adenine has the molecular formula $C_5H_5N_5$, which is exactly five times that of HCN; the chemistry of this and related reactions is described in Chapter 5.

HCN is also the starting point of the reaction network of Patel and co-workers described in Section 3.5, from which precursors of ribonucleotides, amino acids and lipids can all be derived (Patel et al. 2015).

3.6.3. Formaldehyde

Formaldehyde (H_2CO) is the simplest **aldehyde**, a compound containing the group –CHO, in which a carbon atom is double-bonded to oxygen and bonded to hydrogen. It is planar, with an sp^2-hybridised carbon atom (Section 2.3), and boils at 253.85 K (−19.3 °C) (National Institute of Standards and Technology, n.d.b).

Formaldehyde is the feedstock for sugars. In 1861 Butlerov discovered that formaldehyde, treated with a base, is converted into a sugary substance (Boutlerow 1861); the process is now called the **formose reaction**. In 1959 Breslow proposed a mechanism in which glycolaldehyde, the simplest molecule formed from two formaldehyde units, catalyses its own formation, so that the reaction is autocatalytic (Section 2.8) (Breslow 1959). The sugar of RNA, ribose, contains five carbon atoms and can in principle be formed from five formaldehyde units. In practice, however, ribose is difficult to form selectively (Powner, Gerland, and Sutherland 2009), and the formose reaction gives a complex mixture. This problem is examined in Section 4.4.

3.6.4. Cyanamide

Cyanamide (H_2NCN) consists of an amino group, $–NH_2$, bonded to a nitrile group, –C≡N. In the solid state it melts near its triple-point temperature of 318.71 K (45.56 °C) (National Institute of Standards and Technology, n.d.b).

Its importance in prebiotic chemistry was shown in 2009 by Powner, Gerland and Sutherland (Powner, Gerland, and Sutherland 2009). They synthesised activated pyrimidine ribonucleotides, two of the four units from which RNA is built, by a short sequence of reactions that does not pass through free ribose or free nucleobases (Powner, Gerland,

and Sutherland 2009). The starting materials were cyanamide, cyanoacetylene (HC≡C–C≡N), glycolaldehyde, glyceraldehyde and inorganic phosphate, all regarded as plausible prebiotic feedstock molecules (Powner, Gerland, and Sutherland 2009). This synthesis is described in detail in Chapter 5.

3.6.5. Phosphate and the phosphate problem

Every nucleotide of RNA contains a phosphate group, so a source of phosphorus in a reactive form is essential. At the Earth's surface phosphorus occurs mainly as **phosphate**, PO_4^{3-}, an ion in which a phosphorus atom (oxidation number +5) is bonded tetrahedrally to four oxygen atoms. Phosphate is also a component of cell membranes and of energy-transfer molecules such as adenosine triphosphate (ATP) (Toner and Catling 2020).

Two obstacles make phosphate difficult to use in prebiotic chemistry. First, phosphorus is scarce in the crust: phosphorus pentoxide (P_2O_5) makes up only 0.15 percent of the upper continental crust by mass (Table 1.3). Second, phosphate combines with calcium to form minerals of the **apatite** group, which have low solubility. As a result, dissolved phosphate is generally limited to micromolar concentrations in natural waters, whereas laboratory experiments that attach phosphate to organic molecules commonly use molar concentrations, about a million times higher (Toner and Catling 2020). This mismatch between laboratory conditions and environmental constraints is known as the **phosphate problem** (Toner and Catling 2020).

Concentrations here are expressed in moles of solute per litre of solution (**molar**, symbol M) or per kilogram of solvent (**molal**, symbol mol kg^{-1}); for dilute aqueous solutions the two are nearly equal.

Two proposed solutions illustrate the kinds of setting that may have overcome the phosphate problem.

- **Carbonate-rich lakes.** In lakes rich in dissolved carbonate, calcium is removed from solution as carbonate minerals, so phosphate does not precipitate as apatite. Toner and Catling calculated that such lakes could accumulate up to about 0.1 molal phosphate under steady evaporation and inflow, and more than 1 molal during evaporation without inflow, at moderate pH (6.5 to 9) buffered by the CO_2-rich atmosphere (Toner and Catling 2020). Before the origin of life there were no microorganisms to consume phosphate, which would have made such lakes even richer in it (Toner and Catling 2020).
- **Reduced phosphorus from meteorites.** Iron meteorites contain **schreibersite**, $(Fe,Ni)_3P$, a mineral in which phosphorus is bonded to iron and nickel rather than to oxygen. When schreibersite corrodes in water, it releases a mixture of phosphorus-bearing ions, including pyrophosphate and triphosphate (chains of two and three phosphate units) and phosphonate, in which phosphorus is less oxidised

than in phosphate (Pasek and Lauretta 2005). Iron meteorites were therefore probably an important source of reactive phosphorus (Pasek and Lauretta 2005). Gull and co-workers showed that synthetic schreibersite, simply mixed with aqueous solutions of the nucleosides adenosine or uridine and mildly heated under slightly basic conditions (pH about 9), attaches phosphate groups to them (Gull et al. 2015). (A **nucleoside** is a nucleobase joined to ribose; adding a phosphate gives a nucleotide; see Chapter 5.)

Neither solution is established. Both show, however, that the availability of phosphate depended on local conditions, a point taken up in Section 3.7.

3.7. Candidate settings

3.7.1. Requirements of a setting

The preceding sections described where small molecules could come from. Prebiotic chemistry, however, also requires a place where these molecules could accumulate and react. A useful setting must bring feedstock molecules together at concentrations high enough for reactions between them to proceed, supply energy, and protect the products long enough for further steps to occur. Different settings meet these requirements in different ways. Four kinds of setting have been proposed: ponds and hot springs on land, hydrothermal vents on the sea floor, impact craters, and ice. Each is described below, and their main features are compared in Table 3.5.

Before turning to the settings, it is worth recalling the evidence for the earliest delivery of organic matter from space, since it bears on all of them.

Key experiment: organic molecules in the Murchison meteorite

On 28 September 1969, a meteorite fell near the town of Murchison, Victoria, Australia (Meteoritical Society 2026). It is a **carbonaceous chondrite**, a chondrite (Section 1.7) rich in carbon, of the group designated CM2, and about 100 kg of material was recovered (Meteoritical Society 2026).
In 1970 Kvenvolden and co-workers reported evidence for extraterrestrial amino acids and hydrocarbons in the Murchison meteorite (Kvenvolden et al. 1970). Later work extended the list to many other classes of compound. Callahan and co-workers analysed 12 meteorites and found that Murchison and one other meteorite contained a diverse suite of nucleobases, including three nucleobase analogues that are rare on Earth: purine, 2,6-diaminopurine and 6,8-diaminopurine (Callahan et al. 2011). These compounds were not detected in control samples, including a terrestrial soil and Antarctic ice, which supports their extraterrestrial origin (Callahan et al. 2011). The same suite of nucleobases was produced in laboratory reactions

of ammonium cyanide, suggesting that the meteoritic purines formed by cyanide chemistry on the meteorite's parent asteroid (Callahan et al. 2011) (Section 3.6). A non-targeted analysis of the material that can be extracted from Murchison with solvents found tens of thousands of different molecular compositions, corresponding to probably millions of distinct structures (Schmitt-Kopplin et al. 2010). The Murchison meteorite thus shows that chemistry producing some of the molecules of interest to the origin of life took place on small bodies of the early solar system, independently of the Earth, and that its products reached the Earth's surface.

3.7.2. Ponds and hot springs on land

Small bodies of water on land, from shallow ponds to the pools of volcanic hot springs, can dry out and refill repeatedly. These **wet-dry cycles** are their most important feature: they have been found to promote the joining of building blocks into chains (Pearce et al. 2017; Damer and Deamer 2020). During the dry phase dissolved molecules are concentrated, and during the wet phase products are redistributed and new material is supplied. Damer and Deamer have developed a hypothesis in which fluctuating volcanic hot-spring pools play a central role, based on experimental evidence that polymers enclosed in lipid membranes can be synthesised by cycles of hydration and dehydration (Damer and Deamer 2020).

The model of Pearce and co-workers, introduced in Section 3.5, placed meteoritic nucleobases in such ponds (Pearce et al. 2017). Ponds could exist only once land had risen above the early global ocean, but the growing area of land was offset by a declining rate of meteorite delivery (Pearce et al. 2017). Because nucleobases were lost rapidly by seepage and by ultraviolet light, the authors concluded that the formation of nucleotides and their joining into RNA must have occurred within one to a few wet-dry cycles, and that, under these conditions, RNA polymers probably appeared before 4.17 billion years ago (Pearce et al. 2017).

The carbonate-rich lakes of Section 3.6 are a related setting: closed basins in which evaporation concentrates phosphate and other reagents to levels used in laboratory syntheses (Toner and Catling 2020). Surface settings are exposed to ultraviolet light, which can drive reactions such as those of Patel et al. (2015) but can also destroy their products.

3.7.3. Hydrothermal vents

Hydrothermal vents (Section 3.5) place chemistry in the deep ocean, protected from ultraviolet light and from the effects of all but the largest impacts. Many vents, such as the black smokers of mid-ocean ridges, are driven by volcanic heat (Kelley et al. 2005). A different type is represented by the Lost City hydrothermal field, which is not volcanically driven (Kelley et al. 2005). There, reactions between seawater and rocks of

the upper mantle produce fluids rich in methane and hydrogen, at temperatures from below 40 °C to 90 °C and at pH 9 to 11, and the fluids build carbonate chimneys 30 to 60 metres tall (Kelley et al. 2005).

Alkaline vents of this kind provide H_2, a strong reducing agent, in contact with seawater that contained CO_2. The parallels between this H_2-CO_2 chemistry and the core energy metabolism of some microorganisms (Section 3.5) underlie the proposal that life began in such vents (Martin et al. 2008). A difficulty for the synthesis of RNA in the open ocean is dilution: molecules that are formed disperse into a very large volume of water unless they are retained, for example within the porous interiors of chimneys.

3.7.4. Impact craters

Impacts were not only destructive. Osinski and co-workers reviewed the ways in which impacts may have assisted the origin of life (Osinski et al. 2020). The heat deposited by a large impact can drive hydrothermal systems within the crater, on land or beneath the sea, which may persist long after the impact; impacts also create sediment-filled basins and pools, and they can deliver or generate chemical ingredients for life and mineral surfaces such as clays that can act as catalysts (Osinski et al. 2020). An impact crater can thus combine several of the features of the other settings in one place. As noted in Section 3.4, large impacts of iron-rich bodies may in addition have produced transient reducing atmospheres (K. J. Zahnle et al. 2020).

3.7.5. Ice

Low temperatures slow down reactions (Section 2.8), but they also slow down the decomposition of products, and freezing can concentrate solutes: as water freezes, dissolved substances are largely excluded from the growing ice crystals and accumulate in the remaining liquid. Levy and co-workers kept dilute solutions of ammonium cyanide frozen for 25 years at −20 °C and −78 °C (Levy et al. 2000). Adenine and guanine, together with a simple set of amino acids dominated by glycine, were formed in substantial yields (Levy et al. 2000). Cold settings may therefore have been suitable for some prebiotic syntheses, and may have been common if the faint young Sun (Section 3.2) left parts of the early Earth frozen.

3.7.6. Comparison

Table 3.5.: Candidate settings for prebiotic chemistry on the early Earth.

Setting	Main advantages	Main difficulties	Key evidence
Ponds and hot springs on land	Wet-dry cycles concentrate molecules and favour joining of building blocks; access to ultraviolet light	Require exposed land; ultraviolet light also destroys products	Damer and Deamer (2020), Pearce et al. (2017), Toner and Catling (2020)
Hydrothermal vents	Continuous supply of H_2 and chemical energy; protection from ultraviolet light and most impacts	Dilution in the ocean; high temperatures in some vents	Martin et al. (2008), Kelley et al. (2005)
Impact craters	Combine heat, hydrothermal circulation, pools and delivered material	Each event is local and its hydrothermal activity eventually ceases	Osinski et al. (2020)
Ice	Slow decomposition; concentration of solutes on freezing	Slow reactions	Levy et al. (2000)

No single setting has been shown to be the one in which life began, and several may have contributed different steps. What the evidence of this chapter establishes is that, by about 4.4 Ga, the Earth had liquid water (Section 3.3), that its atmosphere was probably neutral for most of the time but may have been reducing for periods after large impacts (Section 3.4), and that organic molecules, including nucleobases, were supplied from both terrestrial and extraterrestrial sources (Section 3.5). The next chapter follows the chemistry by which feedstock molecules such as those of Section 3.6 are transformed into the building blocks of life.

Summary

- Radioactive decay follows $N = N_0\, e^{-\lambda t}$, with half-life $t_{1/2} = \ln 2/\lambda$. The ratio of radiogenic daughter to remaining parent in a closed system gives the time since the system closed.

- The uranium-lead system provides two independent clocks with half-lives of 4.4683 Ga and 703.81 Ma; their agreement tests whether a sample has remained closed. Lead-lead ages date the oldest solids of the solar system to 4.567 to 4.568 Ga.
- Patterson's meteorite lead measurements of 1956 gave an age of 4.55 ± 0.07 Ga for the Earth and meteorites, close to the modern age of the oldest solar-system solids.
- Earth history is divided into four eons; the Hadean ends at 4031 ± 3 Ma and the Archean at 2500 Ma.
- The Moon formed when a Mars-sized body struck the proto-Earth. Zircons from Apollo 14 show that the Moon had solidified by 4.51 Ga, so the impact occurred within about 60 million years of the formation of the solar system.
- After the impact the Earth passed through a silicate-vapour atmosphere and a magma ocean lasting about 2 million years; water then condensed into oceans beneath a dense carbon dioxide atmosphere. Habitable conditions may have been reached 10 to 20 million years after the impact.
- The young Sun was about 30 percent fainter than today, so the early Earth needed more greenhouse gases to keep liquid water. Impacts continued throughout the Hadean, with an episode of elevated impact rate between about 4.2 to 4.0 and 3.5 Ga.
- Detrital zircons from the Jack Hills, Western Australia, are up to 4404 ± 8 million years old. Their high $\delta^{18}O$ values show that by 4.4 Ga surface rocks had interacted with liquid water: the Earth had oceans early in the Hadean.
- Graphite inclusions in a 4.10 Ga zircon have $\delta^{13}C = -24 \pm 5$ ‰, consistent with, but not proof of, a biological origin.
- In a reducing atmosphere carbon and nitrogen are present mainly as CH_4 and NH_3; in a neutral one, as CO_2 and N_2. Cerium in Hadean zircons indicates that magmas were as oxidised as today's, so gases from the interior would have formed a neutral atmosphere.
- Transient reducing atmospheres, lasting up to tens of millions of years, may have followed large impacts of iron-bearing material, and organic synthesis in neutral atmospheres may be more efficient than once thought.
- Organic molecules could be delivered from space, made by impact shocks, or made on the Earth by ultraviolet light and electrical discharges; hydrothermal systems provide a further source. The more oxidised the atmosphere, the smaller the atmospheric production and the more important delivery from space.
- Key feedstock molecules include hydrogen cyanide (a source of nucleobases), formaldehyde (a source of sugars through the formose reaction) and cyanamide. Phosphate is scarce and poorly soluble in the presence of calcium (the phosphate problem); carbonate-rich lakes and the meteoritic mineral schreibersite are two proposed sources of usable phosphorus.
- The Murchison meteorite, which fell in 1969, contains amino acids, a diverse suite of nucleobases and tens of thousands of other organic compounds of extraterrestrial origin.
- Proposed settings for prebiotic chemistry include ponds and hot springs on land

(wet-dry cycles), hydrothermal vents, impact craters and ice. Each offers advantages and difficulties, and none has been established as the site of the origin of life.

Exercises

1. Show, starting from Equation 3.2, that the half-life equals ln $2/\lambda$. Calculate the decay constant of potassium-40 from its half-life in Table 3.1.
2. A zircon crystal has a radiogenic $^{206}Pb/^{238}U$ atom ratio of 0.70. Using Equation 3.5, calculate its ^{238}U-^{206}Pb age.
3. What fraction of the potassium-40 present when the solar system formed remains today? Of the potassium-40 that has decayed, what fraction has become argon-40?
4. Explain why carbon-14 cannot be used to date rocks from the Hadean eon.
5. A uranium-bearing mineral gives a ^{238}U-^{206}Pb age of 3.9 Ga and a ^{235}U-^{207}Pb age of 4.1 Ga. Is it concordant? Which assumption of Equation 3.5 has probably failed?
6. Explain why lead from the troilite of the Canyon Diablo meteorite was important for Patterson's determination of the age of the Earth.
7. Using the data in Section 3.2, calculate the mean density of the Moon from its mass and radius, and compare your result with the tabulated value.
8. What does the similarity of the oxygen isotope compositions of the Earth and Moon imply about the giant impact?
9. Why is the age of 4.51 Ga obtained from lunar zircons a minimum age for the Moon-forming impact?
10. Using Equation 3.6, calculate the luminosity of the Sun, relative to its present value, at 3.8 Ga.
11. List the stages through which the surface of the Earth passed between the giant impact and the first oceans, with the approximate duration of each.
12. A zircon has an $^{18}O/^{16}O$ ratio 0.74 percent higher than that of VSMOW. What is its $\delta^{18}O$ value?
13. Explain why a detrital zircon can be much older than the sedimentary rock in which it is found.
14. Explain the reasoning by which a high $\delta^{18}O$ value in a zircon is taken as evidence for liquid water at the Earth's surface.
15. Why should the graphite in the 4.10 Ga Jack Hills zircon be regarded as potentially, rather than certainly, biological in origin?
16. Assign oxidation numbers to each atom in H_2O, NH_3, HCN and CO_2.
17. Write a balanced equation for the reaction of metallic iron with water to give wüstite (FeO) and hydrogen. Which element is oxidised and which is reduced?
18. Explain why the oxidation state of Hadean magmas determines the composition of the gases they release.

19. Why are the reducing atmospheres proposed by Zahnle and co-workers described as transient?
20. Calculate the energy, in eV, of a photon of wavelength 250 nm. Is it sufficient to break an O–H bond (Table 2.8)?
21. According to Chyba and Sagan, which sources of organic molecules were most important in an atmosphere with $[H_2]/[CO_2]$ of about 0.1? Explain why the answer differs for a strongly reducing atmosphere.
22. Explain why the nucleobases delivered to a pond by a meteorite could have been used only in chemistry that was completed within a short time.
23. What is a redox couple? Identify the oxidising and the reducing member of the couple H_2-CO_2.
24. Assign oxidation numbers to every atom in formaldehyde and in cyanamide.
25. Show that the molecular formula of adenine, $C_5H_5N_5$, corresponds to five molecules of HCN. Do the same for ribose, $C_5H_{10}O_5$, and formaldehyde.
26. A laboratory experiment uses 1 M phosphate, and a natural water contains 2 μM phosphate. By what factor do the concentrations differ?
27. Explain why carbonate-rich lakes can contain much more dissolved phosphate than ordinary natural waters.
28. What evidence indicates that the nucleobases found in the Murchison meteorite are extraterrestrial rather than terrestrial contamination?
29. Explain how wet-dry cycles can help to overcome the problem of low concentrations of feedstock molecules.
30. Compare a land-based hot-spring pool with an alkaline hydrothermal vent as a setting for prebiotic chemistry, considering energy sources, concentration of molecules and exposure to ultraviolet light.
31. Why might freezing promote some prebiotic reactions even though low temperature reduces rate constants?

4. From Simple Molecules to Building Blocks

Learning objectives

After reading this chapter you should be able to:

1. Recognise the functional groups that occur in the building blocks of RNA, and describe the main types of reaction that transform them.
2. Describe the Miller-Urey experiment and later work on the synthesis of organic molecules from simple gases, and evaluate what these experiments show.
3. Explain how sugars, including ribose, can be formed from formaldehyde, and why the selective formation of ribose is difficult.
4. Explain how nucleobases can be formed from hydrogen cyanide and related molecules, and why the activation of phosphate is necessary.
5. Explain why prebiotic reactions generally give mixtures rather than single products, and how selectivity might arise.

4.1. Functional groups

4.1.1. Skeletons and functional groups

The molecules of Chapter 3 contain at most a few atoms. The building blocks of RNA are larger: the sugar ribose has 20 atoms and the nucleobase adenine 15. To describe such molecules, chemists divide them into parts. An organic molecule is regarded as consisting of a relatively unreactive **skeleton**, for example a chain of carbon atoms joined by single bonds, to which one or more **functional groups** are attached. A functional group is an atom, or a group of atoms, that has similar chemical properties whenever it occurs in different compounds, and it determines the characteristic physical and chemical properties of the family of compounds that contain it (IUPAC, n.d.k).

This division is useful because the reactions of a molecule are, to a good approximation, the reactions of its functional groups. Once the behaviour of a few functional groups is known, the behaviour of a large number of molecules can be predicted. In what follows, the letter R stands for the rest of the molecule, a carbon-containing group whose details do not matter for the reaction under discussion.

4.1.2. The functional groups of this book

Table 4.1 lists the functional groups that occur in the building blocks of RNA and in their precursors. Their structures are shown in Figure 4.1.

Table 4.1.: Functional groups of the building blocks of RNA and their precursors.

Family	Functional group	General formula	Example
Alcohol	Hydroxy group, –OH	R–OH	Hydroxy groups of ribose
Aldehyde	Carbonyl group with one H	R–CHO	Formaldehyde, glycolaldehyde
Ketone	Carbonyl group between two carbons	R–CO–R′	Dihydroxyacetone
Carboxylic acid	Carboxy group, –COOH	R–COOH	Glycine (an amino acid)
Amine	Amino group, $–NH_2$	$R–NH_2$	Amino group of adenine
Amide	Carbonyl bonded to nitrogen	$R–CO–NH_2$	Formamide
Nitrile	Cyano group, –C≡N	R–C≡N	Cyanamide, cyanoacetylene
Ester	Acid joined to an alcohol with loss of water	R–CO–O–R′; $(RO)_2P(=O)OH$	Phosphate esters of RNA

The groups are described below.

- **Alcohols** are compounds in which a hydroxy group, –OH, is attached to a carbon atom that forms only single bonds (IUPAC, n.d.a). The O–H bond is polar (Section 2.4), so hydroxy groups form hydrogen bonds with water and with one another (Section 2.5). Ribose carries several hydroxy groups, which is why it dissolves readily in water.
- The **carbonyl group** consists of a carbon atom double-bonded to an oxygen atom, C=O. Because oxygen is more electronegative than carbon, the carbon atom carries a partial positive charge, and this is the site at which carbonyl compounds react with electron-rich species. In an **aldehyde**, the carbonyl carbon is bonded to one hydrogen atom and to one R group, RC(=O)H (IUPAC, n.d.b); formaldehyde, H_2CO, is the special case in which R is a second hydrogen atom. In a **ketone**, the carbonyl carbon is bonded to two carbon atoms.
- A **carboxylic acid** contains the carboxy group, –COOH, in which a carbonyl group and a hydroxy group are attached to the same carbon atom. The hydrogen

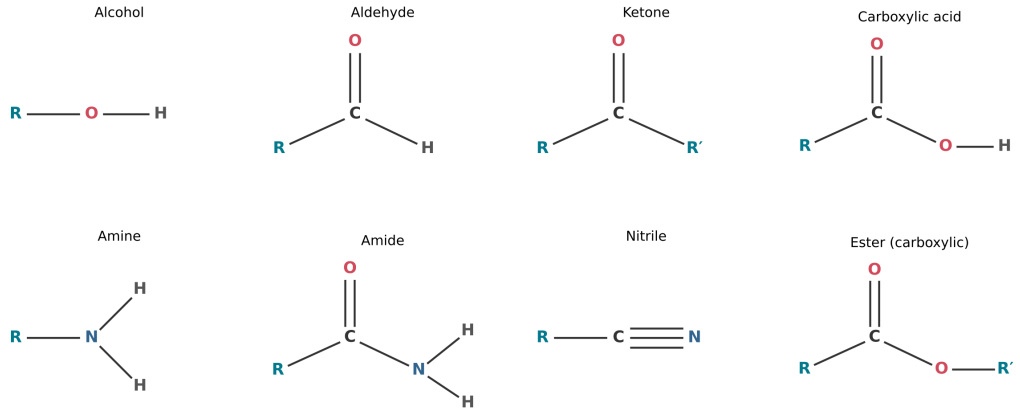


Figure 4.1.: Structures of the functional groups of Table 4.1. R and R′ represent the rest of the molecule. The carbonyl group, a carbon atom double-bonded to an oxygen atom, is common to aldehydes, ketones, carboxylic acids, amides and carboxylic esters.

of the hydroxy group can be released as H^+ in water (Section 2.6), leaving the carboxylate ion $–COO^–$.

- An **amine** contains a nitrogen atom bonded to one or more carbon atoms and carrying a lone pair (Section 2.2); the simplest form is the amino group, $–NH_2$. The lone pair can accept a proton, so amines are bases.
- An **amide** is formally derived from an acid by replacing a hydroxy group of the acid with an amino or substituted amino group (IUPAC, n.d.c). In a carboxylic amide, the carbonyl group is therefore bonded directly to nitrogen, $–CO–NH_2$. Formamide, $HCONH_2$, the simplest amide, is a starting material for the synthesis of nucleobases (Section 4.5).
- A **nitrile** contains a carbon atom triple-bonded to nitrogen, $–C{\equiv}N$, the group present in HCN and cyanamide (Section 3.6).
- An **ester** is formally derived from an acid and an alcohol by linking them with the loss of water, from a hydroxy group of the acid and the hydroxy group of the alcohol (IUPAC, n.d.j). The acid need not be a carboxylic acid: phosphoric acid forms **phosphate esters**, and it is through phosphate ester linkages that the units of RNA are joined into a chain (Chapter 6).

4.1.3. Isomers

Molecules with the same molecular formula can have different structures. **Isomers** are species that have the same atomic composition (molecular formula) but different arrangements of their atoms, and hence different physical or chemical properties (IUPAC, n.d.m). For example, glyceraldehyde and dihydroxyacetone both have the formula $C_3H_6O_3$. In glyceraldehyde the carbonyl group is at the end of the carbon chain, so the

molecule is an aldehyde; in dihydroxyacetone it is in the middle, so the molecule is a ketone (Figure 4.2). Isomers of this kind, which differ in the order in which atoms are bonded, are called **structural isomers**.

Ribose has the formula $C_5H_{10}O_5$, and many other sugars share this formula. The problem of forming ribose rather than one of its isomers is central to Section 4.4.

4.1.4. Chirality

A second kind of isomerism arises from the three-dimensional arrangement of atoms. An object is **chiral** if it cannot be superposed on its mirror image (IUPAC, n.d.f). In molecules, chirality most often arises from a carbon atom bonded to four different groups. Such a carbon atom is called a **stereocentre**. Because the four bonds point to the corners of a tetrahedron (Section 2.3), the four groups can be arranged in two ways that are mirror images of each other and cannot be interconverted without breaking bonds (Figure 4.2).

Two molecules that are non-superposable mirror images of each other are called **enantiomers**. Enantiomers have identical physical properties, such as melting point and solubility, except in their interaction with other chiral objects, including polarised light and other chiral molecules. Glyceraldehyde has one stereocentre, its central carbon atom, and therefore exists as two enantiomers, designated D and L. Ribose has three stereocentres in its open-chain form, and the ribose in RNA is exclusively D-ribose. A chemical synthesis from achiral starting materials such as formaldehyde gives equal amounts of the two enantiomers, a mixture called a **racemate**. How the uniform chirality of biological molecules arose is one of the open questions discussed in Section 4.7.

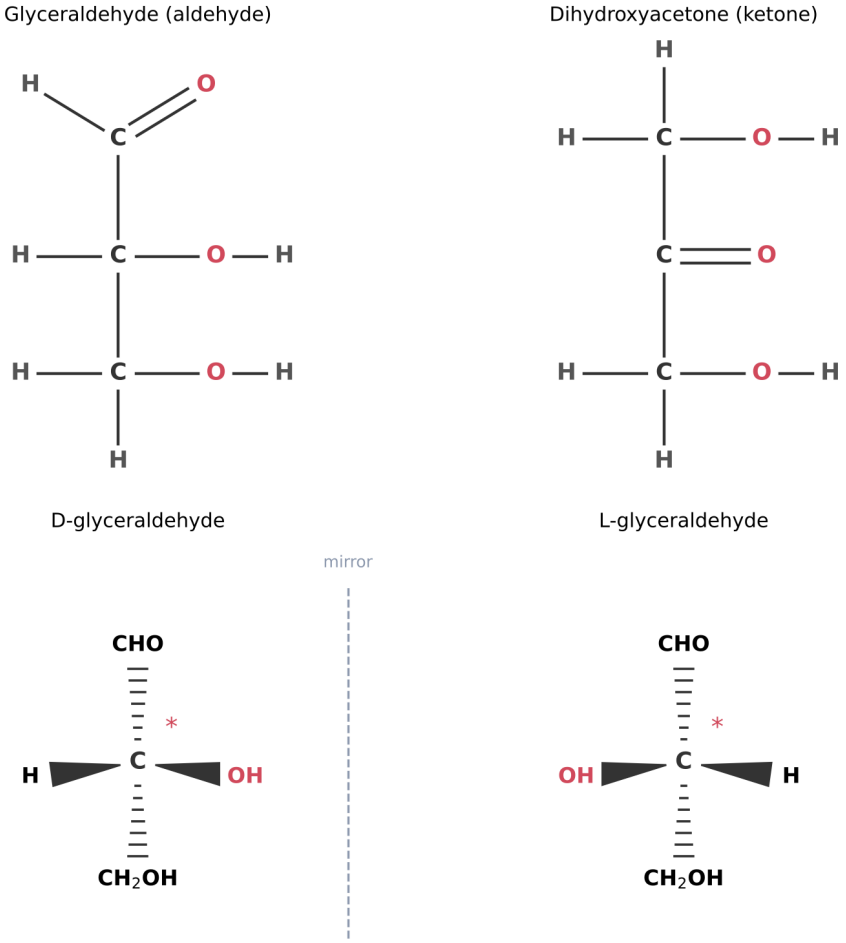


Figure 4.2.: Isomers of $C_3H_6O_3$. Top: the structural isomers glyceraldehyde (an aldehyde) and dihydroxyacetone (a ketone). Bottom: the two enantiomers of glyceraldehyde, D and L, drawn as mirror images about their stereocentre (marked with an asterisk). The enantiomers are drawn so that the horizontal bonds (wedges) point towards the viewer and the vertical bonds (dashed) away from the viewer; in this orientation the hydroxy group of D-glyceraldehyde lies on the right.

4.2. Reaction types

4.2.1. Nucleophiles and electrophiles

Most of the reactions in this book involve the formation of a new covalent bond in which both bonding electrons come from one of the two partners. The partner that supplies the electron pair is the **nucleophile**, and the partner that accepts it is the **electrophile**. An electrophile is a reagent that forms a bond to its reaction partner by accepting both bonding electrons from that partner (IUPAC, n.d.h). Nucleophiles are electron-rich: they typically carry a lone pair or a negative charge, as do the oxygen of a hydroxy group, the nitrogen of an amino group, water and the hydroxide ion. Electrophiles are electron-poor: the most important electrophile in this book is the carbon atom of a carbonyl group, which carries a partial positive charge (Section 4.1). The **nucleophilicity** of a reagent is its reactivity as a nucleophile, measured by comparing the rate constants of its reactions with a common partner (IUPAC, n.d.n).

The rest of this section describes four types of reaction, defined by what happens to the reacting molecules as a whole.

4.2.2. Addition

In an **addition reaction**, two molecules combine to form a single product, without the loss of any atoms. A characteristic example is the addition of a nucleophile to a carbonyl group. The nucleophile bonds to the carbonyl carbon, one of the two bonds between carbon and oxygen is broken, and the oxygen atom picks up a hydrogen atom to become a hydroxy group (Figure 4.3 a). For example, water adds to formaldehyde to give methanediol, $H_2C(OH)_2$, and HCN adds to an aldehyde RCHO to give a **cyanohydrin**, RCH(OH)CN, a compound in which a hydroxy group and a nitrile group are attached to the same carbon atom. Addition to a carbonyl group creates a new bond to carbon, and therefore lengthens the carbon skeleton when the nucleophile contains carbon. Many of the reactions that build sugars from formaldehyde, described in Section 4.4, are additions of this kind.

4.2.3. Condensation and hydrolysis

A **condensation reaction** is a reaction in which two or more reactants, or two reactive sites within the same molecule, yield a single main product with the accompanying formation of water or of some other small molecule (IUPAC, n.d.g). The formation of an ester from an acid and an alcohol is a condensation (Figure 4.3 b):

$$\mathrm{R{-}COOH + HO{-}R' \rightleftharpoons R{-}CO{-}O{-}R' + H_2O.} \tag{4.1}$$

The reverse reaction, in which a bond is broken by reaction with water, is **hydrolysis**. Condensation and hydrolysis are central to this book because every link in an RNA chain, and every bond between a nucleobase, ribose and phosphate in a nucleotide, is formed by a condensation and broken by hydrolysis.

Whether a condensation or a hydrolysis is favoured depends on the Gibbs free energy change of the reaction (Section 2.7). For a reaction at equilibrium, the ratio of the activities (effective concentrations) of products to reactants, each raised to the power of its coefficient in the balanced equation, has a fixed value at a given temperature, called the **equilibrium constant**, K. For Equation 4.1,

$$K = \frac{a_{\text{ester}}\, a_{\text{H}_2\text{O}}}{a_{\text{acid}}\, a_{\text{alcohol}}}, \tag{4.2}$$

where a denotes activity. The equilibrium constant is related to the standard Gibbs free energy change of the reaction, ΔG°, by

$$\Delta G^\circ = -RT \ln K, \tag{4.3}$$

where R is the molar gas constant and T the absolute temperature (Section 2.8). A negative ΔG° corresponds to $K > 1$, which favours products; a positive ΔG° to $K < 1$. At 298.15 K, a reaction with $\Delta G^\circ = +10$ kJ mol^{-1} has $K = 0.018$, and one with +20 kJ mol^{-1} has $K = 3.1 \times 10^{-4}$.

Equation 4.2 explains why condensation reactions are difficult in water. In a dilute aqueous solution, water is present in enormous excess: its concentration is about 55 mol per litre (997.05 g L^{-1} at 25 °C (International Association for the Properties of Water and Steam 2011) divided by its molar mass of 18.015 g mol^{-1}). Every molecule of ester formed must coexist with this large amount of water, which drives the equilibrium of Equation 4.1 back towards hydrolysis. In water, hydrolysis is therefore generally favoured over condensation, even when the condensation itself releases energy in the absence of water. This is one of the central difficulties of prebiotic chemistry, and it explains the interest in settings where water is removed, such as the drying phase of wet-dry cycles (Section 3.7), and in **activated** reactants that carry a built-in source of free energy (Section 4.6).

4.2.4. Oxidation and reduction

Oxidation and reduction were defined in Section 3.4 as the loss and gain of electrons, followed through oxidation numbers. In organic molecules, oxidation usually means the formation of more bonds from carbon to oxygen (or other electronegative atoms), or fewer bonds from carbon to hydrogen; reduction means the reverse. For example, the oxidation number of carbon increases from −2 in methanol (CH_3OH) to 0 in formaldehyde (H_2CO)

and +2 in formic acid (HCOOH), so the sequence methanol → formaldehyde → formic acid is a series of oxidations. A sugar such as ribose, $C_5H_{10}O_5$, has the same ratio of carbon to hydrogen to oxygen as formaldehyde, CH_2O, and its average carbon oxidation number is likewise 0.

Redox reactions matter in prebiotic chemistry because the feedstock molecules are not at the same oxidation level as the building blocks. HCN, for example, contains carbon at oxidation number +2, whereas most of the carbon atoms of ribose are at lower oxidation numbers. Converting one into the other requires a reducing agent, such as the hydrogen sulfide used in the reactions of Patel et al. (2015) (Section 3.5).

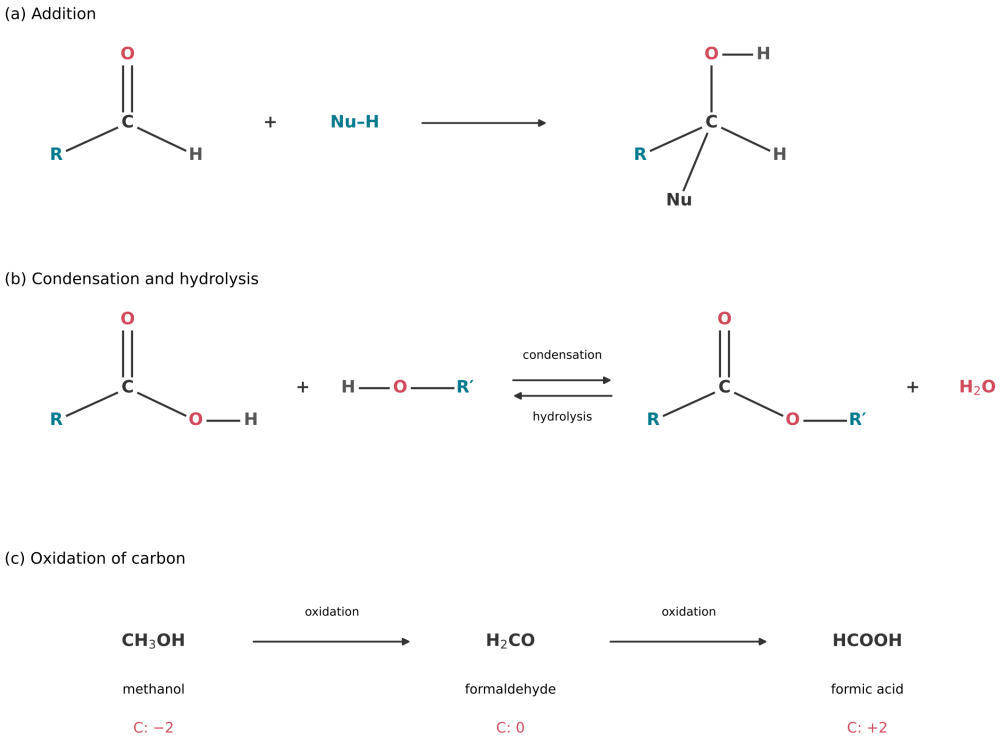


Figure 4.3.: Three reaction types. (a) Addition: a nucleophile (Nu–H) adds to the carbonyl group of an aldehyde, giving a single product in which the carbonyl oxygen has become a hydroxy group. (b) Condensation and hydrolysis: an acid and an alcohol combine to give an ester and water; the reverse reaction is hydrolysis. (c) Oxidation states of carbon in the series methanol, formaldehyde, formic acid.

4.3. The Miller-Urey experiment and its descendants

4.3.1. Amino acids

The first building blocks of life to be made in a laboratory simulation of the early Earth were not components of RNA but **amino acids**, the units from which proteins are built. An amino acid, in the sense used in biochemistry, is a compound that contains both an amino group and a carboxy group (Section 4.1) attached to the same carbon atom. Its general formula is H_2N–CHR–COOH. In the simplest amino acid, **glycine**, R is a hydrogen atom; in **alanine**, R is a methyl group, $–CH_3$. Every amino acid except glycine has a stereocentre and therefore exists as two enantiomers (Section 4.1). Although amino acids are not part of RNA, the experiments that produced them established the methods and the questions that later work on RNA has followed, and they are described here for that reason.

Key experiment: Miller 1953

In 1953 Stanley Miller, a graduate student working with Harold Urey, published the results of an experiment designed to test whether organic compounds could be formed under conditions then thought to resemble those of the early Earth (Miller 1953). A mixture of methane, ammonia and hydrogen, together with water vapour, was sealed in a glass apparatus (Figure 4.4). Water was boiled in a small flask so that its vapour circulated through a larger flask containing the gases, in which a continuous electric spark was passed between two electrodes to simulate lightning. A condenser cooled the gases, and the water that condensed collected in a U-shaped trap, from which it returned to the boiling flask. Products formed in the gas were thus washed into the liquid, a model of an ocean receiving material from the atmosphere.

After about a week of continuous operation, the solution was analysed by paper chromatography. Using this method, Miller identified glycine, alanine (in two isomeric forms, α- and β-alanine) and, less certainly, aspartic acid and α-aminobutyric acid (Miller 1953).

The result showed for the first time that biologically important molecules can be formed from simple gases by a natural energy source, without any biological agency. It transformed the study of the origin of life into an experimental field and established the basis of prebiotic chemistry (Bada 2013).

4.3.2. How the amino acids form

The spark acts on the gas phase, but the amino acids were collected from the water. Products of the reactions in the gas dissolve in the water and react further in solution. Among the small reactive molecules that can give amino acids in water are aldehydes

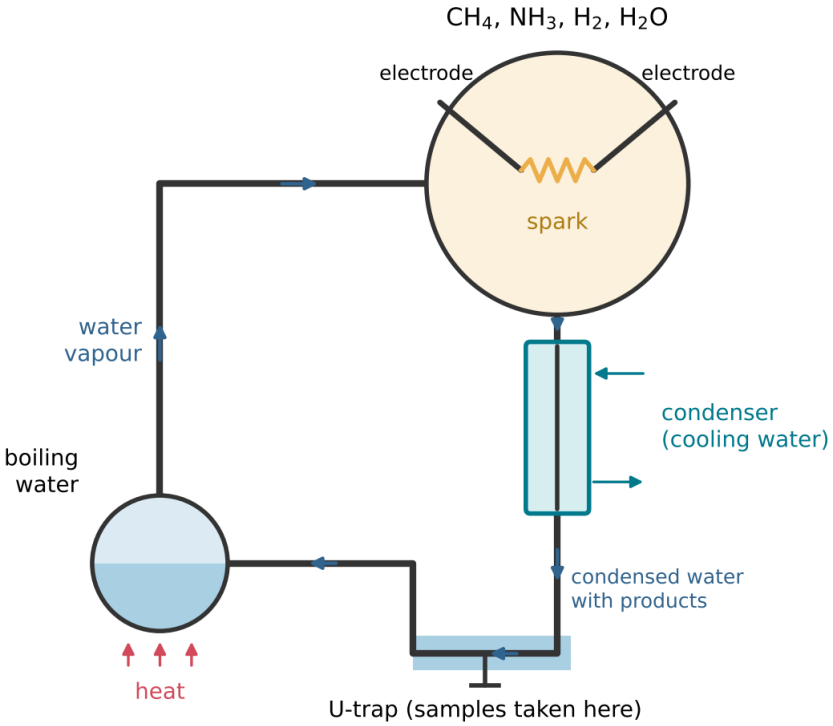


Figure 4.4.: Schematic diagram of the Miller apparatus. Water boiled in the small flask supplies vapour to the large flask, where a spark acts on the gas mixture. The condenser returns water and dissolved products to the trap, from which samples were taken. Arrows show the direction of circulation. The drawing is schematic and not to scale.

and hydrogen cyanide (Section 3.6), together with the ammonia of the gas mixture. The route by which these three combine to give an amino acid is the **Strecker synthesis**, named after Adolph Strecker, who described it in the nineteenth century:

1. Ammonia adds to the carbonyl group of an aldehyde, RCHO, and water is lost, giving an imine, RCH=NH, a compound with a carbon-nitrogen double bond.
2. HCN adds to the imine, giving an **aminonitrile**, $RCH(NH_2)CN$.
3. Hydrolysis of the nitrile group converts the aminonitrile into the amino acid $RCH(NH_2)COOH$.

With formaldehyde (R = H), the product is glycine; with acetaldehyde ($R = CH_3$), alanine. The sequence uses the reaction types of Section 4.2: addition, condensation and hydrolysis. Because the starting aldehyde is achiral and nothing in the solution favours one enantiomer, the amino acids are formed as racemates.

4.3.3. Later experiments

Miller kept the extracts of many of his experiments. Archived samples, some of which had never been analysed, were found in his laboratory material shortly before his death in May 2007, and have since been analysed with modern methods such as high-performance liquid chromatography and mass spectrometry (Bada 2013; Parker et al. 2011).

- One set of experiments from the 1950s used, besides the apparatus of Figure 4.4, one that generated a hot water mist in the spark flask, simulating a volcanic eruption rich in water vapour and accompanied by lightning. Re-analysis of its extracts showed that this "volcanic" apparatus produced a wider variety of amino acids than the classic one (A. P. Johnson et al. 2008).
- Another, carried out in 1958 but not previously reported, used a gas mixture of hydrogen sulfide (H_2S), methane, ammonia and carbon dioxide. Analysis of its residues found 23 amino acids and 4 amines, including 7 compounds containing sulfur; the amino acids with stereocentres were racemic, indicating that they were not contaminants introduced during storage (Parker et al. 2011). The relative abundances of the amino acids were very similar to those in some carbonaceous meteorites (Parker et al. 2011) (Section 3.7).

Taken together, these re-analyses found more than 40 different amino acids and amines, many more than Miller was able to identify with the methods of 1953 (Bada 2013).

4.3.4. What the experiments show

The Miller experiment and its successors show that, given a reducing gas mixture and a source of energy, a variety of organic molecules of biological interest form readily and in the same racemic mixtures that a non-biological origin requires. They also show the

limitations of this approach. The yields depend strongly on the composition of the gas: as described in Section 3.4, neutral mixtures of CO_2 and N_2 give much less, although more than was once thought (Cleaves et al. 2008). And the products are mixtures of many compounds, of which the molecules of interest are only a part. Both points recur in the synthesis of the building blocks of RNA, which is the subject of the remaining sections of this chapter.

4.4. Sugars: the formose reaction and the ribose problem

4.4.1. Sugars and ribose

A **sugar**, or **monosaccharide**, is a compound of formula $C_n(H_2O)_n$, usually with n from 3 to 7, that contains one carbonyl group and a hydroxy group on each of the other carbon atoms. If the carbonyl group is an aldehyde at the end of the chain, the sugar is an **aldose**; if it is a ketone, the sugar is a **ketose**. Sugars are named by the number of carbon atoms: trioses (n = 3), tetroses (4), pentoses (5) and hexoses (6). The smallest compound of this type, glycolaldehyde ($HOCH_2CHO$, n = 2), has no stereocentre and is not usually counted as a sugar, but it is the starting point of sugar chemistry under prebiotic conditions.

Ribose, the sugar of RNA, is an aldopentose, $C_5H_{10}O_5$. In its open-chain form, three of its carbon atoms (C2, C3 and C4) are stereocentres, so that $2^3 = 8$ stereoisomers share its structural formula: D- and L-ribose, and the D and L forms of three other aldopentoses, arabinose, xylose and lyxose. The D/L label refers to the configuration of the stereocentre furthest from the carbonyl group, compared with that of D-glyceraldehyde (Section 4.1). RNA contains only D-ribose. In D-ribose, all three hydroxy groups on the stereocentres lie on the same side of the chain in the Fischer projection (Figure 4.5 b).

In water, a sugar does not remain in the open-chain form. A hydroxy group of the same molecule adds to the carbonyl group (an addition, Section 4.2), closing a ring. When the hydroxy group on C4 of ribose adds to the aldehyde at C1, the product is a five-membered ring of four carbon atoms and one oxygen atom, a **furanose**; addition of the hydroxy group on C5 gives a six-membered ring, a **pyranose**. The forms interconvert in solution. In RNA, ribose is present as the furanose, with its C1 bonded to a nucleobase and its C5 (and C3) bonded to phosphate (Chapter 5).

4.4.2. The formose reaction

The formose reaction, introduced in Section 3.6, converts formaldehyde into sugars in a basic solution. It is built from one type of carbon-carbon bond-forming step, the **aldol reaction**: in a basic solution, the carbon atom next to a carbonyl group loses a proton and becomes a nucleophile, which adds to the carbonyl carbon of a second molecule

(an electrophile), joining the two carbon skeletons. Formaldehyde, which has no carbon atom next to its carbonyl group, can act only as the electrophile.

The mechanism proposed by Breslow (Breslow 1959) explains why the reaction starts slowly and then accelerates (Figure 4.5 a):

1. Glycolaldehyde (C_2) adds to formaldehyde (aldol reaction), giving glyceraldehyde (C_3).
2. Glyceraldehyde isomerises to dihydroxyacetone (C_3), the corresponding ketose.
3. Dihydroxyacetone adds a second formaldehyde, giving a ketotetrose (C_4), which isomerises to an aldotetrose.
4. The aldotetrose splits in a reverse aldol reaction into two molecules of glycolaldehyde.

Each turn of the cycle consumes two molecules of formaldehyde and converts one molecule of glycolaldehyde into two. The cycle is therefore autocatalytic (Section 2.8): once a little glycolaldehyde is present, its amount grows, and the reaction accelerates. The direct combination of two formaldehyde molecules to give glycolaldehyde is very slow, which is why the reaction has an induction period unless glycolaldehyde is added at the start.

The same aldol and isomerisation steps also join the intermediates to one another and to formaldehyde in many other combinations. The formose reaction therefore yields numerous sugars without any selectivity for ribose (Larralde, Robertson, and Miller 1995), together with branched sugars and, at longer reaction times, compounds that are no longer sugars. Ribose is one product among many.

4.4.3. The stability problem

A second difficulty is that ribose, once formed, does not survive for long.

Key experiment: Larralde, Robertson and Miller 1995

Sugars were known to decompose in strong acid or base, but few data existed for neutral solutions. Larralde, Robertson and Miller measured the rate of decomposition of ribose between pH 4 and pH 8 at temperatures from 40 °C to 120 °C (Larralde, Robertson, and Miller 1995). The half-life of ribose at pH 7.0 was 73 minutes at 100 °C, and the value extrapolated to 0 °C was 44 years. The other aldopentoses and aldohexoses, as well as 2-deoxyribose and ribose 5-phosphate, had half-lives within an order of magnitude of these values.
The authors concluded that the backbone of the first genetic material could not have contained ribose or other sugars, because of their instability (Larralde, Robertson, and Miller 1995).

The two half-lives differ by a factor of about 3×10^5. Inserting them into the Arrhenius equation (Section 2.8) gives an apparent activation energy of about 107 kJ mol^{-1}, and, by interpolation, a half-life of the order of one year at 25 °C (Exercise 16). On a geological timescale even the value at 0 °C is short: ribose would have to be used within years to centuries of its formation, or protected.

4.4.4. Proposed solutions

Three types of response to the ribose problem have been proposed.

- **Stabilisation by minerals.** Borate, the anion of boric acid, forms complexes with sugars that have hydroxy groups on adjacent carbon atoms. Ricardo and co-workers reported that borate minerals stabilize ribose (Ricardo et al. 2004). Later experiments by the same group, starting from formaldehyde and glycolaldehyde, showed how borate, molybdate and calcium minerals guide the formation of tetroses, pentoses (including ribose) and heptoses, and suggested that pentoses would have formed as stable borate complexes on the surface of an early Earth (Kim et al. 2011). The stabilisation is not absolute: over longer times, the borate-bound pentoses pass on to other products (Kim et al. 2011). Whether borate minerals were available on the Hadean Earth is itself debated; Kim et al. (2011) argue that they could have formed from hydrated rocks in subduction zones.
- **Different backbones.** The first genetic material may not have contained ribose at all, but a backbone built from a simpler or more stable unit, which was later replaced by ribose-phosphate (Larralde, Robertson, and Miller 1995).
- **Bypassing free ribose.** Ribose need not be formed as a free sugar and then attached to a nucleobase. In the synthesis of Powner, Gerland, and Sutherland (2009), the carbon atoms of the sugar and part of the nucleobase are assembled together, so that free ribose never appears (Chapter 5).

None of these is accepted as a complete solution. The ribose problem illustrates a general difficulty of prebiotic chemistry that recurs in this chapter: reactions that form the right molecule usually form many others as well, and the right molecule is often not stable in the conditions under which it forms.

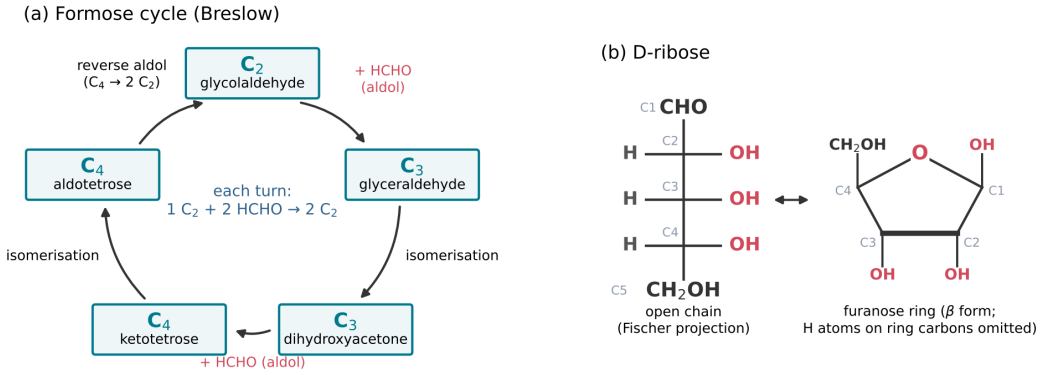


Figure 4.5.: Sugars. (a) The formose cycle proposed by Breslow. Glycolaldehyde (C_2) adds formaldehyde (HCHO) to give glyceraldehyde (C_3), which isomerises to dihydroxyacetone; a second formaldehyde gives a tetrose (C_4), which splits into two glycolaldehyde molecules, so that each turn doubles the glycolaldehyde. (b) D-ribose in the open-chain form (Fischer projection, stereocentres C2 to C4) and as the five-membered furanose ring, the form present in RNA, shown here as the β anomer with the hydrogen atoms on the ring carbons omitted.

4.5. Nucleobases from HCN and formamide

4.5.1. Purines and pyrimidines

The **nucleobases** of RNA are **heterocyclic** compounds: their rings contain atoms of at least one element other than carbon, in this case nitrogen. There are two families (Figure 4.6).

- **Purines** have two fused rings, a six-membered ring and a five-membered ring sharing one bond, with nine ring atoms in total, four of them nitrogen. The purines of RNA are **adenine** (A), $C_5H_5N_5$, and **guanine** (G), $C_5H_5N_5O$.
- **Pyrimidines** have a single six-membered ring containing two nitrogen atoms, separated by one carbon atom. The pyrimidines of RNA are **cytosine** (C), $C_4H_5N_3O$, and **uracil** (U), $C_4H_4N_2O_2$ (National Institute of Standards and Technology, n.d.b). In DNA, uracil is replaced by thymine, which carries an additional methyl group.

In RNA, each nucleobase is bonded through one of its ring nitrogen atoms to C1 of ribose (Section 4.4): N9 in the purines and N1 in the pyrimidines. The edges of the bases carry hydrogen-bond donors (N–H) and acceptors (ring N and C=O) (Section 2.5), and it is through these that adenine pairs with uracil and guanine with cytosine, the basis of the copying of genetic information described in later chapters.

The prebiotic problem is different for the two families. The purines can be formed directly from HCN; the pyrimidines require other routes.

4.5.2. Purines from hydrogen cyanide

Key experiment: Oró 1960-1961

Around 1960 Joan Oró found that adenine is formed from hydrogen cyanide in concentrated solutions of ammonium cyanide. He published a full account with Kimball in 1961 (Oró and Kimball 1961), together with a proposed mechanism (Oró 1961). The result was surprising because adenine, a molecule central to biology, has the formula $C_5H_5N_5$, exactly five times that of HCN: it can be regarded as a pentamer of HCN, assembled without the loss of any atom.
Later re-examination of this chemistry found that the yields are low: about 0.1 percent of adenine after hydrolysis of the soluble products, and about 0.05 percent from the insoluble black polymer that also forms (Borquez et al. 2005) (Section 3.6). Guanine is also formed.

The reaction proceeds through intermediates of increasing size. Four HCN molecules combine to give the HCN tetramer, whose structure was established in 1955 as **diaminomaleonitrile** (DAMN) (Webb, Frank, and Schneider 1955). Further steps, including the incorporation of a fifth HCN unit and the closing of the two rings, lead to adenine; detailed mechanisms have been proposed on the basis of experiments and quantum-chemical calculations (Oró 1961; Roy, Najafian, and Ragué Schleyer 2007).

Because HCN polymerises only when it is concentrated, the reaction requires a mechanism of concentration. One proposal is freezing: when a dilute solution freezes, the dissolved molecules are excluded from the ice and concentrated in the remaining liquid. In the experiment of Levy et al. (2000), introduced in Section 3.7, dilute solutions of ammonium cyanide kept frozen for 25 years gave adenine and guanine in substantial yields.

4.5.3. Pyrimidines

The pyrimidines are not formed in significant amounts from HCN alone. Two routes have been studied, both starting from **cyanoacetylene**, HC≡C–C≡N, a molecule produced by a spark discharge in a mixture of methane and nitrogen (Robertson and Miller 1995):

- Cyanoacetylene reacts with cyanate (NCO^-) to give cytosine. This reaction requires cyanate concentrations above 0.1 mol L^{-1}, which are unlikely in water, because cyanate is hydrolysed rapidly to CO_2 and NH_3 (Robertson and Miller 1995).

- Cyanoacetylene is hydrolysed to **cyanoacetaldehyde**, which reacts with urea, $(NH_2)_2CO$. At low concentrations of urea, no detectable cytosine is formed. Robertson and Miller showed that in concentrated urea solution, such as might be found in an evaporating lagoon or in pools on drying beaches, cyanoacetaldehyde gives cytosine in yields of 30 to 50 percent, and uracil is formed from cytosine by hydrolysis (Robertson and Miller 1995).

The second route illustrates again the importance of concentration by evaporation (Section 3.7). The conversion of cytosine into uracil is itself a hydrolysis (Section 4.2): the amino group of cytosine is replaced by a C=O group, with the release of ammonia.

4.5.4. Formamide

Formamide, $HCONH_2$, is the amide of formic acid (Section 4.1) and the product of the addition of water to HCN. It is a liquid between its melting point of 275.7 K (2.6 °C) (National Institute of Standards and Technology, n.d.b) and a boiling point of about 484 K (211 °C), calculated from vapour-pressure data (National Institute of Standards and Technology, n.d.b). Its liquid range thus extends far above that of water, and reactions can be carried out in it at temperatures at which water would boil.

In 2001 Saladino and co-workers reported the synthesis of purine, adenine, cytosine and 4(3H)-pyrimidinone by heating formamide in the presence of simple mineral catalysts (Saladino et al. 2001). Later work examined many other mineral catalysts and energy sources (Saladino et al. 2012), including high-energy chemistry initiated by laser pulses (Ferus et al. 2015). Formamide is attractive as a solvent and reagent because one compound supplies both purines and pyrimidines. Its weakness, as with HCN, is concentration: formamide mixes with water in all proportions, and how a concentrated formamide environment could have arisen on the early Earth is an open question.

4.5.5. Nucleobases from space

Nucleobases have also been found in carbonaceous meteorites. Callahan and co-workers detected a diverse suite of nucleobases in the Murchison meteorite and another meteorite, together with nucleobase analogues that are rare on Earth (Callahan et al. 2011) (Section 3.7). Meteorites may therefore have delivered small amounts of nucleobases to the early Earth, in addition to those formed on its surface.

4.5.6. What remains

Purines can be formed from HCN, and pyrimidines from cyanoacetylene and urea or from formamide. In each case, the bases are formed in mixtures with many other compounds and in low yield unless the reactants are concentrated. More fundamentally, a free

nucleobase and a free ribose do not combine efficiently: the addition of nucleobases to ribose is inefficient for the purines and does not occur at all for the pyrimidines of RNA (Powner, Gerland, and Sutherland 2009). This problem, and the route that avoids it by building the nucleotide in pieces, are the subject of Chapter 5.

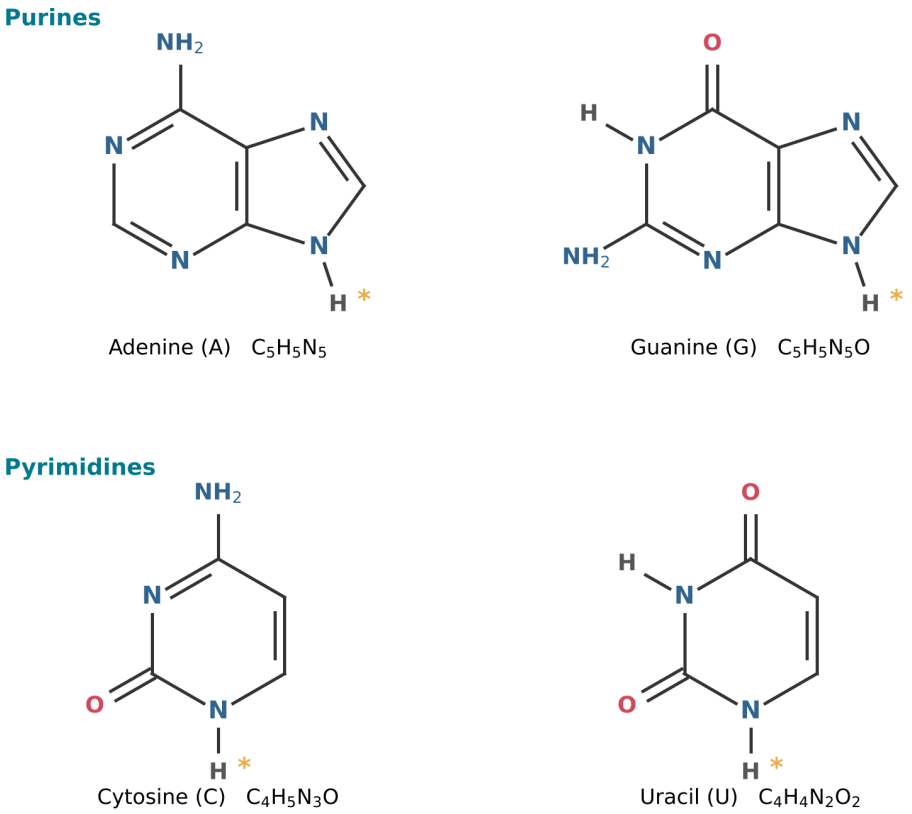


Figure 4.6.: The nucleobases of RNA. Purines (top): adenine and guanine, with a six-membered ring fused to a five-membered ring. Pyrimidines (bottom): cytosine and uracil, with a single six-membered ring. Nitrogen atoms are shown in blue and oxygen atoms in red; the hydrogen atom that is replaced by ribose in RNA (on N9 in purines, N1 in pyrimidines) is marked with an asterisk. Hydrogen atoms bonded to carbon are omitted.

4.6. Phosphate: esters, anhydrides and activation

4.6.1. Phosphoric acid and its esters

The phosphorus of RNA is present as phosphate (Section 3.6). Its parent compound, **phosphoric acid**, H_3PO_4, has a phosphorus atom bonded to one oxygen atom by a

double bond and to three hydroxy groups. Each of the three hydroxy groups can lose a proton (Section 2.6), so that phosphoric acid is a triprotic acid: depending on the pH, it is present as H_3PO_4, $H_2PO_4^-$, HPO_4^{2-} or PO_4^{3-}. In neutral water the dominant forms are $H_2PO_4^-$ and HPO_4^{2-}; the general term **phosphate** covers all of them.

A hydroxy group of phosphoric acid can condense with the hydroxy group of an alcohol, with the loss of water, to form a **phosphate ester** (Section 4.2). One such condensation gives a **phosphate monoester**, R–O–PO_3^{2-} at neutral pH; a second, with another alcohol, gives a **phosphate diester**, R–O–PO_2^-–O–R′ (Figure 4.7). In RNA, each phosphate is a diester that joins the C3 hydroxy group of one ribose to the C5 hydroxy group of the next. This linkage, the phosphodiester bond, is the subject of Chapter 6. The addition of a phosphate group to a molecule is called **phosphorylation**.

Two phosphate groups can also condense with each other. The product, pyrophosphate (diphosphate), contains a P–O–P linkage called a **phosphoanhydride bond**. Longer chains and rings of phosphate units are called polyphosphates; trimetaphosphate is a ring of three. The hydrolysis of a phosphoanhydride bond releases more free energy than the hydrolysis of a phosphate ester, so that a molecule that contains such a bond can transfer a phosphate group to an alcohol. Phosphoanhydride bonds are the source of free energy in adenosine triphosphate (ATP) in modern cells, and they are an example of the activated reactants introduced in Section 4.2.

4.6.2. Why phosphate?

In an influential essay, Westheimer asked why living organisms use phosphate to link the units of their nucleic acids and to carry many of their metabolites (Westheimer 1987). His answer rested on the charge of phosphate. A phosphate diester, such as the linkage in RNA, still carries one negative charge. The charge keeps the molecule dissolved in water and retained within a cell, and it repels negatively charged nucleophiles such as the hydroxide ion, so that the ester is hydrolysed very slowly. Yet the ester can be made and broken by enzymes when required (Westheimer 1987). No other simple acid combines these properties as well; this is one reason to think that phosphate was part of RNA from the beginning.

4.6.3. Phosphorylation in water

The stability that makes phosphate esters suitable for genetic molecules also makes them difficult to form. The formation of a phosphate ester is a condensation, and like the formation of the ester in Equation 4.1 it is disfavoured in water (Equation 4.2): the equilibrium lies on the side of the free phosphate and the alcohol. In addition, phosphate is scarce and poorly soluble in the presence of calcium (the phosphate problem, Section 3.6). Prebiotic phosphorylation therefore requires either the removal of water or

a phosphorus compound that carries its own free energy. Several approaches have been studied.

- **Heating with urea.** Lohrmann and Orgel showed that mixtures of urea and inorganic phosphate can act as prebiotic phosphorylating agents (Lohrmann and Orgel 1971).
- **Polyphosphates.** Volcanic processes can produce polyphosphates, which are activated forms of phosphate (Yamagata et al. 1991).
- **Reduced phosphorus.** The corrosion of schreibersite from meteorites releases reactive phosphorus compounds, including pyrophosphate; schreibersite phosphorylates nucleosides in water (Section 3.6) (Pasek and Lauretta 2005; Gull et al. 2015).
- **Diamidophosphate.** Amidophosphates, in which an amino group replaces a hydroxy group of phosphoric acid, are activated phosphorylating agents (see the key experiment below).

Key experiment: Gibard and co-workers 2018

Earlier prebiotic phosphorylation reactions had limited success or required particular environments that are incompatible with the subsequent formation of oligomers (Gibard et al. 2018). Gibard, Bhowmik, Karki, Kim and Krishnamurthy studied **diamidophosphate** (DAP), $PO_2(NH_2)_2^-$, which can be produced from trimetaphosphate (Gibard et al. 2018). In water, in solution or in a paste, DAP phosphorylated a wide variety of building blocks: nucleosides and nucleotides, amino acids and the precursors of lipids, without the need for an additional condensing agent. Under the same conditions, short chains of nucleotides, short peptides and lipid vesicles were formed (Gibard et al. 2018).
The result showed that a single, plausible phosphorus reagent can phosphorylate the three classes of biological building block in water, and so connect the formation of building blocks with their assembly into larger structures in one environment (Gibard et al. 2018).

4.6.4. What remains

Each approach to phosphorylation solves part of the problem. Dry heating removes water but is hard to combine with the reactions that precede and follow it; polyphosphates, schreibersite and amidophosphates provide activated phosphorus but depend on particular geological sources and settings. Moreover, phosphorylation of a nucleoside can occur at more than one hydroxy group (C2, C3 or C5 of ribose), and RNA requires a specific pattern. This problem of selectivity is the subject of the next section.

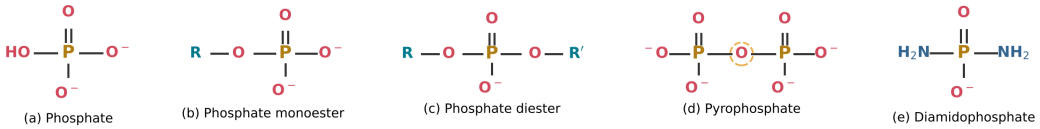


Figure 4.7.: Phosphorus compounds discussed in this section, drawn in the ionisation states that predominate near neutral pH. (a) Phosphate (HPO_4^{2-}). (b) A phosphate monoester. (c) A phosphate diester, the linkage between nucleotides in RNA. (d) Pyrophosphate, with a phosphoanhydride (P–O–P) bond. (e) Diamidophosphate (DAP). R and R′ denote organic groups.

4.7. Selectivity, mixtures and homochirality

4.7.1. Mixtures are the default

The preceding sections share a pattern. The spark discharge experiments gave more than 40 amino acids and amines (Bada 2013); the formose reaction yields numerous sugars without selectivity for ribose (Larralde, Robertson, and Miller 1995); the polymerisation of ammonium cyanide gives adenine in yields of about 0.1 percent, the rest being other products (Borquez et al. 2005). This outcome is not a failure of particular experiments but a consequence of the chemistry. Small, reactive molecules such as formaldehyde and HCN can combine in many ways, the number of possible products grows rapidly with their size, and in the absence of a catalyst that favours one route, the products are formed in proportion to their rates of formation and survival. A mixture is the default outcome of prebiotic chemistry.

The building blocks of RNA must nevertheless be assembled from a restricted set of components: one sugar (D-ribose), four nucleobases and phosphate, joined in specific positions. The question of how such a selection could arise is called the problem of **selectivity**. Three types of answer have been explored.

- **Selective reactions.** Some reactions convert only certain members of a mixture. Springsteen and Joyce showed that ribose can be selectively converted into a derivative and removed from a prebiotic mixture of sugars (Springsteen and Joyce 2004).
- **Sequences of reactions that pass through the right intermediates.** In the synthesis of Powner, Gerland, and Sutherland (2009), the building blocks are assembled in a sequence in which each step uses the product of the previous one; the mixture problem is reduced because the intermediates themselves are formed selectively. The reaction network of Patel et al. (2015), in which precursors of nucleotides, amino acids and lipids all derive from HCN, extends this approach to the three classes of biological building block. These syntheses are described in Chapter 5.

- **Physical separation.** Differences in solubility, volatility, adsorption on minerals or crystallisation can separate compounds formed together. Wet-dry cycles and other physical processes of the settings in Section 3.7 act on mixtures as well as on single compounds.

None of these solves the problem in general. Each requires specific conditions, and each is easier to demonstrate in the laboratory, where the starting materials are pure and their amounts controlled, than in a natural setting where many processes act at once.

4.7.2. Homochirality

A special case of selectivity concerns enantiomers. Syntheses from achiral starting materials give racemates (Section 4.1), yet RNA contains only D-ribose and proteins only L-amino acids. The property of consisting of only one enantiomer is called **homochirality**. Its origin is an open question, which can be divided into two parts: how a small excess of one enantiomer could arise, and how it could then be amplified.

The composition of a mixture of enantiomers is described by its **enantiomeric excess** (ee):

$$\text{ee} = \frac{[\text{L}] - [\text{D}]}{[\text{L}] + [\text{D}]} \tag{4.4}$$

where [L] and [D] are the concentrations of the two enantiomers. A racemate has ee = 0; a homochiral sample has ee = ± 1, often expressed as ± 100 percent.

Small initial excesses. Small excesses of one enantiomer may arise from chance fluctuations, which are significant in small numbers of molecules, or from physical influences that distinguish mirror images, such as circularly polarised light. Cronin and Pizzarello reported enantiomeric excesses, favouring the L form, in amino acids of the Murchison meteorite (Cronin and Pizzarello 1997), which suggests that excesses can arise without biology.

Amplification. A small excess becomes significant only if it is amplified. In 1953 Frank proposed a simple reaction scheme in which this occurs spontaneously (Frank 1953). Each enantiomer catalyses its own formation from an achiral reactant (autocatalysis, Section 2.8), and the two enantiomers react with each other to form an inactive product. The enantiomer present in excess then grows at the expense of the other, and the excess is amplified towards homochirality (Figure 4.8). In 1995 Soai and co-workers discovered a real reaction with this property, an asymmetric autocatalysis in which a product with a small enantiomeric excess is formed with a much larger one (Soai et al. 1995).

Key experiment: Viedma 2005

Some compounds crystallise as separate crystals of the two mirror-image forms. Viedma studied such a system in which a large population of crystals, initially containing equal numbers of the two forms, was kept in contact with its saturated solution and continuously ground (Viedma 2005). Grinding breaks crystals into fragments that serve as seeds for further growth, while small fragments dissolve and their material is recycled through the solution. Over time, the population moved into complete chiral purity, with one of the two forms disappearing entirely (Viedma 2005).

The experiment showed that total symmetry breaking and complete chiral purity can be achieved from a system that initially contains both forms, through a nonlinear process of autocatalysis and recycling (Viedma 2005). It provided a physical counterpart to Frank's chemical model.

These mechanisms show that homochirality is not physically improbable once a mechanism of amplification exists. Which mechanism, if any, operated in the origin of RNA is not known. The question is taken up again in Chapter 5, where the chirality of ribose is

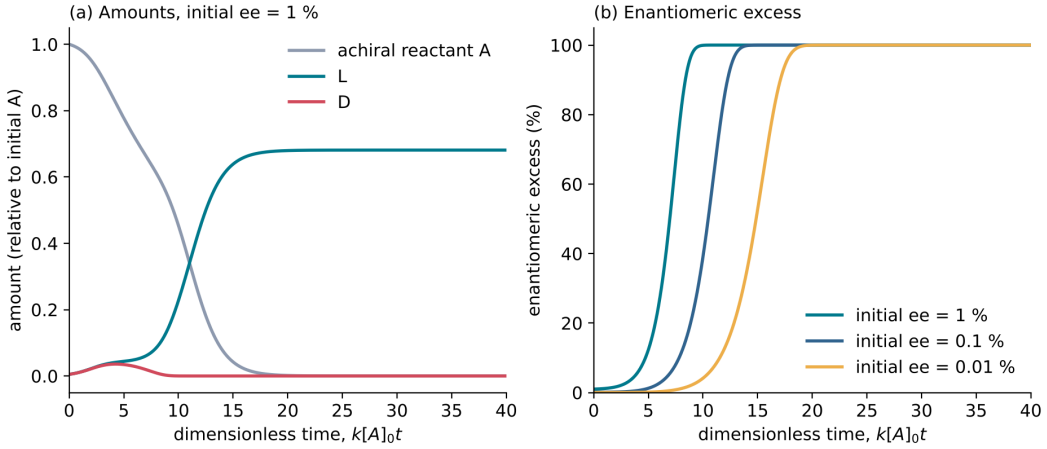


Figure 4.8.: A numerical solution of a Frank-type model of chiral amplification. An achiral reactant A is converted into L or D, each of which catalyses its own formation (rate constant k); L and D react with each other to form an inactive product (rate constant 20k). All amounts are relative to the initial amount of A, and the initial amount of L plus D is 0.01. (a) Amounts of A, L and D for an initial enantiomeric excess of 1 percent. (b) Enantiomeric excess for initial excesses of 1, 0.1 and 0.01 percent; in each case the excess is amplified to 100 percent. The model is illustrative, and its parameters do not represent a specific chemical system.

Summary

- Organic molecules can be described as skeletons carrying functional groups, which determine their chemical properties. The functional groups of the building blocks of RNA are hydroxy, carbonyl (aldehyde and ketone), carboxy, amino, amide, nitrile and ester groups, including phosphate esters.
- Isomers have the same molecular formula but different structures. Enantiomers are non-superposable mirror images; synthesis from achiral starting materials gives racemates, whereas RNA contains only D-ribose.
- Nucleophiles donate an electron pair to electrophiles such as the carbonyl carbon. Addition joins two molecules into one; condensation joins them with the loss of water, and hydrolysis reverses it. In water, hydrolysis is generally favoured, because water is present at about 55 mol per litre.
- Oxidation and reduction change the oxidation numbers of carbon; converting feedstock molecules such as HCN into sugars requires reduction.
- In 1953 Miller showed that an electric spark acting on a mixture of methane, ammonia, hydrogen and water vapour produces amino acids, including glycine and alanine. Re-analyses of his archived samples found more than 40 amino acids and amines, formed as racemates. The Strecker synthesis makes amino acids from aldehydes, ammonia and HCN.
- Ribose is an aldopentose with three stereocentres, one of eight stereoisomers. The formose reaction, an autocatalytic cycle of aldol and isomerisation steps, converts formaldehyde into many sugars without selectivity for ribose, and ribose decomposes with a half-life of 73 minutes at 100 °C and pH 7. Proposed responses include stabilisation by borate minerals, alternative backbones and routes that bypass free ribose.
- The nucleobases are purines (adenine, guanine) and pyrimidines (cytosine, uracil). Adenine, $C_5H_5N_5$, is a pentamer of HCN and forms from concentrated ammonium cyanide, in low yield. Cytosine forms from cyanoacetaldehyde in concentrated urea, and uracil from cytosine by hydrolysis. Heated formamide with mineral catalysts gives both purines and pyrimidines. Free nucleobases and ribose do not combine efficiently.
- Phosphoric acid forms phosphate monoesters and diesters with alcohols, and phosphoanhydride bonds with itself. The negative charge of the phosphate diester makes RNA soluble and resistant to hydrolysis. Phosphorylation in water is disfavoured; proposed solutions include heating with urea, polyphosphates, schreibersite and diamidophosphate.
- Prebiotic reactions give mixtures by default. Selective reactions, reaction sequences that pass through the right intermediates, and physical separation are three responses. Homochirality requires a small initial enantiomeric excess and a mechanism of amplification, such as the autocatalysis of Frank's model, the Soai reaction or Viedma's crystal grinding.

Exercises

1. Identify the functional groups in glycolaldehyde ($HOCH_2CHO$) and in glycine (H_2NCH_2COOH).
2. Draw the structures of the two structural isomers of formula C_2H_6O, and name the family to which each belongs.
3. Explain why the carbon atom of a carbonyl group carries a partial positive charge.
4. How many stereocentres does glycolaldehyde have? Does it have enantiomers?
5. Write the molecular formula of the ester formed from acetic acid (CH_3COOH) and methanol (CH_3OH), and identify the small molecule released.
6. Write the equation for the addition of HCN to formaldehyde, and name the functional groups in the product.
7. A condensation has $\Delta G^\circ = +15$ kJ mol^{-1} at 298.15 K. Calculate its equilibrium constant.
8. Using Equation 4.2, explain why removing water, for example by evaporation, shifts the equilibrium of an esterification towards the ester.
9. Assign oxidation numbers to each carbon atom in glycolaldehyde ($HOCH_2CHO$) and in glyceraldehyde ($HOCH_2CH(OH)CHO$).
10. Identify the nucleophile and the electrophile in the addition of water to formaldehyde.
11. Write the three steps of the Strecker synthesis of alanine from acetaldehyde (CH_3CHO), and classify each step as an addition, a condensation or a hydrolysis.
12. Explain why the alanine formed in a spark discharge experiment is racemic, and why the racemic composition of the amino acids in Miller's archived samples argues against contamination.
13. Which features of Miller's apparatus (Figure 4.4) correspond to an atmosphere, to lightning and to an ocean? Which processes on the early Earth does the apparatus omit?
14. Draw the Fischer projections of D-ribose and L-ribose. Which of the stereocentres determines the D/L label?
15. In the formose cycle (Figure 4.5 a), how many molecules of formaldehyde are consumed for each new molecule of glycolaldehyde? Starting from one molecule of glycolaldehyde, how many are present after 10 turns of the cycle, if every molecule completes each turn and none is lost to side reactions?
16. Using the half-lives of ribose measured by Larralde and co-workers (73 min at 100 °C and 44 years at 0 °C, pH 7.0), estimate the apparent activation energy of ribose decomposition with the Arrhenius equation, and the half-life at 25 °C. State the assumption on which the estimate rests.
17. Explain why a reaction that gives ribose as one product among many is a problem for the origin of RNA even if the total yield of sugars is high.
18. Show that the molecular formulas of adenine and guanine differ by one oxygen atom. Which functional groups distinguish them in Figure 4.6?
19. Using the molar masses of HCN (27.03 g mol^{-1}) and adenine (135.13 g mol^{-1}),

calculate the mass of adenine formed from 1 kg of HCN at a yield of 0.1 percent by mass, and the maximum mass if every HCN molecule were converted into adenine.

20. Write the equation for the hydrolysis of cytosine ($C_4H_5N_3O$) to uracil ($C_4H_4N_2O_2$), and identify the other product.
21. Why does the synthesis of cytosine from cyanoacetaldehyde require a concentrated urea solution? Name two prebiotic settings in which such a solution could form.
22. Write the equation for the condensation of phosphoric acid with methanol to give methyl phosphate, and identify the bond that is formed.
23. How many negative charges does the phosphate diester linkage carry at neutral pH? Explain how this charge affects the rate of attack by hydroxide ions.
24. Using Equation 4.3, calculate the equilibrium constant of a phosphorylation with $\Delta G^\circ = +12$ kJ mol^{-1} at 298.15 K. What does the result imply for phosphorylation in dilute aqueous solution?
25. Explain why a phosphoanhydride bond can be used to drive the formation of a phosphate ester, whereas inorganic phosphate cannot.
26. A sample contains 0.52 mol of L-alanine and 0.48 mol of D-alanine. Calculate its enantiomeric excess with Equation 4.4.
27. In Frank's model (Figure 4.8), what is the role of the reaction between L and D? What would happen to the enantiomeric excess if this reaction were absent?
28. Explain why the formose reaction gives many sugars, using the reaction steps of Section 4.4.
29. Why is selectivity easier to demonstrate in the laboratory than in a natural setting? Give two reasons.

5. Assembling Nucleotides

Learning objectives

After reading this chapter you should be able to:

1. Describe the structure of a ribonucleotide, distinguish a nucleoside from a nucleotide, and use the primed numbering of the ribose ring.
2. Explain why joining a free nucleobase to free ribose is inefficient for purines and fails for the pyrimidines of RNA.
3. Outline the synthesis of activated pyrimidine ribonucleotides by Powner, Gerland and Sutherland, and the routes to purine nucleosides through formamidopyrimidines.
4. Explain where and how phosphate is attached to a nucleoside, and why the 2′,3′-cyclic phosphate is an activated form.
5. Explain how the chirality of ribose affects the copying of nucleotide chains, and how wet-dry cycles shift condensation equilibria.

5.1. Anatomy of a nucleotide

5.1.1. Nucleosides and nucleotides

A **nucleoside** is a nucleobase (Section 4.5) joined to a sugar. In a ribonucleoside, the sugar is D-ribose in its furanose form (Section 4.4), and the bond joins C1 of the ribose to a ring nitrogen of the base: N9 in the purines and N1 in the pyrimidines. This carbon-nitrogen bond is called the **N-glycosidic bond**. A **nucleotide** is a nucleoside that carries one or more phosphate groups (Section 4.6), usually on the hydroxy group at C5 of the ribose. A nucleotide therefore has three parts: a nucleobase, a sugar and a phosphate (Figure 5.1).

To distinguish the atoms of the sugar from those of the base, the carbon atoms of ribose in a nucleoside are numbered with a prime: C1′ to C5′. The N-glycosidic bond joins C1′ to the base; the hydroxy groups of ribose are on C2′, C3′ and C5′; and the ring oxygen links C1′ and C4′.

The four ribonucleosides of RNA are **adenosine**, **guanosine**, **cytidine** and **uridine**. The corresponding nucleotides with one phosphate on C5′ are the nucleoside

5′-monophosphates, abbreviated AMP, GMP, CMP and UMP. Their formulas follow from the condensation reactions that join their parts (Table 5.1). For example, adenine ($C_5H_5N_5$) and ribose ($C_5H_{10}O_5$) combine with the loss of one molecule of water to give adenosine, $C_{10}H_{13}N_5O_4$; adenosine and phosphoric acid (H_3PO_4) combine with the loss of a second molecule of water to give AMP, $C_{10}H_{14}N_5O_7P$.

Table 5.1.: The ribonucleosides of RNA and their 5′-monophosphates, shown as the neutral (fully protonated) acids. Molar masses are calculated from the abridged standard atomic weights H 1.0080, C 12.011, N 14.007, O 15.999 and P 30.974 (CIAAW, n.d.).

Base	Nucleoside	Formula	M (g mol^{-1})	5′-Monophosphate	Formula	M (g mol^{-1})
Adenine	Adenosine	$C_{10}H_{13}N_5O_4$	267.25	AMP	$C_{10}H_{14}N_5O_7P$	347.22
Guanine	Guanosine	$C_{10}H_{13}N_5O_5$	283.24	GMP	$C_{10}H_{14}N_5O_8P$	363.22
Cytosine	Cytidine	$C_9H_{13}N_3O_5$	243.22	CMP	$C_9H_{14}N_3O_8P$	323.20
Uracil	Uridine	$C_9H_{12}N_2O_6$	244.20	UMP	$C_9H_{13}N_2O_9P$	324.18

5.1.2. Configuration at C1′

The formation of the N-glycosidic bond creates a new stereocentre at C1′. The base can lie on the same side of the furanose ring as the CH_2OH group at C4′, or on the opposite side. The two forms are called the β and α **anomers**, respectively. All the nucleosides of RNA are β-D-ribonucleosides: the base is on the same face of the ring as C5′. The α anomers are diastereomers of the natural nucleosides, not enantiomers, and have different properties. A prebiotic synthesis must therefore control not only which sugar and which base are joined, but also the configuration of the new bond.

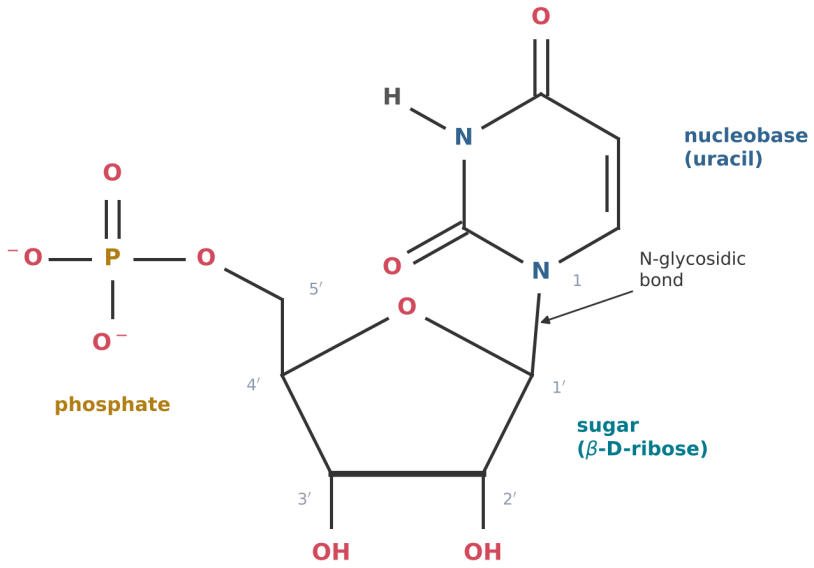


Figure 5.1.: Anatomy of a ribonucleotide, shown for uridine 5′-monophosphate (UMP). The nucleobase (uracil) is joined through N1 to C1′ of β-D-ribofuranose by the N-glycosidic bond; a phosphate group is attached to the oxygen on C5′. The base without the phosphate is the nucleoside uridine. Hydrogen atoms on carbon are omitted.

5.2. The glycosidic bond

5.2.1. A condensation in water

The N-glycosidic bond is formed by a condensation (Section 4.2): the hydroxy group on C1′ of ribose and a hydrogen atom on the ring nitrogen of the base are lost as water. Like other condensations, it is disfavoured in water (Equation 4.2), so that the direct reaction of a base with ribose in dilute solution gives little product. The obvious remedy, the removal of water by drying and heating, was explored in early work.

Fuller, Sanchez and Orgel reported a synthesis of purine nucleosides from purines and ribose (Fuller, Sanchez, and Orgel 1972), but this and later attempts showed the reaction to be inefficient. The underlying difficulty was summarised by Powner, Gerland and Sutherland: ribose is difficult to form selectively, and the addition of nucleobases to ribose is inefficient in the case of the purines and does not occur at all in the case of the canonical pyrimidines, cytosine and uracil (Powner, Gerland, and Sutherland 2009).

5.2.2. Why the pyrimidines fail

The ring nitrogen of a pyrimidine that must form the glycosidic bond, N1, is a poor nucleophile, because its lone pair is delocalised into the adjacent carbonyl group and the ring. The N1–C1′ bond is therefore not formed in detectable amounts (Powner, Gerland, and Sutherland 2009), while ribose itself decomposes over the same period (Section 4.4). This combination of a slow desired reaction and faster competing reactions is a recurring form of the selectivity problem of Section 4.7.

The failure of the direct route led to two different strategies. The first constructs the glycosidic bond before the nucleobase is complete, so that the nitrogen that forms the bond is part of a more reactive intermediate. This strategy underlies the pyrimidine synthesis of Section 5.3. The second modifies the base, as in the purine routes through formamidopyrimidines described in the same section.

5.3. Building the nucleotide in pieces

5.3.1. The strategy

If the complete base and the complete sugar do not combine, the nucleotide can be assembled from smaller pieces, each containing part of the base and part of the sugar. In this approach, the bond that will become the N-glycosidic bond is formed early, between fragments in which the nitrogen atom is still reactive, and the rings of the base and the sugar are completed afterwards. Anastasi and co-workers showed that precursors of nucleosides can be assembled directly from units of two and three carbon

atoms (Anastasi et al. 2006), and this work led to the synthesis of activated pyrimidine ribonucleotides described below.

Key experiment: Powner, Gerland and Sutherland 2009

Powner, Gerland and Sutherland reported a synthesis of activated pyrimidine ribonucleotides that bypasses both free ribose and the free nucleobases (Powner, Gerland, and Sutherland 2009). The starting materials were cyanamide, cyanoacetylene, glycolaldehyde, glyceraldehyde and inorganic phosphate, all regarded as plausible prebiotic feedstock molecules (Section 3.6, Section 4.5).
The sequence passes through two key intermediates (Figure 5.2). Cyanamide and glycolaldehyde give 2-aminooxazole, a small heterocycle that contains the future C1′ and C2′ of the sugar together with the nitrogen and carbon atoms that will become part of the base. Reaction of 2-aminooxazole with glyceraldehyde, which supplies the remaining three carbon atoms of the sugar, gives a pentose amino-oxazoline; the arabinose isomer, arabinose amino-oxazoline, is carried forward. Cyanoacetylene then adds to complete the pyrimidine ring, giving an anhydronucleoside. Finally, phosphate is attached and the anhydronucleoside is converted into the activated pyrimidine nucleotide (Powner, Gerland, and Sutherland 2009). Phosphate is incorporated only at the last stage, but its presence from the start is essential: it controls three of the earlier reactions by acting as a general acid-base catalyst, a nucleophilic catalyst, a pH buffer and a chemical buffer (Powner, Gerland, and Sutherland 2009). The authors concluded that prebiotic syntheses should be studied in mixed chemical systems, in which the reactants for one step can also control others (Powner, Gerland, and Sutherland 2009).

The 2009 synthesis changed the questions asked about the origin of RNA. It showed that the failure of the direct coupling of base and sugar need not exclude RNA as the first genetic material, and that the stepwise order in which chemists assemble a molecule need not be the order in which it is formed prebiotically. It also left problems: it produces the pyrimidine nucleotides only, it requires the reactants to be added in a specific order, and its starting materials must be supplied in relatively pure form.

5.3.2. Routes to the purines

The purines require a different approach. Becker and co-workers described a high-yielding and strictly regioselective route to purine nucleosides (Becker et al. 2016). In this route the purine is not formed first: instead, **formamidopyrimidines**, pyrimidines that carry the atoms of the future five-membered ring in open form, react with the sugar, and the five-membered ring is closed afterwards. The condensation gives the natural N9 nucleosides selectively (Becker et al. 2016).

The same group later reported a synthesis of the pyrimidine nucleosides from small molecules and ribose, driven solely by wet-dry cycles, and compatible with the purine

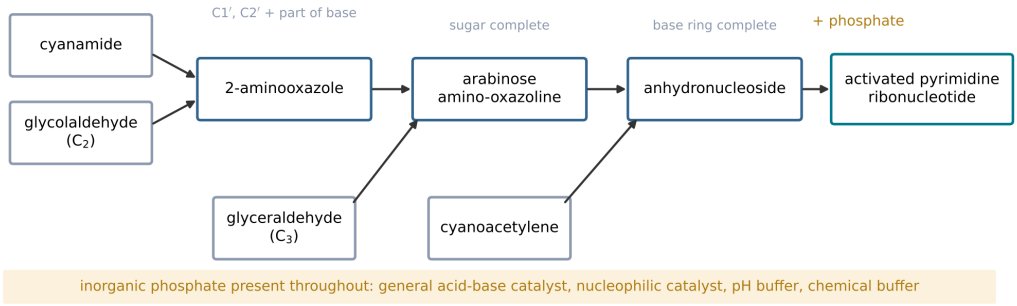


Figure 5.2.: Simplified outline of the synthesis of activated pyrimidine ribonucleotides by Powner, Gerland, and Sutherland (2009). Each box names a starting material or an intermediate; the free sugar ribose and the free nucleobases do not appear. Phosphate is present throughout as a catalyst and buffer and is incorporated in the final step.

synthesis, so that all four of the canonical bases could form under one set of conditions (Becker et al. 2019). A key starting material, hydroxylamine or hydroxylamine disulfonate, can form under plausible conditions in the early atmosphere (Becker et al. 2019). In the presence of minerals containing phosphate, the 5′-monophosphates and 5′-diphosphates of the nucleosides also formed selectively in one-pot reactions (Becker et al. 2019).

These two families of route, the cyanamide chemistry of Sutherland and co-workers and the formamidopyrimidine chemistry of Carell and co-workers, differ in their starting materials and conditions. In their published forms, both involve steps in which the experimenter supplies reagents in a controlled order, and how they could proceed together in one natural setting is an open question.

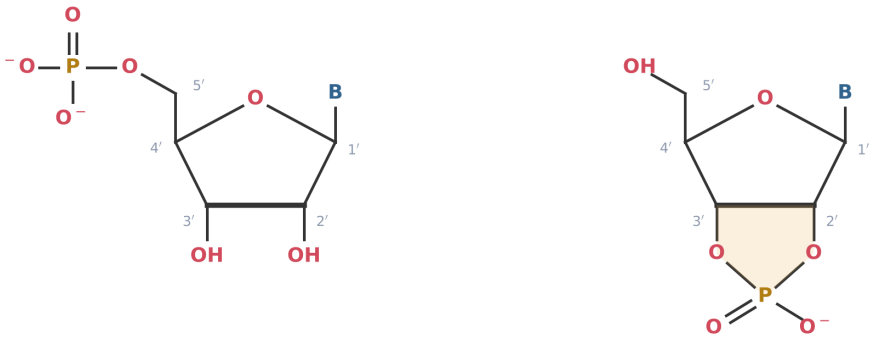
5.4. Phosphorylation of nucleosides

5.4.1. Three hydroxy groups

A ribonucleoside has three hydroxy groups, at C2′, C3′ and C5′, any of which can be phosphorylated. In RNA, the phosphate diester joins C3′ of one nucleotide to C5′ of the next (Section 4.6). A prebiotic phosphorylation that attaches phosphate at random would give a mixture of 2′-, 3′- and 5′-monophosphates and of products with more than one phosphate group. The position of phosphorylation is therefore a further question of selectivity (Section 4.7).

5.4.2. The 2′,3′-cyclic phosphate

When a phosphate group is attached to C2′ or C3′, it can condense a second time with the neighbouring hydroxy group, forming a five-membered ring that contains C2′, C3′, two oxygen atoms and the phosphorus atom. The product is the nucleoside **2′,3′-cyclic phosphate** (Figure 5.3). The ring is strained, and its opening releases free energy. The cyclic phosphate is therefore an activated form of the nucleotide, able to transfer its phosphate to the 5′-hydroxy group of another nucleotide and so to form a phosphodiester bond without an external source of energy. The activated pyrimidine ribonucleotides of Powner, Gerland, and Sutherland (2009) are cyclic phosphates of this kind.



(a) 5′-monophosphate (b) 2′,3′-cyclic phosphate

Figure 5.3.: Phosphate groups on a ribonucleoside. The ribose ring of a nucleoside is shown with its three hydroxy-bearing positions. (a) 5′-Monophosphate, the form found in RNA chains. (b) 2′,3′-Cyclic phosphate, in which one phosphate bridges C2′ and C3′ in a five-membered ring; its opening releases free energy and can form a phosphodiester bond. B denotes the nucleobase.

5.4.3. Sources of the phosphate

The approaches to prebiotic phosphorylation introduced in Section 4.6 apply to nucleosides. Schreibersite attaches phosphate groups to adenosine and uridine in water (Gull et al. 2015). Diamidophosphate phosphorylates nucleosides and nucleotides in water or a paste without an additional condensing agent, and under the same conditions forms short chains of nucleotides (Gibard et al. 2018). In the wet-dry cycle chemistry of Becker et al. (2019), phosphate-containing minerals gave 5′-mono- and diphosphates selectively. In each case, the conditions that favour phosphorylation are those that reduce the activity of water: drying, pastes, or the use of an activated phosphorus reagent.

5.5. Chirality and the copying of nucleotides

5.5.1. D-ribose in every nucleotide

Each ribonucleotide contains four stereocentres in its sugar, at C1′, C2′, C3′ and C4′, and all the nucleotides of an RNA molecule have the same configuration at each of them: they are β-D-nucleotides. The mirror image of a β-D-nucleotide is a β-L-nucleotide. Because syntheses from achiral feedstock molecules give racemates (Section 4.1), a prebiotic nucleotide synthesis that begins with achiral or racemic materials will give equal amounts of D and L nucleotides unless an asymmetric influence acts. In the synthesis of Powner, Gerland, and Sutherland (2009), for example, glyceraldehyde is the only chiral starting material; if it is racemic, so are the products.

5.5.2. Enantiomeric cross-inhibition

The importance of homochirality for RNA appears when nucleotides are joined into chains on a template. A **template** is an existing chain whose sequence directs the order in which monomers are joined; template-directed copying is the subject of Chapter 8. Joyce and co-workers studied the template-directed formation of chains of guanosine nucleotides on a template of polycytidylic acid, poly(C), using activated monomers of either one or both handedness (Joyce et al. 1984). Their results showed chiral selection: when nucleotides of the opposite handedness were present, they were incorporated at the end of a growing chain and blocked further growth (Joyce et al. 1984). This effect, called **enantiomeric cross-inhibition**, means that copying is inefficient in a racemic mixture of nucleotides.

Enantiomeric cross-inhibition is both a problem and a possible clue. It shows that a racemic pool of nucleotides is a poor starting point for RNA. But it is also a form of the mutual inhibition between enantiomers in Frank's model (Section 4.7), and so a possible mechanism of amplification once an excess of one enantiomer exists.

5.6. Condensation in water and wet-dry cycles

5.6.1. Water as a product

Every bond that joins the parts of a nucleotide, and every bond that joins nucleotides into chains, is formed by a condensation that releases water. For a condensation of the form

$$\mathrm{A{-}OH + H{-}B \rightleftharpoons A{-}B + H_2O}$$

the equilibrium constant can be written in terms of the activity of water, a_{w}:

$$K = \frac{[\mathrm{A{-}B}]\, a_{\mathrm{w}}}{[\mathrm{A{-}OH}][\mathrm{H{-}B}]} \tag{5.1}$$

In dilute solution, a_{w} is close to 1, and with K small (Section 4.2) the ratio of product to reactants is small. The **activity** of water measures its effective concentration relative to pure water; it is reduced when water evaporates and solutes become concentrated, or when the reaction takes place in a film, a paste or the pores of a mineral. At constant K, a tenfold reduction in a_{w} increases the equilibrium ratio of product to reactants tenfold. Drying also increases the concentrations of the reactants themselves, which increases the rate of the condensation and, for bimolecular reactions, the fraction of reactants converted.

5.6.2. Wet-dry cycles

A **wet-dry cycle** alternates between a wet phase, in which reactants are dissolved and transported, and a dry phase, in which condensations are favoured. Such cycles occur naturally at the margins of ponds and in hot-spring settings, where water evaporates and is replenished (Section 3.7) (Damer and Deamer 2020). They combine two requirements that cannot be met at the same time: mobility of the reactants, which requires water, and condensation, which requires its removal. The synthesis of nucleosides by Becker et al. (2019) was driven solely by such cycles.

A simple model shows how cycling accumulates a product. Suppose that, in each dry phase, a fraction f of the remaining reactant is converted into a condensation product, and that in each wet phase a fraction h of the product is hydrolysed back to the reactant. If $\mathrm{x_n}$ is the fraction of material present as product after n cycles,

$$x_{n+1} = (1 - h)\,[\,x_n + f(1 - x_n)\,] \tag{5.2}$$

which approaches the steady-state value $\mathrm{x^*} = \mathrm{f}(1 - \mathrm{h})/(\mathrm{f} + \mathrm{h} - \mathrm{fh})$ after many cycles (Figure 5.4). With $\mathrm{f} = 0.2$ and $\mathrm{h} = 0.05$, for example, $\mathrm{x^*} = 0.19/0.24 \approx 0.79$. The product accumulates to a high fraction even though each dry phase converts only a fifth of the reactant, provided that hydrolysis in the wet phase is slow compared with condensation in the dry phase. If hydrolysis is fast, the steady state is low however many cycles occur. The model is schematic: in real cycles the products also include side products and decomposition products, which accumulate as well.

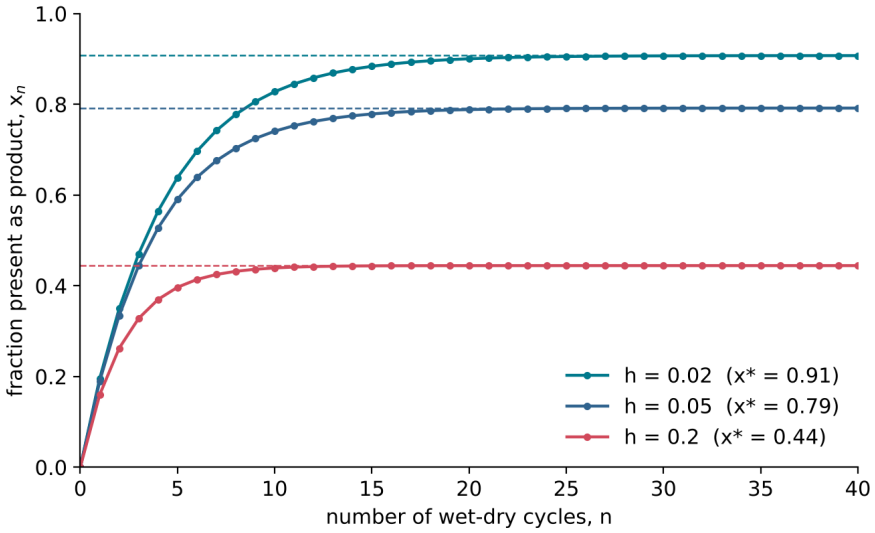


Figure 5.4.: The accumulation of a condensation product in a simple model of wet-dry cycles (Equation 5.2). In each cycle a fraction f = 0.2 of the remaining reactant is converted in the dry phase, and a fraction h of the product is hydrolysed in the wet phase. Curves are shown for h = 0.02, 0.05 and 0.2; dashed lines mark the steady states.

Summary

- A nucleoside is a nucleobase joined by an N-glycosidic bond from N9 (purines) or N1 (pyrimidines) to C1′ of β-D-ribofuranose; a nucleotide carries in addition one or more phosphate groups, usually on C5′.
- The direct condensation of a free base with free ribose is inefficient for the purines and does not occur for the canonical pyrimidines.
- Powner, Gerland and Sutherland (2009) made activated pyrimidine ribonucleotides from cyanamide, cyanoacetylene, glycolaldehyde, glyceraldehyde and phosphate, through 2-aminooxazole, arabinose amino-oxazoline and an anhydronucleoside, bypassing free ribose and free bases. Carell and co-workers developed routes to purine nucleosides through formamidopyrimidines and a wet-dry cycle synthesis compatible with all four bases.
- Phosphorylation can occur at C2′, C3′ or C5′; the 2′,3′-cyclic phosphate is an activated form of the nucleotide.
- Nucleotides of the wrong handedness block template-directed chain growth (enantiomeric cross-inhibition), so copying requires homochiral nucleotides.
- Condensations are favoured when the activity of water is low. Wet-dry cycles combine mobility in the wet phase with condensation in the dry phase, and can accumulate products when hydrolysis is slow.

Exercises

1. Write the equation for the condensation of cytosine with ribose to give cytidine, and verify the formula of cytidine in Table 5.1.
2. Calculate the molar mass of guanosine 5′-monophosphate from the abridged standard atomic weights, and compare it with Table 5.1.
3. How many stereocentres does the ribose of a nucleoside contain? How many stereoisomers of adenosine are possible in principle?
4. Explain the difference between the α and β anomers of a nucleoside. Are they enantiomers?
5. In the synthesis of Powner, Gerland, and Sutherland (2009), which starting materials supply the atoms of the sugar, and which supply the atoms of the base?
6. Phosphate is incorporated into the nucleotide only in the final step of the 2009 synthesis. Explain why its presence from the start was nevertheless essential.
7. Draw the 2′-, 3′- and 5′-monophosphates of uridine, and the 2′,3′-cyclic phosphate. Which of them is activated, and why?
8. Explain, with reference to Equation 5.1, why a paste or a film of reactants favours phosphorylation more than a dilute solution.
9. Explain how enantiomeric cross-inhibition could both hinder and help the origin of homochiral RNA.
10. Using Equation 5.2, derive the expression for the steady-state fraction x* by setting $x_{n+1} = x_n$.
11. For $f = 0.2$, calculate x* for $h = 0.02$ and $h = 0.2$. How many cycles are needed in each case for x to reach 90 percent of x*, starting from $x_0 = 0$?
12. Name two simplifications of the model of Equation 5.2, and explain how each would change the predicted accumulation of product.

6. Joining Nucleotides into Chains

Learning objectives

After reading this chapter you should be able to:

1. Describe the phosphodiester bond and the 5′ to 3′ direction of an RNA chain, and write sequences in the standard notation.
2. Explain why the polymerisation of nucleotides in water requires activated monomers, and describe phosphorimidazolides and cyclic phosphates.
3. Describe the competition between chain growth and hydrolysis, and explain why it limits chain length in dilute solution.
4. Describe how mineral surfaces, ice and wet-dry cycles favour polymerisation, with reference to key experiments.
5. Calculate chain length distributions from a probability of extension, and explain why long chains are rare.

6.1. The phosphodiester bond

6.1.1. Structure of an RNA chain

An RNA molecule is a chain of nucleotides (Section 5.1) joined by **phosphodiester bonds**. Each phosphate group is esterified twice (Section 4.6): once to the 3′-hydroxy group of one nucleotide and once to the 5′-hydroxy group of the next. The alternating sugar and phosphate units form the **backbone** of the chain, and the nucleobases are attached to it as side groups at C1′ of each ribose (Figure 6.1). Each phosphodiester carries one negative charge at neutral pH (Section 4.6).

Because the two ends of each nucleotide unit differ, the chain has a direction. One end carries a free 5′ group (often a phosphate), and the other a free 3′-hydroxy group. By convention, sequences are written from the 5′ end to the 3′ end, using the one-letter abbreviations of the nucleobases: 5′-AGCU-3′ denotes a chain of four nucleotides in which adenosine carries the free 5′ end and uridine the free 3′ end. A short chain is called an **oligonucleotide**, often with the number of units given, as in a 10-mer; a long one a **polynucleotide**.

6.1.2. 3′,5′ and 2′,5′ linkages

Ribose has hydroxy groups on both C2′ and C3′, and a phosphodiester can be formed with either. Natural RNA contains only 3′,5′ linkages. A non-enzymatic polymerisation, in contrast, can form 2′,5′ linkages as well, and the proportion of each depends on the conditions and on the activating group. Ferris reviewed evidence that montmorillonite-catalysed oligomerisation shows selectivity in the linkages formed, as well as in sequence and in the handedness of the monomers incorporated (Ferris 2006). The regiochemistry of the linkage is therefore another dimension of the selectivity problem (Section 4.7).

The 2′-hydroxy group also makes RNA more labile than DNA, which lacks it. In basic solution, the 2′-oxygen can attack the neighbouring phosphorus atom, breaking the chain by transesterification and leaving a 2′,3′-cyclic phosphate at the end of one fragment (Section 5.4) (Y. Li and Breaker 1999). RNA is therefore both formed and broken through the same cyclic intermediate.

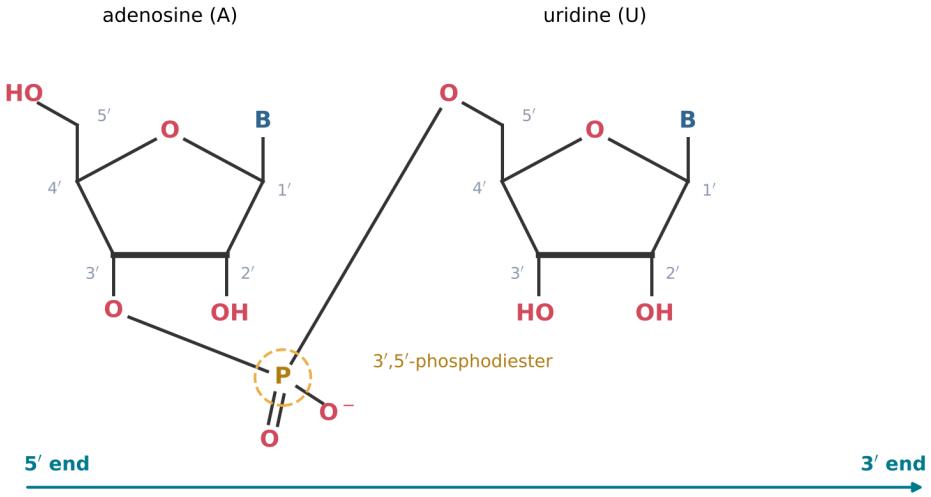


Figure 6.1.: A dinucleotide, 5′-ApU-3′, shown schematically. The phosphodiester bond joins O3′ of adenosine to O5′ of uridine; the backbone runs from the free 5′ end (left) to the free 3′ end (right). The 2′-hydroxy groups, which distinguish RNA from DNA, are shown on each ribose. B denotes a nucleobase.

6.2. Activated nucleotides

6.2.1. Why activation is needed

The formation of a phosphodiester bond from a nucleoside monophosphate and the hydroxy group of another nucleotide is a condensation, and in water its equilibrium lies

on the side of the separated monomers (Section 4.2, Section 5.6). In living cells the energy is supplied by nucleoside triphosphates, whose phosphoanhydride bonds are broken as each nucleotide is added. In prebiotic chemistry, the monomers must be **activated**: the phosphate must carry a group that is easily displaced, a **leaving group**, so that the attack of a hydroxy group on phosphorus is favourable.

6.2.2. Phosphorimidazolides

The most widely used activated monomers in laboratory studies are **nucleoside 5′-phosphorimidazolides**, in which one oxygen of the 5′-phosphate is replaced by a nitrogen atom of imidazole or a substituted imidazole such as 2-methylimidazole. Imidazole is a five-membered aromatic ring containing two nitrogen atoms. Imidazole is a good leaving group, so that the 3′-hydroxy group of a neighbouring nucleotide can attack the phosphorus and displace it, forming a phosphodiester bond.

Activated monomers of this kind made possible the study of template-directed synthesis, the copying of a sequence without enzymes. Inoue and Orgel showed that templates containing mainly cytidine direct the synthesis of oligonucleotides from mixtures of activated mononucleotides (Inoue and Orgel 1983). A nucleotide was incorporated into the products if and only if its complement was present in the template, the reaction had high fidelity, and the products had mean chain lengths of six to ten nucleotides (Inoue and Orgel 1983). This chemistry is the subject of Section 8.2.

6.2.3. Cyclic phosphates

The 2′,3′-cyclic phosphates of Section 5.4 are a second type of activated nucleotide. They are formed directly in some prebiotic nucleotide syntheses (Powner, Gerland, and Sutherland 2009), and their ring strain supplies the energy for phosphodiester formation.

The problem with all activated monomers is that the same reactivity that allows them to form phosphodiester bonds also allows water to attack them. Every activated monomer can either join a chain or be hydrolysed back to an inactive nucleoside monophosphate.

6.3. Polymerisation versus hydrolysis

6.3.1. A kinetic competition

Consider an activated monomer M* in solution together with growing chains. In a short time interval, a given M* can react with the end of a chain (rate constant k_p, proportional to the concentration of chain ends) or with water (rate constant k_h). The probability that it joins a chain rather than being hydrolysed is

$$p = \frac{k_\mathrm{p}\,[\mathrm{ends}]}{k_\mathrm{p}\,[\mathrm{ends}] + k_\mathrm{h}} \tag{6.1}$$

where [ends] is the concentration of chain ends that can accept a monomer. Water is present at about 55 mol L^{-1} (Section 4.2), whereas the concentration of chain ends in a dilute prebiotic solution is small. Even if a chain end is much more reactive than a water molecule, hydrolysis can dominate.

The experimental consequence was stated by Monnard, Kanavarioti and Deamer: in simulations of polymerisation, dilute solutions of activated mononucleotides in the millimolar range hydrolyse extensively, and only trace amounts of dimers and trimers are formed (Monnard, Kanavarioti, and Deamer 2003). Similarly, Ferris and co-workers found that reactions in solution produce only short oligomers, the longest typically being a 10-mer (Ferris et al. 1996).

6.3.2. Ways to win the competition

Equation 6.1 shows two ways to increase p: increase the concentration of chain ends and monomers, or decrease the rate of hydrolysis. The settings discussed in the remainder of this chapter do one or both:

- **Mineral surfaces** adsorb monomers and chains, concentrating them in a thin layer and holding them in favourable orientations (Section 6.4).
- **Ice** excludes solutes as it freezes, concentrating them in liquid channels between ice crystals, and the low temperature slows hydrolysis (Section 6.5).
- **Drying** removes water, reducing its activity and concentrating the reactants (Section 6.6).

6.4. Help from surfaces: montmorillonite

Clay minerals are layered silicates formed by the weathering of rocks in water. **Montmorillonite** is a clay mineral whose layers of aluminium silicate carry a negative charge, balanced by cations held between the layers. Organic molecules, including nucleotides, can enter the spaces between the layers and adsorb on their surfaces. Montmorillonite forms by the alteration of volcanic ash, and clays are expected to have been common on the early Earth wherever rocks were weathered by water.

Key experiment: Ferris, Hill, Liu and Orgel 1996

Most theories of the origin of biological organisation assume that polymers of 30 to 60 monomers are needed to make a genetic system viable, but such polymers

had not been formed by condensation in aqueous solution, because hydrolysis competes with polymerisation (Ferris et al. 1996). Ferris and co-workers studied the oligomerisation of activated monomers, both nucleotides and amino acids, in the presence of minerals: montmorillonite for the nucleotides, and illite and hydroxylapatite for the amino acids (Ferris et al. 1996).
In solution, the reactions produced only short oligomers, the longest typically a 10-mer. In the presence of the mineral surfaces, oligomers up to 55 monomers long were formed. The long chains were obtained by successive "feedings" with fresh monomers: the mineral with its adsorbed chains was repeatedly supplied with activated monomer, and polymerisation took place on the surface in a manner akin to the solid-phase synthesis of biopolymers (Ferris et al. 1996).

The experiment showed that chains in the length range thought necessary for a genetic system can be formed without enzymes. It also showed the price: the monomers were activated, and they were supplied repeatedly. The repeated feeding is a laboratory counterpart of a natural setting in which fresh activated monomers arrive continuously, for example from the processes of Chapter 5, while the growing chains remain bound to a mineral.

6.5. Help from ice: eutectic concentration

When a dilute aqueous solution is cooled below its freezing point, pure ice crystallises first. The solutes are excluded from the ice and accumulate in the remaining liquid, whose freezing point falls as its concentration rises. At a sufficiently low temperature, the system consists of ice crystals and a network of concentrated liquid channels between them, the **eutectic phase**. The concentration of solutes in these channels can exceed that of the original solution by orders of magnitude (Section 3.7).

Key experiment: Monnard, Kanavarioti and Deamer 2003

Monnard, Kanavarioti and Deamer froze dilute solutions of activated mononucleotides together with the metal-ion catalysts Mg(II) and Pb(II), and kept them at −18 °C for periods up to 38 days (Monnard, Kanavarioti, and Deamer 2003). Under these conditions the monomers were concentrated in eutectic phases within the ice matrix. The hydrolysis of the activated monomers was suppressed at the low temperatures, and polymerisation was enhanced, with yields up to 90 percent (Monnard, Kanavarioti, and Deamer 2003).
The products were random RNA analogues with mixed sequences, 5- to 17-mers with traces of longer products, and purine and pyrimidine bases were incorporated at comparable rates and yields (Monnard, Kanavarioti, and Deamer 2003). The authors suggested that ice deposits on the early Earth could have facilitated the

synthesis of short and medium-sized random-sequence RNA analogues (Monnard, Kanavarioti, and Deamer 2003).

Ice thus addresses both terms of Equation 6.1: it concentrates the reactants, increasing k_p[ends], and lowers the temperature, decreasing k_h (Section 2.8). The equal incorporation of purines and pyrimidines is also significant, since a polymerisation that strongly favoured one type of base would produce chains with little diversity of sequence.

6.6. Help from cycles

The wet-dry cycles of Section 5.6 apply to polymerisation as well as to the formation of nucleotides. In the dry phase, the activity of water is low and condensation is favoured; in the wet phase, monomers and chains are redistributed and new material is supplied. Freeze-thaw cycles combine the eutectic concentration of Section 6.5 with periodic mixing, and temperature cycles can separate the strands of double-stranded products, a step needed for copying (Section 8.3). Repeated cycles can accumulate longer chains only if the bonds formed in the favourable phase survive the unfavourable one; in the model of Equation 5.2, this requires the hydrolysis fraction h to be small compared with the conversion fraction f.

Cycles also introduce a form of selection. Chains that are more resistant to hydrolysis, because of their sequence, their folding or their binding to a surface, survive more cycles and so accumulate. This link between chemical stability and persistence is a first step from chemistry towards the selection of sequences discussed in Chapters 7 and 8.

6.7. Chain length distributions

6.7.1. The geometric distribution

Suppose that each time a chain end is offered a monomer, it is extended with probability p and terminated (by hydrolysis of the activated monomer, by blocking of the end, or by any other process) with probability $1 - p$, independently of its length. A chain that has reached length n is then extended to $n + 1$ with probability p. The fraction of chains that reach at least length n is p^{n-1}, and the fraction that have exactly length n is

$$P(n) = (1 - p)\, p^{\,n-1}, \qquad n = 1, 2, 3, \ldots \tag{6.2}$$

This is the **geometric distribution**. Its mean is

$$\langle n \rangle = \frac{1}{1-p} \tag{6.3}$$

so that p = 0.5 gives a mean length of 2, p = 0.9 a mean of 10, and p = 0.95 a mean of 20 (Figure 6.2).

6.7.2. Why long chains are rare

Equation 6.2 decreases exponentially with n. With p = 0.9, the fraction of chains that reach a length of at least 30 is $0.9^{29} \approx 0.047$, and the fraction that reach 55 is $0.9^{54} \approx 0.0034$. With p = 0.5, as in a dilute solution where hydrolysis competes on equal terms, the fraction that reach 30 is $0.5^{29} \approx 1.9 \times 10^{-9}$. Long chains require p close to 1, and a small change in p near 1 has a large effect on the abundance of long chains. This is why the methods of the preceding sections, which raise p, are decisive, and why repeated feeding, as in the experiment of Ferris et al. (1996), can extend chains far beyond the length reached in a single exposure to monomer.

The simple model omits several effects: the rate of extension may depend on chain length and sequence; chains may be broken by hydrolysis of the backbone as well as terminated; and chains may join end to end. More detailed models treat each reaction as a random event occurring at a rate, and follow the populations of chains of each length by stochastic simulation. The qualitative conclusion is the same: without a mechanism that strongly favours extension, the products are dominated by short chains.

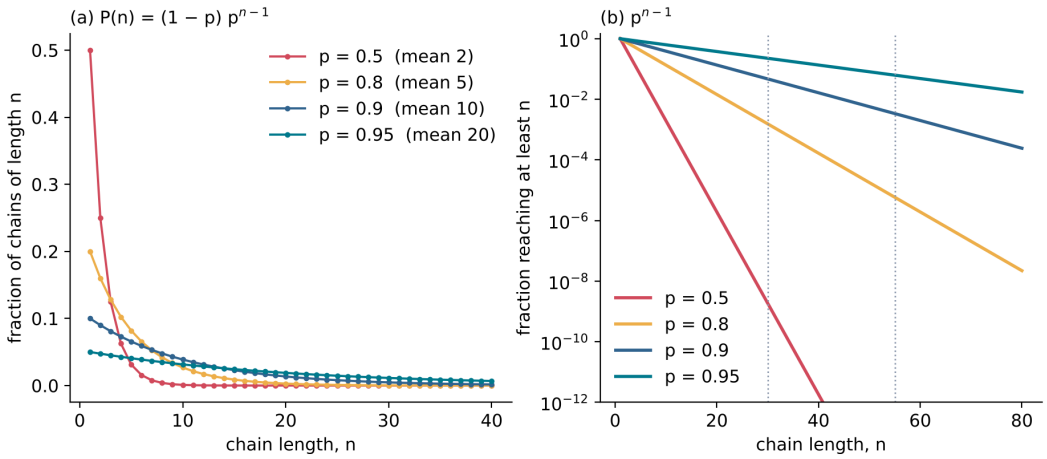


Figure 6.2.: Chain length distributions predicted by the geometric model (Equation 6.2). (a) Fraction of chains of exactly length n, for probabilities of extension p = 0.5, 0.8, 0.9 and 0.95. (b) Fraction of chains that reach at least length n, on a logarithmic scale; dotted lines mark lengths 30 and 55.

Summary

- In RNA, phosphodiester bonds join the 3′-hydroxy group of one nucleotide to the 5′-hydroxy group of the next; chains have a 5′ to 3′ direction. Non-enzymatic polymerisation can also form 2′,5′ linkages, and the 2′-hydroxy group makes RNA susceptible to cleavage.
- Polymerisation in water requires activated monomers, such as nucleoside 5′-phosphorimidazolides or 2′,3′-cyclic phosphates. Templates direct the incorporation of complementary nucleotides with high fidelity.
- Activated monomers are also hydrolysed. In dilute millimolar solutions hydrolysis dominates, and only short oligomers form.
- Montmorillonite with repeated feeding of activated monomers gives oligomers up to 55 units long; ice eutectic phases at −18 °C give 5- to 17-mers in yields up to 90 percent; wet-dry and freeze-thaw cycles combine concentration with mixing.
- If each extension succeeds with probability p, chain lengths follow a geometric distribution with mean $1/(1-\mathrm{p})$; long chains require p close to 1.

Exercises

1. Write the sequence of the chain complementary to 5′-AGGCU-3′, using the base pairs A–U and G–C, in the 5′ to 3′ direction.
2. How many phosphodiester bonds does a linear 20-mer contain? How many negative charges does its backbone carry at neutral pH if both ends carry free hydroxy groups?
3. Explain why RNA is more easily cleaved in basic solution than DNA.
4. In a phosphorimidazolide, which bond is broken when a phosphodiester bond forms? What is the leaving group?
5. Using Equation 6.1, calculate p when $k_p[\text{ends}] = k_h$, and when $k_p[\text{ends}] = 9k_h$. What mean chain lengths do these values give according to Equation 6.3?
6. Explain why freezing a dilute solution can both concentrate the reactants and slow hydrolysis.
7. In the experiment of Ferris et al. (1996), why were successive “feedings” with monomers needed to obtain chains of 55 units?
8. Using Equation 6.2, calculate the fraction of chains of length at least 10 for $\mathrm{p} = 0.8$ and for $\mathrm{p} = 0.95$.
9. Show that the mean of the geometric distribution is $1/(1-\mathrm{p})$. (Hint: use $\sum_{n=1}^{\infty} n x^{n-1} = 1/(1-x)^2$ for $|x| < 1$.)
10. Name two effects omitted from the geometric model, and state for each whether including it would increase or decrease the abundance of long chains.

7. Folding into Shapes

Learning objectives

After reading this chapter you should be able to:

1. Describe the Watson-Crick base pairs and the G · U wobble pair, and explain why paired strands are antiparallel.
2. Explain how base stacking and base pairing together stabilise a double helix, and calculate the free energy of a short helix with the nearest-neighbour model.
3. Relate the free energy of a structure to the fraction of molecules that adopt it, and define the melting temperature.
4. Name and recognise the elements of RNA secondary structure, and write a structure in dot-bracket notation.
5. Explain the principle of minimum free energy structure prediction.
6. Define a ribozyme, describe the discovery of catalytic RNA, and explain how in vitro selection isolates functional molecules from random sequences.
7. Estimate the size of sequence space, and discuss what is known about how common folded and functional sequences are within it.

7.1. Base pairing

7.1.1. Complementary bases

A single RNA chain (Section 6.1) is flexible. What gives it a defined shape is the ability of its nucleobases (Section 4.5) to form specific hydrogen-bonded pairs (Section 2.5). Two nucleobases joined in this way form a **base pair**.

The pairs found in double helices of nucleic acids were proposed by Watson and Crick in their model of DNA (Watson and Crick 1953): adenine pairs with thymine, and guanine with cytosine. In RNA, uracil takes the place of thymine. The **Watson-Crick base pairs** of RNA are therefore A–U, held by two hydrogen bonds, and G–C, held by three (Figure 7.1 a). In each pair a purine faces a pyrimidine, so that both pairs have nearly the same width, and a helix built from any sequence of them has a regular shape. Two sequences that can form Watson-Crick pairs along their whole length are called **complementary**.

RNA also forms a third pair, between guanine and uracil, held by two hydrogen bonds. In this pair the two bases are shifted relative to each other compared with a Watson-Crick pair; it is called the **G · U wobble pair** (Figure 7.1 a). G · U pairs occur frequently within RNA helices and are included in the energy parameters used below (Turner and Mathews 2010). In the notation of this book, a dash (A–U) marks a Watson-Crick pair and a centred dot (G · U) a wobble pair.

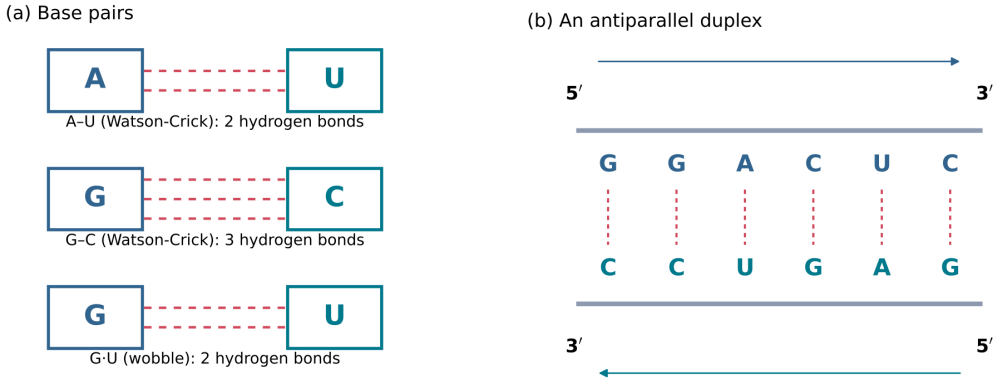


Figure 7.1.: (a) The three base pairs of RNA helices and the number of hydrogen bonds in each: the Watson-Crick pairs A–U and G–C, and the G · U wobble pair. (b) A duplex of 5′-GGACUC-3′ with its complementary strand. The two strands run in opposite directions.

7.1.2. Antiparallel strands

Base pairs form only when the two strands are **antiparallel**: one runs from 5′ to 3′ in one direction, the other from 5′ to 3′ in the opposite direction (Figure 7.1 b). Thus 5′-GGACUC-3′ pairs with 5′-GAGUCC-3′, which is its complement read in the 5′ to 3′ direction. When a single chain folds back on itself, its two paired segments are antiparallel for the same reason. The paired region winds into a right-handed **double helix**. A molecule made of two separate strands held together by base pairs is called a **duplex**.

7.2. Duplex stability

7.2.1. Stacking and pairing

Hydrogen bonds alone do not explain why a double helix is stable. When a helix forms, the hydrogen bonds that each base made with water before pairing are replaced by hydrogen bonds to its partner, so the net gain in hydrogen bonding is small. A large

part of the stability of a helix comes from **base stacking**: the flat aromatic rings of neighbouring base pairs lie on top of each other, face to face, along the axis of the helix. Stacking is favoured by dispersion forces between the rings and by the hydrophobic effect (Section 2.5), which drives the non-polar faces of the bases out of contact with water. The contribution of a base pair therefore depends on its neighbours, not only on its own identity.

7.2.2. The nearest-neighbour model

The **nearest-neighbour model** expresses this dependence quantitatively. The standard Gibbs energy change (Section 2.7) for forming a helix is written as a sum of terms, one for each pair of adjacent base pairs, called a **stack**, plus a term for starting the helix:

$$\Delta G^\circ_{\mathrm{helix}} = \Delta G^\circ_{\mathrm{init}} + \sum_{\mathrm{stacks}} \Delta G^\circ_{\mathrm{stack}} \tag{7.1}$$

The stack terms have been measured by optical melting experiments on many short duplexes and are collected in the Nearest Neighbor Database (Turner and Mathews 2010). The current set of RNA parameters is the 2004 set of Turner and co-workers (Mathews et al. 2004). Table 7.1 lists the values for the ten distinct Watson-Crick stacks at 37 °C. The parameters are given in the literature in kcal mol^{-1}; they are converted here to kJ mol^{-1} (1 kcal = 4.184 kJ).

Table 7.1.: Free energies of the ten Watson-Crick stacks of RNA at 37 °C, Turner 2004 parameters (Mathews et al. 2004; Turner and Mathews 2010), as distributed with the ViennaRNA package (Lorenz et al. 2011). Each stack is written as the top strand 5′ to 3′ over the bottom strand 3′ to 5′; a stack read from the other strand is identical (for example 5′-GG-3′/3′-CC-5′ is the same as 5′-CC-3′/3′-GG-5′).

Stack	ΔG°_{37} (kcal mol^{-1})	ΔG°_{37} (kJ mol^{-1})
5′-GC-3′ / 3′-CG-5′	−3.4	−14.2
5′-CC-3′ / 3′-GG-5′	−3.3	−13.8
5′-CG-3′ / 3′-GC-5′	−2.4	−10.0
5′-GA-3′ / 3′-CU-5′	−2.4	−10.0
5′-AC-3′ / 3′-UG-5′	−2.2	−9.2
5′-AG-3′ / 3′-UC-5′	−2.1	−8.8
5′-CA-3′ / 3′-GU-5′	−2.1	−8.8
5′-UA-3′ / 3′-AU-5′	−1.3	−5.4
5′-AU-3′ / 3′-UA-5′	−1.1	−4.6
5′-AA-3′ / 3′-UU-5′	−0.9	−3.8

Three features of the table are important for what follows. All stacks have negative ΔG°, so every additional base pair stabilises a helix. Stacks containing two G–C pairs are the most stable, and stacks of two A–U pairs the least; the difference is up to a factor of about four. And the order of the bases matters: 5′-GC-3′/3′-CG-5′ (-14.2 kJ mol^{-1}) is more stable than 5′-CG-3′/3′-GC-5′ (-10.0 kJ mol^{-1}), although both contain the same two pairs.

For two separate strands, the initiation term is positive. It reflects the cost of bringing two independent molecules together into one complex. In the Turner 2004 set it is +4.09 kcal mol^{-1} (+17.1 kJ mol^{-1}) (Turner and Mathews 2010). As an example, the duplex of Figure 7.1 b contains five stacks, whose values from Table 7.1 sum to -12.4 kcal mol^{-1} (-51.9 kJ mol^{-1}). With the initiation term, Equation 7.1 gives $\Delta G^\circ_{37} \approx -8.3$ kcal mol^{-1}, or -34.8 kJ mol^{-1}. The same value is obtained with the duplex routine of ViennaRNA.

7.2.3. From free energy to populations

A free energy determines what fraction of molecules adopt a structure. For a molecule that is either folded (F) or unfolded (U), with $\Delta G = G_F - G_U$, the ratio of the two populations at equilibrium is $[F]/[U] = \exp(-\Delta G/RT)$, where R is the molar gas constant and T the absolute temperature (Section 2.8). The fraction folded, θ, is therefore

$$\theta = \frac{1}{1 + \exp(\Delta G/RT)} \tag{7.2}$$

At 37 °C, RT = 8.314 × 310.15 J mol^{-1} = 2.58 kJ mol^{-1}. A structure that is 10 kJ mol^{-1} more stable than the unfolded state is present in about 98 percent of molecules; one that is 20 kJ mol^{-1} more stable, in more than 99.9 percent.

7.2.4. Melting temperature

ΔG depends on temperature through $\Delta G = \Delta H - T\Delta S$ (Section 2.7). Forming a helix releases enthalpy ($\Delta H < 0$, from stacking and pairing) and lowers the entropy ($\Delta S < 0$, because the chain becomes ordered). At low temperature the enthalpy term dominates and the helix is stable; at high temperature the entropy term dominates and it falls apart. The temperature at which $\Delta G = 0$, so that half of the molecules are folded in the two-state model, is the **melting temperature**, $T_m = \Delta H/\Delta S$.

Figure 7.2 shows this behaviour for the 12-nucleotide chain 5′-GGACUUCGGUCC-3′, which folds back on itself into a stem of four base pairs closed by the four-nucleotide loop UUCG. The stem contributes three stacks (-13.8, -10.0 and -9.2 kJ mol^{-1} from Table 7.1, total -33.1 kJ mol^{-1}) and the loop a destabilising +15.5 kJ mol^{-1}, giving $\Delta G^\circ_{37} = -17.6$ kJ mol^{-1}. Because the two halves belong to one molecule, there is no

intermolecular initiation term. From the computed values at 25 °C (-25.4 kJ mol^{-1}) and 60 °C (-2.6 kJ mol^{-1}), the line of Figure 7.2 a gives $\Delta S \approx -0.65$ kJ K^{-1} mol^{-1} and $\Delta H \approx -220$ kJ mol^{-1}, so that $T_m = \Delta H/\Delta S \approx 337$ K, or 64 °C. The full calculation, which also includes partly paired alternative structures, gives the melting curve of Figure 7.2 b, in which half of the molecules are completely unpaired at 65 °C.

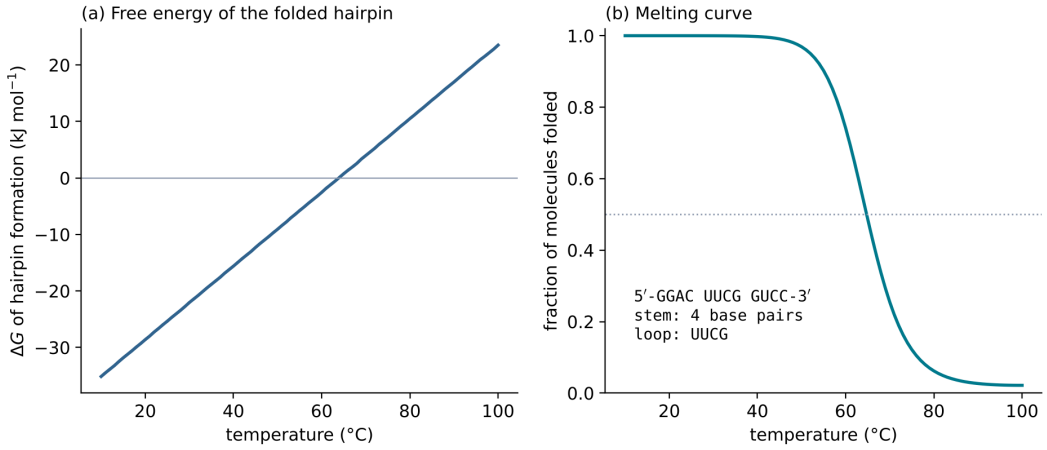


Figure 7.2.: Melting of the hairpin 5′-GGACUUCGGUCC-3′, calculated with the ViennaRNA package 2.7.2 (Lorenz et al. 2011) and the Turner 2004 parameters (Mathews et al. 2004). (a) Gibbs energy of the hairpin structure relative to the unfolded chain. (b) Fraction of molecules with at least one base pair, from the partition function (Section 7.4); the dotted line marks one half.

Two consequences matter for the origin of life. First, short helices are only marginally stable: a stem of four base pairs melts at about 64 °C, and one of two or three base pairs does not form at all at ordinary temperatures. Second, the melting temperature rises with the length and G–C content of a helix. A hot environment therefore favours single strands, and a cool one favours folded structures and duplexes; this trade-off returns in Section 8.3, where two strands must be separated in order to be copied.

7.3. Secondary structure

7.3.1. Definitions

The set of base pairs formed by an RNA molecule is its **secondary structure**. The three-dimensional arrangement of its atoms is its **tertiary structure**. The secondary structure is usually restricted to pairs that are **nested**: if nucleotide i pairs with j, and k pairs with l, with $i < k$, then either $i < k < l < j$ (one pair lies inside the other) or $i < j < k < l$ (the pairs lie side by side). Pairs that cross, with $i < k < j < l$, form a **pseudoknot**; pseudoknots occur in natural RNAs but are excluded from the standard

prediction methods described below. A hairpin loop must contain at least three unpaired nucleotides, because a shorter loop cannot span the distance between the two strands of a helix (Turner and Mathews 2010).

A nested secondary structure can be written in **dot-bracket notation**: a string of the same length as the sequence, in which each paired nucleotide is shown by a bracket, opening for the 5′ partner and closing for the 3′ partner, and each unpaired nucleotide by a dot. The hairpin of Figure 7.2 is written

```
GGACUUCGGUCC
((((....))))
```

7.3.2. Structural elements

A nested secondary structure can be decomposed uniquely into a small set of elements (Figure 7.3):

- A **helix**, or **stem**, is a run of consecutive stacked base pairs.
- A **hairpin loop** is a loop of unpaired nucleotides closed by a single base pair at the end of a helix.
- A **bulge** is a loop between two helices with unpaired nucleotides on one strand only.
- An **internal loop** is a loop between two helices with unpaired nucleotides on both strands.
- A **multibranch loop**, or junction, is a loop from which three or more helices radiate.
- The **exterior loop** contains the unpaired nucleotides that are not enclosed by any base pair, including the two ends of the chain.

In the energy model, every stack contributes the negative free energy of Table 7.1, and every loop contributes a positive free energy, which accounts for the entropy lost when the unpaired nucleotides are held in a closed ring. The loop terms depend on the size and type of the loop and, for some small loops, on their sequence (Turner and Mathews 2010). The stability of a structure is the sum of all these terms.

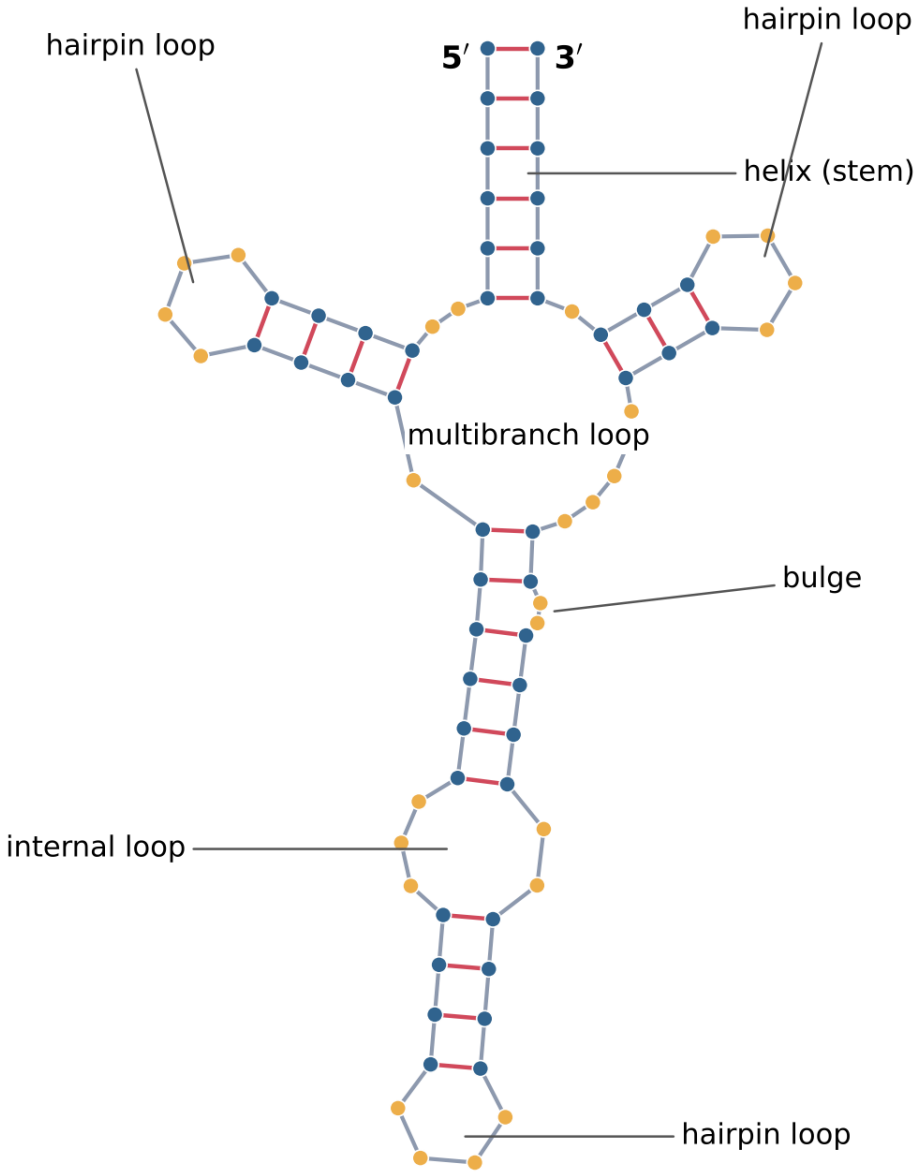


Figure 7.3.: Elements of RNA secondary structure in a 74-nucleotide example structure, drawn with the coordinate layout of the ViennaRNA package (Lorenz et al. 2011). Paired nucleotides are connected by short lines across each helix.

7.4. Predicting structure

7.4.1. The number of possible structures

How many secondary structures can a chain form? If any two nucleotides could pair, subject only to nesting and the minimum hairpin loop of three, a simple recursion gives 65 possible structures for a chain of 10 nucleotides, about 1.1×10^5 for 20, 6.3×10^{11} for 40 and 5.6×10^{25} for 80 (author's calculation). The number grows roughly by a constant factor with each added nucleotide, that is, exponentially. Base-pairing rules exclude most of these structures for a given sequence, but far too many remain to be examined one by one.

7.4.2. Minimum free energy folding

Under the conditions of a folding experiment, a molecule is expected to spend most of its time in the structure of lowest Gibbs energy, the **minimum free energy (MFE) structure**, if that structure is clearly more stable than all others (Equation 7.2). Zuker and Stiegler showed in 1981 that the MFE structure can be found without listing all structures, by **dynamic programming** (Zuker and Stiegler 1981). The method exploits nesting: the best structure of a segment from nucleotide i to nucleotide j can be built from the best structures of shorter segments, which are computed first and stored. The computing time grows only as the cube of the length of the chain, so that molecules of hundreds of nucleotides can be folded routinely on an ordinary computer.

The same recursions can be used to calculate the sum of exp(−ΔG/RT) over all structures, the **partition function** Z. The probability of any single structure s is then

$$p(s) = \frac{\exp(-\Delta G_s / RT)}{Z} \tag{7.3}$$

where ΔG_s is its free energy relative to the unfolded chain. Equation 7.2 is the special case of Equation 7.3 with two states. These calculations are implemented in the ViennaRNA package (Lorenz et al. 2011), used for all folding calculations in this chapter, which applies the Turner 2004 parameters (Mathews et al. 2004).

Predicted structures are models. The energy parameters were measured for short model duplexes and loops, in solutions of defined salt content; tertiary interactions, pseudoknots and bound metal ions are not included. Predictions are most reliable for short molecules with stable helices, and less reliable for long molecules whose function depends on tertiary structure.

7.5. From shape to function: ribozymes

7.5.1. Enzymes made of RNA

An **enzyme** is a biological catalyst (Section 2.8). Until the early 1980s, all known enzymes were proteins. A **ribozyme** is an RNA molecule that acts as a catalyst. The existence of ribozymes was demonstrated by two independent discoveries.

Key experiment: Kruger et al. 1982; Guerrier-Takada et al. 1983

In the ciliate *Tetrahymena*, the precursor of a ribosomal RNA contains an intervening sequence that must be removed, and the flanking pieces joined, in a process called **splicing**. The group of Thomas Cech found that the purified precursor RNA, incubated with monovalent and divalent cations and a guanosine factor, spliced itself: the intervening sequence was cut out, with guanosine attached to its 5′ end, and the flanking RNA was joined. The authors concluded that "splicing activity is intrinsic to the structure of the RNA, and that enzymes, small nuclear RNAs and folding of the pre-rRNA into an RNP are unnecessary for these reactions" (Kruger et al. 1982).
Ribonuclease P is an enzyme of RNA and protein that cuts the precursors of transfer RNAs. The groups of Sidney Altman and Norman Pace showed that the RNA components of the enzyme from two bacteria, *Escherichia coli* and *Bacillus subtilis*, cleave the precursors on their own in solutions containing 60 mM Mg^{2+}, or 10 mM Mg^{2+} with 1 mM spermidine. Under these conditions, "the RNA acts as a true catalyst … whereas the protein moieties of the enzymes alone show no catalytic activity"; at lower Mg^{2+} concentrations without spermidine both subunits were required (Guerrier-Takada et al. 1983).
Altman and Cech received the Nobel Prize in Chemistry 1989 "for their discovery of catalytic properties of RNA" (Nobel Prize Outreach, n.d.).

Catalysis by RNA depends on folding. A ribozyme positions the reacting groups of its substrate next to each other, and next to bound metal ions or to functional groups of its own nucleotides, so that the transition state (Section 2.8) is stabilised. Only a molecule with a defined three-dimensional structure can do this. A ribozyme is therefore both a sequence, which can in principle be copied, and a folded catalyst. This dual role is the basis of the RNA world hypothesis, introduced in Section 8.1.

7.5.2. In vitro selection

The natural ribozymes were discovered in living cells. In 1990, two groups showed that RNA molecules with a desired property can also be isolated from a large collection of synthetic sequences, by **in vitro selection**. The procedure has three steps, repeated in cycles:

1. A **pool** of RNA molecules is prepared in which part of the sequence is random, so that the pool contains a very large number of different sequences.
2. The molecules that have the desired property, for example binding to a target molecule fixed on a solid support, are separated from those that do not.
3. The selected molecules are copied and amplified with enzymes to form the pool for the next cycle.

Because each cycle enriches the pool in active molecules, a few cycles can isolate sequences that were initially extremely rare. RNA molecules selected to bind a specific target are called **aptamers**.

Key experiment: Ellington and Szostak 1990; Tuerk and Gold 1990

Ellington and Szostak isolated, from a population of random-sequence RNA molecules, subpopulations that bind specifically to a variety of organic dyes. They estimated that "roughly one in 10^{10} random sequence RNA molecules folds in such a way as to create a specific binding site for small ligands" (Ellington and Szostak 1990).
Tuerk and Gold randomised an eight-nucleotide region of an RNA that binds the DNA polymerase of bacteriophage T4, giving a pool of $4^8 = 65\ 536$ sequences. Cycles of selection for binding and amplification yielded two sequences: the natural sequence, and a variant differing from it at four positions, with equivalent binding constants. The title of their paper names the method: systematic evolution of ligands by exponential enrichment, abbreviated SELEX (Tuerk and Gold 1990).

In vitro selection was soon applied to catalysis. Bartel and Szostak isolated new ribozymes from a large pool of random-sequence RNA. The ribozymes ligate two RNA molecules aligned on a template, by catalysing the attack of a 3′-hydroxy group on an adjacent 5′-triphosphate. This forms a 3′,5′-phosphodiester bond, the reaction used by the protein enzymes that synthesise RNA in cells. After further cycles of selection, ribozymes emerged whose reaction rates were 7 million times faster than the uncatalysed reaction (Bartel and Szostak 1993). Ribozymes that join RNA strands are the starting point for the ribozyme polymerases discussed in Section 8.5.

7.6. How common is function?

7.6.1. Sequence space

The set of all sequences of a given length is called **sequence space**. With four nucleotides, there are 4^n sequences of length n. For n = 40 this is $4^{40} \approx 1.2 \times 10^{24}$, or 2.0 mol of molecules. The mean mass of an RNA residue is about 321 g mol^{-1} (the mean molar mass of the four nucleoside monophosphates in Table 5.1, 339.5 g mol^{-1}, minus one water molecule, 18.0 g mol^{-1}), so a single copy of every 40-mer would have a mass

of about 26 kg. For n = 100, $4^{100} \approx 1.6 \times 10^{60}$. This is far more than the number of atoms in the Earth, which is less than 4×10^{51}: that is the number the Earth would contain if its whole mass, 5.9722×10^{24} kg (Section 3.2), were made of hydrogen, the lightest atom, with a mass of 1.67×10^{-27} kg (Section 1.1). Any real population of RNA molecules, whether in a laboratory or on the early Earth, contains only a vanishingly small fraction of sequence space for chains of this length.

The question for the origin of life is therefore not whether a given functional sequence was present, but how many functional sequences there are, and whether a random sample can be expected to contain any of them. A pool of 10^{15} molecules, for example, corresponds to 1.7 nmol, about 21 μg of 40-mers. If one sequence in 10^{10} binds a given small molecule, as estimated by Ellington and Szostak (1990), such a pool would contain about 10^5 binders.

7.6.2. Most random sequences fold

A necessary condition for function is folding. Figure 7.4 shows the MFE structures of random RNA sequences, computed with ViennaRNA at 37 °C (author's calculation; 4000 sequences each of length 20 and 40, and 2000 of length 80, for each alphabet). With all four nucleotides, 35 percent of random 20-mers have no stable structure, but only 0.9 percent of random 40-mers and none of the 80-mers; the mean MFE is −5.9, −26.4 and −72.5 kJ mol^{-1} for the three lengths, and about half of the nucleotides of an 80-mer are paired. The structures are also diverse: the 4000 random 40-mers gave 3882 different MFE structures, and the 4000 random 20-mers gave 682.

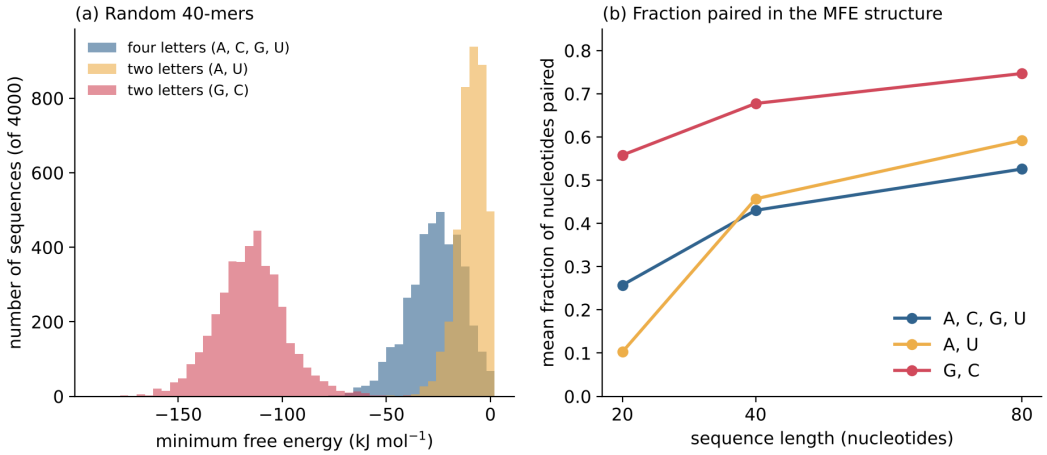


Figure 7.4.: Folding of random RNA sequences at 37 °C, calculated with ViennaRNA and the Turner 2004 parameters (author's calculation). (a) Distribution of the minimum free energy of 4000 random 40-mers made of four nucleotides, or of only A and U, or of only G and C. (b) Mean fraction of nucleotides paired in the MFE structure as a function of length.

Folding depends strongly on the alphabet. Sequences made only of A and U fold weakly, because A–U stacks are the least stable in Table 7.1: 84 percent of random 20-mers of A and U have no stable structure, and the mean MFE of the 40-mers is only −9.4 kJ mol^{-1}. Sequences made only of G and C fold very strongly (mean MFE of the 40-mers −115.3 kJ mol^{-1}), but they can form many alternative structures of similar stability, and their helices are hard to melt. The four-letter alphabet lies between these extremes.

7.6.3. A ribozyme with two letters

Must a functional RNA contain all four nucleotides? Reader and Joyce used in vitro evolution to obtain ligase ribozymes composed of only two nucleotides, 2,6-diaminopurine and uracil. 2,6-Diaminopurine is an adenine analogue with an additional amino group, which allows it to form three hydrogen bonds with uracil instead of the two of an A–U pair. These ribozymes catalyse the template-directed joining of two RNA molecules, one bearing a 5′-triphosphate and the other a 3′-hydroxy group, and form specifically 3′,5′-phosphodiester bonds. The fastest had a catalytic rate about 36 000 times the uncatalysed rate (Reader and Joyce 2002). This is far below the rate enhancement of the four-letter ligases of Bartel and Szostak (1993), but it shows that a binary alphabet can encode catalysis.

These results establish two facts and leave one open question. Established: most random RNA sequences longer than a few tens of nucleotides fold into some structure, and functional sequences, such as binders and ligases, can be isolated from random-sequence pools in the laboratory. Open question: how frequent functional sequences are for the activities that matter most for the origin of life, above all the copying of RNA, and whether the short chains that prebiotic chemistry can make (Section 6.7) are long enough to contain them. Chapter 8 takes up this question.

Summary

- RNA forms Watson-Crick pairs, A–U (two hydrogen bonds) and G–C (three), and G · U wobble pairs. Paired strands are antiparallel and wind into a right-handed double helix.
- Helix stability comes from base stacking as well as pairing. In the nearest-neighbour model the free energy of a helix is the sum of stack terms plus an initiation term; G–C stacks are the most stable (down to −14.2 kJ mol^{-1} at 37 °C), A–U stacks the least.
- The fraction of folded molecules is $1/(1 + \exp(\Delta G/RT))$. Because both ΔH and ΔS of helix formation are negative, helices melt on heating; a four-base-pair hairpin with a UUCG loop melts at about 64 °C.

- Secondary structure is the set of nested base pairs, written in dot-bracket notation and composed of helices, hairpin loops, bulges, internal loops, multibranch loops and the exterior loop. The minimum free energy structure is found by dynamic programming.
- Ribozymes are catalytic RNA molecules, discovered in self-splicing and in ribonuclease P (Nobel Prize 1989). In vitro selection isolates binders and ribozymes, including RNA ligases, from random-sequence pools.
- Sequence space is vast ($4^{40} \approx 1.2 \times 10^{24}$ sequences of length 40). Most random sequences of 40 or more nucleotides fold; about one in 10^{10} binds a given small molecule; a ribozyme can be built from only two nucleotides. How common copying activity is remains an open question.

Exercises

1. Write the sequence complementary to 5′-GCAUUG-3′ in the 5′ to 3′ direction. Which base pairs does the duplex contain, and how many hydrogen bonds hold it together?
2. Explain why a G · U pair can occur within a helix, whereas an A · C or a G · A pair usually cannot form there in the same way.
3. Using Table 7.1, calculate the sum of the stack free energies of the duplex 5′-GGACUC-3′/3′-CCUGAG-5′ and check the value given in Section 7.2.
4. Using Table 7.1, compare the stack free energies of 5′-GCGC-3′ and 5′-CGCG-3′, each paired with its complement. Which helix is more stable?
5. Using Equation 7.2, calculate the fraction folded at 37 °C for structures with $\Delta G = -5$, -10 and -20 kJ mol^{-1}.
6. The hairpin of Figure 7.2 has $\Delta G = -9.1$ kJ mol^{-1} at 50 °C. Calculate the fraction folded at this temperature with Equation 7.2.
7. Using $\Delta H \approx -220$ kJ mol^{-1} and $\Delta S \approx -0.65$ kJ K^{-1} mol^{-1}, calculate ΔG at 37 °C and T_m for the hairpin of Figure 7.2.
8. Explain, in terms of enthalpy and entropy, why a helix of two base pairs closed by a hairpin loop does not form at 37 °C.
9. Write the dot-bracket notation for a 20-nucleotide chain in which nucleotides 1 to 5 pair with 16 to 20 and nucleotides 6 to 15 are unpaired. Which structural element do nucleotides 6 to 15 form?
10. Which of the following pairs of base pairs are nested and which form a pseudoknot: (2, 10) and (4, 8); (2, 10) and (12, 18); (2, 10) and (6, 15)?
11. Calculate the number of different RNA sequences of length 30, and the mass of one copy of each, using a residue mass of 321 g mol^{-1}.
12. In the experiment of Tuerk and Gold (1990), how many cycles of 10-fold enrichment would be needed to raise a sequence from a frequency of 1 in 65 536 to more than half of the pool, assuming the other sequences are not enriched?

13. Explain why random sequences made only of A and U fold more weakly than sequences of all four nucleotides, and why the two-letter ribozyme of Reader and Joyce (2002) used 2,6-diaminopurine instead of adenine.

8. RNA Copying Itself

Learning objectives

After reading this chapter you should be able to:

1. State the RNA world hypothesis and explain why template-directed copying is central to it.
2. Describe non-enzymatic template-directed primer extension, the improvements made to it, and its present limits.
3. Explain the strand-separation problem, and use the melting temperature of a duplex to show how it depends on length and concentration.
4. Derive the error threshold of a replicating sequence and calculate the longest sequence that a given copying accuracy can maintain.
5. Describe the main ribozyme polymerases made by in vitro evolution and what they can and cannot yet do.
6. Explain how selection arises among replicating molecules, with reference to self-sustained replication of RNA enzymes.
7. Compare measured copying rates with the rates that heredity requires, and list the main open questions.

8.1. Templating

8.1.1. The RNA world

In all known organisms, genetic information is stored in DNA, and most catalysis is carried out by proteins. Each of these two kinds of molecule is made with the help of the other. The discovery of ribozymes (Section 7.5) suggested a way out of this circularity. In Joyce's words, "there are strong reasons to conclude that DNA- and protein-based life was preceded by a simpler life form based primarily on RNA. This earlier era is referred to as the 'RNA world', during which the genetic information resided in the sequence of RNA molecules and the phenotype derived from the catalytic properties of RNA" (Joyce 2002). The term was introduced by Gilbert in 1986 (Gilbert 1986). Orgel regarded the evidence as strong: "The demonstration that ribosomal peptide synthesis is a ribozyme-catalyzed reaction makes it almost certain that there was once an RNA

World. The central problem for origin-of-life studies, therefore, is to understand how a protein-free RNA World became established on the primitive Earth" (Orgel 2004).

The **RNA world hypothesis** is the hypothesis that, at an early stage of life, RNA served both as the genetic material and as the main catalyst. That such a stage existed is a hypothesis with substantial support; how it arose from the chemistry of Chapters 4 to 7 is an open question. Its central requirement is that RNA sequences can be copied.

8.1.2. Copying by complementarity

The rules of base pairing (Section 7.1) provide a copying mechanism. A **template** is a strand whose sequence directs the order in which nucleotides are joined (Section 5.5). A short strand bound to the template, the **primer**, is extended at its 3′ end: a nucleotide that pairs with the next template base is held in position and joined to the primer (Figure 8.1 a). The product is the complement of the template, not an identical copy. A second round of copying, with the complement as template, restores the original sequence. **Replication** of a sequence therefore requires two rounds of template-directed synthesis, and between the rounds the product must be separated from its template (Figure 8.1 b).

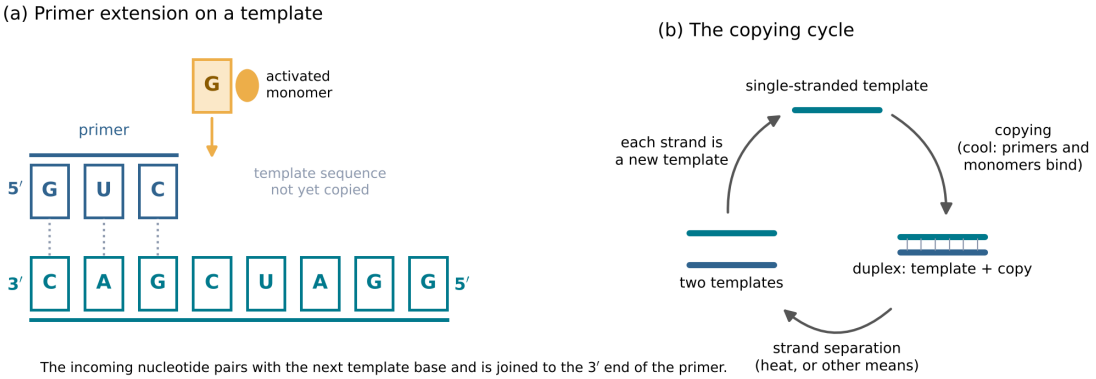


Figure 8.1.: (a) Template-directed primer extension. The primer (blue) is paired with the 3′ end of the template (teal); an activated nucleotide that pairs with the next template base is joined to the 3′ end of the primer. (b) The copying cycle. Copying produces a duplex of template and complement; only after the two strands are separated can each serve as a template again.

8.2. Non-enzymatic primer extension

8.2.1. What has been achieved

Template-directed synthesis without enzymes was introduced in Section 6.2: activated nucleotides, such as phosphorimidazolides, join on a template in the order specified by base pairing.

> Key experiment: Inoue and Orgel 1983
>
> Inoue and Orgel showed that templates containing mainly cytidine direct the synthesis of oligonucleotides from mixtures of activated mononucleotides. A nucleotide was incorporated into the products if and only if its complement was present in the template, the reaction had high fidelity, and the products had mean chain lengths of six to ten nucleotides (Inoue and Orgel 1983). The experiment established that sequence information can be transferred by base pairing alone, without enzymes, and that the transfer is selective.

The successful early experiments used templates rich in C, which direct the incorporation of G (Inoue and Orgel 1983). Templates containing A and U are copied far more slowly, by a factor of at least 1000 without helper oligonucleotides (Prywes et al. 2016; Wahiro 2026). Several improvements have been made since.

- **Better leaving groups.** Li and co-workers showed that "activating nucleotides with 2-aminoimidazole results in superior reaction kinetics and improved yields of primer extension reaction products"; among the leaving groups they compared, the rates differed by up to 200 times, with 2-aminoimidazole giving the fastest (L. Li et al. 2017).
- **Activated helper oligonucleotides.** Prywes and co-workers found that "short 5′-activated oligonucleotides act as catalysts that accelerate primer extension, and allow for the one-pot copying of mixed sequence RNA templates"; with this method they synthesised a significant part of an active hammerhead ribozyme (Prywes et al. 2016).

Chemical self-replication has also been demonstrated with short synthetic templates: in 1986, von Kiedrowski reported a self-replicating hexadeoxynucleotide, a DNA chain of six nucleotides (Kiedrowski 1986).

8.2.2. Present limits

Three limits remain. First, copying is slow: under the most favourable conditions reported, adding one nucleotide takes from tens of minutes to about a day (Section 8.7). Second, an incorrect nucleotide slows further extension. Rajamani and co-workers found

"that most mismatches decrease the rate of primer extension by more than 2 orders of magnitude relative to a matched (Watson-Crick) control" (Rajamani et al. 2010). This **stalling** reduces the number of complete copies that carry errors, but it also wastes time and template. Third, no non-enzymatic system has yet copied a sequence of useful length through repeated cycles of replication. The second round of copying requires the product to be separated from its template, which is the subject of the next section.

8.3. The strand-separation problem

8.3.1. Duplexes are too stable, primers too weak

For a duplex formed from two different strands, each present at total strand concentration $C_T/2$, the melting temperature is the temperature at which half of the strands are paired. It depends on concentration:

$$T_m = \frac{\Delta H^\circ}{\Delta S^\circ + R \ln(C_T/4)} \tag{8.1}$$

where ΔH° and ΔS° are the standard enthalpy and entropy of duplex formation (Section 7.2) and R is the molar gas constant. Because ΔS° is negative and $\ln(C_T/4)$ is negative for concentrations below 4 mol L^{-1}, lowering the concentration lowers T_m.

Figure 8.2 shows T_m calculated in this way for random RNA duplexes of 4 to 30 base pairs, at total strand concentrations of 1 μM and 1 mM (author's calculation, with ΔH° and ΔS° from the Turner 2004 parameters in ViennaRNA (Mathews et al. 2004; Lorenz et al. 2011)). At 1 μM, the median melting temperature is −33 °C for 4 base pairs, 17 °C for 6, 41 °C for 8, 56 °C for 10, 85 °C for 20 and 95 °C for 30. At 1 mM, it is 3 °C for 4 base pairs and 44 °C for 6.

These numbers define the **strand-separation problem**. Short primers and single nucleotides bind weakly: a primer of 4 to 6 nucleotides stays on its template at ordinary temperatures only at high concentration. The finished product, however, is bound too strongly: a duplex of 20 base pairs or more melts only above about 85 °C at micromolar concentration. A system that copies at low temperature must therefore be heated almost to boiling to release the copy, and at such temperatures RNA is cleaved much faster (Section 6.3). After the strands have separated, they can pair with each other again on cooling, in competition with the primers and monomers that would copy them.

8.3.2. Possible solutions

Several ways around this problem have been proposed; all are hypotheses.

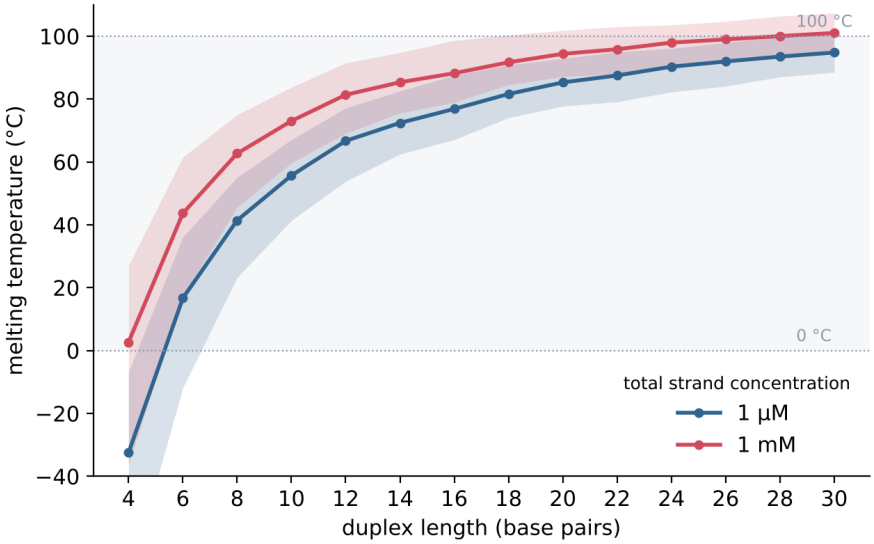


Figure 8.2.: Melting temperature of random RNA duplexes as a function of length, from Equation 8.1 with nearest-neighbour parameters (Turner 2004, ViennaRNA 2.7.2; author's calculation, 300 random sequences per length). Lines: median; shaded bands: 10th to 90th percentile; blue, total strand concentration 1 μM; red, 1 mM. Values above 100 °C are extrapolations of the parameter set.

- **Temperature cycles** of short duration, such as those in thermal gradients, could separate strands briefly without prolonged exposure to high temperature.
- **Lower duplex stability**, through changes in salt or pH, or through modified nucleotides that pair more weakly, would lower T_m, at the cost of also weakening primer binding.
- **Copying with short oligonucleotides** rather than single nucleotides joins pieces that bind more strongly, so that copying can take place at higher temperatures, closer to T_m of the product.

None of these has yet produced repeated cycles of non-enzymatic replication of a long sequence.

8.4. Errors and the error threshold

8.4.1. Fidelity

Every copying process makes errors. Let u be the probability that a nucleotide is copied incorrectly, the **error rate per nucleotide**; q = 1 − u is the **fidelity**. If errors at different positions are independent, the probability that a sequence of length L is copied without any error is

$$Q = q^L = (1-u)^L \approx e^{-uL} \tag{8.2}$$

The approximation holds for small u. The mean number of errors per copy is uL. For example, at u = 0.01, a copy of 100 nucleotides contains on average one error, and only $e^{-1} \approx 37$ percent of copies are error-free.

8.4.2. The error threshold

Consider a population in which one sequence, the **master sequence**, is replicated σ times faster than all its mutants, and in which mutants rarely mutate back to the master. Only the error-free copies of the master are masters, so the master population grows at a relative rate proportional to σQ, and all mutants at a rate proportional to 1. The master can persist only if

$$\sigma Q > 1, \quad \text{that is,} \quad L < \frac{\ln \sigma}{-\ln q} \approx \frac{\ln \sigma}{u} \tag{8.3}$$

In the steady state its fraction in the population is

$$x_0 = \frac{\sigma Q - 1}{\sigma - 1} \tag{8.4}$$

which falls to zero at $\sigma Q = 1$. The quantity σ is called the **selective superiority** of the master. The maximum length $L_{max} \approx \ln \sigma / u$ is the **error threshold**. Above it, the information of the master sequence is lost in a cloud of mutants, an outcome called the **error catastrophe**. The concept was introduced by Eigen in his theory of the self-organisation of biological macromolecules (Eigen 1971); the population of a master sequence and its mutants is called a **quasispecies**.

Figure 8.3 a shows the fraction of the master in a population of sequences of length 50 with $\sigma = 10$ (author's calculation). The full calculation, which includes back-mutation for a four-letter alphabet, agrees with Equation 8.4 to within the width of the line. The master fraction falls to zero at $u_c = 1 - 10^{-1/50} = 0.045$.

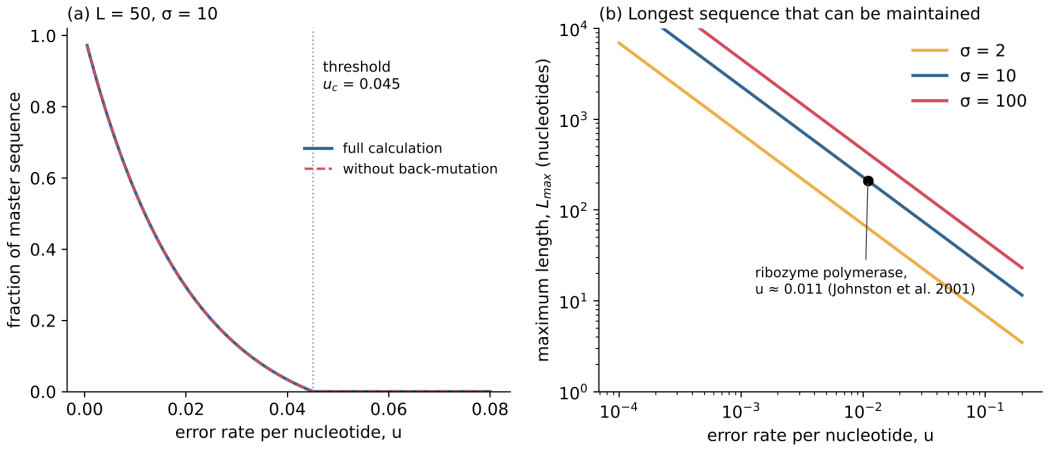


Figure 8.3.: The error threshold. (a) Fraction of the master sequence in the steady state for $L = 50$ and $\sigma = 10$, as a function of the error rate per nucleotide; solid line, full calculation with back-mutation (four-letter alphabet); dashed line, Equation 8.4. (b) Maximum length that can be maintained, $L_{max} = \ln \sigma / u$, for $\sigma = 2$, 10 and 100; the point marks the error rate of the ribozyme polymerase of Johnston et al. (2001) (author's calculation).

Because σ enters only through its logarithm, the error threshold depends mainly on the error rate. For $\sigma = 10$, an error rate of 0.01 allows sequences of about 230 nucleotides, and an error rate of 0.1 only about 23. A system that maintains a ribozyme of 200 nucleotides with $\sigma = 10$ must have u below $\ln 10/200 \approx 0.012$, that is, fewer than about one error in 90 nucleotides. This is a quantitative statement of the requirement that copying be accurate as well as fast.

8.5. Ribozyme polymerases

8.5.1. Laboratory evolution of RNA polymerases

If RNA was once copied by RNA, an RNA molecule must have catalysed the polymerisation of RNA on an RNA template. No such ribozyme is known in nature. Laboratory evolution, starting from the ligase ribozymes of Bartel and Szostak (1993), has produced ribozymes with this activity.

Key experiment: Johnston, Unrau, Lawrence, Glasner and Bartel 2001

Johnston and co-workers engineered a ribozyme that "uses nucleoside triphosphates and the coding information of an RNA template to extend an RNA primer by the successive addition of up to 14 nucleotides—more than a complete turn of an RNA helix". Its polymerisation was accurate: "when primers extended by 11 nucleotides were cloned and sequenced, 1088 of 1100 sequenced nucleotides matched the template" (Johnston et al. 2001). This corresponds to an error rate of 12/1100 ≈ 0.011 per nucleotide. With $\sigma = 10$, Equation 8.3 gives $L_{max} \approx 2.30/0.0109 \approx 211$ nucleotides, which exceeds the length of the ribozyme (Figure 8.3 b). The obstacle was not accuracy but the length of the product, far shorter than a ribozyme.

Later work extended the length of the products:

- Wochner and co-workers evolved and engineered "an RNA polymerase ribozyme capable of synthesizing RNAs of up to 95 nucleotides in length", and used it to synthesise an active ribozyme from an RNA template (Wochner et al. 2011).
- Attwater, Wochner and Holliger evolved polymerase ribozymes in water-ice. These were "able to synthesise RNA in ices at temperatures as low as −19°C", and led to "a first polymerase ribozyme capable of catalysing the accurate synthesis of an RNA sequence longer than itself (adding up to 206 nucleotides)" (Attwater, Wochner, and Holliger 2013).
- Horning and Joyce obtained a polymerase ribozyme able "to synthesize a variety of complex structured RNAs, including aptamers, ribozymes, and, in low yield, even tRNA". It amplified "short RNA templates by more than 10,000-fold in an RNA-catalyzed form of the PCR" (Horning and Joyce 2016).
- Attwater and co-workers reported RNA-catalysed RNA synthesis on structured templates using trinucleotide triphosphates as substrates (Attwater et al. 2018). Building blocks of three nucleotides bind more strongly than single nucleotides and can copy templates that are folded.

Horning and Joyce concluded: "the two prerequisites of Darwinian life—the replication of genetic information and its conversion into functional molecules—can now be accomplished with RNA in the complete absence of proteins" (Horning and Joyce 2016).

8.5.2. What remains to be done

A ribozyme that copies itself, and its complement, through repeated cycles without human intervention has not yet been made. The polymerase ribozymes are large folded molecules, the products of extensive laboratory evolution and engineering, and they use nucleoside triphosphates or trinucleotide triphosphates as substrates, whose prebiotic availability is uncertain. A folded RNA is also a difficult template, because its own base pairs must be opened before it can be copied; the use of triplet building blocks was one response to this difficulty (Attwater et al. 2018). The gap between these molecules and the short chains of Chapter 6 (Section 6.7) is large.

8.6. Selection

8.6.1. Replication and competition

Replicators grow exponentially when their substrates are not limiting. If a population doubles every t_d, its size at time t is

$$N(t) = N_0 \, 2^{t/t_d} \tag{8.5}$$

Substrates are always limited eventually. Then different replicators compete for the same substrates, and those that replicate faster, more accurately or for longer increase in frequency at the expense of others. This is **natural selection** at the molecular level. It requires only three things: replication, variation (introduced by copying errors or recombination) and differences in replication among the variants.

Key experiment: Lincoln and Joyce 2009

Lincoln and Joyce converted an RNA ligase ribozyme into a pair of enzymes that "catalyze each other's synthesis from a total of four oligonucleotide substrates". These cross-replicating enzymes underwent "self-sustained exponential amplification in the absence of proteins or other biological materials", with "a doubling time of about 1 hour", and amplification "can be continued indefinitely". When populations of different cross-replicating enzymes competed for a common pool of substrates, "recombinant replicators arose and grew to dominate the population" (Lincoln and Joyce 2009).

Each cross-replicating enzyme catalyses the formation of its partner, and so, over two steps, of copies of itself: the system is autocatalytic in the sense of Section 2.8. With a doubling time of 1 hour, Equation 8.5 gives a $2^{24} \approx 1.7 \times 10^7$-fold increase in a day, as long as substrates last. The experiment demonstrates exponential replication and selection among RNA enzymes. In this system, however, the replicators are assembled from

four supplied oligonucleotides, so new variants arise by recombination of the supplied pieces, not by copying sequences one nucleotide at a time.

8.7. How fast is fast enough?

8.7.1. A race against loss

A template is useful only if it is copied before it is destroyed. If a template of L nucleotides is copied at r nucleotides per unit time, one copy takes L/r. If the template survives for a mean time τ, set by hydrolysis (Section 6.3), outflow or adsorption, the mean number of complete copies it can make is about $r\tau/L$, and a lineage can persist only if

$$\frac{r\tau}{L} > 1 \tag{8.6}$$

This inequality is necessary but not sufficient: each copy must also be separated from its template (Section 8.3) and be accurate enough (Section 8.4), and the copies must in turn be copied.

8.7.2. Measured rates against required rates

The author's own simulations illustrate how strongly the outcome depends on r (Wahiro 2026). That study built a bottom-up model chemistry, from feedstock through activated monomers, chains, catalysts and template copying to vesicles, and searched 13 environmental variables, with ranges taken from the literature, for environments in which heredity arises. Heredity, defined as a sequence lineage that outlives its molecules because it is copied, appeared in a narrow class of cool environments with slow flushing and temperature cycles across the melting temperature of short duplexes.

A later check showed that this result depends on the copying rate. In the model, template-directed extension was about 10^9 to 10^{11} times faster than measured non-enzymatic copying, which, with activated nucleotides at 20 to 50 mM and 100 to 200 mM Mg^{2+}, takes from 30 minutes to about a day per added nucleotide even with activated helper oligonucleotides (Prywes et al. 2016; Wahiro 2026). When the extension rate was set to a value anchored in that literature, about one nucleotide per hour, template copying stopped entirely: short duplexes dissociated about 10^6 times faster than a nucleotide was added. Two other routes, templated ligation of chance oligomers and chance long primers, also failed to start heredity from scratch within the ranges studied. A second, independent limit appeared when the model's constants were translated into real base pairs with nearest-neighbour parameters: the environments where heredity arose combined the weak pairing of A–U with the accuracy and copying speed found experimentally only for G–C-rich templates, and no measured base pair has both (Wahiro

2026). A supplementary study found, in addition, that heredity at the confirmed points depended in part on a numerical rule of the simulation for sharing out scarce chain slots (Wahiro 2026).

These results are not a proof that heredity cannot arise by non-enzymatic chemistry. They are a quantitative statement of the gap: with measured rates and measured base-pair stabilities, a short template is released from its partner long before it is copied, and nothing in the environments studied closes this gap. A path to heredity therefore needs either copying chemistry many orders of magnitude faster than any yet measured, a mechanism that holds templates and building blocks together for much longer (for example in ice, on surfaces or in compartments), or a first replicator that was not a template copied one nucleotide at a time.

8.8. What remains unknown

The path described in this book, from atoms to self-copying RNA, has been established in parts, not as a whole. The following are open questions, in the order of the chapters.

1. **Feedstock and building blocks.** Which environments supplied the nucleotides of Chapters 4 and 5, in sufficient purity and concentration, together with a means of activating them repeatedly?
2. **Homochirality.** How did a single handedness of ribose arise and persist (Section 5.5)?
3. **Chains.** How were chains of 30 or more nucleotides made in useful amounts, given the competition between polymerisation and hydrolysis (Section 6.7)?
4. **Function.** How common are sequences with copying activity among short random sequences (Section 7.6)?
5. **Copying.** Can any chemistry, with or without a ribozyme, copy RNA fast and accurately enough, and separate the strands, to sustain repeated cycles of replication (Section 8.3, Section 8.7)?
6. **Selection and compartments.** How did replicating sequences become linked to the functions they encode, so that selection could act on them? This question leads to protocells, the subject of the Epilogue.

Summary

- The RNA world hypothesis holds that RNA once served as both genetic material and catalyst. It requires that RNA can be copied. Copying by base pairing produces the complement; replication requires two rounds and strand separation.

- Non-enzymatic primer extension with activated nucleotides copies sequences selectively. Better leaving groups (2-aminoimidazole) and activated helper oligonucleotides have extended it to mixed sequences, but it remains slow, and mismatches stall extension by more than two orders of magnitude.
- Short primers bind weakly and long products bind too strongly: the median T_m of a random 6-base-pair duplex at 1 μM is 17 °C, that of a 20-base-pair duplex 85 °C. This is the strand-separation problem.
- A master sequence of length L with selective superiority σ is maintained only if $\sigma(1-u)^L > 1$, so that $L_{max} \approx \ln \sigma / u$. An error rate of 0.01 per nucleotide allows a few hundred nucleotides.
- Ribozyme polymerases made in the laboratory extend primers by up to 206 nucleotides, synthesise functional RNAs and amplify RNA exponentially, but none yet replicates itself. Cross-replicating ligases replicate exponentially with a doubling time of about 1 hour and show selection.
- Measured non-enzymatic copying is far slower than heredity from scratch requires; in the author's simulations, heredity appeared only with copying about 10^9 to 10^{11} times faster than measured.

Exercises

1. Write the product of copying the template 5′-GGCAU-3′, and the product of copying that product. Why does replication require two rounds of copying?
2. Explain why non-enzymatic copying of templates rich in C is easier than copying of templates rich in A and U, using the stack free energies of Table 7.1.
3. A duplex has $\Delta H^\circ = -250$ kJ mol^{-1} and $\Delta S^\circ = -0.70$ kJ K^{-1} mol^{-1}. Using Equation 8.1, calculate T_m at $C_T = 1$ μM and at 1 mM.
4. Using Figure 8.2, estimate the shortest primer that remains bound at 20 °C at a total strand concentration of 1 mM.
5. Explain why heating a solution to separate a 30-base-pair duplex is harmful to the RNA, with reference to Section 6.3.
6. Using Equation 8.2, calculate the fraction of error-free copies of a 50-nucleotide sequence at u = 0.001, 0.01 and 0.05.
7. Using Equation 8.3, calculate L_{max} for $\sigma = 2$, 10 and 100 at u = 0.01. By what factor must u fall to double L_{max} at fixed σ?
8. Show that Equation 8.4 gives $x_0 = 1$ when u = 0 and $x_0 = 0$ when $\sigma Q = 1$.
9. The ribozyme of Johnston et al. (2001) copied 1088 of 1100 nucleotides correctly. Calculate its error rate, and the fraction of error-free copies it would make of a sequence of 200 nucleotides.
10. Using Equation 8.5, calculate the time needed for a population with a doubling time of 1 hour to increase 10^6-fold.
11. A template of 30 nucleotides is copied at one nucleotide per hour and has a mean lifetime of 10 hours. Using Equation 8.6, can the lineage persist? What lifetime

would be needed?

12. List the three conditions for natural selection among molecules and explain how each is met in the experiment of Lincoln and Joyce (2009).
13. Choose one of the open questions in Section 8.8 and describe an experiment or a calculation that would help to answer it.

Epilogue: From RNA to Cells

This book has followed one chemical path: from the formation of the elements, through small molecules and nucleotides, to chains of RNA that fold, catalyse and, under laboratory conditions, copy other RNA. A self-copying molecule is not yet a living cell. This epilogue outlines the next steps and states where the frontier of knowledge lies.

Compartments

A replicating molecule in open solution shares its products with every other molecule nearby. A ribozyme that improves replication helps its competitors as much as itself, and selection cannot favour it. Selection for function requires that the products of a sequence stay with that sequence. A **compartment**, a closed boundary that holds a set of molecules together, provides this link. A compartment that contains a replicating genetic system is called a **protocell**.

Models of protocell membranes are made from **fatty acids**: molecules with a carboxylic acid head group (Section 4.1) and a long hydrocarbon tail. Such molecules are **amphiphilic**, having a polar part that interacts favourably with water and a non-polar part that does not. In water, above a critical concentration, they assemble into **micelles**, small aggregates with the tails inside and the head groups facing the water, or into **vesicles**: closed membranes, made of a double layer of molecules, that enclose a volume of water. The hydrophobic effect (Section 2.5) drives this assembly. Fatty acid membranes are simpler than the phospholipid membranes of modern cells, and are more permeable to small molecules.

Laboratory work has shown that such vesicles can carry out several functions of a cell. Hanczyc, Fujikawa and Szostak found that the clay montmorillonite, known to catalyse the polymerisation of RNA (Section 6.4), "accelerates the spontaneous conversion of fatty acid micelles into vesicles"; RNA adsorbed to clay can be encapsulated within vesicles; and "once formed, such vesicles can grow by incorporating fatty acid supplied as micelles and can divide without dilution of their contents by extrusion through small pores" (Hanczyc, Fujikawa, and Szostak 2003). Zhu and Szostak described a simpler pathway: large multilamellar vesicles fed with fatty acid micelles grow into long thread-like vesicles, and "modest shear forces are then sufficient to cause the thread-like vesicles to divide into multiple daughter vesicles without loss of internal contents"; encapsulated RNA is distributed to the daughters, and the protocells "can proceed through multiple cycles of reproduction" (Zhu and Szostak 2009).

Coupling genome and compartment

> Key experiment: Chen, Roberts and Szostak 2004
>
> Chen, Roberts and Szostak showed that "RNA, encapsulated in fatty acid vesicles, exerts an osmotic pressure on the vesicle membrane that drives the uptake of additional membrane components, leading to membrane growth at the expense of relaxed vesicles, which shrink." The authors concluded: "Thus, more efficient RNA replication could cause faster cell growth, leading to the emergence of Darwinian evolution at the cellular level" (Chen, Roberts, and Szostak 2004). The coupling requires no function of the RNA other than its presence, and its replication.

Copying chemistry and membranes must also be compatible. Fatty acid membranes allow small charged molecules, such as nucleotides, to pass, and activated nucleotides added outside a model protocell can cross the membrane and take part in template copying inside it (Mansy et al. 2008). The Mg^{2+} ions needed for non-enzymatic copying disrupt fatty acid membranes, but Adamala and Szostak found that citrate protects the membranes from high Mg^{2+} concentrations while allowing RNA copying to proceed (Adamala and Szostak 2013).

From RNA to proteins

Modern cells make proteins on the ribosome, a large complex of RNA and protein. The structure of the large ribosomal subunit showed that the site at which peptide bonds are formed consists of RNA: the ribosome is a ribozyme (Nissen et al. 2000). As noted in Section 8.1, Orgel regarded this as making it "almost certain that there was once an RNA World" (Orgel 2004). How an RNA world began to make proteins, how the genetic code arose, and how DNA took over the storage of genetic information are questions beyond the scope of this book.

The open frontier

Each chapter of this book has separated what is established from what is hypothesised. The established parts are substantial: the elements of RNA other than hydrogen are made in stars (Chapter 1); its building blocks form in plausible prebiotic reactions (Chapters 4 and 5); activated nucleotides join into chains on minerals and in ice (Chapter 6); random RNA sequences fold, and some of them bind and catalyse (Chapter 7); RNA enzymes made in the laboratory copy RNA and replicate exponentially (Chapter 8); and vesicles grow, divide and compete (this epilogue).

The open questions lie at the joins between these parts. Each step has been demonstrated under its own conditions, often with purified reagents and controlled cycles, and the conditions differ from step to step. No experiment has yet started from simple feedstock and arrived at a replicating, evolving system. The quantitative gaps are largest at copying: measured non-enzymatic copying is far too slow to sustain heredity from scratch (Section 8.7), and laboratory ribozyme polymerases are far longer and more complex than the chains that prebiotic chemistry produces.

A further requirement became clear in the author's simulations of protocells. Where heredity and vesicles coexisted, vesicle families carrying copied sequences won a single selective sweep, after which copying refilled every vesicle to the same load, and no heritable variation remained for selection to act on (Wahiro 2026). In that model, sustained evolution would need a sequence-dependent trait that copying does not erase, a sequence that does something, for example by catalysing its own copying, building its own membrane or protecting its compartment. This is the link between Chapters 7 and 8 that remains to be found: a sequence that is both copied and useful, in a chemistry and an environment where both happen at once.

The origin of life is therefore not a solved problem. It is a problem whose parts can now be stated precisely, measured in the laboratory and tested in models, which is the condition for solving it.

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A. Glossary

This glossary lists the technical terms defined in the book, in alphabetical order. Each entry gives the term, its definition, and the chapter in which it is first used (or the Epilogue).

Table A.1.: Terms defined in this book, with the chapter of first use.

Term	Definition	First used
5′ to 3′ direction	The direction of a nucleic acid chain from the end with a free 5′ group to the end with a free 3′-hydroxy group; sequences are written in this direction.	6
Abundance (cosmic)	The relative amount of an element in a defined body or region, expressed as the number of atoms relative to hydrogen or as a mass fraction.	1
Acid	A substance that donates a proton (H^+) to another molecule.	2
Activated reactant	A reactant that carries a built-in source of free energy, allowing a condensation to proceed in water.	4
Activation energy (E_a)	The energy barrier that reactants must overcome to react; defined by IUPAC through the temperature dependence of the rate constant, $E_a = RT^2\ d(\ln k)/dT$.	2
Activity	The effective concentration of a dissolved species, corrected for interactions between particles, expressed relative to a standard concentration of 1 mol per litre.	2
Activity (of water)	The effective concentration of water relative to pure water; it is lowered by evaporation and by concentration of solutes.	5
Addition reaction	A reaction in which two molecules combine to form a single product without the loss of any atoms.	4
Adenine	The purine $C_5H_5N_5$; formally a pentamer of HCN.	4
Alanine	The amino acid $H_2NCH(CH_3)COOH$ (R = CH_3); β-alanine, $H_2NCH_2CH_2COOH$, is a structural isomer.	4

Term	Definition	First used
Alcohol	A compound in which a hydroxy group, –OH, is attached to a carbon atom that forms only single bonds.	4
Aldehyde	A compound containing the group –CHO, a carbon atom double-bonded to oxygen and bonded to hydrogen.	3
Aldol reaction	Addition of the carbon atom next to the carbonyl group of one molecule, made nucleophilic in basic solution, to the carbonyl carbon of another, forming a carbon-carbon bond.	4
Aldose	A sugar whose carbonyl group is an aldehyde at the end of the carbon chain.	4
Alkaline hydrothermal vent	A hydrothermal vent, not driven directly by volcanic heat, that emits warm, alkaline fluids rich in H_2 and CH_4, such as the Lost City field.	3
Alpha particle	A helium-4 nucleus, consisting of two protons and two neutrons.	1
Amide	A compound derived from an acid by replacing a hydroxy group with an amino or substituted amino group; in carboxylic amides, –CO–NH_2.	4
Amine	A compound containing a nitrogen atom bonded to one or more carbon atoms and carrying a lone pair; the simplest group is the amino group, –NH_2.	4
Amino acid	A compound containing an amino group and a carboxy group; in the α-amino acids of proteins both are attached to the same carbon atom, with general formula H_2N–CHR–COOH.	4
Aminonitrile	A compound containing an amino group and a nitrile group on the same carbon atom, RCH(NH_2)CN; the intermediate of the Strecker synthesis.	4
2-Aminooxazole	A small heterocycle formed from cyanamide and glycolaldehyde; an intermediate in the synthesis of pyrimidine nucleotides by Powner and co-workers.	5
Amphiphilic	Having a polar part that interacts favourably with water and a non-polar part that does not.	Epilogue
Ångström (Å)	A unit of length equal to 10^{-10} m, convenient for bond lengths.	2
Anhydronucleoside	A nucleoside in which an additional bond links the base to the sugar; an intermediate in the 2009 synthesis of pyrimidine nucleotides.	5

Term	Definition	First used
Annum (a)	The year as a unit of time in geochronology; multiples are written ka (10^3 a), Ma (10^6 a) and Ga (10^9 a).	3
Anomer	One of two stereoisomers of a ring-form sugar or nucleoside that differ only in the configuration at C1 (α or β).	5
Antiparallel	Of two paired nucleic acid strands: running in opposite 5′ to 3′ directions, the arrangement required for base pairing.	7
Apatite	A group of calcium phosphate minerals of low solubility.	3
Aptamer	A nucleic acid molecule, usually obtained by in vitro selection, that binds a specific target molecule.	7
Archean	The second eon of Earth history, from 4031 ± 3 Ma to 2500 Ma.	3
Arrhenius equation	$k = A\ e^{-E_a/RT}$: the dependence of a rate constant k on absolute temperature T, with pre-exponential factor A and activation energy E_a.	2
Asymmetric autocatalysis	Autocatalysis in which a chiral product catalyses the formation of itself rather than its enantiomer, so that an enantiomeric excess can be amplified.	4
Atom	The smallest unit of a chemical element: a positively charged nucleus surrounded by electrons.	1
Atom ratio	The number of atoms of one element in a sample divided by the number of atoms of another element in the same sample.	1
Atomic mass unit (u)	One twelfth of the mass of a carbon-12 atom; equivalent to 931.494 MeV of energy.	1
Atomic number (Z)	The number of protons in the nucleus of an atom; it defines the element.	1
Autocatalysis	Catalysis of a reaction by one of its own products.	2
Autotroph	An organism that makes its organic matter from CO_2.	3
Backbone	The alternating sugar-phosphate chain of a nucleic acid, to which the nucleobases are attached.	6
Bar	A unit of pressure equal to 10^5 Pa, close to the mean atmospheric pressure at sea level.	3
Base	A substance that accepts a proton (H^+) from another molecule.	2
Base pair	Two nucleobases on the same or different strands joined by specific hydrogen bonds.	7

Term	Definition	First used
Base stacking	The face-to-face packing of the flat rings of adjacent nucleobases along a helix, stabilised by dispersion forces and the hydrophobic effect.	7
Beta-plus decay	Radioactive decay in which a proton in a nucleus becomes a neutron, emitting a positron and a neutrino.	1
Big Bang	The model in which the Universe evolved from an extremely hot, dense initial state about 13.8 billion years ago and has been expanding and cooling since.	1
Big Bang nucleosynthesis	The formation of the lightest nuclei (mainly hydrogen and helium-4, with traces of deuterium, helium-3 and lithium-7) in the first minutes after the Big Bang.	1
Binding energy	The energy required to separate a nucleus completely into its protons and neutrons; equal to the mass difference between the free nucleons and the nucleus, multiplied by c^2.	1
Bond angle	The angle between two bonds that share an atom.	2
Bond enthalpy	The enthalpy needed to break one mole of a given bond in molecules in the gas phase; e.g. 436.0 kJ mol^{-1} for H–H.	2
Bond length	The distance between the nuclei of two atoms joined by a chemical bond.	2
Bond order	The number of electron pairs shared between two bonded atoms: 1 for a single, 2 for a double and 3 for a triple bond.	2
Borate	The anion of boric acid; it forms complexes with sugars that have hydroxy groups on adjacent carbon atoms.	4
Buffer (oxygen)	An assemblage of minerals that fixes the oxygen fugacity at a known value at each temperature, such as iron-wüstite or fayalite-magnetite-quartz.	3
Bulge	A loop between two helices with unpaired nucleotides on one strand only.	7
Calcium-aluminium-rich inclusion (CAI)	A rare component of chondritic meteorites and the oldest dated solid of the solar system, 4.567-4.568 billion years old.	1
Carbonaceous chondrite	A chondrite rich in carbon, often containing a wide variety of organic compounds; the Murchison meteorite is an example.	3

Term	Definition	First used
Carbonyl group	A carbon atom double-bonded to an oxygen atom, C=O.	4
Carboxylic acid	A compound containing the carboxy group, –COOH.	4
Catalyst	A substance that increases the rate of a reaction without changing its overall standard Gibbs energy change; it is regenerated in the reaction.	2
Chandrasekhar mass	The largest mass that the pressure of electrons can support against gravity in a stellar core; when an iron core exceeds it, the core collapses.	1
Chirality	The property of an object of being non-superposable on its mirror image.	4
Chondrite	A meteorite consisting of unprocessed early solar-system material: CAIs, chondrules and iron-nickel metal in a fine-grained matrix.	1
Chondrule	An abundant, rounded, magnesium- and silicon-rich component of chondritic meteorites, formed up to about 4 million years after CAIs.	1
Clay mineral	A layered silicate formed by the weathering of rocks in water.	6
Closed system (geochronology)	A sample to which no parent or daughter atoms have been added and from which none have been removed since the event being dated.	3
CNO cycle	A cycle of nuclear reactions in which carbon, nitrogen and oxygen nuclei catalyse the conversion of four protons into one helium-4 nucleus; the main stellar source of nitrogen-14.	1
Compartment	A closed boundary that holds a set of molecules together and separates them from their surroundings.	Epilogue
Complementary	Of two sequences: able to form Watson-Crick base pairs along their whole length when aligned antiparallel.	7
Concordant age	A uranium-lead age for which the ^{238}U-^{206}Pb and ^{235}U-^{207}Pb ages agree within uncertainty.	3
Condensation reaction	A reaction in which two reactants (or two sites in one molecule) yield one main product together with water or another small molecule.	4
Continental crust	The outer layer of rock that forms the continents; its upper part is the most accessible region of the solid Earth.	1

Term	Definition	First used
Core electrons	The electrons in the inner, completely filled shells of an atom; they are tightly bound and do not take part in bonding.	2
Core-collapse supernova	The explosion of a star of more than about 8 solar masses, triggered by the collapse of its iron core.	1
Cosmic microwave background	Nearly uniform microwave radiation filling the sky, released when the Universe first became transparent at recombination and now cooled to about 2.7 K.	1
Covalent bond	A chemical bond formed by a pair of electrons shared between two atoms.	2
Cyanamide	H_2NCN, an amino group bonded to a nitrile group; a prebiotic feedstock molecule.	3
Cyanoacetylene	HC≡C–C≡N, a product of spark discharge in CH_4/N_2 mixtures and a precursor of pyrimidines.	4
Cyanohydrin	A compound in which a hydroxy group and a nitrile group are attached to the same carbon atom, formed by addition of HCN to a carbonyl group.	4
2′,3′-Cyclic phosphate	A nucleotide in which one phosphate bridges C2′ and C3′ in a five-membered ring; an activated form that can form phosphodiester bonds.	5
Cytosine	The pyrimidine $C_4H_5N_3O$; it is converted to uracil by hydrolysis.	4
Daughter (nuclide)	The nuclide formed by the radioactive decay of a parent nuclide.	3
Debye (D)	The customary unit of molecular dipole moment; 1 D = 3.335 64 × 10^{-30} C m.	2
Decay (radioactive)	The spontaneous transformation of an unstable nucleus into a different nucleus, with emission of radiation.	1
Decay constant (λ)	The probability per unit time that a nucleus of a given radioactive nuclide decays; related to the half-life by $t_{1/2} = \ln 2/\lambda$.	3
Delta notation (δ)	The deviation of an isotope ratio in a sample from that of a standard, $(R_{\text{sample}}/R_{\text{standard}} - 1) \times 1000$, expressed in per mil (‰).	3
Detrital	Describes a mineral grain eroded from its original rock and incorporated into a sediment.	3
Deuterium bottleneck	The delay in Big Bang nucleosynthesis caused by the destruction of newly formed deuterons by energetic photons, lasting until the Universe cooled to about 10^9 K.	1

Term	Definition	First used
Deuteron	The nucleus of deuterium, consisting of one proton and one neutron.	1
Diamidophosphate (DAP)	$PO_2(NH_2)_2^-$, a phosphorylating agent that acts in water without an additional condensing agent.	4
Diaminomaleonitrile (DAMN)	The tetramer of HCN, an intermediate in the formation of adenine from HCN.	4
Differentiation (planetary)	The melting of a planetary body and its internal separation into layers, such as a metallic core and a rocky mantle.	1
Dipole moment (μ)	For charges +q and −q separated by a distance d, the product $\mu = qd$; a measure of the separation of charge in a molecule.	2
Dipole-dipole interaction	The attraction between the positive end of one polar molecule and the negative end of another.	2
Dispersion force	A weak attraction between temporary, mutually induced dipoles in neighbouring atoms or molecules; it acts between all molecules and grows with molecular size.	2
Dot-bracket notation	A notation for nested secondary structure: a string with a bracket for each paired nucleotide (opening for the 5′ partner, closing for the 3′ partner) and a dot for each unpaired nucleotide.	7
Double bond	A covalent bond in which two atoms share two pairs of electrons.	2
Double helix	The right-handed helical structure formed by two antiparallel, base-paired nucleic acid strands or segments.	7
Doubling time (t_d)	The time in which an exponentially growing population doubles in size.	8
Dry mass	The mass of biological material remaining after all water has been removed.	1
Duplex	A complex of two separate nucleic acid strands held together by base pairs.	7
Dynamic programming	A method that solves a problem by combining stored solutions of smaller sub-problems; used to find minimum free energy RNA structures.	7
Electrical discharge	The passage of electric current through a gas, as in lightning, which heats the gas and breaks its molecules into fragments.	3

Term	Definition	First used
Electron	A subatomic particle with one negative elementary charge; electrons occupy orbitals around the nucleus and determine chemical behaviour.	1
Electron configuration	A list of the number of electrons in each subshell of an atom, such as $1s^2\ 2s^2\ 2p^2$ for carbon.	2
Electronegativity (χ)	A measure of the tendency of an atom in a bond to attract the shared electrons towards itself; on the Pauling scale, H 2.20, C 2.55, N 3.04, O 3.44, P 2.19.	2
Electronvolt (eV)	The energy gained by a particle with one elementary charge moving through a potential difference of one volt; exactly $1.602\ 176\ 634 \times 10^{-19}$ J.	2
Electrophile	A reagent that forms a bond by accepting both bonding electrons from its reaction partner.	4
Element	A substance consisting of atoms that all have the same atomic number.	1
Enantiomeric cross-inhibition	Inhibition of template-directed chain growth by monomers of the opposite handedness, which are incorporated and block further extension.	5
Enantiomeric excess (ee)	$([L] - [D])/([L] + [D])$, the fractional excess of one enantiomer in a mixture; 0 for a racemate, ± 1 for a homochiral sample.	4
Enantiomers	Two molecules that are non-superposable mirror images of each other.	4
Endothermic	Describes a process that absorbs heat from its surroundings ($\Delta H > 0$).	2
Energy level	One of the discrete values of energy that an electron in an atom or molecule can have.	2
Enthalpy (H)	Internal energy of a system plus the product of its pressure and volume; its change equals the heat exchanged at constant pressure.	2
Entropy (S)	A thermodynamic quantity measuring the number of microscopic arrangements consistent with the macroscopic state of a system.	2
Enzyme	A biological catalyst; a protein or, in the case of a ribozyme, an RNA molecule.	7
Eon	The largest unit of geological time; Earth history is divided into the Hadean, Archean, Proterozoic and Phanerozoic eons.	3

Term	Definition	First used
Equilibrium (chemical)	The state of a reversible reaction in which the forward and reverse reactions proceed at equal rates, so that the amounts of reactants and products no longer change.	2
Equilibrium constant (K)	The ratio, at equilibrium, of the activities of products to those of reactants, each raised to the power of its coefficient; related to ΔG° by $\Delta G^\circ = -RT \ln K$.	4
Error catastrophe	The loss of the information of a master sequence among its mutants when the error rate exceeds the error threshold.	8
Error rate per nucleotide (u)	The probability that a nucleotide is copied incorrectly; the fidelity is q = 1 − u.	8
Error threshold	The greatest sequence length, $L_{max} \approx \ln \sigma / u$, that can be maintained by replication with error rate u and selective superiority σ.	8
Ester	A compound formally derived from an acid and an alcohol by linking them with the loss of water.	4
Eutectic phase	The concentrated liquid that remains between ice crystals when a solution is cooled below its freezing point.	6
Excited state	A state of a nucleus, atom or molecule with more energy than its lowest-energy state.	1
Exogenous delivery	The arrival on the Earth of organic molecules already present in extraterrestrial objects.	3
Exothermic	Describes a process that releases heat to its surroundings ($\Delta H < 0$).	2
Exterior loop	The unpaired nucleotides of a secondary structure that are not enclosed by any base pair, including the two chain ends.	7
Faint young Sun problem	The difficulty of reconciling the lower luminosity of the young Sun, about 30 percent below the present value, with evidence for liquid water on the early Earth.	3
Fatty acid	A carboxylic acid with a long hydrocarbon chain; an amphiphilic molecule used in models of protocell membranes.	Epilogue
Feedstock molecule	A small, simple molecule from which more complex molecules can be built in a sequence of reactions.	3
Fidelity (q)	The probability that a nucleotide is copied correctly.	8

Term	Definition	First used
Formaldehyde	H_2CO, the simplest aldehyde; the feedstock of the formose reaction.	3
Formamide	$HCONH_2$, the amide of formic acid; a liquid from 2.6 °C to about 211 °C, from which nucleobases form on heating with mineral catalysts.	4
Formamidopyrimidine	A pyrimidine carrying a formamido group, whose atoms can close a second ring to give a purine; an intermediate in prebiotic purine nucleoside syntheses.	5
Formose reaction	The conversion of formaldehyde, in the presence of a base, into a mixture of sugars; it is autocatalytic.	3
Freeze-out	The point at which a reaction becomes too slow, relative to the expansion of the Universe, to keep the amounts of its products in equilibrium.	1
Functional group	An atom or group of atoms that has similar chemical properties whenever it occurs in different compounds.	4
Furanose	The five-membered ring form of a sugar (four carbon atoms and one oxygen atom), formed by addition of a hydroxy group to the carbonyl group of the same molecule.	4
G·U wobble pair	A base pair between guanine and uracil held by two hydrogen bonds, with the bases shifted relative to a Watson-Crick pair; common in RNA helices.	7
Gamma ray (γ)	A photon of very high energy, typically emitted in nuclear reactions.	1
Geometric distribution	The distribution $P(n) = (1 - p)p^{n-1}$ of chain lengths when each extension succeeds with probability p; its mean is $1/(1 - p)$.	6
Giant impact hypothesis	The hypothesis that the Moon formed from debris thrown into orbit when a body about the size of Mars collided with the proto-Earth.	3
Gibbs free energy (G)	G = H - TS. At constant temperature and pressure, a process occurs spontaneously only if it lowers G.	2
Glycine	The simplest amino acid, H_2NCH_2COOH (R = H); the only common amino acid without a stereocentre.	4
Graphite	A crystalline form of pure carbon.	3
Ground state	The lowest-energy state of a nucleus, atom or molecule.	1
Guanine	The purine $C_5H_5N_5O$.	4

Term	Definition	First used
Hadean	The first eon of Earth history, from the formation of the solar system to 4031 ± 3 Ma.	3
Hairpin loop	A loop of unpaired nucleotides closed by a single base pair at the end of a helix; it contains at least three nucleotides.	7
Half-life	The time required for half of the nuclei in a sample of a radioactive isotope to decay.	1
Helioseismology	The study of the interior of the Sun through its oscillations, observed at its surface.	1
Helium burning	Nuclear fusion of helium in a stellar core, producing mainly carbon-12 and oxygen-16; it requires temperatures near 10^8 K.	1
Helium mass fraction (Y)	The fraction of the total mass of ordinary matter that is in the form of helium-4.	1
Helix (stem)	In secondary structure, a run of consecutive stacked base pairs.	7
Hess's law	The enthalpy change of a reaction is independent of the route taken, so it can be calculated from tabulated enthalpies of formation.	2
Heterocyclic compound	A cyclic compound whose ring contains atoms of at least one element other than carbon.	4
Homochirality	The property of consisting of only one of two enantiomers, as RNA consists only of D-ribose units.	4
Hoyle state	The excited state of carbon-12 at 7.65 MeV, through which the triple-alpha process proceeds.	1
Hund's rule	In a partly filled subshell, electrons occupy separate orbitals with parallel spins before any orbital receives a second electron.	2
Hybridisation	The description of a bonded atom's valence orbitals as mixtures of its s and p orbitals (sp^3, sp^2, sp) that point in the directions of its bonds.	2
Hydration shell	The layer of oriented water molecules surrounding a dissolved ion or molecule.	2
Hydrogen bond	An attractive interaction between a hydrogen atom of a group X–H, in which X is more electronegative than H, and an atom or group of atoms in the same or another molecule, in which there is evidence of bond formation (IUPAC 2011); written X–H $\cdot\cdot\cdot$ Y.	2
Hydrogen burning	Nuclear fusion of hydrogen into helium in a star; a nuclear, not a chemical, process.	1

Term	Definition	First used
Hydrogen cyanide	HCN, a linear molecule with a carbon-nitrogen triple bond; a feedstock for nucleobases.	3
Hydrogen-bond acceptor	The atom Y, usually O or N with a lone pair, that receives a hydrogen bond X–H $\cdot\cdot\cdot$ Y.	2
Hydrogen-bond donor	The atom X to which the hydrogen of a hydrogen bond X–H $\cdot\cdot\cdot$ Y is covalently bonded.	2
Hydrolysis	The breaking of a bond by reaction with water; the reverse of a condensation.	4
Hydronium ion	The ion H_3O^+, formed when a water molecule accepts a proton.	2
Hydrophilic	Interacting favourably with water; describes polar and charged molecules or groups.	2
Hydrophobic effect	The tendency of non-polar molecules or groups to cluster together in water, reducing their contact with water.	2
Hydrothermal vent	A site on the sea floor where water heated within the crust flows out into the ocean.	3
Hydroxide ion	The ion OH^-, formed when a water molecule loses a proton.	2
Imidazole	A five-membered aromatic ring containing two nitrogen atoms, $C_3H_4N_2$.	6
Imine	A compound containing a carbon-nitrogen double bond, $R_2C{=}NR$; formed by condensation of an aldehyde or ketone with ammonia or an amine.	4
Impact-shock synthesis	The formation of organic molecules by shock heating during the entry or impact of extraterrestrial objects.	3
In vitro selection	Repeated cycles of selecting molecules with a desired property from a pool of random sequences and amplifying them.	7
Intermolecular force	An attraction between molecules, much weaker than a covalent bond, such as dipole-dipole, dispersion and hydrogen-bond interactions.	2
Internal loop	A loop between two helices with unpaired nucleotides on both strands.	7
Interplanetary dust particle (IDP)	A small particle of extraterrestrial dust that falls on the Earth from space.	3
Ion	An atom or molecule with a net electric charge, resulting from an unequal number of protons and electrons.	1

Term	Definition	First used
Ion product of water (K_w)	The product of the activities of H_3O^+ and OH^- in water at equilibrium; $pK_w = 13.99$ at 25 °C.	2
Ionic bond	A bond formed by the electrostatic attraction between ions of opposite charge, resulting from the transfer of electrons between atoms of very different electronegativity.	2
Ionisation energy (first)	The minimum energy required to remove one electron from a neutral atom in its ground state.	2
Isomers	Species with the same molecular formula but different structures.	4
Isotope	Atoms of the same element (same atomic number) that differ in their number of neutrons.	1
Isotope fractionation	The partial separation of isotopes of different mass by physical or chemical processes.	3
Ketone	A compound in which a carbonyl group is bonded to two carbon atoms.	4
Ketose	A sugar whose carbonyl group is a ketone.	4
Late Heavy Bombardment	Impacts on the planets and the Moon after their outer layers became rigid enough to preserve craters and basins; in particular, an episode of elevated impact rate between about 4.2 to 4.0 Ga and 3.5 Ga.	3
Late veneer	Material accreted by the Earth after the Moon-forming impact.	3
Lead-lead age	An age calculated from the ratio of radiogenic ^{207}Pb to radiogenic ^{206}Pb, which depends on the time elapsed but not on the amount of uranium in the sample.	3
Leaving group	The group that departs with an electron pair when a nucleophile attacks; a good leaving group makes the reaction easier.	6
Lewis structure	A diagram of a molecule in which shared electron pairs are drawn as lines and lone pairs as pairs of dots.	2
Log epsilon ($\log \varepsilon$)	An abundance scale defined as $\log_{10}(N_X/N_H) + 12$, where N_X and N_H are the numbers of atoms of element X and of hydrogen; hydrogen has 12 by definition.	1
Lone pair	A pair of valence electrons that belongs to one atom and is not shared in a bond.	2

Term	Definition	First used
Luminosity	The total power emitted by a star, integrated over all wavelengths; the present luminosity of the Sun is 3.85×10^{26} W.	3
Magma ocean	A global layer of molten rock at the surface of a planet or moon.	3
Mass fraction	The mass of one component of a sample divided by the total mass of the sample.	1
Mass number (A)	The total number of protons and neutrons in a nucleus.	1
Master sequence	In quasispecies theory, the sequence that replicates fastest; its mutants replicate more slowly.	8
Melting temperature (T_m)	The temperature at which half of the molecules are folded or paired; in the two-state model $T_m = \Delta H/\Delta S$.	7
Meteorite	An extraterrestrial body that survives passage through the atmosphere and reaches the ground.	3
Micelle	A small aggregate of amphiphilic molecules in water, with the non-polar parts inside and the polar parts facing the water.	Epilogue
Minimum free energy (MFE) structure	The secondary structure of lowest Gibbs energy that a sequence can form under a given energy model.	7
Molal (mol kg^{-1})	A concentration expressed as moles of solute per kilogram of solvent.	3
Molar (M)	A concentration expressed as moles of solute per litre of solution.	3
Mole	The amount of substance containing exactly $6.022\,140\,76 \times 10^{23}$ elementary entities (atoms, molecules, ions).	2
Molecular cloud	A cold, dense cloud of interstellar gas, in which hydrogen is present mainly as H_2 molecules, together with dust; stars form in such clouds.	1
Molecule	Two or more atoms held together by covalent bonds and acting as a single unit.	2
Monoisotopic element	An element that has only one stable isotope, such as phosphorus (^{31}P).	1
Monosaccharide (sugar)	A compound $C_n(H_2O)_n$, usually with n from 3 to 7, containing one carbonyl group and a hydroxy group on each other carbon atom.	4

Term	Definition	First used
Montmorillonite	A layered clay mineral whose negatively charged aluminium silicate layers adsorb organic molecules; it catalyses the oligomerisation of activated nucleotides.	6
Multibranch loop	A loop from which three or more helices radiate; also called a junction.	7
N-glycosidic bond	The bond between C1′ of the sugar and a ring nitrogen of the base (N9 in purines, N1 in pyrimidines) in a nucleoside.	5
Natural selection	The increase in frequency of variants that replicate faster, more accurately or for longer, among replicators competing for limited resources.	8
Nearest-neighbour model	An energy model in which the Gibbs energy of a helix is the sum of terms for each pair of adjacent base pairs plus an initiation term.	7
Neutral atmosphere	An atmosphere in which carbon and nitrogen are present mainly as CO_2 and N_2, without much H_2 or free O_2.	3
Neutrino	An electrically neutral elementary particle of very small mass that interacts only weakly with matter.	1
Neutron	An electrically neutral subatomic particle found in the nucleus.	1
Nitrile	A compound containing the group –C≡N.	4
Nuclear fusion	A nuclear reaction in which two light nuclei combine to form a heavier nucleus. For nuclei lighter than iron, fusion releases energy.	1
Nucleobase	A nitrogen-containing heterocyclic compound that, bonded to ribose, forms part of a nucleotide; in RNA, adenine, guanine, cytosine and uracil.	4
Nucleon	A proton or a neutron.	1
Nucleophile	A reagent that forms a bond by donating an electron pair to its reaction partner.	4
Nucleoside	A nucleobase joined to a sugar (ribose in RNA), without a phosphate group.	3
Nucleosynthesis	The formation of atomic nuclei by nuclear reactions.	1
Nucleotide	A nucleoside carrying one or more phosphate groups, usually on C5′ of the sugar; the monomer unit of RNA.	5
Nucleus (atomic)	The small, dense, positively charged centre of an atom, made of protons and neutrons.	1

Term	Definition	First used
Octet rule	The tendency of carbon, nitrogen and oxygen to form bonds until each has eight valence electrons, the configuration of neon.	2
Oligonucleotide	A short chain of nucleotides, often described by its length, as in a 10-mer; a long chain is a polynucleotide.	6
Orbital	A region around a nucleus, described by a wavefunction, in which an electron of a given energy is likely to be found; each orbital holds at most two electrons.	2
Oxidation	The loss of electrons by an atom, ion or molecule; an increase in oxidation number.	3
Oxidation number	The charge an atom would carry if the shared electrons of each of its covalent bonds were assigned to the more electronegative atom.	3
Oxygen fugacity	The effective pressure of O_2 in equilibrium with a magma or other system; a measure of how oxidising it is.	3
Parent (nuclide)	A radioactive nuclide that decays into a daughter nuclide.	3
Partition function (Z)	The sum of $\exp(-\Delta G/RT)$ over all structures of a molecule; the probability of a structure is its term divided by Z.	7
Pauli exclusion principle	No two electrons in an atom can be in the same quantum state; hence an orbital holds at most two electrons, with opposite spins.	2
Pentose	A sugar with five carbon atoms; ribose is an aldopentose.	4
Per mil (‰)	Parts per thousand.	3
pH	The negative base-10 logarithm of the activity of hydronium ions in a solution; neutral pH equals $pK_w/2$, which is 7.00 at 25 °C.	2
Phosphate	The ion PO_4^{3-}, in which phosphorus is bonded tetrahedrally to four oxygen atoms; also used for groups derived from it.	3
Phosphate ester	An ester formed from phosphoric acid and an alcohol; the units of RNA are joined by phosphate ester linkages.	4
Phosphate problem	The mismatch between the low concentrations of phosphate in natural waters and the high concentrations used in prebiotic experiments.	3

Term	Definition	First used
Phosphoanhydride bond	A P–O–P linkage formed by condensation of two phosphate groups, as in pyrophosphate and ATP; its hydrolysis releases more free energy than that of a phosphate ester.	4
Phosphodiester bond	The phosphate diester linkage that joins the C3 hydroxy group of one ribose to the C5 hydroxy group of the next in RNA.	4
Phosphoric acid	H_3PO_4, a triprotic acid; its anions are collectively called phosphate.	4
Phosphorimidazolide	A nucleotide in which one oxygen of the phosphate is replaced by a nitrogen atom of imidazole or a substituted imidazole; an activated monomer.	6
Phosphorylation	The attachment of a phosphate group to a molecule.	4
Photochemistry	Chemical reactions initiated by the absorption of light.	3
Photosphere	The layer of a star from which its visible light escapes into space; stellar abundances are measured from absorption lines formed there.	1
Phytoplankton	Microscopic photosynthetic organisms living in the surface waters of oceans and lakes.	1
Pi (π) bond	A covalent bond formed by side-by-side overlap of two parallel p orbitals, with electron density above and below the line joining the nuclei.	2
Plasma	A gas in which a significant fraction of atoms are ionised, so that it contains free electrons and positively charged nuclei or ions.	1
Polar bond	A covalent bond in which the shared electrons are unequally distributed because the two atoms differ in electronegativity; the ends carry partial charges $\delta+$ and $\delta-$.	2
Polar molecule	A molecule with a net dipole moment, because the dipoles of its polar bonds do not cancel.	2
Polyphosphate	A chain or ring of phosphate units joined by phosphoanhydride bonds, such as pyrophosphate or trimetaphosphate.	4
Positron	The antiparticle of the electron, with the same mass and opposite (positive) charge.	1
Primer	A short strand bound to a template and extended at its $3'$ end during copying.	8

Term	Definition	First used
Principal quantum number (n)	The number (1, 2, 3, …) that labels an electron shell; higher n means electrons that are on average farther from the nucleus.	2
Protocell	A compartment, such as a vesicle, that contains a replicating genetic system; a model of an early cell.	Epilogue
Proton	A subatomic particle with one positive elementary charge, found in the nucleus.	1
Proton-proton chain	The sequence of fusion reactions that converts four protons into one helium-4 nucleus, two positrons and two neutrinos; the main energy source of the Sun.	1
Protoplanetary disc	A flattened, rotating disc of gas and dust around a young star, from which planets form.	1
Pseudoknot	A structure containing base pairs that cross rather than nest: pairs (i, j) and (k, l) with $i < k < j < l$.	7
Purine	A nucleobase with a six-membered ring fused to a five-membered ring, nine ring atoms of which four are nitrogen; adenine and guanine.	4
Pyranose	The six-membered ring form of a sugar (five carbon atoms and one oxygen atom).	4
Pyrimidine	A nucleobase with a single six-membered ring containing two nitrogen atoms; cytosine and uracil (thymine in DNA).	4
Quantum mechanics	The physical theory that describes the behaviour of matter and energy on the scale of atoms and subatomic particles.	2
Quasispecies	The population formed by a master sequence and the mutants generated by its error-prone replication.	8
Racemate	A mixture containing equal amounts of two enantiomers.	4
Radiogenic	Produced by radioactive decay; used of daughter atoms formed within a sample.	3
Radiometric age	The age of a sample calculated from the measured ratio of radiogenic daughter to parent atoms and the decay constant of the parent.	3
Rate constant (k)	The proportionality factor between the rate of a reaction and the concentrations of its reactants; it depends on temperature but not on concentration.	2
Rate of reaction	The rate of change of the concentration of a reactant or product with time, divided by its coefficient in the reaction equation.	2

Term	Definition	First used
Recombination	The epoch, about 380,000 years after the Big Bang, when electrons first bound permanently to nuclei to form neutral atoms.	1
Redfield ratio	The average atom ratio C:N:P = 106:16:1 used as a reference for the composition of marine organisms.	1
Redox couple	A reducing and an oxidising substance that can react with each other by electron transfer, such as H_2 and CO_2.	3
Redox reaction	A reaction in which one species is oxidised and another reduced.	3
Reducing atmosphere	An atmosphere in which carbon and nitrogen are present mainly in low oxidation states, as CH_4 and NH_3, together with H_2.	3
Reduction	The gain of electrons by an atom, ion or molecule; a decrease in oxidation number.	3
Relative atomic mass (A_r)	The average mass of the atoms of an element in a given sample, expressed in atomic mass units; the standard atomic weight is its recommended value for normal materials.	1
Relative permittivity (ε_r)	The factor by which a medium reduces the electrostatic force between charges compared with a vacuum; 78.4 for water at 25 °C.	2
Replication	The production of a copy of a sequence; for nucleic acids, two rounds of template-directed synthesis separated by strand separation.	8
Resonance (nuclear)	A strong enhancement of a nuclear reaction rate when the energy of the colliding nuclei matches the energy of an excited state of the product nucleus.	1
Ribose	The aldopentose $C_5H_{10}O_5$ of RNA; RNA contains only D-ribose, in the furanose form.	4
Ribosome	The complex of RNA and protein on which cells make proteins; its peptide-bond-forming site consists of RNA.	Epilogue
Ribozyme	An RNA molecule that acts as a catalyst.	7
Ribozyme polymerase	A ribozyme that extends an RNA primer on an RNA template; so far obtained only by laboratory evolution.	8
RNA world hypothesis	The hypothesis that, at an early stage of life, RNA served both as genetic material and as the main catalyst.	8

Term	Definition	First used
Schreibersite	The meteoritic mineral $(Fe,Ni)_3P$, in which phosphorus is bonded to metal atoms; it releases reactive phosphorus compounds when it corrodes in water.	3
Second law of thermodynamics	In any spontaneous process, the total entropy of a system and its surroundings increases.	2
Secondary structure	The set of base pairs formed by a nucleic acid molecule.	7
Selective superiority (σ)	The ratio of the replication rate of the master sequence to that of its mutants.	8
Selectivity	The extent to which a reaction or process forms one product, or acts on one compound, in preference to others.	4
SELEX	Systematic evolution of ligands by exponential enrichment: the in vitro selection procedure of Tuerk and Gold (1990).	7
Sequence space	The set of all sequences of a given length; for RNA of length n it contains 4^n sequences.	7
Shell (electron)	The set of orbitals in an atom with the same principal quantum number n; it holds at most $2n^2$ electrons.	2
Short-lived radionuclide	A radioactive isotope with a half-life of less than 200 million years that was present in the early solar system and is now extinct.	1
Sigma (σ) bond	A covalent bond formed by head-on overlap of orbitals, with electron density concentrated along the line joining the nuclei.	2
Snow line	The distance from a young star beyond which ice of a given substance can form; for water the corresponding temperature is about 150 K.	1
Spark discharge experiment	A laboratory simulation in which an electric discharge acts on a gas mixture in contact with water, as in the experiment of Miller (1953).	4
Spin	An intrinsic property of the electron that can take one of two values, conventionally called up and down.	2
Splicing	The removal of an intervening sequence from an RNA precursor and the joining of the flanking pieces.	7
Stack	In the nearest-neighbour model, a pair of adjacent base pairs in a helix.	7

Term	Definition	First used
Stalling	The slowing of primer extension after a mismatched nucleotide has been incorporated.	8
Standard enthalpy of formation ($\Delta_f H^\circ$)	The enthalpy change when one mole of a compound is formed from its elements in their most stable forms, all in their standard states.	2
Star	A self-gravitating ball of gas in which nuclear fusion in the core supplies the energy radiated from the surface.	1
Stereocentre	An atom, usually carbon bonded to four different groups, at which exchanging two groups gives a stereoisomer.	4
Strand-separation problem	The difficulty that the product of template copying is bound to its template more strongly than primers and monomers, so that releasing it requires conditions that damage RNA.	8
Strecker synthesis	Formation of an amino acid from an aldehyde, ammonia and hydrogen cyanide, through an imine and an aminonitrile that is then hydrolysed.	4
Strong nuclear force	The fundamental force that binds protons and neutrons together in a nucleus; it is much stronger than the electrical repulsion between protons but acts only over distances of about 10^{-15} m.	1
Structural isomers	Isomers that differ in the order in which their atoms are bonded.	4
Subshell	A group of orbitals within a shell with the same shape, labelled s, p, d or f and containing 1, 3, 5 or 7 orbitals.	2
Supernova	The explosive death of a star, which ejects newly synthesised elements into interstellar space.	1
Supernova remnant	The expanding cloud of gas ejected by a supernova explosion, together with the interstellar gas it sweeps up.	1
Template	An existing nucleotide chain whose sequence directs the order in which monomers are joined.	5
Tertiary structure	The three-dimensional arrangement of the atoms of a folded molecule.	7
Tetrahedral angle	The angle between lines from the centre of a regular tetrahedron to two of its corners, $\arccos(-1/3)$, approximately 109.47°.	2
Transition state	The arrangement of atoms at the point of highest energy along the path from reactants to products.	2

Term	Definition	First used
Triple bond	A covalent bond in which two atoms share three pairs of electrons.	2
Triple-alpha process	The fusion of three helium-4 nuclei into carbon-12 inside stars, proceeding via unstable beryllium-8.	1
Ultraviolet light	Electromagnetic radiation with wavelengths shorter than those of visible light, below about 400 nm.	3
Uracil	The pyrimidine $C_4H_4N_2O_2$; it pairs with adenine in RNA.	4
Valence	The number of covalent bonds an atom usually forms: 1 for H, 4 for C, 3 for N and 2 for O.	2
Valence electrons	The electrons in the outermost shell of an atom, which take part in chemical bonding.	2
Van der Waals forces	The collective name for dipole-dipole, dipole-induced dipole and dispersion interactions between molecules.	2
Vesicle	A closed membrane, made of a double layer of amphiphilic molecules, that encloses a volume of water.	Epilogue
VSEPR	Valence-shell electron-pair repulsion: a model in which the electron pairs around an atom arrange themselves as far apart as possible, determining molecular shape.	2
Watson-Crick base pair	The base pairs A–U (A–T in DNA), with two hydrogen bonds, and G–C, with three.	7
Wet-dry cycle	Repeated drying and rewetting of a small body of water, which concentrates dissolved molecules and promotes the joining of building blocks into chains.	3
Zircon	The mineral zirconium silicate, $ZrSiO_4$; it contains trace uranium, resists weathering, and is the main source of information about the Hadean Earth.	3

B. Units, Constants and Orders of Magnitude

This appendix collects the units, physical constants and conversion factors used in the book, and lists characteristic magnitudes that appear in the chapters.

B.1. SI units and prefixes

The book uses the International System of Units (SI). Table B.1 lists the base and derived units that occur most often, and Table B.2 the prefixes used to form multiples.

Table B.1.: SI units used in this book.

Quantity	Unit	Symbol	Expressed in other SI units
Length	metre	m	base unit
Mass	kilogram	kg	base unit
Time	second	s	base unit
Temperature	kelvin	K	base unit
Amount of substance	mole	mol	base unit
Energy	joule	J	$\mathrm{kg\ m^2\ s^{-2}}$
Pressure	pascal	Pa	$\mathrm{J\ m^{-3}} = \mathrm{kg\ m^{-1}\ s^{-2}}$
Electric charge	coulomb	C	A s
Concentration (amount)	mole per litre	$\mathrm{mol\ L^{-1}}$ (also written M)	$10^3\ \mathrm{mol\ m^{-3}}$

Table B.2.: SI prefixes.

Prefix	Symbol	Factor	Prefix	Symbol	Factor
kilo	k	10^3	milli	m	10^{-3}
mega	M	10^6	micro	μ	10^{-6}
giga	G	10^9	nano	n	10^{-9}

Prefix	Symbol	Factor	Prefix	Symbol	Factor
tera	T	10^{12}	pico	p	10^{-12}
peta	P	10^{15}	femto	f	10^{-15}

Several units outside the SI are used in the book because they are standard in their fields:

- **Degree Celsius** (°C). A temperature t in °C corresponds to the absolute temperature T = t + 273.15 K. A temperature difference of 1 °C equals a difference of 1 K.
- **Litre** (L), equal to 10^{-3} m^3.
- **Bar**, equal to 10^5 Pa; the **standard atmosphere** (atm) is 101 325 Pa exactly (National Institute of Standards and Technology, n.d.a).
- **Electronvolt** (eV), the energy gained by an elementary charge across a potential difference of 1 V, equal to 1.602 176 634 × 10^{-19} J exactly (National Institute of Standards and Technology, n.d.a).
- **Annum** (a), the year used in geology and astronomy, with the multiples ka, Ma and Ga (Section 3.1).
- **Kilocalorie** (kcal). The nearest-neighbour parameters of Chapter 7 are published in kcal mol^{-1}; the conversion used in this book is 1 kcal = 4.184 kJ (the thermochemical calorie).
- **Unified atomic mass unit** (u), also called the **dalton** (Da), equal to one twelfth of the mass of an atom of carbon-12; its value is the atomic mass constant in Table B.3.

B.2. Physical constants

Table B.3 lists the constants used in the book, with their values from the 2022 CODATA adjustment (National Institute of Standards and Technology, n.d.a). Values marked exact follow from the definition of the SI; for the Boltzmann constant in eV K^{-1}, the molar gas constant and the Faraday constant, which are products or quotients of exact constants, the digits shown are truncated. The last five values are measured, and the digits in parentheses give the standard uncertainty in the last digits.

Table B.3.: Physical constants, CODATA 2022 (National Institute of Standards and Technology, n.d.a).

Constant	Symbol	Value
Speed of light in vacuum	c	299 792 458 m s^{-1} (exact)
Planck constant	h	6.626 070 15 × 10^{-34} J Hz^{-1} (exact)

Constant	Symbol	Value
Elementary charge	e	$1.602\ 176\ 634 \times 10^{-19}$ C (exact)
Boltzmann constant	k_B	$1.380\ 649 \times 10^{-23}$ J K^{-1} (exact) $8.617\ 333\ 262 \times 10^{-5}$ eV K^{-1} (exact)
Avogadro constant	N_A	$6.022\ 140\ 76 \times 10^{23}$ mol^{-1} (exact)
Molar gas constant	$R = N_A k_B$	8.314 462 618 J mol^{-1} K^{-1} (exact)
Faraday constant	$F = N_A e$	96 485.332 12 C mol^{-1} (exact)
Atomic mass constant	m_u	$1.660\ 539\ 068\ 92(52) \times 10^{-27}$ kg
Proton mass	m_p	$1.672\ 621\ 925\ 95(52) \times 10^{-27}$ kg
Neutron mass	m_n	$1.674\ 927\ 500\ 56(85) \times 10^{-27}$ kg
Electron mass	m_e	$9.109\ 383\ 7139(28) \times 10^{-31}$ kg
Bohr radius	a_0	$5.291\ 772\ 105\ 44(82) \times 10^{-11}$ m

B.3. Useful conversions

Chemical energies are given in the book per mole, physical energies per particle. The two are related by the Avogadro constant:

$$E_{\text{molar}} = N_A\, E_{\text{particle}} \tag{B.1}$$

From the values of Table B.3 follow the conversions and reference values of Table B.4.

Table B.4.: Conversions and reference values derived from Table B.3.

Quantity	Value
1 eV per particle	96.485 kJ mol^{-1}
1 kcal mol^{-1}	4.184 kJ mol^{-1}
$k_B T$ at 25 °C (298.15 K)	0.025 69 eV
RT at 25 °C (298.15 K)	2.479 kJ mol^{-1}
RT at 37 °C (310.15 K)	2.579 kJ mol^{-1}

Quantity	Value
RT ln 10 at 25 °C (change in ΔG for a tenfold change in an equilibrium constant)	5.708 kJ mol^{-1}
0 °C	273.15 K

The quantity RT sets the scale of thermal energy per mole. By Equation 7.2, a free-energy difference of a few RT changes the population of a state by an order of magnitude or more: a difference of RT ln 10 changes it tenfold.

B.4. Orders of magnitude in this book

Table B.5 collects characteristic quantities from the chapters, with the section in which each is discussed.

Table B.5.: Characteristic magnitudes discussed in this book.

Quantity	Value	Section
Time		
Age of the Universe	about 13.8 Ga	Section 1.2
Oldest solids of the solar system (CAIs)	4.567 to 4.568 Ga	Section 1.7
Oldest known zircon from the Jack Hills	4404 ± 8 Ma	Section 3.3
Half-life of ribose at 100 °C, pH 7	73 min	Section 4.4
Doubling time of cross-replicating RNA enzymes	about 1 h	Section 8.6
Non-enzymatic copying, per added nucleotide	30 min to about 1 day	Section 8.7
Energy		
Thermal energy k_BT at 25 °C	0.026 eV per particle; RT = 2.5 kJ mol^{-1}	Section 2.1
Hydrogen bond (IUPAC 1994)	usually less than 20 to 25 kJ mol^{-1}	Section 2.5
Watson-Crick stack, 37 °C	−3.8 to −14.2 kJ mol^{-1}	Section 7.2
O–H bond enthalpy in water	463.5 kJ mol^{-1}	Section 2.7
Concentration and number		
Water in liquid water at 25 °C	about 55 mol L^{-1}	Section 4.2

Quantity	Value	Section
Sequences of 40 nucleotides	$4^{40} \approx 1.2 \times 10^{24}$	Section 7.6
Random RNA molecules binding a given small molecule	roughly 1 in 10^{10}	Section 7.6

C. How to Read a Scientific Paper

Every key experiment in this book is taken from a published research paper, and every claim is linked to its source in the References. This appendix describes how such papers are organised and how to read them critically.

C.1. Kinds of publication

- A **research article** reports new results: an experiment, an observation, a calculation or a model. It is the primary source for a claim.
- A **review** summarises and evaluates the research articles on a topic. Reviews are the best entry point into an unfamiliar field, but a number quoted in a review should be checked in the research article it cites.
- A **preprint** is a manuscript made public before, or without, formal peer review, on a server such as arXiv, bioRxiv or Zenodo.
- **Peer review** is the evaluation of a manuscript by independent experts before publication in a journal. It reduces errors but does not guarantee that a result is correct; the test of a result is whether other laboratories can reproduce it.

Each publication is identified by a **digital object identifier** (DOI), a persistent string such as 10.1126/science.1060786. Prefixing it with https://doi.org/ gives a link that resolves to the publication. The DOIs of the works cited in this book are listed in the References.

C.2. The structure of a research article

Most research articles in chemistry and biology follow the same order:

1. **Title and abstract.** The abstract states the question, the approach, the main result and its significance, usually in a single paragraph. It is the part of a paper most often quoted, and the key experiment boxes of this book quote abstracts directly wherever possible.
2. **Introduction.** The background, the open problem and the aim of the work.

3. **Methods** (or Materials and Methods, sometimes placed at the end or in supplementary material). The exact conditions: concentrations, temperatures, pH, times, instruments and analysis. Whether a result is relevant to the early Earth often depends on these details.
4. **Results.** The observations, usually organised around figures and tables.
5. **Discussion.** The interpretation of the results, their limitations and their relation to earlier work.
6. **References** and **supplementary information**, which often contain the full data.

C.3. Reading critically

A paper is not read once from beginning to end. A practical order is: title and abstract, then the figures and their legends, then the results and methods for the parts that matter, and finally the discussion. The following questions help to evaluate a claim.

- **What exactly was measured?** Distinguish the observation (for example, the products detected after a reaction) from the interpretation (for example, the mechanism proposed for their formation).
- **Under what conditions?** Concentrations, temperature, pH, purity of reagents and duration determine whether a result could apply to a natural environment. Many experiments in Chapters 4 to 8 use concentrations or purified starting materials that are chosen to make a single step work.
- **How large is the effect, and how certain?** Look for the numbers, their units and their uncertainties. A rate enhancement of 36 000 and one of 7 million (Section 7.6, Section 7.5) are both catalysis, but they differ by more than two orders of magnitude.
- **What controls were run?** A control is an experiment identical to the main one except for the factor under test, such as a reaction without the template or without the catalyst. Without controls, an effect cannot be attributed to its proposed cause.
- **Has the result been reproduced?** Results confirmed by independent groups are more secure than single reports.
- **Do the conclusions follow from the data?** A statement in the discussion may go beyond the results; the abstract may state the most favourable interpretation.

C.4. Established, hypothesis, open question

The book distinguishes three kinds of statement, as described in the Preface. When reading a paper, the same distinction is useful:

- A result is **established** when it rests on reproducible measurements under stated conditions.

- An explanation of how the result fits into the origin of life, such as a proposed prebiotic setting, is usually a **hypothesis**.
- What the paper does not address, or names as a limitation, often marks an **open question**.

For example, the experiment of Powner, Gerland, and Sutherland (2009) established a chemical route to activated pyrimidine nucleotides under stated laboratory conditions; whether the starting materials were available together on the early Earth is a hypothesis that the paper discusses but cannot settle by itself.

C.5. Numbers and models

Some results in this book are calculations rather than experiments: nearest-neighbour energies, melting curves, chain-length distributions and error thresholds. A calculation is only as good as its inputs. For each one, check which parameters were used, where they came from, under what conditions they were measured, and which effects the model omits. Calculations made for this book are marked as the author's calculations, and their parameters and sources are given where they appear.